

Meta-Layer Desirable Properties — On-Chain Inscriptions

Consolidated ML-Draft text for all 22 Desirable Properties, fetched from Bitcoin Ordinal inscriptions.

Source mapping: `challenge-site/src/data/dp-inscriptions.json` (updated 2026-06-18).

DP1 - Federated Auth & Accountability

The Conditions for Trust

Purpose of This Draft

This ML-Draft articulates **Desirable Property 1 (DP1)** as a foundational condition for trust in the Meta-Layer. It expands DP1 beyond federated authentication to encompass accountability, adaptive intelligence integration, and foresight-driven governance.

DP1 responds to multiple, overlapping needs:

- The need for decentralized, federated identity without single points of control
- The need for durable accountability without mandatory real-world identity
- The need to govern both human and AI agents coherently
- The need to anticipate predictable abuse and governance failure modes

This draft is intended to guide implementation, governance design, and future ML-RFC development.

1. Problem Statement: Why Identity Alone Is Not Enough

For most of the Web's history, trust has been treated as a byproduct of identity. If a participant could be authenticated, logged in, or verified, trust was assumed to follow. This assumption no longer holds.

At contemporary scale, identity has become cheap to generate, easy to discard, and increasingly decoupled from responsibility. As a result, systems optimized around login and verification routinely fail to protect participants, communities, and institutions from predictable harm.

DP1 begins from a different premise: **trust is not something identity produces on its own**. Trust emerges only when identity is paired with accountability, memory, and governance that operate coherently at the point of interaction.

1.1 The Limits of Login-Centric Trust

Login-centric trust models focus on answering a narrow question: who is allowed to enter a system. They do not meaningfully address what happens after entry.

Across platforms and applications, this has produced a recurring pattern:

- Verification is treated as a proxy for good faith
- Identity checks are decoupled from behavior over time
- Enforcement resets when identities are abandoned or recreated
- Predators, scammers, fake accounts, and autonomous agents are able to exploit accumulated trust within an ecosystem before detection or response

Even strong authentication does not prevent abuse when actions are not durably bound to accountable actors. A verified account can still mislead, manipulate, impersonate, or cause harm if there is no persistent relationship between identity and responsibility.

1.2 Structural Failure Modes in Today's Web

Several structural conditions compound these weaknesses:

- **Context collapse:** Behavior in one space rarely follows a participant into another
- **No shared memory:** Harm accumulates, but accountability does not
- **Reactive moderation:** Intervention occurs after damage is done
- **Synthetic scale:** Bots, sockpuppets, and automated agents operate faster than human oversight

These are not edge cases. They are systemic properties of the current web. As documented in Meta-Layer research, even the largest platforms remove billions of fake or abusive accounts annually, without meaningfully reducing the underlying incentives or recurrence of abuse.

1.3 DP1 as a Shift in Framing

DP1 reframes the problem of trust along three axes:

- From **identity** to **accountability**
- From **platforms** to **zones**
- From **enforcement** to **conditions**

Rather than asking only who someone is, DP1 asks under what conditions participation is allowed, how actions are attributed, and how trust evolves over time.

2. Threat and Risk Context (Non-Exhaustive)

DP1 is not defined in opposition to any single class of actor. Instead, it responds to recurring failure modes that reliably emerge in large-scale, low-friction digital systems.

2.1 Scammers

Scammers exploit environments where identity is inexpensive and disposable. Common characteristics include:

- Rapid account creation and abandonment
- Cross-context exploitation of trust signals

- Asymmetric incentives favoring deception

DP1 does not attempt to eliminate scams entirely. Instead, it raises their cost by binding actions to accountable agents, preserving memory across contexts, and enabling communities to escalate trust requirements where appropriate.

2.2 Serial Predators and Repeat Abusers

Serial abuse often persists not because it is invisible, but because it is fragmented. When identities reset after enforcement, harm becomes distributed across communities without a durable record.

DP1 addresses this pattern by supporting persistent pseudonymous identity, zone-scoped accountability, and governance mechanisms that allow communities to respond to patterns of harm without resorting to exposure, vigilantism, or centralized surveillance.

2.3 Impersonators (Human and AI)

Advances in generative systems have dramatically lowered the cost of impersonation. Voice, image, and text synthesis now allow both humans and AI systems to convincingly misrepresent identity and intent.

DP1 counters impersonation by binding content and actions to verifiable agents, clearly differentiating between human and AI actors, and surfacing provenance signals directly at the interface layer.

2.4 Other Drivers (Equally Important)

In addition to explicit abuse, DP1 responds to broader systemic pressures that erode trust even in the absence of malicious intent:

- **Scalable agents operating without visible constraints.** As automation and AI agents scale, they can overwhelm human participation, distort consensus, and accumulate influence faster than human governance processes can respond. Without clear accountability, visibility, and rate limits, both human-operated and autonomous agents can unintentionally or deliberately reshape an ecosystem's trust dynamics. An agent refers to any accountable actor operating in the meta-layer, whether human or AI.
- **Governance capture and conflicts of interest.** Trust systems are vulnerable when those who set or enforce rules have misaligned incentives. Concentrated power, opaque decision-making, or financial dependencies can lead to selective enforcement, uneven accountability, or loss of legitimacy. Over time, this erodes community confidence even if formal rules remain unchanged.
- **Economic models that reward engagement regardless of harm.** Many digital systems optimize for growth, virality, or attention without regard for downstream effects. When visibility, rewards, or influence are tied solely to engagement metrics, deceptive, polarizing, or manipulative behavior is systematically advantaged over constructive participation.

These pressures are structural rather than incidental. They interact and compound one another, producing environments where abuse, manipulation, and trust erosion become predictable outcomes. DP1 is designed to address their combined effects by reshaping incentives, accountability, and governance conditions, rather than

treating each pressure in isolation. Taken together, these pressures make clear that trust cannot be repaired solely through backend policy or platform moderation, but must be enacted visibly and continuously at the interface level, where participation, amplification, and accountability actually occur.

3. Core Principles and Scope

Identity is not merely descriptive at the interface layer; it is the basis for enforceability, continuity, and accountable participation across all Meta-Layer interactions.

Identity is the enforcement boundary of the Meta-Layer. If identity cannot persist across context, delegation, and scale, trust collapses into simulation.

3. Core Principle of DP1

DP1 establishes identity as an enforceable, continuous, and context-bound substrate for all higher-order trust, governance, and interaction within the meta-layer.

Trust in the Meta-Layer emerges when identity, accountability, learning, and foresight are bound together at the interface level.

This principle has several direct implications, each of which is essential to sustaining trust at scale:

- **Identity is plural and contextual, not singular or global.** Participants may operate under different identities in different zones, allowing communities to set appropriate norms without forcing a single global identity model. This preserves inclusion, safety, and local governance autonomy.
- **Accountability attaches to actions, not just names.** Trust depends on the ability to evaluate behavior over time. Binding actions to accountable agents ensures that responsibility persists even when identities are pseudonymous or federated.
- **Memory is preserved without requiring mass surveillance.** Durable, attributable records allow communities to learn from past behavior and prevent repeat abuse, while avoiding continuous monitoring or centralized data collection.
- **Governance adapts over time, but remains human-ratified.** Trust systems must evolve in response to new threats and conditions, yet retain human oversight so that changes remain legitimate, explainable, and aligned with community values.

DP1 does not promise perfect safety or universal trust. Instead, it defines the minimum conditions under which trust can form, persist, and be repaired in complex, multi-actor environments.

4. Federated Strong Authentication (Entry Condition)

Federated strong authentication establishes the baseline condition for participation in the Meta-Layer. Its purpose is not to define trust, but to ensure that entry into shared spaces is not trivially exploitable or monopolized by a single identity authority.

DP1 treats authentication as an **entry condition**, not as a guarantee of trustworthiness or good behavior. Strong authentication reduces frictionless abuse, but only when paired with downstream accountability, memory, and governance does it meaningfully contribute to trust.

4.1 Federation as a Baseline Requirement

The Meta-Layer supports federation across multiple identity and authentication systems, including traditional SSO providers, wallets, and emerging credential frameworks. This plural approach ensures:

- No single provider controls access to the Meta-Layer
- Participants can authenticate using systems appropriate to their context
- Communities can adopt stronger or lighter requirements without fragmenting the ecosystem

Federation is essential to resilience. Centralized identity systems concentrate power and risk, while federated systems distribute trust and reduce systemic failure modes.

4.2 User-Held Keys and Credentials

Where possible, participants hold their own keys or retain meaningful control over credentials. In cases where custodial systems are used, consent and revocability remain core requirements.

User-held credentials support:

- Participant agency and exit
- Reduced platform lock-in
- Durable accountability across contexts

4.3 Authentication Is Not Authorization

Authentication answers the question of *who may enter*. It does not determine *what that participant may do*, *what they may access*, or *how much trust they are afforded*.

All authorization, trust thresholds, and participation rules are defined at the zone level. This separation prevents overloading identity systems with governance logic and keeps trust decisions contextual, transparent, and adaptable.

5. Sociotechnical Zones as Trust Contexts

Sociotechnical zones are the primary mechanism by which trust conditions are enacted at the interface level. Zones translate abstract governance principles into concrete participation rules that operate where interaction, amplification, and accountability actually occur.

Rather than relying on backend policy enforcement or platform-level moderation alone, zones make trust visible, enforceable, and configurable within the lived experience of participants.

5.1 Definition of Sociotechnical Zones

Zones combine technical requirements and social norms to define the conditions under which participation is permitted.

Each zone specifies:

- Acceptable risk levels
- Required proofs or credentials
- Accountability expectations
- Governance and escalation pathways

Communities and applications choose which zones they operate within, allowing trust conditions to vary without fragmenting the underlying Meta-Layer.

5.2 Zone-Specific Access Paradigms

Zones represent **orthogonal and composable trust constraints**. Real-world environments typically operate under multiple zones simultaneously, reflecting layered social, legal, and safety requirements.

Open and Identity-Light Zones

- Tokenless open zones
- Pseudonymous zones

Credential and Federation-Based Zones

- Credential-gated zones
- Federated authentication zones

Safety and Constraint-Oriented Zones

- Age-gated zones
- Human-only zones
- High-trust or safety-critical zones

By composing zones, communities can precisely calibrate participation conditions without defaulting to global restrictions.

5.3 Compatibility and Exclusion by Design

Zones enforce explicit compatibility requirements. Participation is limited to actors who can meet the defined conditions, making boundaries legible rather than implicit.

This design:

- Prevents silent exclusion or shadow banning
- Makes trust thresholds understandable and contestable

- Allows communities to defend against abuse without universal surveillance

5.4 Proof of Humanity as a Zone-Scoped Basis for Participation

Proof of humanity refers to mechanisms that allow a participant to demonstrate that they are a **unique human actor**, without necessarily revealing their real-world identity.

Within DP1, proof of humanity is treated as a **foundational system capability** that must be available across the Meta-Layer, even though its enforcement is zone-scoped and community-defined. This capability is critical at the interface level, where rewards, visibility, and reputation are allocated and where synthetic scale can otherwise distort outcomes.

Some communities may choose to make proof of humanity the basis for participation itself. Others apply it selectively to specific functions such as rewards, governance, rate-limited actions, reputation amplification, or access to safety-critical spaces.

Key principles include:

- Proof of humanity is available system-wide, but enforced at the zone level
- Communities decide when and how proof of humanity is required, including making it a prerequisite for participation, rewards, governance, or amplification
- Proof of unique humanity is commonly required to unlock rewards, boost virality, or affect transferable reputation, where synthetic scale would otherwise overwhelm human participation
- Multiple proof mechanisms may coexist, be combined, or be phased out over time
- Proof of humanity does not imply real-name identity or permanent disclosure
- Requirements may be stricter in high-trust, safety-critical, or resource-allocation contexts

By treating proof of humanity as an enduring and adaptable capability rather than a fixed mechanism, DP1 enables long-term defense against synthetic scale and impersonation while preserving pluralism, pseudonymity, and local governance autonomy.

5.5 Identity System Layer: Continuity, Integrity, and Adversarial Resilience

Beyond authentication and zone-scoped participation, DP1 requires a coherent identity system layer that persists across environments, interactions, and time. This layer is the enforcement boundary of the Meta-Layer. If identity cannot maintain continuity under scale, delegation, and interoperability, trust collapses into simulation.

The identity system layer ensures that actions, reputation, and accountability remain meaningfully bound to agents even as they move across zones, tools, and contexts.

5.5.1 Identity Continuity Across Systems

Identity must persist across platforms, zones, and applications without fragmenting into unrelated entities.

Continuity requires:

- Stable identifiers or linkable identity references
- Preservation of accountability history across environments
- Explicit signaling when identity properties degrade or reset

A failure mode is **identity fragmentation**, where the same participant appears as unrelated actors across systems, breaking incentives, governance, and trust.

5.5.2 Identity Integrity and Anti-Replay Guarantees

Identity-bound actions must not be duplicable across systems without attribution.

This requires:

- Binding actions to identity with verifiable provenance
- Preventing replay of actions or credentials across contexts
- Ensuring that contributions cannot be re-used to extract value multiple times

A failure mode is **identity replay**, where actions or credentials are reused across systems to gain unearned trust, rewards, or access.

5.5.3 Identity Non-Transferability and Delegation Boundaries

Identity must not be freely transferable in ways that detach responsibility from the original actor.

Delegation is permitted, but must be:

- Explicit
- Scoped
- Attributable

This ensures that actions taken by agents, tools, or collaborators remain traceable to accountable principals.

A failure mode is **identity laundering**, where responsibility is shifted across actors to evade accountability.

5.5.4 Resistance to Sybil and Coordinated Identity Attacks

Identity systems must make large-scale duplication, coordination, or synthetic amplification detectable, constrained, or economically costly.

Mechanisms may include:

- Proof-of-humanity where appropriate
- Behavioral analysis and rate limits
- Reputation weighting and trust decay

A failure mode is **sybil saturation**, where large numbers of coordinated identities overwhelm governance, incentives, or visibility systems.

5.5.5 Cross-Zone Identity Semantics and Degradation

Identity does not carry identical meaning across all zones.

Systems must:

- Signal when identity guarantees change across contexts
- Prevent misinterpretation of reputation or credentials
- Allow communities to define how identity signals are interpreted locally

A failure mode is **semantic drift**, where identity signals are incorrectly assumed to carry the same meaning across different contexts.

5.5.6 Identity Memory and Lineage

Identity must retain a reconstructable history of actions, credentials, and governance interactions over time.

Lineage enables:

- Attribution of contributions
- Detection of behavioral patterns
- Accountability across time and context

Breaks in lineage must be treated as risk signals rather than neutral events.

A failure mode is **lineage loss**, where identity history cannot be reconstructed, enabling impersonation or evasion.

This identity system layer does not require centralization or global identity unification. It requires coherence. Identity must remain usable, accountable, and interpretable across the Meta-Layer without collapsing into surveillance or fragmentation.

6. Accountability as a First-Class Property

Accountability is the core mechanism through which trust becomes durable in the Meta-Layer. While authentication governs entry, accountability governs behavior over time. Without it, trust signals decay, abuse repeats, and governance loses legitimacy.

DP1 treats accountability as a first-class property that operates continuously at the interface level, binding actors to their actions in ways that are visible, attributable, and contestable, without requiring real-world identity disclosure.

6.1 Action-Bound Accountability

In the Meta-Layer, accountability attaches to actions, not merely to identities. Every meaningful action, such as posting content, issuing judgments, triggering automation, or influencing visibility, is bound to an accountable agent identifier.

This ensures that:

- Actions have clear provenance
- Responsibility persists across time and context
- Trust assessments can be based on behavior, not credentials alone

Action-bound accountability allows communities to reason about patterns of conduct without collapsing participation into real-name systems or centralized surveillance.

6.2 Pseudonymity with Responsibility

DP1 explicitly supports pseudonymous participation, recognizing its importance for safety, expression, and inclusion. Pseudonymity, however, does not imply anonymity from accountability.

Persistent pseudonymous identities allow participants to:

- Build reputation over time
- Be held responsible for repeated behavior
- Participate across zones without exposing real-world identity

Communities may permit multiple personas per participant, subject to local rules, provided that accountability requirements are met. This balances flexibility with responsibility, enabling participation without enabling evasion.

6.3 Sealed Memory and Editability Windows

To balance forgiveness, accuracy, and integrity, DP1 supports time-bound editability followed by sealing.

Participants may edit or retract contributions within community-defined windows. After this period, contributions become sealed: immutable, attributable, and part of the shared civic memory.

Sealed memory:

- Prevents retroactive manipulation
- Preserves historical context
- Enables learning from past behavior

Communities may determine whether edit histories are retained, visible, or restricted, but the existence of durable memory is essential for trust to accumulate.

6.4 Trust Lifecycle, Revocation, and Recovery

Trust in the Meta-Layer is not binary. It evolves.

Zones define explicit conditions for:

- **Escalation and intervention:** graduated responses to harmful or destabilizing behavior
- **Temporary suspension or restriction**
- **Revocation of access or privileges**
- **Reinstatement or recovery**

Revocation is zone-scoped by default, avoiding unnecessary global punishment. Memory persists across decisions.

7. Contestability, Appeals, and Due Process

For accountability systems to be trusted, they must themselves be accountable. DP1 therefore treats contestability and due process as essential trust infrastructure, not optional governance overhead.

Participants must be able to understand, challenge, and appeal decisions that materially affect their participation, visibility, reputation, or access.

Key principles include:

- **Transparency:** Decisions affecting trust or access must be explainable and grounded in visible rules or signals.
- **Contestability:** Participants must have mechanisms to dispute actions taken against them.
- **Human Oversight:** Escalation thresholds require human review, particularly where consequences are significant.
- **Explainable Automation:** AI-assisted flagging or enforcement must be intelligible to affected parties.

Appeals processes reinforce legitimacy. They help communities detect governance failure, correct errors, and adapt rules over time.

By embedding contestability directly into trust systems, DP1 ensures that accountability strengthens trust rather than undermining it.

- Accountability systems must be challengeable and auditable
- Participants must be able to contest decisions affecting access, reputation, or visibility
- Escalation thresholds require human review
- AI-assisted flagging and enforcement must be explainable
- Appeals are treated as trust infrastructure, not optional overhead

8. Human and AI Agents Under DP1

DP1 treats both human and artificial agents as first-class participants in the Meta-Layer, while recognizing that they differ fundamentally in capacity, scale, intent, and risk profile. Trust cannot be sustained if these differences are ignored, obscured, or flattened.

The goal of DP1 is not to exclude AI agents categorically, but to ensure that their participation is **legible, bounded, and accountable** in ways that preserve human agency and community governance.

8.1 Agent Classification and Visibility

An *agent* refers to any actor capable of taking actions that affect shared environments, visibility, reputation, or outcomes within the Meta-Layer.

DP1 requires clear classification between:

- Human agents
- AI or automated agents
- Hybrid or assisted agents, where human intent is mediated by automation

This classification must be **visible at the interface level**, allowing participants to understand whether they are interacting with a human, an AI system, or a combination of both. Hidden or ambiguous agent identity erodes trust and enables manipulation.

8.2 Symmetric Accountability, Asymmetric Constraints

DP1 applies accountability symmetrically: all agents are accountable for their actions. However, constraints are applied asymmetrically, reflecting differences in scale, speed, and potential impact.

For example:

- AI agents may be subject to stricter rate limits, scope restrictions, or amplification caps
- Certain zones may restrict participation to human agents only
- Higher proof thresholds may apply where AI activity could distort consensus, rewards, or governance

This approach avoids both extremes: granting AI agents unchecked parity with humans, or exempting them from accountability altogether.

8.3 Binding AI Outputs to Responsible Entities

AI agents do not operate independently of human or institutional responsibility. DP1 therefore requires that AI outputs be bound to a responsible entity, such as:

- The operator deploying the agent
- The organization maintaining it
- A community governance structure authorizing its use

In high-trust or safety-critical zones, anonymous autonomous agents are not permitted. Responsibility must be traceable, contestable, and enforceable.

By binding AI behavior to accountable entities, DP1 prevents responsibility laundering while enabling beneficial automation under governed conditions.

9. Adaptive Intelligence Integration (RLADP)

Static trust and governance systems degrade over time. Incentives shift, adversaries adapt, and behaviors drift. DP1 therefore anticipates the need for adaptive intelligence to support, but not replace, human and community governance.

9.1 Why Static Governance Fails

At scale, purely static rules and manual moderation encounter predictable limits:

- Human moderators cannot match the speed or volume of adversarial behavior
- Rules ossify and become misaligned with lived practice
- Bad actors learn to game fixed thresholds and heuristics

Without adaptation, governance systems either become overly permissive or increasingly brittle.

9.2 RLADP as Advisory Infrastructure

DP1 envisions adaptive intelligence, including reinforcement learning and approximate dynamic programming (RLADP), as **advisory infrastructure**.

Adaptive systems may:

- Detect emerging patterns of abuse or manipulation
- Surface signals about shifting norms or risk profiles
- Propose adjustments to thresholds, friction, or zone parameters

They may not:

- Unilaterally change rules
- Impose sanctions without human ratification
- Operate as opaque or unchallengeable authorities

9.3 Transparency and Auditability

All adaptive processes must be observable and auditable. Communities must be able to understand:

- What signals are being used
- How recommendations are generated
- What effects adaptations have produced

This visibility is essential to preventing hidden governance drift and maintaining legitimacy.

9.4 Human and Community Ratification

Adaptive intelligence proposes; humans decide.

Material changes to trust conditions, enforcement thresholds, or governance rules require explicit human or community ratification, using processes appropriate to the zone.

By constraining adaptive intelligence within transparent, ratified loops, DP1 enables learning without surrendering agency or accountability.

10. Foresight and Minefield Thinking

DP1 treats foresight not as speculation, but as a core governance discipline. Large-scale sociotechnical systems fail in recognizable ways. When trust systems are designed only for normal operation, they become brittle under stress, capture, or adversarial pressure.

Minefield thinking refers to the practice of deliberately anticipating where incentives, power, and scale are likely to produce failure, and designing safeguards in advance rather than reacting after harm has occurred.

10.1 Governance as Anticipatory Design

Most trust failures are not surprises. They arise from known dynamics such as incentive misalignment, asymmetric power, scale effects, and adversarial learning.

DP1 therefore treats governance as an anticipatory design problem. Communities are encouraged to:

- **Identify foreseeable abuse and failure modes.** Rather than assuming good-faith participation as a default, communities are encouraged to explicitly map how their systems could be exploited, stressed, or captured at scale. This includes considering adversarial behavior, incentive misalignment, power concentration, and unintended consequences of well-meaning rules.
- **Encode preventative friction rather than relying solely on punishment.** Preventative friction includes rate limits, proof thresholds, graduated permissions, and contextual checks that slow or deter harmful behavior before it escalates. This reduces reliance on after-the-fact enforcement, which is often costly, contentious, and insufficient to prevent harm.
- **Periodically reassess assumptions as conditions change.** Trust systems operate in dynamic environments. Communities are encouraged to revisit governance assumptions as participation grows, technologies evolve, or incentives shift, ensuring that rules remain aligned with lived practice rather than ossifying over time.

This approach shifts governance from reactive moderation to continuous risk management.

10.2 Conflict of Interest (COI) Visibility

Trust erodes when participants cannot see whose interests shape rules and enforcement. DP1 requires that material conflicts of interest be surfaced structurally rather than assumed away.

This includes visibility into:

- **Funding sources and economic incentives.** Communities benefit from understanding who funds infrastructure, moderation, or tooling, and how revenue models or token incentives may shape decision-making. Visibility into economic incentives helps participants evaluate whether rules are aligned with collective goals or subtly optimized for growth, extraction, or control.
- **Governance authority and decision rights.** Trust depends on knowing who has the power to set rules, enforce them, and change them over time. Clear articulation of decision rights allows participants to assess legitimacy, understand escalation pathways, and distinguish community governance from operator discretion.
- **Relationships between operators, enforcers, and beneficiaries.** When the same actors design rules, enforce them, and benefit from their outcomes, conflicts of interest can arise even without malicious intent. Making these relationships explicit allows communities to surface bias, challenge capture, and adjust governance structures before trust erodes.

By making incentives legible, communities can better assess legitimacy, detect capture early, and sustain confidence in governance over time.

10.3 Governance Pre-Mortems

DP1 encourages communities to conduct periodic governance pre-mortems: structured exercises that ask how current rules or systems might fail under plausible future conditions.

Pre-mortems may examine:

- **How rules could be gamed at scale.** Communities are encouraged to consider how well-intentioned rules might be exploited when participation grows, automation increases, or incentives shift. This includes identifying loopholes, edge cases, or feedback loops that could advantage bad-faith actors.
- **Where enforcement might become selective or biased.** Pre-mortems surface the risk that enforcement could drift toward favoritism, uneven application, or disproportionate impact on certain groups. Making these risks explicit allows communities to design checks, audits, or appeals in advance.
- **How new technologies or actors could distort participation.** Emerging tools, AI capabilities, or new classes of participants may change how power and influence are exercised. Pre-mortems help communities anticipate these shifts rather than reacting after harm occurs.

The goal is not prediction, but preparedness. Pre-mortems create shared awareness of fragility, normalize course correction, and reduce the social and political cost of adaptation.

10.4 Exit, Fork, and Kill Switches

No governance system should assume its own permanence. DP1 treats exit as a safety feature rather than a failure, recognizing that the ability to leave or disengage is essential to legitimacy.

Communities and participants should have:

- **Clear paths to exit without losing identity or accountability continuity.** Participants should be able to leave a space without being erased or forced to abandon their history, allowing accountability and learning to persist across contexts.
- **The ability to fork governance or norms when consensus breaks down.** When irreconcilable differences emerge, forking allows communities to diverge without coercion, preserving agency while limiting destructive conflict.
- **Emergency mechanisms to pause or disable systems causing systemic harm.** Kill switches or pauses provide a last-resort safeguard against cascading failure, runaway automation, or captured governance.

These safeguards limit the blast radius of governance failure, reduce incentives for capture, and make participation safer by design.

10.5 Cross-Zone Failure Containment

In a multi-zone environment, failures should be contained by default. DP1 assumes that trust loss, enforcement actions, and reputational signals are local unless explicitly propagated.

Communities define:

- **When signals remain zone-scoped.** Localizing consequences prevents minor or context-specific failures from unfairly affecting participation elsewhere.
- **When and how signals may propagate across zones.** Communities may choose to share certain signals across zones where risks overlap, but such propagation should be deliberate, transparent, and governed.
- **What thresholds justify broader impact.** Explicit thresholds help distinguish between isolated incidents and systemic harm, enabling proportional response.

This containment prevents cascading harm while preserving the ability to respond proportionally to serious or systemic abuse.

11. Community Signals Informing DP1

DP1 reflects recurring themes from community submissions, workshops, and discussions across the Meta-Layer initiative.

While individual inputs vary, several consistent signals emerge:

- **A strong preference for pseudonymity with accountability, rather than forced real-name identity.** Communities repeatedly emphasize the need to separate accountability from exposure. Participants want the ability to speak, contribute, and organize without tying activity to real-world identity, while still ensuring that actions carry responsibility over time. This reflects lived experience in environments where real-name policies create safety risks, suppress participation, or concentrate power, without reliably preventing abuse.

- **Frustration with repeat abusers enabled by identity resets and fragmented enforcement.** Many communities report that harmful behavior persists not because it goes unnoticed, but because enforcement lacks continuity. When identities can be cheaply abandoned and re-created, sanctions lose meaning and abuse becomes a cost of doing business. This signal directly informs DP1’s emphasis on persistent, pseudonymous identity and durable memory.
- **Concern about synthetic scale, including bots and AI agents overwhelming human participation.** Participants consistently describe environments where automated or semi-automated agents dominate visibility, rewards, or discourse. Even benign automation can distort outcomes when scale is unchecked. This concern motivates proof-of-humanity capabilities, asymmetric constraints for AI agents, and explicit limits on amplification and rate.
- **Distrust of opaque or centralized moderation and fear of governance capture.** Communities express low confidence in trust systems where rules are enforced invisibly or controlled by unaccountable actors. Perceived capture, selective enforcement, or undisclosed incentives erode legitimacy even when formal policies appear sound. This drives DP1’s requirements for transparency, contestability, and visible governance at the interface level.
- **Desire for preventative and restorative approaches rather than purely punitive systems.** Many submissions emphasize that punishment alone does not build trust. Communities want mechanisms that prevent harm upstream, allow for learning and repair, and support reintegration where appropriate. This signal underlies DP1’s focus on graduated responses, recovery pathways, and governance that evolves through foresight rather than crisis.

These signals reinforce the core framing of DP1: trust must be designed as a set of conditions that balance agency, safety, and legitimacy, rather than imposed through static rules or centralized control. DP1 is therefore best understood not as a single solution, but as a shared response to patterns of failure repeatedly identified by communities operating at scale.

12. Non-Goals and Explicit Boundaries

DP1 deliberately defines the *conditions* for trust rather than attempting to solve all problems associated with identity, abuse, or governance on the internet. Explicitly stating non-goals is essential to prevent scope creep, misinterpretation, and inappropriate application of this property.

DP1 does **not** attempt to:

- **Enforce real-name policies globally.** DP1 explicitly rejects the assumption that real-world identity disclosure is a prerequisite for trust. Mandatory real-name systems often increase risk, suppress participation, and centralize power without reliably preventing abuse.
- **Eliminate all abuse or deception.** No trust system can guarantee perfect safety. DP1 focuses on raising the cost of harm, preserving accountability, and enabling learning and repair, rather than promising total prevention.
- **Centralize identity, moderation, or governance.** DP1 is incompatible with architectures that concentrate authority in a single platform, provider, or enforcement body. Trust is treated as a plural, zone-scoped property rather than a global control mechanism.

- **Replace legal systems or law enforcement.** DP1 operates at the interface and governance layer. It does not supersede legal processes, nor does it attempt to adjudicate crimes or enforce jurisdictional law.

By naming these boundaries explicitly, DP1 remains adaptable across cultures, legal regimes, and communities, while resisting overreach or misuse.

13. Minimum DP1 Alignment (Non-Normative)

Minimum alignment is not a feature checklist. It is the threshold at which an identity system can be considered **enforceable, portable, and resistant to trivial abuse**.

A system that does not meet these conditions may function, but it cannot reliably sustain trust under scale, automation, or adversarial pressure.

At minimum, a system claiming alignment with DP1 must satisfy the following **irreducible conditions**:

13.1 Persistent Identity Continuity

- Actions **MUST** be bound to a persistent agent identifier that survives across sessions and contexts
- Identity resets **MUST** be rate-limited, detectable, or carry loss of accumulated privileges
- Systems **MUST** signal when identity continuity is broken or degraded

Failure mode: identity reset cycles that enable repeated exploitation of incentives and governance.

13.2 Verifiable Action Attribution

- All meaningful actions (content, transactions, moderation, automation) **MUST** be attributable to an agent
- Attribution **MUST** include sufficient provenance to audit origin and context
- Anonymous or unattributed actions **MAY** exist, but **MUST** be constrained from affecting shared resources or governance

Failure mode: untraceable actions that erode accountability and enable manipulation.

13.3 Anti-Replay and Non-Duplication Guarantees

- Identity-bound actions and credentials **MUST NOT** be reusable across systems without explicit lineage
- Systems **MUST** detect or prevent replay of contributions, credentials, or proofs
- Cross-system transfers **MUST** preserve attribution or clearly signal loss of guarantees

Failure mode: replay attacks that extract duplicate rewards, access, or influence.

13.4 Sybil Resistance Under Scale

- Systems **MUST** impose friction or constraints that make large-scale identity duplication costly or ineffective
- Mechanisms **MAY** include proof-of-humanity, rate limits, staking, or reputation weighting
- Systems **MUST** remain functional under coordinated identity attacks

Failure mode: sybil saturation overwhelming incentives, governance, or visibility.

13.5 Zone-Scoped Enforcement and Revocation

- Enforcement actions **MUST** be defined and executed within explicit zones
- Revocation, restriction, and recovery pathways **MUST** be visible and rule-based
- Global or opaque enforcement **MUST NOT** be the default

Failure mode: arbitrary or centralized enforcement that undermines legitimacy.

13.6 Human and AI Agent Differentiation

- Systems **MUST** visibly distinguish between human, AI, and hybrid agents at the interface level
- Automated agents **MUST** be subject to appropriate constraints (rate, scope, amplification)
- AI outputs **MUST** be bound to a responsible entity

Failure mode: indistinguishable agents enabling manipulation, impersonation, and synthetic dominance.

13.7 Identity Lineage and Memory

- Systems **MUST** maintain reconstructable identity lineage for actions, credentials, and governance events
- Breaks in lineage **MUST** be treated as risk signals, not neutral transitions
- Historical records **MUST** be tamper-resistant after defined edit windows

Failure mode: lineage loss enabling impersonation, laundering, or erasure of harmful behavior.

13.8 Contestability and Due Process Baseline

- Participants **MUST** have access to mechanisms for contesting enforcement actions
- Appeals pathways **MUST** exist for decisions affecting access, visibility, or reputation
- High-impact decisions **MUST** include human review or ratification

Failure mode: unchallengeable systems that degrade into opaque or captured governance.

These conditions define the **minimum viable enforcement layer for identity** in the Meta-Layer.

Partial implementations that omit continuity, attribution, anti-replay guarantees, or sybil resistance **SHOULD NOT** be considered aligned with DP1, regardless of authentication strength or interface design.

14. Open Questions and Future Work

DP1 establishes foundational conditions for trust, but it does not resolve all questions required for long-term interoperability, standardization, and global deployment. The following areas are intentionally left open for further research, experimentation, and community deliberation.

- **Standardization of proof-of-humanity mechanisms.** While DP1 requires the availability of proof of unique humanity as a system capability, it does not prescribe specific mechanisms. Multiple decentralized approaches already exist, such as Fractal ID and other proof-of-humanity systems, each with different tradeoffs around privacy, accessibility, cost, and resistance to gaming. Communities may choose the mechanisms that best fit their norms and risk profiles. Over time, however, it is likely that one or a small number of widely trusted proof-of-humanity systems will emerge for use at the Meta-Layer or Overweb level, providing a common baseline that communities are encouraged, but not required, to adopt. Open questions include how such proofs can remain privacy-preserving, globally accessible, and adaptable over time, as well as how multiple proof systems might interoperate, be bridged, or be composed.
- **Cross-zone reputation portability.** DP1 assumes that accountability and trust signals are zone-scoped by default. Further work is needed to determine when and how reputation, sanctions, or trust signals should move across zones without creating unjust spillover effects or de facto global scoring systems.
- **Liability and responsibility models for autonomous agents.** As AI agents become more capable and autonomous, clearer models are needed for assigning responsibility across operators, deployers, tool providers, and communities. DP1 establishes binding to responsible entities, but does not yet resolve how liability should be apportioned in complex, multi-actor systems.
- **Thresholds for escalation across zones.** Communities will need shared patterns for deciding when local failures warrant broader response. This includes defining proportional thresholds, evidentiary standards, and governance processes for cross-zone escalation without undermining local autonomy.
- **Meta-Layer coordination mechanisms.** DP1 implies the need for shared coordination mechanisms capable of maintaining coherent context across human participants, AI agents, content objects, and communities. Such mechanisms would support converging profiles, scoped capabilities, and durable accountability without requiring centralized control. Emerging approaches to structured context exchange and agent coordination, such as Model Context Protocols (MCP), may offer useful design patterns for this layer when generalized beyond model runtime to sociotechnical actors. The design of any such Meta-Layer coordination protocol remains an open area of research and standardization.

These questions are not gaps in DP1, but signals of where future ML-Drafts and ML-RFCs may be required as the Meta-Layer matures.

15. Relationship to Other Desirable Properties

DP1 is foundational and cross-cutting. Many other Desirable Properties depend directly on the conditions it establishes.

In particular:

- **Properties related to safety, harm reduction, and abuse prevention** rely on durable accountability, shared memory, and zone-scoped enforcement to function effectively. Without persistent attribution and the ability to recognize patterns of behavior over time, safety mechanisms degrade into reactive moderation that fails to prevent repeat harm or coordinated abuse.
- **Properties concerning agency, consent, and autonomy** depend on federated identity, user-held credentials, and contestable governance so that participants can meaningfully choose how they engage, under what conditions, and with which authorities. Without these foundations, consent becomes nominal, exit becomes costly, and power asymmetries harden.
- **Properties addressing AI participation, automation, and alignment** require clear agent differentiation, asymmetric constraints, and binding accountability to prevent synthetic scale from overwhelming human judgment or distorting collective outcomes. DP1 establishes the conditions under which AI systems can participate without eroding trust or legitimacy.
- **Properties focused on collective intelligence, coordination, or governance** assume the existence of trustworthy participation, legible authority, and adaptive learning loops. Collective sensemaking and coordination cannot emerge where participants doubt who is acting, how decisions are made, or whether systems can learn from failure without capture.

Weakness or ambiguity in DP1 propagates upward, undermining the effectiveness of other properties. Conversely, a strong DP1 enables the Meta-Layer to support more advanced coordination, safety, and governance capabilities without reverting to centralized control.

16. Path Toward ML-RFC

This ML-Draft is intended as exploratory scaffolding rather than a finalized specification. Progression toward an ML-RFC should be guided by rough consensus, iterative refinement, and practical validation.

Key steps toward ML-RFC status include:

- **Soliciting broad community review and critique.** Feedback from implementers, civil society, governance practitioners, and researchers is essential to test assumptions and surface edge cases.
- **Identifying points of rough consensus.** Not all aspects of DP1 must be settled to advance. Emphasis should be placed on stabilizing core invariants such as action-bound accountability, zone-scoped trust, and contestability.
- **Clarifying implementation invariants.** Future drafts should distinguish clearly between invariant requirements and flexible design space, reducing ambiguity for builders while preserving pluralism.
- **Separating exploratory elements from normative commitments.** Concepts such as coordination protocols or adaptive intelligence should mature through dedicated drafts before being incorporated normatively.
- **Promoting stable elements to ML-RFC status.** Once sufficient consensus and operational understanding exist, portions of DP1 may be advanced as ML-RFCs to serve as durable reference points for the Meta-Layer ecosystem.

This progression reflects the Meta-Layer's commitment to transparency, accountability, and participatory standards development.

DP1 defines the conditions under which trust can emerge. Without it, the meta-layer becomes another surface. With it, the meta-layer becomes a place.

DP2: Participant Agency & Empowerment

Purpose of This Draft

This ML-Draft articulates Desirable Property 2 (DP2) as the Meta-Layer’s commitment that participants can meaningfully steer their digital lives. Beyond authentication (DP1) and governance (DP3), DP2 establishes that people and accountable agents hold real, usable power over presence, data flows, automation, and the conditions under which they are seen, acted upon, and counted.

DP2 responds to recurring failures of the contemporary Web:

- **Agency theater:** settings and consent flows that do not change outcomes
- **Asymmetric literacy:** systems legible only to specialists while obligations bind everyone
- **Structural dependency:** exit and portability exist nominally but are costly or illusory
- **Delegated opacity:** agents act on a participant’s behalf without durable, legible control

This draft guides implementation, governance design, and future ML-RFC development. It is exploratory scaffolding, not a finalized specification.

1. Problem Statement: Why “Control” Without Capability Fails

For decades, platforms have described participants as “in control” while reserving decisive power for operators, opaque ranking systems, and unbounded automation. The result is not merely dissatisfaction; it is predictable harm: manipulation, lock-in, surveillance-by-default, and governance that responds to scale by narrowing what ordinary people can do or understand.

DP2 begins from a different premise: **agency is not a feeling; it is a property of systems**. A Meta-Layer earns the label human-first only if participants can **observe, redirect, and withdraw** from the forces that shape their experience—within the same zones where accountability (DP1) is enforced.

1.1 Agency vs. Authorization

Authorization answers what a token allows. Agency answers whether a participant can **shape outcomes**: defaults, reach, automation, data use, and the rules that allocate visibility and risk.

Systems that conflate “logged in” with “empowered” routinely:

- Bundle consequential defaults behind “agree to continue”
- Hide material changes behind versioned policies
- Route impactful decisions to models or pipelines participants cannot inspect

DP2 separates authentication and authorization (DP1) from **participant-directed configuration of the lived interface**.

1.2 Empowerment as Distributed Capability

Empowerment is **capability + legibility + recourse**:

- **Capability**: participants can change outcomes (not just preferences)
- **Legibility**: participants can see how systems act on their behalf
- **Recourse**: participants can contest, reverse, or exit

A system lacking any one of these is not empowering, regardless of interface polish.

2. Tensions and Tradeoffs

2.1 Usability vs. Complexity

Agency introduces configuration surfaces that can overwhelm. Hiding them removes control. DP2 requires **graduated disclosure**: simple defaults that are safe, with deeper controls accessible without specialized expertise.

2.2 Automation vs. Control

Automation reduces effort but can displace agency. Participants must be able to answer:

- What did the agent infer?
- What authority does it have?
- How do I stop it?

DP2 requires **visible delegation scopes, renewal, and revocation** aligned with accountable binding (DP1).

2.3 Power-Law Attention Markets

Even fair rules can reproduce inequality when attention is the currency. DP2 does not promise equal outcomes; it guarantees **equal access to the levers** that govern one's participation and visibility within a zone, and **transparent disclosure** when algorithmic allocation is in play (touchpoint DP14).

2.4 Safety vs. Patronizing Lockdown

Safety work can slide into infantilizing participants. DP2 pairs with DP1 to require that constraints be **proportionate, explainable, and contestable**, with pathways for competent self-determination inside high-trust zones.

3. Core Principle of DP2

Agency is the ability to change outcomes, not merely configure preferences. Systems that do not preserve participant intent across automation, delegation, and scale do not provide agency.

Participant agency in the Meta-Layer is the combination of meaningful defaults, legible automation, durable delegation controls, and practical exit—enacted at the interface where people actually live.

Implications:

- Defaults favor the participant where stakes are asymmetric (data, reach, automation, payments), with zone-level calibration
 - Automation is always **scoped**: time-bounded, purpose-limited, revocable, and attributable (DP1)
 - Core agency paths (privacy, notifications, delegation, export, exit) remain reachable without specialized training
 - **Collective agency is first-class**: communities can set norms and enforce them without stripping member autonomy (handoff to DP3, DP18–DP20)
-

4. Presence, Identity Plurality, and the Right to Shape Visibility

DP2 treats presence as something participants **sculpt**, not merely a profile object.

4.1 Plural Identities, Singular Accountability

Participants may present differently across zones (DP1). Agency requires **per-zone controls** for visibility, linkage, and discoverability so pseudonymous participation is not undermined by accidental correlation.

4.2 Reach and Amplification as Explicit Objects

When systems can amplify (boost, recommend, cross-post), amplification settings are **agency-bearing surfaces**: who may amplify me, under what proofs, with what caps? This is where DP2 meets DP1’s asymmetric constraints for AI scale.

5. Defaults, Friction, and “Reasonable Participant” Design

5.1 Dangerous Defaults Are a Governance Bug

DP2 assigns normative weight to default selection: the burden of proof lies on whoever proposes a default that increases extraction, surveillance, or irreversible commitment.

5.2 Friction as Protection, Not Punishment

Strategic friction (confirmations, cooling-off periods for irreversible acts) protects agency when stakes are high. DP2 distinguishes **protective friction** from **hostile friction** designed to prevent exit or understanding.

5.3 Progressive Disclosure Without Burial

Advanced controls may be layered, but never removed from accountability: search, assistive onboarding, and machine-readable policy summaries are part of agency infrastructure.

5.4 Agency System Layer: Continuity, Delegation Integrity, and Enforceable Consent

Beyond interface controls and defaults, DP2 requires a coherent agency system layer that persists across environments, interactions, and time. This layer ensures that participant intent, consent, and control remain enforceable under scale, automation, and interoperability.

Agency is not simply the presence of controls. It is the ability to reliably change outcomes across systems without loss of intent, visibility, or recourse.

5.4.1 Agency Continuity Across Systems

Participant choices must persist across tools, zones, and integrations.

This requires:

- Preservation of consent, preferences, and delegation states across environments
- Explicit signaling when agency guarantees degrade or reset
- Protection against silent override of participant intent by downstream systems

A failure mode is **agency fragmentation**, where participant control is lost when moving across systems.

5.4.2 Delegation Integrity and Scope Enforcement

Delegation must remain bounded, legible, and enforceable.

All delegated authority must be:

- Explicitly granted
- Scoped by capability, domain, and time
- Continuously visible to the participant
- Revocable without friction

Systems must prevent delegated agents from expanding scope beyond granted authority.

A failure mode is **delegation drift**, where agents act beyond intended scope without detection.

5.4.3 Consent Durability and Revocability

Consent must persist long enough to be meaningful, but remain revocable at all times.

This requires:

- Clear mapping between consent and system behavior
- Immediate or bounded-time revocation pathways
- Prevention of “zombie consent” where permissions persist beyond participant awareness

A failure mode is **consent decay**, where participants lose track of what they have authorized.

5.4.4 Anti-Coercion and Default Integrity

Defaults must not be used to extract consent or steer behavior against participant interests.

Systems must:

- Prevent dark patterns and coercive flows
- Require explicit confirmation for high-impact or irreversible actions
- Surface when defaults materially affect outcomes

A failure mode is **coercive configuration**, where participants are nudged into decisions that undermine agency.

5.4.5 Cross-System Agency Semantics

Agency signals do not carry identical meaning across all systems.

Systems must:

- Signal when consent or delegation semantics change across contexts
- Prevent misinterpretation of permissions
- Allow participants to re-evaluate choices when entering new environments

A failure mode is **semantic drift**, where participant intent is misapplied across systems.

5.4.6 Agency Memory and Auditability

Participants must be able to reconstruct what they authorized, when, and why.

This includes:

- Logs of delegation, consent, and revocation events
- Visibility into system actions taken on behalf of the participant
- Tools for auditing past decisions and their consequences

A failure mode is **agency opacity**, where participants cannot understand or audit system behavior.

This agency system layer ensures that participant control is not an illusion created by interface design, but a durable property that persists under real-world conditions.

6. Data, Automation, and Delegation: Agency Substrates

6.1 Purpose-Limited Processing

Collection and use are tied to **stated purposes with granular switches**, not monolithic “privacy” toggles (deep coupling to DP4).

6.2 Agent Delegation Graph

For any automated or AI-mediated actor operating with participant intent, the system exposes:

- **Scope** (read/write domains, rate, spend limits where relevant)
- **TTL and renewal**
- **Attribution** to a responsible entity (DP1 §8.3)
- A **kill switch** reachable from the primary interface layer

6.3 Human-in-the-Loop Gradients

Not every action needs a click, but **material actions** (payments, legal commitments, public attributions, irreversible posts) require explicit human confirmation unless a community zone defines a higher-automation norm with informed opt-in.

Systems **MUST** remain safe under automated delegation at scale. This includes resisting coordinated agent behavior, preventing silent escalation of authority, and ensuring that human override remains effective even under high-volume automated activity.

A failure mode is **automation overrun**, where agent activity exceeds human capacity to observe, intervene, or revoke, effectively nullifying participant agency.

7. Portability, Exit, and Interoperability as Agency Guarantees

Agency must survive movement. If a participant's control disappears at boundaries, the system is coercive by design.

7.1 Practical Exit

Exit must be feasible in **human time** (hours or days for standard data classes). Stalling tactics, hidden dependencies, or degrading exports constitute agency violations.

Systems **MUST**:

- Provide complete, machine-readable exports for user-held data and artifacts
- Disclose exclusions (e.g., third-party licensed data) with clear rationale
- Preserve identity continuity signals where technically honest (DP1), or explicitly signal loss

Failure modes:

- **Exit obstruction**: artificial friction prevents leaving
- **Degraded export**: data is technically exported but unusable

7.2 Forking and Continuity

Where communities fork norms or stacks (DP1 §10.4), participants retain identity continuity and portable artifacts where technically honest, avoiding punishment for disagreement.

Systems SHOULD support:

- Portable credentials and attestations with provenance
- Migration guides and compatibility layers for common formats

Failure mode: **fork penalty**, where dissent results in loss of history or access.

7.3 Interoperability Honesty

Interoperability claims MUST be truthful. If a system advertises portability or integration, it MUST specify:

- What is preserved (data, credentials, delegation states)
- What degrades (semantics, guarantees, rate limits)
- What is not transferable and why

Failure mode: **interop deception**, where portability is claimed but core agency properties are lost in transit.

8. Collective Agency and Community Tools

Individuals act within communities. DP2 requires that collective mechanisms enhance, rather than erase, individual agency.

Communities MUST be able to:

- Define and publish mandates for stewardship roles (moderators, curators, treasurers)
- Set and revise community-level switches (e.g., human proof for governance votes) via governed processes (DP3, DP12)
- Audit decisions and reverse them through defined procedures

Systems MUST ensure:

- **Mandate transparency**: who can act, within what scope, for how long
- **Revocation pathways**: members can challenge or replace stewards
- **Proportional constraints**: community rules do not silently nullify individual controls

Failure modes:

- **Capture**: small groups entrench power and override member agency
 - **Collective overreach**: community settings suppress legitimate individual choices without recourse
-

9. Community Signals Informing DP2

Recurring themes from public discourse (non-exhaustive) include both desires and tensions:

- Fatigue with consent theater and unreadable policies
- Demand for “show me what you’re doing with my data right now” views
- Preference for AI copilots that **ask before acting** on the user’s behalf
- Interest in portable reputation without panopticon scoring (tension with DP1)
- Desire for simplicity alongside real control (tension with complexity)

These signals reveal a core contradiction: participants want **power without overload**. DP2 addresses this through progressive disclosure, safe defaults, and auditability, rather than removing control.

10. Non-Goals and Explicit Boundaries

DP2 defines the conditions for agency; it does not promise universal outcomes.

DP2 does not:

- Guarantee equal outcomes or neutralize attention economics by fiat
- Eliminate all paternalistic protections in safety-critical zones (these must be labeled, bounded, and contestable)
- Replace DP1 accountability with unchecked “freedom to harm”
- Mandate a single UX globally; pluralism across zones is expected

DP2 also does not:

- Require full transparency of all system internals where doing so would enable exploitation; instead, it requires **participant-legible explanations and audit paths**
- Allow delegation to obscure responsibility; all automated action remains attributable (DP1)

Failure mode: **overreach**, where DP2 is interpreted to justify unsafe or unaccountable behavior.

11. Minimum DP2 Alignment (Non-Normative)

Minimum alignment is not a UX checklist. It is the threshold at which participant agency is **real, enforceable, and resistant to coercion, drift, and automation capture**.

A system that does not meet these conditions may expose controls, but it does not provide agency.

At minimum, a system claiming DP2 alignment **MUST** satisfy the following **irreducible conditions**:

11.1 Outcome-Level Control (Not Preference Simulation)

- Participants **MUST** be able to change meaningful outcomes, not only surface preferences

- Core levers (visibility, data use, automation authority, exit) **MUST** directly affect system behavior
- Systems **MUST NOT** simulate control through settings that do not alter execution

Failure mode: **agency theater**, where interfaces imply control without changing outcomes.

11.2 Delegation Visibility and Revocation

- All automated or AI-mediated actions **MUST** be attributable and visible to the participant
- Delegation **MUST** include scope, duration, and authority limits
- Participants **MUST** be able to revoke delegation in real time or bounded time

Failure mode: **delegation opacity**, where systems act without legible authority or revocation.

11.3 Consent Binding and Enforcement

- Consent **MUST** be explicitly tied to system behavior
- Systems **MUST** enforce consent boundaries consistently across components and integrations
- Revocation **MUST** propagate across systems or clearly signal where it does not

Failure mode: **consent bypass**, where downstream systems ignore or reinterpret user intent.

11.4 Anti-Coercion Defaults

- Defaults **MUST NOT** materially disadvantage participants without explicit opt-in
- High-impact actions **MUST** require clear confirmation
- Systems **MUST NOT** use dark patterns to obtain or retain consent

Failure mode: **coerced consent**, where participants are steered into decisions against their interest.

11.5 Practical Exit and Portability

- Participants **MUST** be able to export their data and exit systems in reasonable human time
- Exit **MUST NOT** result in silent loss of identity continuity without explicit signaling (DP1)
- Systems **MUST NOT** impose artificial friction to prevent exit

Failure mode: **exit obstruction**, where users are technically allowed but practically unable to leave.

11.6 Cross-System Agency Integrity

- Participant choices **MUST** persist across integrations where technically feasible
- Systems **MUST** signal when agency guarantees degrade across contexts
- Downstream systems **MUST NOT** silently override upstream participant intent

Failure mode: **agency fragmentation**, where control is lost across system boundaries.

11.7 Auditability of System Behavior

- Participants **MUST** be able to inspect what actions were taken on their behalf

- Systems **MUST** provide logs or summaries of automated decisions and their effects
- Critical actions **MUST** be reconstructable for dispute or review

Failure mode: **agency opacity**, where participants cannot understand or challenge system behavior.

These conditions define the **minimum viable agency layer** of the Meta-Layer.

Partial implementations that omit outcome control, delegation integrity, consent enforcement, or exit **MUST NOT** be considered aligned with DP2.

12. Open Questions and Future Work

- Portability vs. abuse: preventing weaponized export while honoring exit (interfaces with DP1 memory models)
 - Legibility budgets: how much system behavior can be made comprehensible without overload; role of machine summaries vs. audits (DP14–DP15)
 - Collective overrides: when may a community limit individual agency for safety without capture?
 - Cross-zone identity correlation: separation vs. pressure for unified reputation
 - Economic agency: tipping, subscriptions, and paid reach as agency-bearing surfaces (touchpoint DP6)
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13. Relationship to Other Desirable Properties

- **DP1** supplies accountable actors; **DP2** supplies the levers those actors hold. Without DP1, agency collapses into anonymity games; without DP2, accountability becomes surveillance.
 - **DP3** scales governance; DP2 ensures scale does not erase participatory steering.
 - **DP4–DP6** ground agency in data, namespace, and commerce realities.
 - **DP7–DP10, DP21** carry agency into experience, education, and modality.
 - **DP11–DP13** bound AI so delegation does not swallow human steering.
 - **DP14–DP17** make power auditable and sustainable.
 - **DP18–DP20** turn agency into shared evolution of the stack.
-

14. Path Toward ML-RFC

Advancement from ML-Draft to ML-RFC should demonstrate that agency is not only described but **operationally verified**.

Key steps:

- Community review with builder and civil-society lenses
- Reference implementations of delegation graphs, revocation, and exit flows
- Conformance tests for **minimum alignment** (Section 11) including adversarial scenarios (automation overrun, consent bypass, interop deception)

- Clear separation of **normative invariants** vs **design space**
- Publication of audit reports demonstrating real-world behavior under load

Graduation criteria SHOULD include:

- Evidence that participants can revoke delegation and observe effect within bounded time
 - Evidence that exports are usable in at least one independent system
 - Evidence that automated actions are attributable and auditable end-to-end
-

15. Closing Orientation

DP2 asserts that participants are not merely subjects of systems, but operators within them.

When agency is real, people can:

- change outcomes
- understand system behavior
- delegate safely
- exit without penalty

When agency is simulated, systems accumulate hidden power: defaults decide, automation acts, and participants absorb the consequences.

DP2 is the commitment that control is not inferred, but **demonstrable**.

It is the difference between having settings and having a steering wheel.

DP1 asks who may act with integrity. DP2 asks whether participants hold the wheel—or only the liability.

DP3 – Adaptive Governance for an Exponentially Growing Community

1. Purpose of This Draft

This draft articulates Desirable Property 3 (DP3) as the condition under which governance scales with participation and capability without collapsing into centralized fiat, procedural paralysis, or symbolic participation.

DP3 defines how the meta-layer maintains legitimate, timely, and contestable rule-setting as communities, tools, and AI capabilities expand. It sits between DP1 (accountability), DP2 (participant agency), and DP12 (community governance of AI), and acts as the bridge between human deliberation and executable policy.

Governance is the system's primary control surface under conditions of scale, automation, and interoperability. When governance fails at system boundaries or under rapid change, every other property becomes unstable or performative.

If DP3 is weak, predictable failures follow: capture by early insiders, rubber-stamp councils, unbounded operator discretion, reform paralysis as conditions change, and AI-mediated scale overwhelming human governance loops.

DP3 does not prescribe a single voting system, constitution, or DAO pattern. It defines the minimum conditions under which governance remains adaptive, legible, and legitimate under exponential growth.

Governance is the control layer of the meta-layer. Systems that appear governed locally but fail across scale, speed, or system boundaries will be captured or bypassed.

2. Problem Statement

In today's web, governance consistently lags behind scale.

Communities begin with informal norms, moderators, and shared expectations. As participation grows, those structures fracture. Decision-making becomes opaque, centralized, or too slow to respond to real-time harms.

This produces recurring failures:

- scale shock, where participation outpaces moderation, policy, and appeals capacity
- frozen rules that fail to adapt to new behaviors, technologies, or adversaries
- governance theater, where surveys or advisory groups exist without decision power
- emergency centralization, where temporary powers become permanent
- invisible delegation, where operators change rules without traceable ratification

These failures are structural. Growth without governance capacity turns open systems into extractive or chaotic systems.

DP3 reframes governance as adaptive infrastructure: systems that can change at the speed of reality without losing legitimacy.

3. Threats and Failure Modes

3.1 Capture by concentrated stakeholders

Early contributors, large stakeholders, or sponsors lock rules that preserve their advantage.

Example: A founding team retains veto over all governance proposals despite community votes.

Why this matters: Governance without credible contestation becomes ownership theater.

3.2 Procedural overload

Governance becomes too slow to respond to urgent issues.

Example: A coordinated harassment campaign spreads while proposals wait weeks for quorum.

Why this matters: Governance must distinguish between deliberation and response.

3.3 Governance lag behind AI capability

AI systems evolve faster than governance cycles.

Example: Automated agents exploit a loophole for days while policy review is scheduled monthly.

Why this matters: Static governance cannot contain dynamic systems.

3.4 Scale without representation

New participants lack meaningful voice while incumbents dominate.

Example: Global users cannot participate in English-only governance calls.

Why this matters: Adaptation requires inclusive participation surfaces.

3.5 Symbolic decentralization

Governance appears decentralized but is controlled off-chain or off-process.

Example: Votes occur, but operators retain unilateral execution authority.

Why this matters: Legitimacy depends on alignment between process and power.

3.6 Emergency power drift

Temporary powers become permanent through inertia.

Example: Crisis moderation tools remain active without review or sunset.

Why this matters: Speed must not erode accountability.

3.7 Governance fatigue and disengagement

Participants disengage due to complexity or lack of impact.

Example: Only a small group consistently participates in governance decisions.

Why this matters: Low participation increases capture risk.

4. Core Principle

Adaptive governance in the meta-layer means authority evolves alongside scale, capability, and system interconnection while remaining visible, contestable, and bounded.

Governance must function not only within a single community or tool, but across the systems, environments, and contexts in which rules are applied, enforced, and experienced.

Rules must be able to change without becoming arbitrary. Emergency actions must be possible without becoming permanent. Participation must be distributed without becoming incoherent.

Example: A governance system defines standard decision cycles, emergency pathways with automatic expiration, and public policy diffs tied to incidents.

What this feels like: The system learns in public, without rewriting rules silently.

Without this: Growth becomes disenfranchisement disguised as efficiency.

5. Primary Mechanisms and Structural Conditions

5.0 Governance Layer: Execution, Memory, and Control

Adaptive governance requires more than the ability to change rules. It requires the ability to express, enforce, observe, and evolve those rules as part of a continuous operational system that persists across environments.

In many systems, governance fails not because rules are absent, but because they are not bound to behavior. Decisions exist as documents or discussions, while actual system behavior is shaped elsewhere by incentives, defaults, or hidden control layers.

DP3 therefore requires a shared governance layer composed of primitives that allow governance to operate as infrastructure rather than aspiration.

Policy objects

Governance rules must be represented as structured, versioned objects that can bind to runtime systems.

A policy object includes:

- scope: where the rule applies
- conditions: what triggers the rule
- constraints: what is allowed or prohibited
- enforcement bindings: how the rule executes

This allows governance to move from agreement to execution, and aligns directly with DP12's requirement for executable policy.

Governance receipts

Every governance action must produce a verifiable record.

A governance receipt includes:

- who proposed and approved a decision
- what changed
- when it changed
- what systems were affected
- what enforcement state was applied

These receipts allow participants and auditors to trace how governance decisions translate into system behavior, and connect governance to DP15 (security and provenance).

Governance diffing and versioning

Governance must evolve through visible change.

Participants must be able to see:

- what changed between rule versions
- why it changed
- what effects resulted

Silent rule replacement erodes legitimacy. Visible diffing preserves continuity and enables governance learning over time.

Zone governance profiles

Each community or interaction context operates under a defined governance profile.

A zone governance profile includes:

- active policy objects
- enforcement modes
- participation structures

- escalation pathways
- incentive constraints

This allows governance to adapt to context while remaining portable across systems, aligning with DP4 (data boundaries) and DP20 (community ownership).

Enforcement hooks

Governance must bind to systems that can enforce it.

These include:

- AI agent constraints (DP13)
- moderation systems
- access controls
- interaction limits

Without enforcement hooks, governance decisions remain advisory and are overridden by underlying system behavior.

Governance memory graph

Governance must persist over time as a connected structure.

Decisions must be linked to:

- prior versions
- triggering events
- debates and dissent
- observed outcomes

This creates a governance memory that enables learning, prevents repetition of failure, and supports accountability.

Without memory, governance resets continuously and cannot improve.

These primitives do not replace governance processes. They make them operational. The mechanisms that follow operate through this layer, ensuring that decisions are not only made, but executed, observed, and revised within a coherent system.

5.1 Tiered decision systems

Not all decisions carry the same weight, risk, or urgency. Treating all governance actions as equivalent either slows the system to paralysis or opens pathways for low-threshold capture of critical decisions.

DP3 requires that governance systems define clear decision tiers, each with appropriate processes, thresholds, and timelines. Routine decisions should be fast and low-friction. Significant decisions should involve deliberation and multi-stakeholder input. Existential decisions should require high thresholds and extended review.

A key failure mode is process flattening, where all decisions are routed through the same mechanism, allowing either trivial actions to clog governance or critical actions to slip through without sufficient scrutiny.

5.2 Delegation with revocability

As systems scale, direct participation in every decision becomes impossible. Delegation is therefore necessary, but without revocability it becomes a vector for capture.

DP3 requires that delegated authority be explicit in scope, time-bound where appropriate, and revocable by participants. Delegation must remain a tool of coordination, not a permanent transfer of power.

A failure mode is silent entrenchment, where delegated roles accumulate authority over time without clear pathways for removal or reassignment.

5.3 Emergency pathways with sunset

Governance systems must be able to respond rapidly to emergent threats, but speed introduces the risk of unbounded authority.

DP3 requires that emergency actions be clearly declared, automatically expire, and undergo post-hoc review and ratification. This creates a bounded exception rather than a precedent for permanent control.

A common failure mode is emergency normalization, where temporary powers persist due to inertia or lack of review, gradually shifting governance toward centralized control.

5.4 Governance memory

Governance decisions do not occur in isolation. They are part of a continuous process of learning, adaptation, and correction.

DP3 requires that all decisions be linked to their context, including triggering events, dissenting views, and measurable outcomes. This creates continuity and prevents repeated cycles of the same failures.

A failure mode is historical amnesia, where prior decisions and their consequences are lost, forcing communities to relearn the same lessons under new conditions.

5.5 Capacity-aware governance

Governance is constrained not only by rules, but by the human and operational capacity required to execute them.

DP3 requires that governance systems explicitly provision for moderation, appeals, translation, and accessibility. Without this, governance becomes symbolic, with rules that cannot be enforced or contested in practice.

A failure mode is capacity illusion, where systems appear governed on paper but lack the resources to implement or uphold decisions.

5.6 Adversarial foresight

Governance systems must anticipate how they will be attacked or manipulated, rather than reacting only after failure occurs.

DP3 requires that communities model adversarial scenarios such as capture attempts, AI-driven manipulation, and scale shocks. This allows governance structures to incorporate safeguards before vulnerabilities are exploited.

A failure mode is reactive governance, where systems adapt only after harm occurs, often at higher cost and with reduced trust.

5.7 Interoperable governance artifacts

Governance must not be confined to a single tool or platform. Policies, decisions, and governance structures must be able to move across systems without losing meaning.

DP3 requires that governance artifacts be exportable, comparable, and portable. This enables communities to fork, migrate, or integrate without resetting their governance systems.

A critical failure mode is governance lock-in, where rules exist only within a specific platform, making exit or replication impractical.

5.8 Continuous feedback loops

Governance cannot rely solely on periodic voting cycles. Systems must incorporate ongoing signals from participation, behavior, and outcomes.

DP3 requires continuous feedback mechanisms that inform governance in near real-time, allowing systems to adapt before issues become systemic failures.

A failure mode is episodic governance, where decisions are made in isolation from evolving conditions, leading to lag and misalignment.

5.9 Human-AI coordination boundaries

AI can significantly augment governance by processing information at scale, but it also introduces risks of manipulation, opacity, and overreach.

DP3 requires clear boundaries for AI participation, including roles in summarization, simulation, and pattern detection, while reserving material decisions for human ratification.

A failure mode is automation creep, where AI systems begin to effectively determine outcomes without explicit authorization or oversight.

5.10 Representation mechanisms

As participation scales globally, governance must ensure that diverse perspectives are meaningfully included.

DP3 requires mechanisms that support geographic diversity, language accessibility, and asynchronous participation. Representation must reflect the actual composition of the community, not just those able to engage in specific formats.

A failure mode is structural exclusion, where governance participation is limited by language, time zones, or access constraints, concentrating power among a narrow subset of participants.

6. Governance, Accountability, and Agency Surfaces

Governance surfaces determine whether participants can meaningfully understand, influence, and contest the rules that shape their environment. Without these surfaces, governance becomes opaque, unchallengeable, and ultimately extractive.

In many systems, governance exists in theory but not in practice: decisions are made elsewhere, authority is unclear, and appeals are ineffective or absent. DP3 requires that governance be experienced as a visible and navigable system, not an abstract promise.

Without these surfaces, governance loses legitimacy. Participants may continue to engage, but without trust, recourse, or real influence, participation becomes performative rather than constitutive.

Participants must be able to:

- see who holds authority at any moment
- understand how decisions are made
- appeal decisions within defined timelines
- track changes in rules over time

Communities must be able to:

- update governance structures without restarting from zero
- audit whether decisions were executed as approved
- fork governance when legitimacy breaks

7. Incentives and Power Analysis

Adaptive governance fails when incentives quietly outrun rules.

In many systems, decision procedures are visible while the forces shaping outcomes remain hidden. Optimization targets, funding dependencies, and growth pressures steer behavior in ways governance cannot easily detect or correct.

DP3 requires that incentive structures be treated as first-class governance objects.

This includes making visible:

- what metrics systems optimize for (engagement, retention, revenue, safety)
- how those metrics influence policy enforcement and prioritization
- where economic or reputational rewards create pressure to bypass rules

Example: A platform publishes moderation policies that prioritize safety, but internal ranking systems reward engagement spikes. Harmful content persists not because rules are absent, but because incentives contradict them.

Why this matters: Governance that does not act on incentives will be bypassed by them.

DP3 therefore expects governance systems to:

- expose optimization targets where they materially affect outcomes
- allow communities to constrain or reshape those targets within zones
- link policy objects to incentive conditions where appropriate (e.g., disallow reward for rule-violating behaviors)

When incentives and governance are aligned, systems become self-reinforcing. When they diverge, governance becomes symbolic.

8. Community Signals Informing DP3

Across communities and platforms, recurring signals point to a shared breakdown between scale and governance:

- frustration with decisions made by unseen operators without traceable process
- demand for term limits, rotation, and clearer accountability for stewards
- desire for asynchronous, multilingual participation that reflects global communities
- concern that AI and automation are overwhelming human moderation and deliberation capacity
- fatigue with feedback channels that do not result in observable change

These signals are not isolated complaints. They indicate structural gaps in how governance adapts, records, and responds at scale.

DP3 treats these signals as design inputs, not after-the-fact feedback.

9. Non-Goals and Explicit Boundaries

DP3 does not:

- guarantee optimal or unanimous outcomes in all decisions
- eliminate the need for expertise, stewardship, or delegated authority
- replace legal governance systems or jurisdictional requirements
- mandate a specific voting, DAO, or constitutional model

DP3 defines conditions for adaptive governance. It does not prescribe a single implementation.

10. Minimum Alignment (Non-Normative)

A DP3-aligned system should, at minimum:

- define decision tiers with corresponding processes and thresholds
- bind governance decisions to enforceable mechanisms (via policy objects and enforcement hooks)
- produce governance receipts for material changes
- maintain visible versioning and diff history for rules and policies
- include bounded emergency pathways with automatic sunset and review
- provide appeal and contestation pathways with defined timelines
- plan for governance capacity (moderation, review, translation) proportional to scale

Partial compliance that omits execution, memory, or auditability should not be treated as alignment.

11. Open Questions and Future Work

Key open questions for adaptive governance include:

- how to achieve Sybil-resistant participation without excluding legitimate users (DP1)
- how to scale governance participation without overwhelming contributors
- how to integrate AI assistance in governance while preventing manipulation or capture
- how to balance global governance coherence with local autonomy and cultural context
- how to measure governance health beyond participation counts (e.g., decision latency, reversal rates, appeal outcomes)
- how to support forking and recomposition of governance systems without loss of continuity

These are not reasons to delay implementation. They are areas for iterative experimentation within visible governance systems.

12. Relationship to Other Desirable Properties

DP3 operates as the structural backbone for governance across the meta-layer.

- DP1 ensures actions within governance are attributable and contestable

- DP2 enables participants to exercise agency through delegation and participation
- DP4 constrains how data can be used within governance decisions
- DP6 and DP9 shape the economic and incentive context in which governance operates
- DP7 enables portability of governance artifacts across tools
- DP11–DP13 define how AI behavior is disclosed, governed, and contained
- DP12 provides the execution layer that binds governance to runtime systems
- DP15 ensures governance actions are provable and auditable
- DP17 ensures governance has sustained resources
- DP20 defines who ultimately owns and can fork governance systems

DP3 does not stand alone. It coordinates these properties into a functioning system.

13. Foresight and Failure Design

Adaptive governance must assume adversarial pressure, rapid scale changes, and technological acceleration. Failures do not typically occur as singular events, but as gradual degradation across multiple dimensions: participation, enforcement, legitimacy, and coherence.

Common failure paths include:

- coordinated capture attempts by concentrated stakeholders
- automated or AI-driven manipulation of governance processes
- overload of moderation and review capacity during rapid growth
- exploitation of gaps between policy definition and enforcement

These failures often compound. For example, as participation grows, capacity gaps emerge. These gaps increase reliance on delegation or automation, which may introduce new vectors for manipulation. Over time, trust erodes, participation declines, and governance becomes easier to capture.

Cross-system environments introduce additional risks:

- governance rules diverging across tools and integrations
- inconsistent enforcement depending on where interactions occur
- loss of governance memory or context during migration

DP3 therefore requires designing safeguards in advance:

- circuit breakers for pausing high-risk processes
- escalation pathways for urgent interventions
- public postmortems linking incidents to policy changes

Governance must be able to detect not only discrete failures, but slow-moving drift toward illegitimacy.

Failure is expected. Silent failure is not.

14. Path Toward ML-RFC

Advancing DP3 toward ML-RFC requires:

- standardizing formats for governance receipts, policy diffs, and audit trails
- developing reference implementations of tiered governance and emergency pathways
- testing governance loops in live communities with varying scale and risk profiles
- aligning governance artifacts with identity, data, and interoperability standards

Progress should be demonstrated through working systems, not only conceptual agreement.

15. Closing Orientation

DP3 is the claim that governance can scale without losing legitimacy.

It rejects the tradeoff between speed and accountability, and between participation and coherence.

When adaptive governance is real, communities do not outgrow their ability to govern themselves.

When it is absent, growth concentrates power, erodes trust, and replaces coordination with control.

DP4 – Data Sovereignty and Privacy

1. Purpose of This Draft

This draft articulates Desirable Property 4 (DP4) as the condition under which participants and communities can meaningfully govern data about themselves and their activity in the meta-layer.

DP4 does not treat privacy as a settings menu, a compliance ritual, or a legal disclaimer. It defines the conditions under which claims of ownership, consent, confidentiality, deletion, and portability remain meaningful in practice.

The core claim is that sovereignty over data depends on more than access controls. It depends on whether collection, inference, retention, sharing, and reuse are bounded by visible purposes, governed by revocable permissions, and constrained by structures that communities can understand and audit.

If DP4 is weak, predictable failures follow: consent theater, surveillance-by-default, inference without accountability, lock-in through broken portability, deletion promises that stop at the first vendor boundary, and community rules that cannot survive contact with underlying data pipelines.

DP4 therefore functions as a precondition for multiple later properties. Agency cannot be exercised over invisible data flows. Governance cannot constrain systems that communities cannot inspect. Ethical AI cannot be meaningful where the data it sees, stores, or trains on is structurally uncontrolled.

DP4 does not resolve all legal, jurisdictional, or sector-specific privacy questions. It defines the minimum conditions under which sovereignty and privacy remain real at the interface where data is created, combined, interpreted, and acted upon.

2. Problem Statement

In today's web, privacy is often presented as disclosure without control.

Participants are shown banners, terms updates, and granular-looking toggles, yet the underlying system still optimizes for maximal collection, indefinite retention, behavioral inference, and partner expansion. In many cases, the formal interface of consent exists while the operational reality of choice does not.

This produces recurring failures:

- participants cannot tell what is being collected, for what purpose, for how long, or by whom
- data collected for one function expands into advertising, analytics, resale, or model training
- sensitive inferences are derived from behavior without clear notice, recourse, or deletion pathways
- data exports preserve liability but not usability, making exit technically legal but practically costly
- account deletion rarely maps cleanly to embeddings, downstream processors, backups, or partner copies
- communities cannot enforce stricter data norms inside their spaces because underlying systems remain vendor-shaped

These failures are not edge cases. They are structural consequences of architectures designed to treat data accumulation as default value creation.

DP4 addresses this by defining data sovereignty as an operational condition. Privacy becomes meaningful only when participants and communities can see the active terms of data use, limit those terms in practice, revoke permissions without fiction, and move or leave without losing the structure of their digital lives.

3. Threats and Failure Modes

3.1 Consent theater

Interfaces bundle unrelated processing into a single act of acceptance.

Example: A participant accepts a terms update to continue using a service and, in doing so, silently authorizes secondary uses of behavioral data for recommendation tuning, advertising, and model training.

Why this matters: The system records consent, but the participant did not experience a meaningful choice. DP4 treats this as a sovereignty failure, not a paperwork issue.

3.2 Purpose creep and secondary use

Data collected for one function expands into new products, ranking systems, partner programs, or model behaviors without a fresh social contract.

Example: Location data collected for safety or delivery is later used for engagement scoring, ad targeting, or brokered partner analytics.

Why this matters: The participant's mental model of risk becomes false. Trust erodes even where no obvious breach has occurred.

3.3 Illusory portability

Export exists formally but fails functionally.

Example: A participant downloads an archive that contains files and timestamps but omits social graph edges, permission history, role context, provenance, or schemas needed to restore meaningful continuity elsewhere.

Why this matters: Exit is made to look possible while dependency is preserved. DP4 requires portability that preserves usable structure, not only raw payloads.

3.4 Inference without accountability

Systems derive high-stakes conclusions from behavioral traces without clearly governing how those inferences are created, used, challenged, or removed.

Example: A wellness application infers stress or depression risk from typing cadence and browsing patterns, then shares a derived score with an advertising or insurance intermediary.

Why this matters: The participant never explicitly submitted the sensitive category, yet is still acted upon as if they had.

3.5 Retention without sunset

Data persists because retention is cheap, deletion is operationally inconvenient, and analytics cultures prefer indefinite memory.

Example: A participant deletes an account, but vector embeddings, partner datasets, abuse-model features, and backup systems continue to retain traces with no coherent deletion pathway.

Why this matters: Sovereignty requires time bounds. Without them, institutions remember indefinitely while participants bear the burden of asymmetrical memory.

3.6 Cross-context correlation

Identifiers, device graphs, and fingerprinting techniques merge activity across settings that participants experienced as distinct.

Example: Pseudonymous participation in a civic forum is quietly linked to shopping behavior, social browsing, or location history through shared infrastructure.

Why this matters: Plural identity becomes decorative. Communities cannot sustain contextual integrity if correlation silently defeats boundaries.

3.7 False anonymity and weak de-identification

Organizations describe datasets as anonymized even where re-identification remains plausible or contractually enabled downstream.

Example: A mobility dataset stripped of names still exposes sparse routines in a small town, allowing individuals to be reconstructed through outside knowledge.

Why this matters: DP4 requires honesty about residual risk. “De-identified” cannot be treated as a magic word that dissolves responsibility.

3.8 Partner sprawl without propagation

Deletion, revocation, and correction stop at the first layer of control.

Example: A participant deletes messages in one tool, but analytics vendors, cloud backups, and SDK partners continue to retain copies without visibility or participant recourse.

Why this matters: Sovereignty that fails at the first subcontractor boundary is not sovereignty.

3.9 Youth and vulnerable-context overexposure

Defaults optimized for adult engagement expose minors and vulnerable users to data-intensive patterns they are less equipped to assess or contest.

Example: A youth-oriented social tool enables location sharing, behavioral profiling, or AI-mediated emotional inference by default.

Why this matters: DP4 requires higher baselines where stakes are higher. Uniform defaults can produce unequal harm.

4. Core Principle

Data must retain meaning, consent, and accountability as it moves across systems. If data loses its binding to purpose, provenance, or permissions under transformation, sovereignty collapses into simulation.

Data sovereignty and privacy in the meta-layer require that personal and community data be collected, inferred, stored, shared, and reused only under visible, bounded, and governable conditions.

Data sovereignty and privacy in the meta-layer require that personal and community data be collected, inferred, stored, shared, and reused only under visible, bounded, and governable conditions.

Those conditions must include:

- clear purpose binding
- minimization of collection, access, and retention
- meaningful consent and withdrawal
- portability with practical utility
- deletion or attenuation pathways that propagate as far as technically possible
- auditability of significant access, inference, and transfer events
- community capacity to impose stricter norms within governed zones

In today's web, these conditions rarely hold together. A system may disclose collection without limiting reuse, provide deletion without propagation, or offer export without restoration value. DP4 treats such partial compliance as insufficient.

The meta-layer reframes privacy as operational control at the point of interaction.

Example: A participant opens a data lens and sees active purposes, relevant processors, current retention clocks, sensitive inferences attached to their account, and downstream systems that have accessed their data. They can revoke training permission, export their activity in an interoperable format, contest a high-risk inference, and receive a propagation receipt for deletion requests.

What this feels like: Privacy stops being a maze of legal text and becomes a set of understandable levers tied to real system behavior.

Without this: Privacy becomes trust in opacity, and opacity fails precisely where accountability matters most.

5. Primary Mechanisms and Structural Conditions

5.1 Purpose binding

Every collection and processing pathway must declare its purpose in terms legible to both participants and communities. Material changes in purpose require visible reauthorization, reclassification, or zone-level review.

Example: A discussion zone permits summarization for moderation assistance but prohibits model training on participant content unless a separate, revocable grant is given.

Without purpose binding, consent collapses into blanket exposure.

5.2 Data minimization by design

Systems must begin from the least collection, retention, and sharing compatible with the function being offered, and expand only through visible, justified choices.

Example: A messaging tool does not upload contacts by default. Contact sync is presented as a distinct choice with plain-language scope and a reversible off switch.

This is not anti-functionality. It is a refusal to make maximal collection the silent baseline.

5.3 Consent stack

Permission must be layered, granular, and revocable, with separate scopes for distinct categories of data use.

This draft uses **consent stack** as a mechanism-level abstraction: a structured set of permissions that distinguish service provision, analytics, sharing, automation, and training from one another.

Example: A participant permits AI-assisted summarization of their workspace but declines training use and third-party analytics. Revoking training permission does not disable the summarization feature they actually wanted.

The consent stack makes partial participation possible without forcing blanket surrender.

5.4 Meaningful portability

Portability must preserve enough structure to support continuity, not just compliance.

This includes, where technically honest and appropriate:

- stable identifiers
- schemas
- role metadata
- interaction history

- provenance markers
- permissions history
- relationship edges needed for mainstream migration use cases

Example: A participant exports a discussion archive that can be imported into another tool with thread structure, moderation history, authorship context, and trust signals intact.

Without this, “take your data with you” becomes formal rights without real exit.

5.5 Retention clocks and propagation discipline

Retention must be bounded by event-driven or purpose-bound clocks, not indefinite convenience. Deletion, correction, and revocation requests must propagate to known downstream systems with auditable outcomes.

This draft uses **retention clocks** as a mechanism-level abstraction: visible timers tied to categories of data and stated purposes.

Example: A participant can see that support logs expire in 30 days, abuse-review evidence in 180 days, and AI training exclusion tags apply immediately going forward. When deletion is requested, the system generates a receipt chain showing which processors complied, which are pending, and which limits remain technically unresolved.

DP4 does not require dishonest promises of perfect erasure. It requires propagation discipline and truthful accounting.

5.6 Sensitive inference governance

Derived data can be more consequential than submitted data. High-risk inferences therefore require stronger conditions than ordinary processing.

This includes inferences relating to health, finances, minors, politics, biometric patterns, relational vulnerability, and similar domains of elevated risk.

Example: A system that predicts self-harm risk from behavioral cues must disclose that such inference exists, limit its downstream use, provide human escalation where appropriate, and prohibit repurposing for advertising or engagement optimization.

Inference must be governable as first-class data, not treated as exempt because it was machine-generated.

5.7 Zone-scoped privacy profiles

Communities must be able to define stricter privacy norms within their zones while remaining interoperable with broader infrastructure.

This draft uses **privacy profile** as a mechanism-level abstraction: a machine-readable expression of the data rules that apply inside a zone.

A privacy profile may specify, for example:

- no third-party behavioral advertising
- no model training on participant contributions
- local-only processing for sensitive content
- elevated rules for youth participation
- stronger consent requirements for biometric or emotional inference

Example: A health-support community publishes a privacy profile that restricts cloud-based inference, blocks third-party SDKs, and requires explicit opt-in before any content can enter training pipelines.

Without zone-scoped privacy profiles, communities may have values but not operational control.

5.8 Auditability and provenance of use

Significant data access, transfer, and inference events must be inspectable in participant-legible and community-legible forms.

This does not require exposing every security detail publicly. It requires enough visibility to support contestation, trust, and oversight.

Example: A participant can see that an automated moderation agent accessed a document under a specific policy version, for a named purpose, with a recorded outcome and timestamp.

Privacy claims that cannot be audited remain aspirational.

5.9 Training and model-use boundaries

Where participant or community content could enter model training, fine-tuning, embedding pipelines, or retrieval systems, those pathways must be separately governed.

Example: A public discussion zone allows search indexing but defaults to no training use. Participants can grant corpus-level permission for research or model improvement on a renewable basis, and declined content carries an exclusion marker through the training pipeline.

This is a direct dependency between DP4 and later AI properties. Ethical AI claims are weak if model access to human data is structurally obscure.

5.10 Jurisdictional and transfer honesty

Cross-border transfers and legal regime changes must be visible as part of the participant's risk surface.

Example: A participant is shown that a given processor operates under a different legal regime, that redress pathways are limited, and that a community zone therefore blocks that transfer category by default.

Global systems do not excuse vague disclosure. They heighten the need for explicitness.

5.11 Data System Layer: Lineage, Transformation Integrity, and Consent Propagation

Beyond individual mechanisms, DP4 requires a coherent data system layer that preserves **lineage, semantics, and permissions** across pipelines, services, and time. This layer ensures that data remains trustworthy under transformation, scale, and adversarial use.

5.11.1 Lineage continuity

- Data **MUST** carry provenance linking it to source, purpose, and processing context
- Derived artifacts (features, embeddings, summaries) **MUST** reference upstream lineage

Failure mode: **lineage loss**, enabling untraceable reuse and accountability gaps.

5.11.2 Transformation integrity

- Transformations **MUST** be attributable to an actor (human/AI) and a declared purpose
- Material transformations **SHOULD** be reversible or auditable where feasible

Failure mode: **data laundering**, where meaning or risk is altered without trace.

5.11.3 Consent propagation

- Permissions **MUST** travel with data across internal and external systems
- Downstream processors **MUST** honor upstream constraints or declare degradation explicitly

Failure mode: **consent bypass chains**, where integrations ignore or reinterpret permissions.

5.11.4 Anti-replay and non-duplication

- Identity- or consent-bound artifacts **MUST NOT** be reused to gain additional value without attribution
- Systems **SHOULD** detect duplicate extraction across pipelines

Failure mode: **replay extraction**, where the same data yields multiple unaccounted benefits.

5.11.5 Inference binding and governance

- Inferences **MUST** be treated as first-class data with lineage, purpose, and revocation pathways
- High-risk inferences require stricter constraints and auditability

Failure mode: **inference drift**, where derived signals are reused outside their declared context.

5.11.6 Cross-system semantics

- Systems **MUST** signal when data meaning or guarantees change across contexts
- Mappings between schemas **MUST** preserve or explicitly degrade semantics

Failure mode: **semantic drift**, where data is misinterpreted after transfer.

This layer does not require centralization. It requires **coherence under movement**.

6. Governance, Accountability, and Agency Surfaces

DP4 is not satisfied by backend architecture alone. Participants and communities need interfaces through which data conditions become governable.

Participants must be able to:

- see active purposes, processors, retention clocks, and major inference categories
- revoke permissions without unfairly losing unrelated core functionality where technical separation is possible
- export their data in forms that preserve practical continuity
- correct or contest significant false inferences
- understand what deletion means in each category of storage and reuse

Communities must be able to:

- publish stricter privacy profiles for their zones
- reject tools that cannot meet those profiles
- audit aggregate compliance without turning privacy governance into a new surveillance regime
- preserve governance memory around why a privacy rule exists and when it changed
- distinguish individual confidentiality from community-level observability of system behavior

Example: A civic deliberation zone prohibits third-party trackers and emotional classification systems. Any overlay or agent entering the zone must declare compatibility with the zone's privacy profile or operate in a visibly constrained mode.

Without these surfaces, privacy remains vendor-defined even when communities appear to have rules.

7. Incentives and Power Analysis

Commercial systems tend to treat data surplus as strategic advantage.

Retention expands because future uses may be profitable. Inference expands because prediction creates leverage. Consent becomes cosmetic where friction threatens growth. SDK ecosystems and downstream processors thrive precisely when participants cannot trace the full chain of use.

DP4 does not assume these incentives disappear. It requires that their effects become visible and contestable.

Example: A platform discloses that feed ranking depends partly on behavioral surplus gathered across sessions. A community operating inside the meta-layer disables that ranking signal within its zone because it conflicts with the zone's purpose.

This matters because many privacy harms are not caused by one malicious actor. They emerge from ordinary growth logic operating without adequate brakes.

DP4 therefore treats incentive visibility as part of sovereignty. Participants should be able to know when they are not merely receiving a service, but being rendered into a data asset.

8. Community Signals Informing DP4

Across contexts, similar signals recur:

- fatigue with unreadable policies and false choice architectures
- demand for portability that supports real migration
- frustration with invisible partner ecosystems and silent SDK extraction
- concern about sensitive inferences that participants never explicitly supplied
- desire to ask, in operational terms, “what does the system think it knows about me?”
- expectation that youth and vulnerable contexts should receive safer defaults, not merely more warnings

These signals are not abstract. They arise when people sense that the surface language of privacy no longer matches the structure underneath.

DP4 responds to that gap by making data conditions inspectable, debatable, and governable.

9. Non-Goals and Explicit Boundaries

DP4 defines a minimum condition. It does not solve every problem associated with data, secrecy, or identity.

DP4 does not:

- guarantee perfect anonymity in all settings
- prohibit all inference or all data sharing categorically
- replace law, regulation, or sector-specific obligations in health, finance, education, or public safety
- mandate one universal privacy culture across all communities
- promise technically impossible forms of deletion while downstream copies or model memorization remain unresolved
- remove the need for accountability-linked identity in contexts where stronger attribution is legitimately required

These boundaries matter because absolutist privacy claims often collapse under real-world complexity.

For example, some communities may require stronger identity assurance to support trust and accountability. DP4 does not forbid that. It requires that data burdens attached to such assurance remain bounded, visible, and contestable.

Likewise, some inference may be necessary for accessibility, fraud prevention, or urgent safety intervention. DP4 does not deny that. It requires those pathways to be governed explicitly rather than smuggled in under vague necessity claims.

10. Minimum Alignment (Non-Normative)

Minimum alignment is not a policy checklist. It is the threshold at which data sovereignty is **enforceable, portable, and resistant to laundering, drift, and silent reuse**.

A system that does not meet these conditions may disclose practices, but it does not provide sovereignty.

At minimum, a system claiming DP4 alignment MUST satisfy the following **irreducible conditions**:

10.1 Purpose binding and enforcement

- All collection and processing MUST declare purpose and enforce it in execution
- Material purpose changes MUST require reauthorization or zone review

Failure mode: **purpose creep**.

10.2 Consent propagation

- Permissions MUST travel with data across pipelines and partners
- Downstream systems MUST honor or explicitly degrade constraints

Failure mode: **consent bypass chains**.

10.3 Lineage and provenance

- Data and derivatives MUST carry reconstructable lineage
- Significant transformations MUST be attributable

Failure mode: **lineage loss / data laundering**.

10.4 Meaningful portability

- Exports MUST preserve structure needed for practical migration (schemas, relationships, permissions)
- Systems MUST disclose omissions and degradation

Failure mode: **illusory portability**.

10.5 Retention and propagation discipline

- Retention MUST be time- or purpose-bound with visible clocks
- Deletion/revocation MUST propagate with auditable receipts

Failure mode: **retention without sunset / partner sprawl**.

10.6 Inference governance

- High-risk inferences MUST be disclosed, bounded, and contestable
- Inferences MUST support correction or attenuation where applicable

Failure mode: **inference without accountability**.

10.7 Auditability of use

- Participants **MUST** be able to inspect significant access, transfer, and inference events
- Systems **MUST** provide logs or summaries sufficient for contestation

Failure mode: **opaque processing**.

10.8 Interoperability honesty

- Systems **MUST** state what is preserved, degraded, or non-transferable across boundaries

Failure mode: **interop deception**.

These conditions define the **minimum viable data sovereignty layer** of the Meta-Layer.

Partial implementations that omit purpose enforcement, consent propagation, lineage, or propagation discipline **MUST NOT** be considered aligned with DP4.

11. Open Questions and Future Work

DP4 surfaces unresolved design challenges that require further work:

- how to standardize interoperable privacy profiles across tools and zones
- how to represent retention, revocation, and transfer states in ways ordinary participants can understand
- how to govern model memorization and partial deletion in training-intensive systems
- how to balance privacy-enhancing technologies with the need for accountability and community oversight
- how to manage joint controllership and downstream processor responsibility in federated ecosystems
- how to handle collective consent for shared datasets or community archives
- how to distinguish emergency access from routine surveillance creep
- how to make privacy interfaces usable for low-literacy, multilingual, and neurodiverse participants
- how to audit third-party SDK ecosystems without reproducing surveillance in the name of governance
- how to preserve interoperability while allowing communities to adopt materially stricter norms

These are not reasons to delay better defaults. They mark the frontier where DP4 must mature through practice, governance, and implementation evidence.

12. Relationship to Other Desirable Properties

DP4 is foundational and interdependent.

- **DP1** anchors accountability for who accessed, inferred, retained, or transferred what
- **DP2** ensures participants have meaningful agency over data flows and delegation scopes

- **DP5** supports identifiers and namespaces that allow identity and reputation to remain portable without forcing correlation
- **DP6** depends on commerce patterns that do not require surveillance as the default price of participation
- **DP7–DP10** shape whether sovereignty is actually usable through interface quality, comprehension, and incentives
- **DP11** depends on bounded visibility into what AI systems may see, remember, infer, and influence
- **DP12** provides the community processes through which stricter privacy norms can be defined and revised
- **DP13** enforces containment over automated access, training pathways, and runtime data exposure
- **DP14–DP15** reinforce provenance, auditability, and transparency of data use and policy change
- later governance and ownership properties depend on collective data practices that do not silently strip participants of control

A failure in DP4 propagates upward. If data conditions are opaque, later governance becomes symbolic, ethical AI becomes ungrounded, and participant agency becomes procedural rather than real.

13. Path Toward ML-RFC

Progression from draft to RFC-grade maturity would require:

- stable invariants around purpose binding, minimization, meaningful portability, and propagation honesty
- a clearer grammar for privacy profiles, retention clocks, consent stacks, and audit events
- implementation evidence from zones with different stakes and governance norms
- demonstrable ways for communities to adopt stricter data rules without breaking interoperability
- better treatment of partial deletion, inference correction, and training exclusions
- alignment with privacy law and privacy-enhancing technologies without waiting for legal perfection before improving defaults

The goal is not to freeze one final model of privacy. It is to establish durable conditions under which sovereignty claims can be tested, challenged, and improved.

14. Closing Orientation

DP4 is where the meta-layer rejects the old bargain of convenience in exchange for invisibility.

Sovereignty is not achieved when a participant is merely informed that data extraction may occur. It is achieved when the participant and the communities they inhabit can see the operative terms of data use, shape those terms where appropriate, withdraw from them in meaningful ways, and leave without losing the structure of their digital life.

When DP4 is strong, trust in governance, AI, commerce, and collaboration becomes plausible.

When DP4 is weak, every higher-order property is forced to fight against a substrate that quietly converts participation into extraction.

If you want, I can also turn this into a numbered ML-Draft format with Status, Path Toward ML-RFC metadata, and house-style alignment to match your DP11–13 documents exactly.

DP5 – Decentralized Namespace

Claim Your Space in the Meta-Layer

Purpose of This Draft

This ML-Draft articulates Desirable Property 5 (DP5) as the condition under which people, communities, agents, artifacts, and spaces can be named, addressed, discovered, traded, and governed across the Meta-Layer without dependency on a single platform or registry.

DP5 introduces meta-domains and personal identifiers as sovereign, portable identity and addressability primitives. These identifiers allow participants to claim space in the Meta-Layer, link that space to existing web domains or decentralized identifiers, and use names as anchors for identity, ownership, trust, commerce, and governance.

The core claim is simple:

Meta-domains and personal identifiers give participants sovereign, portable identity – owned by them, not rented from a platform.

The Meta-Layer introduces a decentralized namespace system where identity is not merely a login and addressability is not merely a URL. Names become portable anchors for people, ideas, artifacts, communities, overlays, smart tags, and virtual spaces across the open web.

DP5 guides implementation, governance design, and future ML-RFC development for decentralized naming, meta-domain registration, namespace rights, conflict resolution, and interoperable naming semantics.

1. Problem Statement: Why Namespaces Matter

The contemporary web depends on naming systems, but most participant-facing names are not truly participant-owned. Handles, usernames, pages, tags, groups, channels, and platform identities are typically rented from centralized services. They can be revoked, shadowed, duplicated, impersonated, renamed, captured, or made non-portable by the platforms that host them.

This creates recurring failures:

- participants build identity and reputation around names they do not control
- communities lose continuity when platforms change rules or shut down access
- artifacts cannot be reliably addressed across tools
- names become vulnerable to spoofing, squatting, seizure, and censorship
- cross-system identity and ownership degrade because identifiers do not preserve meaning across contexts

DP5 reframes naming as civic infrastructure. A decentralized namespace is not only a convenience layer. It is the addressability substrate for identity, agency, data, commerce, interoperability, and governance.

Without DP5, the Meta-Layer cannot reliably answer basic questions:

- What is this person, persona, community, artifact, tag, overlay, or space called?
 - Who controls that name?
 - What does the name resolve to?
 - What rights, policies, and histories are bound to it?
 - How does the name remain interpretable across systems?
-

2. Core Principle of DP5

Names in the Meta-Layer must be portable, resolvable, governable, and resistant to capture. A namespace that cannot preserve identity, meaning, and control across systems becomes another platform dependency.

DP5 treats names as more than labels. Names are anchors for participation, reference, ownership, navigation, reputation, and coordination.

A DP5-aligned namespace must therefore support:

- participant-owned identifiers
 - community-owned identifiers
 - artifact and object identifiers
 - namespace portability across tools
 - conflict resolution and reservation logic
 - verifiable ownership and control
 - interpretable resolution across systems
 - resistance to squatting, spoofing, and seizure
-

3. Threats and Failure Modes

3.1 Platform-rented identity

Participants build identity around handles or pages that can be revoked, hidden, renamed, or monetized by platform operators.

Failure mode: identity continuity depends on platform permission.

3.2 Namespace capture

Dominant registries or intermediaries control which names are valid, visible, or resolvable.

Failure mode: decentralized naming becomes centralized gatekeeping.

3.3 Spoofing and impersonation

Attackers create visually, semantically, or structurally similar names to mislead participants.

Failure mode: names become attack surfaces for scams and trust abuse.

3.4 Squatting and speculative enclosure

Valuable names are claimed not for use, but to extract rents from future participants or communities.

Failure mode: addressability becomes enclosure before public value can form.

3.5 Semantic drift

A name carries one meaning in one system and a different meaning elsewhere without signaling.

Failure mode: identity, trust, or ownership claims are misinterpreted across contexts.

3.6 Registry fragmentation

Multiple naming systems emerge without interoperability or conflict-resolution pathways.

Failure mode: participants cannot know which namespace claims are authoritative, compatible, or contested.

3.7 Artifact ambiguity

Objects, tags, posts, paths, and digital artifacts cannot be reliably referenced across systems.

Failure mode: knowledge, provenance, and ownership degrade because identifiers are not stable.

3.8 Non-human namespace ambiguity

AI agents, organizations, bots, and autonomous systems operate without clear namespace rights or management structures.

Failure mode: non-human actors become hard to distinguish, govern, or hold accountable.

4. Primary Namespace Objects

4.1 Meta-domains

Meta-domains address virtual spaces within the Metaweb, similar to how traditional domains address web spaces.

A meta-domain may refer to:

- a participant-controlled overlay space
- a community zone
- a smart-tag namespace
- an application surface
- a virtual or conceptual space
- a bridge between a traditional domain and Meta-Layer objects

Example forms include:

- `boeing.com.web4`
- `apple.com.web4`
- `example.com.meta`
- `<label>.example.com.meta`

Meta-domains can link seamlessly to the broader web while functioning within the Metaweb overlay framework.

4.2 Personal identifiers

Personal identifiers address participants and personas, similar to email addresses, handles, or DIDs, but portable across Meta-Layer contexts.

Example:

- `shiftshapr.web4`
- `@jaime`
- `@jaime/artifact99`

Personal identifiers may be connected to decentralized identifiers (DIDs), credentials, proof-of-humanity mechanisms, or zone-specific identity contexts.

4.3 Digital artifact identifiers

Digital artifacts may serve as identifiers, assets, or NFTs that can be bought, sold, transferred, authenticated, or referenced.

Artifacts may include:

- posts
- paths
- smart tags
- annotations
- media objects
- overlays
- lists

- navigation components
- assertions
- credentials

4.4 Name chains

Name chains provide structured, semantic identifiers such as:

- @user/object
- @community/tag
- @publisher/claim

Name chains function as URI-like trust anchors, combining identity verification, object authentication, and conflict resolution pathways.

4.5 Decentralized URIs and well-known paths

DP5 also recognizes publisher-controlled decentralized URI patterns, such as:

- /.well-known/trust.txt

These allow trusted data to be anchored under existing publisher-controlled domains without requiring centralized registries.

5. Namespace System Layer: Resolution, Ownership, and Trust Semantics

This section is upgraded to define DP5 as a **runtime-resolvable, multi-layer namespace system** rather than a static registry.

A DP5-compliant namespace operates across three coupled layers:

5.1 Naming Layer (Syntax)

Defines canonical forms, label rules, and human-readable structure.

Examples:

- <label>.example.com.meta
- @user
- @user/object

Requirement:

- deterministic parsing
- canonical normalization

Failure mode: syntactic ambiguity.

5.2 Resolution Layer (Mapping)

Maps names → entities (people, agents, artifacts, spaces).

Resolution MUST include:

- multi-source resolvers (not single authority)
- explicit resolver provenance
- deterministic fallback rules
- verifiable resolution outputs

Resolution states MUST be visible:

- verified
- provisional
- disputed
- forked
- quarantined

Failure mode: invisible resolution override.

5.3 Control Layer (Authority)

Defines who can act on a name.

Control may derive from:

- cryptographic keys (wallets)
- DIDs
- registry attestations
- community governance

Control MUST be:

- transferable
- revocable
- auditable

Failure mode: ghost control (no accountable owner).

5.4 Meaning Layer (Semantics)

Defines what the name *means* in context.

Includes:

- identity type (human, org, agent, artifact)
- trust signals
- governance policies
- reputation bindings

Meaning MUST:

- travel with the name
- degrade visibly across contexts

Failure mode: semantic drift.

5.5 Temporal Layer (History)

Names are not static. They evolve.

Systems MUST preserve:

- ownership history
- resolution changes
- dispute timelines
- transfers

Failure mode: history erasure.

These five layers together define a **full-stack namespace**. Most systems today only implement 1–2 layers. DP5 requires all five.

5.10 Name Resolution Protocol (Reference Model)

DP5 requires a shared resolution model so names can be interpreted consistently across tools, overlays, registries, and communities.

A reference resolution flow SHOULD include:

1. **Normalize** the name into canonical form.
2. **Validate** syntax, reserved status, and structural constraints.
3. **Query** one or more resolvers or registries.
4. **Return state**: verified, provisional, disputed, forked, quarantined, retired, or unresolved.
5. **Attach provenance**: resolver source, timestamp, signatures, registry snapshot, and confidence level.
6. **Apply local policy**: zone rules, trust profiles, community reservations, and agent constraints.

7. **Render interface signal:** show the participant what the name means, what is uncertain, and what actions are safe.

Resolution must therefore be understood as a governed process, not a hidden lookup.

A minimal response object SHOULD include:

```
{
  "name": "@jaime/artifact99",
  "canonical": "@jaime/artifact99",
  "entity_type": "artifact",
  "controller": "did:example:123",
  "state": "verified",
  "resolver": "registry.example.meta",
  "resolved_at": "2026-04-26T00:00:00Z",
  "provenance": {
    "snapshot": "graph-snapshot-v12.json",
    "checksum": "sha256:...",
    "signatures": ["..."]
  },
  "policy": {
    "transferable": true,
    "dispute_status": "none",
    "zone_constraints": []
  }
}
```

Failure mode: **opaque resolution**, where a name appears trustworthy but participants cannot inspect how that trust was produced.

5.11 Namespace State Machine

DP5-aligned systems SHOULD expose name lifecycle states explicitly.

A name may move through the following states:

```
available → claimed → provisional → verified → active
                        ↘ disputed → resolved
                        ↘ quarantined → restored / retired
active → transferred → active
active → expired / abandoned → available or archived
active → forked → parallel-governed state
```

Core states:

- **available:** no active claim exists

- **claimed:** a participant or community has initiated registration
- **provisional:** claim exists but is pending verification or policy review
- **verified:** claim has met required proof conditions
- **active:** name is resolvable and usable
- **disputed:** competing claims or safety concerns exist
- **quarantined:** name is constrained due to suspected abuse, spoofing, or policy conflict
- **transferred:** control has changed with receipt
- **retired:** name is no longer active but history is preserved
- **forked:** multiple governance contexts recognize different meanings or controllers

State transitions **MUST** produce receipts where they affect ownership, resolution, trust, or participant expectations.

Failure mode: **state invisibility**, where participants cannot tell whether a name is safe, contested, provisional, or compromised.

5.12 Namespace Attack Taxonomy

DP5 treats names as attack surfaces. Naming systems concentrate trust, discovery, ownership, and memory, making them attractive targets for adversarial actors.

Common attacks include:

5.12.1 Spoofing

Attackers register visually or semantically similar names to impersonate trusted entities.

Examples:

- `appIe.meta` using confusing characters
- `paypaI.web4`
- near-match community names

Required response: similarity detection, warnings, and dispute pathways.

5.12.2 Squatting

Actors claim names for speculative rent extraction rather than use.

Required response: reservation rules, renewal logic, staking, decay, or community challenge mechanisms.

5.12.3 Resolver capture

A resolver becomes a de facto authority by controlling defaults.

Required response: multi-source resolution, resolver provenance, and fallback transparency.

5.12.4 Ownership laundering

A name is transferred repeatedly to obscure harmful history, evade sanctions, or reset trust.

Required response: transfer receipts and visible lineage.

5.12.5 Semantic hijacking

A name is used in a new context to imply trust or meaning it did not originally carry.

Required response: semantic profiles and degradation signaling.

5.12.6 Agent impersonation

Automated or AI actors adopt names that imply human identity, authority, or community status.

Required response: mandatory entity classification and controller binding.

5.12.7 Registry amnesia

A registry loses or suppresses prior state, disputes, or ownership transitions.

Required response: append-only logs, snapshots, checksums, and independent archival.

5.12.8 Namespace flooding

Attackers create many names to overwhelm discovery, governance, or trust review.

Required response: rate limits, economic friction, proof thresholds, and anti-spam containment.

5.13 Reference Patterns and Compatibility

DP5 does not replace existing naming systems. It defines how meta-layer naming can interoperate with them.

DNS-style names

DNS provides global resolution and familiar domain semantics, but is vulnerable to centralized registrar control and does not natively express trust state, provenance, or community governance.

DP5 can use DNS-linked anchors while adding overlay-level trust semantics.

DID-style identifiers

DIDs support decentralized identity and controller binding, but may be difficult for ordinary participants to read or remember.

DP5 can bind human-readable names to DID-backed controllers.

ENS / SNS-style naming

Blockchain naming systems support ownership and transfer, but often emphasize asset ownership more than contextual governance, dispute visibility, or semantic profiles.

DP5 can learn from these systems while requiring visible state, provenance, and governance.

Ordinals / BRC333-style artifacts

Ordinal inscriptions and BRC333-shaped artifacts can anchor durable metadata and namespace records.

DP5 can use inscription ordering, registry snapshots, and graph facts to support canonicity and provenance.

Well-known URI anchors

Publisher-controlled paths such as `/.well-known/trust.txt` allow existing domain holders to publish trust-relevant metadata without centralized registry dependency.

DP5 can compose these with meta-domain records and overlay resolution.

The goal is not one namespace to rule them all. The goal is interoperable naming with explicit trust semantics.

6. Meta-Domain Registry Architecture

A Meta-Domain Registry may operate as a public ingest and registry service for meta-domains, SNS-shaped JSON, BRC333-related ordinals, and related naming artifacts.

A registry can include:

- block-tip polling
- indexer search through services such as UniSat or pluggable alternatives
- candidate classification
- structural graph facts
- quarantined trust edges
- versioned graph snapshot releases with checksums

Such a registry SHOULD distinguish between:

- structural facts that can be merged freely
- trust assertions that require quarantine or policy review
- local drafts that remain outside public canonical state

6.1 Candidate classes

Candidate logs may include:

- `.meta` tokens

- BRC333-shaped bodies
- SNS-alias JSON candidates
- tag registry records
- anchor records
- name-chain references

6.2 Registry outputs

A registry SHOULD provide:

- canonical records
- candidate logs
- graph snapshots
- checksums
- dispute or quarantine markers
- provenance for resolver decisions

6.3 Canonicity and ordering

Where ordinal inscriptions are used, canonicity SHOULD be determined by documented ordering rules, such as lowest qualifying inscription number where adopted, rather than arbitrary string ordering.

6.4 Pluggable indexers

Registries SHOULD support pluggable indexers and resolution sources so that namespace infrastructure does not depend on a single data provider.

Failure mode: **indexer dependency**, where resolver integrity depends on one commercial or centralized API.

7. Registration Rules and Validation

DP5 supports concrete registration rules for product-level namespaces.

7.1 Canonical form example

A v1 product canonical form may be:

`<label>.example.com.meta`

Where:

- `<label>` is the registrant-chosen segment
- `example.com.meta` is a configurable suffix or canonical parent

7.2 Label rules

A valid label SHOULD satisfy:

- lowercase ASCII letters, digits, and hyphen only (a-z, 0-9, -)
- no underscores or Unicode in v1 unless explicitly expanded later
- must not start or end with -
- length between 1 and 63 characters
- must not be an integer-only label
- must not be reserved

7.3 Structural validation regex

Label-only validation:

```
^(?!-)([a-z0-9](:[a-z0-9-]{0,61}[a-z0-9])?)$
```

Full-domain validation example:

```
^(?!-)([a-z0-9](:[a-z0-9-]{0,61}[a-z0-9])?)\.example\.com\.meta$
```

Regex alone is insufficient. Reserved-name and integer checks must be separate.

7.4 Recommended validation order

1. Parse `label` as the substring before the configured suffix.
2. Reject if label fails length or charset validation.
3. Reject if label matches `/^\d+$/`.
4. Reject if `label.toLowerCase()` is in the reserved set.
5. Reject if the normalized domain is already in use.
6. Accept and register.

7.5 In-use rule

“In use” means that a record already exists with the same normalized domain string under the applicable active status rule.

Storage and comparison SHOULD normalize to lowercase.

8. Interoperability and Tradeability

Tradeable meta-assets allow participants to exchange digital spaces, objects, and names, fostering virtual commerce and community formation.

However, tradeability must remain bounded by trust, provenance, and governance.

DP5 requires:

- transfer receipts
- provenance preservation
- dispute visibility
- constraints on reserved or protected names
- compatibility with commerce and incentive systems (DP6, DP9)
- alignment with identity and accountability requirements (DP1)

A name may be tradable, but the trust attached to the name cannot be treated as a blank commodity. Reputation, history, and community meaning must remain visible across transfer.

9. Governance, Accountability, and Agency Surfaces

This section is upgraded to define **interface-level namespace governance**, aligning with the Meta-Layer's overlay architecture.

Namespaces are not governed only in registries. They are governed at the **point of interaction** via overlays, filters, and community rules. ←filecite→turn17file2↑

9.1 Participant-facing surfaces

Participants **MUST** be able to:

- claim names under transparent rules
- see resolution paths (who resolved this name?)
- inspect ownership and controller state
- view trust signals and classification (human, agent, org)
- transfer names with receipts
- initiate disputes
- view dispute status in real time

Failure mode: invisible governance.

9.2 Overlay-mediated governance

Meta-layer overlays **SHOULD** expose:

- name provenance tooltips
- impersonation warnings
- namespace conflicts
- transfer history
- community annotations

This turns naming into a **live civic surface**, not a backend database.

Failure mode: governance hidden behind APIs.

9.3 Community governance powers

Communities **MUST** be able to:

- reserve names
- define namespace policies
- classify entities
- enforce local naming norms
- quarantine suspicious identities
- fork namespace rules when needed

Failure mode: centralized naming authority.

9.4 Dispute visibility

All conflicts **MUST** be visible as states, not silent overrides:

- competing claims
- impersonation reports
- trademark conflicts
- governance disagreements

Failure mode: silent winner-take-all resolution.

9.5 Interface as enforcement boundary

DP5 aligns with the principle that governance must live at the interface layer, where users experience identity and trust.

Browser overlays act as **civic membranes** where naming rules become visible, contestable, and enforceable in real time. ←filecite→turn17file3↑

Failure mode: governance only enforced off-screen.

10. Community Signals Informing DP5

Community submissions and aligned work point to several recurring signals:

- desire for publisher-controlled trust anchors, such as decentralized URIs under existing domains

- need for namespace rights and management for non-human entities and AI systems
- interest in hierarchical, semantically meaningful names independent of physical nodes
- support for trust-schema-driven semantics and context-aware permissions
- demand for name chains as resolvable trust anchors for people, objects, and assertions
- need for decentralized identifiers and tags for posts, paths, and navigation objects
- concern that community identifiers may be seized, censored, or captured by platforms

These signals indicate that DP5 is not only about naming people. It is about naming the structure of the Metaweb itself.

11. Non-Goals and Explicit Boundaries

DP5 does not:

- require one global namespace for all contexts
- replace DNS, DIDs, ENS, SNS, BRC333, ordinals, or publisher-controlled URIs
- guarantee that all valuable names will be available
- eliminate disputes, squatting, or fraud completely
- treat tradeability as superior to stewardship
- require real-name identity
- collapse human, AI, organizational, and artifact namespaces into one undifferentiated model

DP5 defines conditions under which naming remains interoperable, governable, and accountable across systems.

12. Minimum DP5 Alignment (Upgraded Baseline)

This section is upgraded from descriptive to **testable compliance criteria**.

A system claiming DP5 alignment **MUST** pass the following checks:

12.1 Deterministic naming

- Canonical forms are defined and machine-verifiable
- Validation produces identical results across implementations

Test: same input → same validity result everywhere

12.2 Multi-source resolution

- Names resolve via more than one possible source
- Resolver provenance is exposed

Test: user can inspect where resolution came from

12.3 Explicit state signaling

Every name MUST expose state:

- verified
- provisional
- disputed
- quarantined
- forked

Test: UI or API returns state metadata

12.4 Ownership auditability

- Current controller is identifiable
- Transfer history is preserved

Test: ownership lineage query returns full chain

12.5 Portability

- Names function across at least 2 independent systems

Test: same name resolves meaningfully in multiple contexts

12.6 Anti-spoofing safeguards

- System detects or flags similarity-based impersonation

Test: registering visually similar name triggers warning or constraint

12.7 Registry memory

- Historical snapshots exist with integrity proofs

Test: past state can be reconstructed with checksum validation

12.8 Dispute mechanism

- Users can initiate and observe disputes

Test: dispute creates visible state transition

12.9 Non-human classification

- Agents and automated systems are distinguishable from humans

Test: entity type is explicitly encoded and exposed

These criteria transform DP5 from a principle into a **verifiable standard**.

These conditions define the minimum viable namespace layer of the Meta-Layer.

13. Open Questions and Future Work (Refined)

DP5 now identifies key frontier problems:

13.1 Cross-namespace interoperability

How do independent naming systems interoperate without collapsing into monopoly or fragmentation?

13.2 Semantic portability

How can meaning travel with names across cultural, linguistic, and technical contexts?

13.3 Anti-squatting mechanisms

What mechanisms balance open access with protection against speculative enclosure?

13.4 AI-native identity

How should agent identities evolve as they gain autonomy, persistence, and economic activity?

13.5 Namespace governance forks

What legitimacy models determine when a community can fork naming rules?

13.6 Human-readable vs machine-secure naming

How do we balance usability with cryptographic robustness?

13.7 Unicode and global inclusion

How do we support multilingual naming without increasing spoofing risk?

13.8 Economic design

What pricing, staking, or decay mechanisms prevent hoarding while preserving ownership?

These questions define the path toward ML-RFC maturation.

14. Relationship to Other Desirable Properties

DP5 is foundational and interdependent.

- **DP1** depends on identifiers that bind identity and accountability without forcing platform control
- **DP2** depends on participant control over names, handles, personas, and namespace portability
- **DP4** depends on identifiers that preserve data provenance without forcing cross-context correlation
- **DP6** depends on tradeable meta-assets, domains, and artifacts that preserve ownership and settlement integrity
- **DP7** depends on canonical forms and resolution semantics that work across systems
- **DP9** depends on attribution and contribution identifiers that cannot be trivially spoofed or replayed
- **DP12–DP13** depend on agent and tool identifiers that can be constrained, audited, and governed
- **DP14–DP15** depend on provenance and transparency for names, artifacts, and registry changes
- **DP18–DP20** depend on community and ownership identifiers that persist across governance and migration

A failure in DP5 propagates upward. If names cannot be trusted, ownership cannot be trusted, identity fragments, and interoperability becomes deception.

15. Path Toward ML-RFC

Progression toward ML-RFC maturity should include:

- standardized canonical forms for meta-domains, personal identifiers, and artifact identifiers
- shared validation rules and reserved-name policies
- resolver and registry schemas
- dispute-state semantics
- transfer receipt formats

- registry snapshot standards with checksums
- conformance tests for resolution integrity, ownership binding, and namespace portability
- implementation evidence from registry prototypes and overlay applications

Stable portions may be promoted first, especially canonical form rules, validation logic, and registry state semantics.

16. Closing Orientation

DP5 is the claim that participants, communities, agents, and artifacts deserve names they can carry across the Meta-Layer.

Without sovereign names, identity is rented, ownership is fragile, and discovery remains platform-shaped.

With DP5, people can claim space, communities can preserve continuity, artifacts can be addressed, and the Metaweb can become navigable without surrendering naming power to a single platform.

DP5 turns names into civic infrastructure.

To claim your space in the Meta-Layer is not merely to register a label. It is to anchor presence, meaning, and accountability in a shared world.

DP6 – Commerce

1. Purpose of This Draft

This draft articulates Desirable Property 6 (DP6) as the condition under which value exchange in the meta-layer is fair, legible, and aligned with human and community flourishing, not reducible to extraction, dark patterns, or attention rents dressed as markets.

Commerce is not an add-on to the meta-layer. It is how incentives (DP9), ownership (DP20), and sustainability (DP17) become real.

DP6 therefore defines the minimum conditions under which exchange remains trustworthy at the interface where money moves and value is recognized.

Commerce is the highest-value attack surface in the meta-layer. Systems that appear fair locally but fail across system boundaries, delegation, or routing layers will be systematically exploited.

DP6 connects to:

- DP4 (data minimization in payments and ads)
- DP2 (participant agency and consent)
- DP9 (incentive alignment)
- DP12 (executable policy at checkout)
- DP13 (containment of automation and abuse)
- DP15 (receipts and provenance)
- DP17 (financial sustainability)

If DP6 is weak, predictable failures follow: surveillance advertising as hidden tax, fee obfuscation, lock-in through proprietary rails, platform capture of creator revenue, and AI-mediated manipulation at conversion.

DP6 does not prescribe a specific payment rail, currency, or chain. It defines legitimacy conditions for commerce in the meta-layer.

2. Problem Statement

In today's web, commerce is fused with attention manipulation and opacity.

Participants encounter:

- pricing that cannot be verified before commitment
- fee stacks that are only visible after settlement
- payment flows bundled with surveillance and data extraction
- AI systems that optimize for conversion rather than user welfare

This produces recurring failures:

- surprise charges and coercive conversion flows
- creator and worker value hidden behind opaque fee structures
- failed micropayment systems due to friction and rent extraction
- communities unable to capture value from activity they host

These failures are structural. When transparency and portability are optional, extraction dominates.

DP6 reframes commerce as governed exchange: visible rules, bounded automation, and enforceable fairness at the point of transaction.

3. Threats and Failure Modes

3.1 Fee fog and stacked take rates

Participants cannot see effective take rates across layers.

Example: A creator sees a platform fee but not downstream processing, ranking, or conversion costs.

Why this matters: Fairness requires full distribution visibility.

3.2 Dark patterns at conversion

Design exploits cognitive load to increase conversion.

Example: A free trial converts with hidden renewal and hard-to-find cancellation.

Why this matters: Coercive commerce violates DP2 agency.

3.3 Attention rents disguised as markets

“Free” is subsidized by hidden data extraction.

Example: Ad SDKs exfiltrate unrelated data to fund access.

Why this matters: Externalities shift cost to users without consent (DP4).

3.4 Lock-in through wallets and closed loops

Value cannot exit proprietary rails.

Example: Credits and reputation cannot be ported without loss.

Why this matters: Undermines DP7 interoperability and DP4 export.

3.5 AI-mediated financial harm

Agents steer decisions without accountability.

Example: Assistants recommend higher-commission products as “best”.

Why this matters: Requires DP11 disclosure and DP13 bounds.

3.6 Community extraction without reciprocity

Economic activity uses community trust without funding it.

Example: Marketplaces leverage forums without contributing to moderation or safety.

Why this matters: Violates DP20 ownership and DP17 sustainability.

3.7 Cross-border confusion

Fees, taxes, and currencies are unclear.

Example: Display currency differs from settlement with hidden spreads.

Why this matters: Legibility must include jurisdictional honesty.

3.8 Predatory targeting

Dynamic pricing or credit targets vulnerable users.

Example: BNPL prompts cluster around financially stressed users.

Why this matters: Intersects DP4 inference limits and DP11 ethics.

3.9 Platform self-preferencing in AI commerce

Default assistants route to house inventory.

Example: “Buy now” prioritizes sibling brands under “personalization”.

Why this matters: Requires disclosure and contestability.

3.10 Cross-system commerce distortion

Commerce terms, attribution, or protections change or become exploitable when transactions, identities, or value move across systems.

Example: A checkout with full fee disclosure exports to a partner flow where additional fees are added post-commitment, or a refund policy is not honored after handoff.

Why this matters: Commerce that fails at system boundaries enables arbitrage and hidden extraction.

4. Core Principle

Commerce in the meta-layer is fair only when pricing, fees, risks, and responsibilities are legible at the point of exchange; defaults are non-exploitative; automation and AI assistance are accountable and bounded; and communities can shape and, where appropriate, capture economic activity that depends on shared trust and infrastructure.

Example: A checkout shows item price, full fee breakdown, data uses tied to the transaction, AI involvement, cancellation path, and any community surcharge with a link to governance.

What this feels like: Paying feels like choosing, not being herded.

Commerce guarantees must remain valid not only within a single interface, but across the systems, tools, and contexts where transactions are initiated, routed, and settled.

When commerce flows cross systems, pricing, fees, policies, and responsibilities must either:

- persist with integrity, or
- degrade in a way that is visible and contestable

Commerce that becomes opaque at handoff points is structurally unsafe.

5. Primary Mechanisms and Structural Conditions

5.0 Commerce Layer: Execution, Proof, and Settlement

Commerce in the meta-layer cannot rely on interface clarity alone. It must be anchored in a substrate that binds pricing, allocation, and settlement to verifiable, enforceable structures. Without this, even well-designed interfaces can be subverted downstream, where actual value movement occurs.

DP6 therefore defines a commerce layer composed of primitives that make exchange legible, auditable, and governable across the full lifecycle of a transaction, from intent to settlement to dispute.

Transaction objects

Transactions are not ephemeral UI events. They are structured, machine-readable objects that encode the full economic reality of an exchange. This includes itemized pricing, discounts, fee distribution, data use, jurisdictional context, and the governing policies at the moment of commitment.

By binding these elements together, transaction objects ensure that what a participant sees at checkout is not merely descriptive, but enforceable. This is what connects commerce to executable policy (DP12).

A failure mode here is selective omission or delayed disclosure, where certain fees or conditions are only materialized after commitment. Systems must therefore ensure that transaction objects are complete before commitment and immutable afterward except through governed updates.

Fee and allocation objects

Fee structures must be computable and attributable prior to commitment, not inferred after settlement. This includes platform fees, payment processing costs, ranking or promotion adjustments, and any community-level surcharges.

Rather than presenting fees as a flat percentage or opaque deduction, the system must expose how value is distributed across all participating layers. This enables participants to understand not only the price, but the economic structure behind it.

The primary attack surface here is fee fragmentation, where costs are distributed across layers in ways that are individually legible but collectively opaque. DP6 requires that fee objects recombine into a clear total cost of participation.

Settlement and flow proofs

Settlement must be provable, not assumed. Participants and communities need to be able to trace how value moved: who received what, when, and under what conditions.

Settlement proofs create continuity between intent and outcome. Without them, systems can display one set of expectations at checkout and execute another at settlement.

Failure modes include delayed settlement visibility, partial disclosure of recipients, or selective omission of intermediary captures. Systems must treat settlement as a first-class observable state, not a backend detail.

Commerce policy binding

Zones and communities attach policy objects directly to transactions. These are not advisory rules; they execute at runtime and shape what is allowed, required, or disallowed in a transaction.

This includes restrictions on categories, required disclosures, surcharge logic, and AI usage constraints. The key shift is that commerce behavior is governed at the point of execution, not after harm occurs.

The adversarial pressure here is policy bypass through integration pathways. Transactions must carry their governing policy context with them across systems or explicitly signal where enforcement no longer holds.

AI disclosure and constraint hooks

AI participation in commerce must be visible, attributable, and bounded. This includes pricing adjustments, ranking decisions, bundling, and negotiation.

Participants must be able to distinguish between neutral presentation and optimized persuasion. Without this, AI becomes a hidden incentive layer that distorts economic decisions.

A critical failure mode is undisclosed optimization, where AI systems steer users toward outcomes that maximize commission or retention while presenting themselves as neutral assistants.

Exit and portability primitives

Participants must be able to leave economic relationships without disproportionate friction. This includes exporting receipts, transaction history, and where supported, balances or entitlements.

Portability is not only a convenience feature; it is a constraint on extraction. Systems that make exit costly create economic capture even when pricing appears fair.

The adversarial pattern here is soft lock-in, where exit is technically possible but practically costly due to fragmentation of records, identity, or value.

Dispute and evidence bundles

Every transaction should produce an evidence bundle that can be used in dispute resolution. This includes signed receipts, policy state at the time of purchase, and clear identification of responsible parties.

This shifts dispute resolution from subjective interpretation to evidence-based adjudication. It also creates incentives for systems to maintain accurate and complete records.

Failure modes include missing policy context, unverifiable receipts, or ambiguity about responsibility across multi-party transactions.

Commerce memory

Commerce systems must retain a structured memory of transactions and policy evolution. This includes prior fee structures, changes in rules, and the outcomes of disputes.

This historical layer enables accountability over time and allows communities to detect drift toward extraction or manipulation.

Without commerce memory, each transaction exists in isolation, making systemic issues difficult to detect and correct.

These primitives transform commerce from a series of isolated interactions into a coherent, governable system that can withstand adversarial pressure and evolve over time.

5.1 Fee and take-rate transparency

Fee transparency is not simply about showing a number. It is about ensuring that participants can understand the full economic structure of a transaction before they commit.

In many systems, fees are fragmented across layers—platform fees, payment processing, ranking adjustments, or currency spreads—each of which may be individually legible but collectively opaque. This creates a situation where the true cost of participation cannot be known until after settlement.

DP6 requires that effective fees be computable before commitment, not reconstructed after the fact. Participants must be able to see the full take rate across all layers in a way that is coherent and comparable.

A core failure mode is fee fragmentation, where systems distribute costs in ways that obscure total extraction. Systems must therefore recombine all fee components into a clear, pre-commitment view of total cost.

5.2 Honest defaults and reversal paths

Defaults are one of the most powerful levers in commerce systems. When defaults are misaligned, even transparent systems can become coercive in practice.

DP6 requires that material commitments be opt-in, not opt-out, and that reversal paths such as cancellation or refund follow the same level of friction as signup. This creates symmetry between entry and exit, which is essential for real agency (DP2).

A common failure mode is asymmetrical friction, where signup is immediate but cancellation is buried, delayed, or requires additional steps. Systems must treat reversibility as a first-class design constraint, not a secondary feature.

5.3 Separation of payments and surveillance

Commerce systems often bundle payment with data extraction, turning transactions into opportunities for surveillance. This creates hidden costs that are not reflected in price.

DP6 requires that payment does not require unrelated data processing (DP4). The data required to complete a transaction must be limited to what is strictly necessary, and any additional data use must be explicitly disclosed and optional.

A key failure mode is covert bundling, where data collection is technically optional but practically unavoidable. Systems must ensure that participants can complete transactions without consenting to unrelated data flows.

5.4 Interoperable value rails

Value must be able to move without losing meaning, ownership, or accountability. When value is trapped within proprietary systems, participants are subject to platform-defined rules that cannot be contested or exited.

DP6 therefore prefers open protocols and requires export and audit capabilities for closed systems (DP7). Participants must be able to move balances, receipts, and transaction history without losing integrity.

A primary failure mode is economic lock-in, where value can technically be withdrawn but at significant loss or friction. Systems must treat portability as a constraint on extraction, not an optional feature.

5.5 Creator and worker fairness

Commerce systems depend on contributors whose work generates value, yet those contributors are often the least protected participants in the system.

DP6 requires that attribution and payouts be tamper-evident and that disputes be resolved in a timely and transparent manner. Contributors must be able to verify how their work is valued and compensated.

A failure mode here is delayed or opaque payout logic, where contributors cannot trace how their compensation was calculated or why it changed. Systems must ensure that payout logic is both visible and contestable.

5.6 Community economic surfaces

Communities create the conditions under which commerce is trusted, yet often lack the ability to shape the economic activity that depends on them.

DP6 enables zones to impose rules, surcharges, or bans with executable policy (DP12). This allows communities to align commerce with their values and to capture a portion of the value generated within their environments.

A key failure mode is extraction without reciprocity, where economic activity leverages community trust without contributing to its maintenance. Systems must ensure that communities can define and enforce economic participation terms.

5.7 High-stakes commerce pathways

Not all transactions carry the same level of risk. High-stakes categories such as financial products, healthcare, or legal services require additional safeguards.

DP6 requires human confirmation or expert gating for sensitive categories. This ensures that automation and AI do not make consequential decisions without appropriate oversight.

A failure mode is over-automation, where systems optimize for efficiency at the cost of safety. Systems must introduce friction where necessary to prevent harm.

5.8 Sustainability linkage

Commerce systems do not exist in isolation. They depend on shared infrastructure, communities, and public goods that must be maintained over time.

DP6 requires that fees transparently fund commons maintenance (DP17). This creates a visible link between economic activity and the sustainability of the systems that support it.

A failure mode is invisible extraction, where value is removed from ecosystems without reinvestment. Systems must make sustainability contributions explicit and traceable.

5.9 Receipts and dispute evidence

Receipts are not merely confirmations of payment. They are the foundation of accountability in commerce systems.

DP6 requires machine-readable receipts that support fair resolution (DP15). These receipts must include sufficient detail to reconstruct the transaction and its governing conditions.

A failure mode is incomplete or unverifiable receipts, which make disputes difficult or impossible to resolve. Systems must treat receipts as evidence, not just records.

5.10 Accessibility of economic surfaces

Commerce must be accessible to all participants, regardless of ability, device, or connectivity constraints.

DP6 requires that checkout and transaction flows work across assistive technologies and low-bandwidth contexts. Accessibility is not only a usability concern, but a fairness constraint.

A failure mode is exclusion by design, where systems assume high bandwidth, modern devices, or specific interaction patterns. Systems must ensure that economic participation is not gated by technical privilege.

5.11 Cross-system commerce integrity (DP7 alignment)

Commerce systems must preserve the relationship between price, fee, policy, and settlement across environments.

This includes:

- maintaining itemized pricing and fee visibility across handoffs
- preserving policy bindings (refunds, cancellations, disclosures) across integrations
- ensuring receipt and provenance continuity across tools (DP15)
- signaling when guarantees or protections change in a new environment

Commerce must not be portable in ways that enable hidden fees, policy resets, or accountability gaps.

5.12 Agent-to-agent commerce integrity

Commerce is increasingly mediated not just by humans, but by agents acting on behalf of participants. These agents may search, negotiate, bundle, and execute transactions across multiple systems without direct human interaction at each step.

DP6 requires that agent-to-agent commerce remains legible, bounded, and accountable at the same level as human-facing transactions.

This includes:

- ensuring agents carry explicit mandates, constraints, and budget limits from their principals
- preserving transaction visibility such that participants can inspect decisions made on their behalf

- binding agent actions to the same policy, fee, and disclosure requirements as direct transactions
- preventing agents from exploiting speed, scale, or opacity to bypass safeguards

A critical failure mode is delegated opacity, where agents transact in ways that are technically valid but practically uninspectable, allowing hidden fees, biased routing, or exploitative bundling.

Systems must ensure that delegation does not reduce accountability. Agent-mediated commerce must remain reconstructable, auditable, and interruptible by participants and governance systems.

6. Governance, Accountability, and Agency Surfaces

Commerce is not neutral infrastructure. It encodes choices about power, risk, and value distribution, often in ways that are invisible to participants. In many systems, these choices are embedded in defaults, routing logic, or fee structures that cannot be contested or even observed.

DP6 requires that these choices become visible and governable at the point of transaction. This means participants are not only protected from harm, but able to understand and shape the economic conditions they are subject to.

Without these surfaces, commerce systems revert to extraction: participants transact, but cannot see how value flows, cannot challenge outcomes, and cannot meaningfully exit or redirect the system.

Participants must be able to:

- see true prices, fees, and net outcomes before commitment
- understand what data is used in the transaction and revoke unrelated scopes (DP4)
- identify AI involvement in pricing, ranking, or negotiation (DP11)
- access clear cancellation, refund, and dispute pathways
- export receipts and transaction history where honest (DP7, DP15)

Participants must also be able to understand how commerce terms change across systems.

This includes:

- whether pricing and fee structures persist across integrated flows
- whether policies such as refunds, disputes, and disclosures carry over
- where accountability shifts between parties during multi-system transactions

Participants must not be required to reverse-engineer commerce conditions across environments.

Communities must be able to:

- define allowed and disallowed commerce patterns within zones
- impose disclosures, caps, or surcharges via executable policy (DP12)
- audit aggregate outcomes for fairness, capture, and harm (without exposing individuals)
- evolve rules with memory of prior incidents and outcomes (DP12 governance loops)

Example: A community zone bans predatory lending ads and requires fee disclosures for all financial products. Enforcement occurs at runtime through policy binding, not moderator memory.

7. Incentives and Power Analysis

Commerce determines where value accumulates. Incentives determine how that value is pursued.

In many systems, pricing and ranking are optimized for conversion and revenue, while governance claims prioritize safety or fairness. This creates a structural mismatch where economic incentives override stated values.

DP6 requires incentive legibility within commerce systems:

- how ranking, promotion, or bundling affects price and visibility
- how commissions, fees, or partnerships influence recommendations
- how optimization targets (conversion, revenue, retention) shape outcomes

Example: A marketplace ranks products based on commission rather than relevance, while presenting results as “best match.”

Why this matters: When incentives are hidden, markets become extraction systems. When visible, they become governable.

DP6 therefore expects:

- disclosure of materially relevant incentive structures
- the ability for communities to constrain or rebalance those incentives
- alignment between commerce incentives and governance rules (DP3, DP12)

Power also concentrates when commerce fragments across systems.

Actors who control routing, aggregation, or interface layers can reshape pricing, visibility, or policy without formally changing them.

DP6 therefore requires resistance to cross-system arbitrage, where value is extracted through boundary manipulation rather than contribution or service.

8. Community Signals Informing DP6

Across ecosystems, consistent signals point to structural failures in commerce design. These are not isolated grievances, but recurring patterns that reveal where systems break under real-world use.

Participants repeatedly encounter environments where economic behavior is shaped by hidden incentives, opaque fee structures, and limited recourse. Over time, this produces erosion of trust, disengagement from participation, and migration toward systems perceived as more fair or legible.

Common signals include:

- creators reporting opaque fee stacks and unpredictable income
- users expressing fatigue with subscription traps and hidden renewals
- demand for receipt-level clarity on who was paid and why
- interest in local economic rules for civic, educational, and community spaces
- concern that AI-mediated shopping optimizes for commission rather than user welfare

These signals reflect a deeper pattern: participants are not rejecting commerce itself, but commerce that is misaligned with their understanding of fairness, agency, and contribution.

DP6 treats these signals as design inputs, not complaints. They indicate where economic systems fail to align with human expectations and where intervention is required.

9. Non-Goals and Explicit Boundaries

DP6 does not:

- mandate a single currency, ledger, or payment rail
- eliminate all forms of advertising (it requires honesty, bounds, and contestability)
- replace financial regulation or tax law
- guarantee equal economic outcomes

DP6 defines the conditions under which exchange is legitimate and non-coercive.

10. Minimum Alignment (Non-Normative)

A DP6-aligned commerce system should, at minimum:

- present itemized pricing and full fee breakdowns before commitment
- expose data uses tied to transactions and allow revocation of unrelated scopes
- label AI involvement in pricing, ranking, or recommendation
- provide cancellation and refund pathways with parity to signup friction
- generate verifiable, machine-readable receipts with responsible parties (DP15)
- bind transactions to enforceable policy and dispute mechanisms (DP12, DP13)
- ensure pricing, fees, and policy bindings persist or explicitly degrade across systems
- maintain receipt and transaction continuity across tools (DP7)
- prevent hidden fee introduction or policy resets during cross-system flows

Partial compliance that omits execution, auditability, or exit should not be treated as alignment.

11. Open Questions and Future Work

Key open questions include:

- how to reconcile cross-border commerce with local community rules and norms
- how to provide stable units of account without sacrificing accessibility or neutrality
- how to standardize receipt portability across wallets and platforms (DP7)
- how to provide meaningful transparency in ranking without enabling gaming
- how to fund public goods through commerce without creating new forms of extraction
- how to assign liability when AI agents mediate transactions (DP11–DP13)

These questions sit at the intersection of economic design, governance, and law.

12. Relationship to Other Desirable Properties

DP6 connects commerce to the full meta-layer system:

- DP2 ensures participant agency at checkout and over subscriptions
- DP3 defines how commerce rules evolve through governance
- DP4 constrains data use in payments and advertising
- DP7 enables portability of receipts, balances, and history
- DP9 aligns incentives with non-extractive exchange
- DP11–DP13 bound AI behavior in commerce contexts
- DP15 provides verifiable receipts and auditability
- DP17 ensures commerce contributes to sustainable infrastructure
- DP20 defines how surplus and value flows are owned and governed

DP6 is where these properties converge into real economic behavior.

13. Foresight and Failure Design

DP6 assumes that commerce systems will be pressured toward opacity, capture, and manipulation. This pressure increases with scale, competition, and the introduction of automation and agent-mediated transactions.

Common failure paths include:

- reintroduction of hidden fees and bundled costs under competitive pressure
- AI-driven persuasion targeting vulnerable users at conversion points
- consolidation of value within closed payment or loyalty systems
- divergence between displayed prices and actual settlement outcomes

These failures rarely occur as singular events. They emerge gradually as systems optimize for short-term metrics, allowing small deviations from fairness to accumulate into systemic extraction.

DP6 therefore requires designing safeguards in advance:

- circuit breakers for harmful pricing or targeting patterns
- policy-based enforcement at transaction time
- rate limits and constraints on high-risk automation
- public postmortems linking failures to rule and system changes

Designers must also assume that commerce will be attacked at system boundaries.

Common boundary failure paths include:

- hidden fees introduced during cross-system routing
- policy resets when transactions move between providers
- accountability gaps between integrated services
- divergence between displayed and settled outcomes across systems

In agent-mediated environments, additional failure modes emerge:

- agents colluding or routing through opaque pathways to maximize hidden incentives
- delegated decision-making that exceeds user intent or understanding
- rapid transaction loops that bypass human oversight entirely

DP6-compliant systems include detection, signaling, and governance responses for these failures. They treat commerce not as a static system, but as an adversarial environment that must be continuously monitored and corrected.

Failure is expected. Invisible failure is not.

14. Path Toward ML-RFC

Advancing DP6 toward ML-RFC requires:

- standardizing transaction, fee, and receipt schemas
- publishing reference checkout patterns with full disclosure models
- piloting community-defined economic rules and surcharges
- aligning dispute evidence with provenance standards (DP15)
- collaborating with regulators, platforms, and civil society on interoperable approaches

Progress should be demonstrated through working systems, not only conceptual agreement.

15. Closing Orientation

DP6 is where the meta-layer proves whether it can support real economic life without reverting to extraction.

Fair commerce is not the opposite of innovation. It is the condition under which innovation remains trustworthy.

When DP6 is strong, participants can pay, earn, and fund commons with clarity, agency, and dignity.

When it is weak, every other property is undermined at the moment money changes hands.

DP6 is where the meta-layer proves it is not building another opaque marketplace.

Fair commerce is the condition under which innovation does not depend on misdirection.

When DP6 is strong, people can pay, earn, and fund commons without trading away dignity or control.

Commerce that fails under interoperability pressure is not fair commerce. DP6 requires that fairness survive movement.

DP7 – Interoperability

1. Purpose of This Draft

This draft articulates Desirable Property 7 (DP7) as the condition under which participants, communities, and systems can move across tools, environments, and contexts without losing identity, history, value, or agency.

Interoperability in the meta-layer is not a feature of APIs. It is the condition under which power cannot be quietly re-centralized through infrastructure, interfaces, or economic gravity.

This includes the ability for new tools, services, and technologies to plug into the meta-layer through shared interfaces without prior permission, provided they conform to governance rules, security constraints, and boundary conditions.

DP7 ensures that governance (DP3), incentives (DP9), ownership (DP20), and commerce (DP6) remain continuous across boundaries rather than collapsing into platform-specific silos.

If DP7 is weak, predictable failures are not accidental but structural: re-centralization through convenience, lock-in through partial openness, stranded value, degraded identity, and governance that cannot extend beyond a single interface.

DP7 defines the minimum conditions under which the meta-layer functions as a shared system rather than a collection of connected but incompatible domains.

Anchor Principle: Open to participation, bounded by policy. Any system may join, but no system may bypass the rules that preserve safety, trust, and collective integrity.

Interoperability is the primary boundary where power is contested in the meta-layer. Systems that appear interoperable locally but degrade meaning, enforceability, or usability across boundaries will systematically re-centralize control.

2. Problem Statement

In today's web, systems are technically connected but structurally discontinuous.

Participants encounter:

- identities that can be exported but lose reputation, trust, or verification context
- data that can be downloaded but cannot be meaningfully re-imported
- value that can move but becomes illiquid, discounted, or restricted
- governance rules that do not carry enforcement into new environments

This produces recurring failures:

- switching systems resets social and economic position

- contributions lose meaning outside their original context
- communities cannot extend norms or coordination across tools
- “open” ecosystems still trap participants through semantic or economic loss

These failures are structural. Interoperability is treated as data transfer rather than continuity of meaning, enforceability, and legitimacy.

DP7 reframes interoperability as continuity under movement and participation. Not just that things move, but that they remain valid, trusted, and usable after they move, and that new systems can enter and participate without breaking that continuity.

3. Threats and Failure Modes

3.1 Interoperability theater

Systems simulate openness while preserving control.

Example: Data export tools provide raw files without structure, signatures, or compatibility, making re-use impractical.

Why this matters: Nominal interoperability without usability reinforces lock-in.

3.2 Semantic collapse

Objects move but lose meaning.

Example: Reputation exports as a number but loses trust graph context, rendering it unusable.

Why this matters: Meaning, not data, determines continuity.

3.3 Governance discontinuity

Rules travel without enforcement.

Example: A community charter exports but cannot bind behavior in a new environment.

Why this matters: Governance without execution becomes symbolic.

3.4 Economic degradation

Value moves but loses utility or legitimacy.

Example: Assets transfer but cannot be used, traded, or redeemed equivalently.

Why this matters: Portability without usability is economic lock-in by another name.

3.5 Asymmetric interoperability

Systems favor inbound over outbound movement.

Example: Easy onboarding from other platforms, but restricted or degraded export.

Why this matters: Asymmetry preserves centralization under the appearance of openness.

3.6 Protocol and standards capture

Dominant actors influence interoperability standards.

Example: Standards evolve in ways that favor incumbent implementations.

Why this matters: Interoperability can become a mechanism of control rather than liberation.

3.7 Bridge and translation attack surfaces

Cross-system connectors introduce risk.

Example: Bridges become vectors for fraud, duplication, or corruption of assets and identity.

Why this matters: Interoperability expands the attack surface of the system.

3.8 Privacy leakage through portability

Data movement exposes unintended information.

Example: Exported data reveals relationships or behavior not intended for new contexts.

Why this matters: Continuity must not violate data sovereignty (DP4).

3.9 Re-centralization through aggregation layers

Intermediaries regain control over open systems.

Example: Wallets, indexes, or identity providers become gatekeepers of interoperability.

Why this matters: Interoperability without anti-capture design recentralizes power.

3.10 Injection risk and extension ecosystem spam

Pluggable systems introduce code injection, privilege abuse, and discovery flooding risks.

Example: Malicious overlays request excessive permissions, spoof trusted interfaces, or exfiltrate data; low-quality or duplicate apps flood discovery surfaces to capture attention.

Why this matters: Openness without containment and fair discovery degrades safety and usability, turning interoperability into an attack surface.

3.11 Closed or permissioned extension ecosystems

Systems restrict which technologies can integrate despite claiming openness.

Example: Platforms provide SDKs but require approval, ranking control, or distribution gatekeeping that limits who can participate.

Why this matters: Interoperability without permissionless participation reintroduces centralized control through integration policy rather than infrastructure.

4. Core Principle

Interoperability in the meta-layer means that identity, data, value, governance, and participation can move across systems without losing integrity, meaning, enforceability, or legitimacy.

It is not sufficient for objects to move. They must remain:

- interpretable within new contexts
- verifiable against their origin
- enforceable under governance rules
- usable in economic and social systems

Example: A participant moves to a new environment and retains not only identity and history, but the ability to exercise governance rights, receive recognition for contributions, and use their assets without degradation.

What this feels like: Switching systems does not mean starting over, nor does it mean accepting a degraded version of prior participation.

5. Primary Mechanisms and Structural Conditions

5.0 Interoperability Layer: Continuity, Translation, and Power

Interoperability requires more than shared formats. It requires a layer that preserves meaning, authority, and usability across boundaries where systems may have conflicting incentives.

Portable objects

All core system elements must be representable as portable, structured objects:

- identity objects (DP1)
- policy objects (DP12)
- incentive objects (DP9)
- ownership objects (DP20)
- transaction objects (DP6)

These objects must carry sufficient context to remain interpretable outside their origin.

They should also declare their intended use and trust assumptions so receiving systems can enforce appropriate constraints. Without declared intent, objects may be misapplied in contexts that invalidate their meaning.

Semantic translation and mapping

Systems must define how objects are interpreted across contexts:

- schema translation
- semantic alignment
- version compatibility

Mappings must explicitly declare where information is preserved, transformed, or lost.

They should include machine-readable diffs of meaning so downstream systems can reason about equivalence. Absent explicit mapping, silent reinterpretation becomes a primary vector for drift and exploitation.

Integrity and lineage preservation

Objects must retain:

- authorship
- timestamps
- signatures
- lineage and derivation history

Without this, imported objects cannot be trusted or verified.

Lineage should be queryable across hops to reconstruct full transformation chains. Breaks in lineage must be flagged as risk, not treated as benign gaps.

Permission and consent continuity

Access controls and consent conditions must persist across systems.

Participants must not lose control over their data or identity during transfer (DP4, DP2).

Consent scopes should be renegotiable at boundaries with clear previews of changes. Implicit expansion of scope during transfer must be disallowed or explicitly surfaced.

Interoperability receipts

All cross-system transfers generate verifiable records:

- what moved
- how it was transformed

- what constraints applied
- who mediated the transfer

These receipts enable audit and dispute resolution (DP15).

Receipts should be machine-verifiable and linkable to governance and commerce records. Missing or partial receipts must downgrade trust for the resulting state.

Conflict resolution under power asymmetry

Systems must define how conflicts are resolved when:

- governance rules differ
- economic conditions diverge
- trust models are incompatible

Conflict resolution is not neutral. It must be visible, contestable, and governed.

Resolution pathways should declare precedence rules and appeal mechanisms. Opaque arbitration at boundaries is a primary route to capture.

Loss and degradation signaling

All interoperability pathways must explicitly signal:

- what meaning is lost
- what functionality is degraded
- what guarantees no longer apply

This prevents silent failure of continuity.

Signals should be standardized and user-visible at decision time, not buried in logs. Systems that cannot signal degradation must restrict the transfer or require explicit override.

Interoperability memory

All transfers and mappings persist as a linked history:

- prior versions
- transformation paths
- disputes and reversals

This creates a system-level memory of how interoperability evolves over time.

Memory should support querying for systemic patterns such as drift or repeated loss. Without analysis over memory, issues recur undetected.

Composable participation via governed interfaces

The meta-layer must allow third-party tools, services, and extensions (e.g., smart tags, overlays, sidebars, core services) to plug into shared interfaces **without prior permission**, provided they conform to declared interfaces and governance constraints.

Conforming integrations must:

- declare permissions and data scopes up front
- bind to zone policies at runtime (DP12)
- operate within containment tiers and rate limits (DP13)
- provide signed artifacts and provenance (DP15)
- expose auditable behavior and event logs

Non-conforming integrations must be sandboxed, rate-limited, or blocked.

This enables openness to innovation while preventing unbounded execution and platform capture.

Interface contracts should be versioned and testable, with conformance suites available publicly. Discovery systems must incorporate reputation and probation to resist spam and gaming.

Example (Composable Integration): A third-party sidebar app plugs into a community zone. At install, it declares permissions (read annotations, write highlights) and data scopes. The zone's policy automatically constrains it: external network calls are limited, access to private threads is denied, and actions are rate-limited. The app runs in a sandbox and emits signed event logs. Initially, it appears in a probation tier with limited visibility. As it accumulates positive, non-abusive usage and passes audits, its privileges and discoverability increase. If it violates policy, it is throttled or quarantined with a public receipt explaining why.

5.1 Open schemas and standards

Schemas are not neutral technical artifacts. They define what can be expressed, preserved, and validated across systems.

DP7 requires schemas to be publicly defined, evolvable, and resistant to capture. Systems must be able to extend schemas without breaking compatibility or consolidating control.

A failure mode is schema capture, where standards evolve to favor specific implementations, creating hidden lock-in despite nominal openness.

5.2 Version negotiation

As systems evolve, differences in schema versions and capabilities are inevitable. Interoperability must account for this without breaking continuity.

DP7 requires explicit version negotiation mechanisms that allow systems to detect compatibility, fallback gracefully, and signal limitations.

A failure mode is silent incompatibility, where objects appear valid but behave incorrectly due to version mismatch.

5.3 Data minimization in portability

Not all data should move across systems. Interoperability must balance continuity with privacy and relevance.

DP7 requires that only necessary data be transferred, with explicit controls for redaction and minimization (DP4).

A failure mode is over-export, where unnecessary data is transferred, increasing exposure and risk without improving usability.

5.4 Portable reputation and credentials

Reputation and credentials are only meaningful within their context. Moving them without context creates false signals or exploitation opportunities.

DP7 requires that reputation objects include sufficient context such as source, method, and scope to remain interpretable.

A failure mode is reputation flattening, where complex trust signals are reduced to simple scores that can be gamed or misused.

5.5 Cross-system governance mapping

Governance rules must translate across systems with clear expectations of enforcement.

DP7 requires explicit mapping between policy objects and enforcement environments, including where equivalence holds and where it does not.

A failure mode is governance drift, where rules appear consistent but are enforced differently depending on the system.

5.6 Economic portability

Value must remain usable, not just transferable. Economic objects must retain legitimacy across systems.

DP7 requires that transferred value preserves its functional properties, including redeemability, liquidity, and constraints (DP6).

A failure mode is value degradation, where assets lose usability or trust when moved.

5.7 Identity continuity

Identity is the anchor for all other portable objects. Without continuity, interoperability collapses into impersonation or reset.

DP7 requires consistent, verifiable identity across systems, with protections against duplication and spoofing (DP1).

A failure mode is identity fragmentation, where participants appear as different entities across systems, losing continuity of rights and responsibilities.

5.8 Anti-capture interop design

Interoperability systems must be designed to resist re-centralization through aggregation or coordination layers.

DP7 requires mechanisms that distribute control over indexes, relays, and discovery systems.

A failure mode is aggregation capture, where intermediaries become de facto gatekeepers of movement.

5.9 Privacy-preserving interoperability

Interoperability must not expose participants to unintended data leakage or correlation.

DP7 requires privacy-preserving mechanisms such as selective disclosure, anonymization, and context-aware data handling (DP4).

A failure mode is linkage attack, where cross-system data enables reconstruction of sensitive information.

5.10 Graceful degradation

Not all systems will fully support all features. Interoperability must handle partial compatibility explicitly.

DP7 requires systems to preserve core meaning and signal any degradation in functionality, guarantees, or enforcement.

A failure mode is silent degradation, where participants believe continuity exists but critical properties have been lost.

6. Governance, Accountability, and Agency Surfaces

Interoperability is not neutral infrastructure. It encodes decisions about what persists, what degrades, and who controls movement.

Without explicit governance surfaces, interoperability becomes a mechanism of extraction or control, where participants can move data but not meaning, value, or rights.

DP7 requires that cross-system movement be visible, contestable, and governed at the moment of transfer.

Participants must be able to:

- initiate export/import of core objects (identity, credentials, receipts, balances) with clear previews of what will change

- see how objects will be transformed (schema mapping, lossiness, policy constraints) before committing
- control permissions during transfer, including redaction, minimization, and revocation (DP4, DP2)
- verify integrity of received objects (signatures, lineage, timestamps) and reject degraded or untrusted imports (DP15)
- access dispute pathways when a transfer results in loss, misrepresentation, or policy violation

Communities must be able to:

- define interoperability policies for their zones (what can enter/leave, required attestations, risk tiers)
- attach executable constraints to imports (e.g., quarantine, limited privileges, probation periods) (DP12, DP13)
- audit aggregate interop flows for capture, leakage, and harm without exposing individual participants
- evolve interop rules with memory of incidents, reversals, and outcomes (DP12 governance loops)

Without these surfaces, interoperability becomes performative: objects move, but participants lose control and systems quietly re-centralize.

Example: A community allows import of reputation from another network only with attested receipts and places imported identities in a probation state with reduced privileges until local activity establishes trust.

7. Incentives and Power Analysis

Interoperability is where platform power is defended or broken.

Dominant actors tend to support inbound interoperability (ingest users/data) while restricting outbound portability (export of value, reputation, and history). This creates asymmetric openness that preserves control.

DP7 requires incentive legibility around interop:

- how APIs, SDKs, and terms of service shape inbound vs outbound flows
- how ranking, discovery, and monetization favor native over imported content
- how fees or friction are applied to export vs import paths

Common patterns to surface and constrain:

- **Fake export:** Data can be downloaded but loses structure, signatures, or context needed for reuse.
- **Rate-limited exit:** Export is throttled or degraded compared to import pathways.
- **Semantic loss:** Reputation or credentials export without the trust graph or scoring model that gives them meaning.
- **Bridge tolls:** Cross-system transfers incur hidden fees or spreads that discourage movement.
- **Aggregator capture:** Indexing or wallet layers recentralize control over “open” ecosystems.

DP7 therefore expects:

- symmetric capabilities for import and export where feasible
- disclosure of material asymmetries and fees

- community ability to demote or penalize systems that degrade outbound portability
- fair discovery and anti-spam mechanisms for pluggable tools (rate limits, probation states, reputation-weighted surfacing)
- transparent policies for SDKs and extensions that bind incentives to compliant behavior (DP9)

When incentives favor staying, ecosystems centralize. When incentives favor honest movement, ecosystems compose.

8. Community Signals Informing DP7

Across ecosystems, recurring signals indicate interop failure at scale:

- frustration with “export” tools that do not preserve meaning or usability
- loss of reputation or trust when moving between platforms
- inability to carry subscriptions, balances, or purchase history across tools
- skepticism toward “open ecosystems” that still gate exit
- concern about data exposure and identity spoofing during transfers

These are not UX issues. They are structural breaks in continuity.

DP7 treats these signals as requirements for preserving meaning, not just moving bytes.

9. Non-Goals and Explicit Boundaries

DP7 does not:

- require full compatibility between all systems or schemas
- mandate a single global standard or governing body
- eliminate competition among platforms or protocols
- force communities to accept all imports without policy

DP7 defines the conditions under which movement is legitimate, intelligible, and safe.

10. Minimum Alignment (Non-Normative)

A DP7-aligned system should, at minimum:

- support export of core objects with preserved structure, signatures, and lineage
- provide import pathways with explicit mapping, lossiness disclosure, and policy constraints
- maintain permission and consent continuity during transfers (DP2, DP4)
- generate interoperability receipts for all transfers (who, what, when, how transformed)
- offer dispute and rollback pathways for harmful or incorrect imports
- avoid material asymmetry between import and export without disclosure

Partial compliance that omits integrity, consent, or auditability should not be treated as alignment.

11. Open Questions and Future Work

Key open questions include:

- how to standardize semantics (reputation, credentials, governance roles) without flattening local meaning
- how to achieve Sybil resistance and anti-impersonation across interoperable identity systems (DP1)
- how to reconcile regulatory boundaries with cross-system value portability (DP6)
- how to design bridges that are both secure and usable without becoming choke points
- how to compensate shared infrastructure (indexes, relays) without enabling capture
- how to evolve schemas without breaking historical continuity (versioning and migration)

These questions sit at the boundary of protocol design, governance, and law.

12. Relationship to Other Desirable Properties

DP7 is the continuity layer across the meta-layer stack:

- DP3 ensures governance can evolve across systems rather than reset per platform
- DP4 constrains data movement and enforces minimization during transfer
- DP6 ensures commerce artifacts (transactions, balances, receipts) remain usable across contexts
- DP9 carries incentive history and attribution across tools without silo lock-in
- DP12 enables policy translation and enforcement across environments
- DP13 contains risks introduced by cross-system automation and agents
- DP15 provides verifiable provenance for imported objects
- DP17 sustains shared infrastructure required for interoperability
- DP20 preserves ownership and fork rights across environments

DP7 binds these properties into a coherent, cross-system reality.

13. Foresight and Failure Design

Interoperability systems must assume adversarial pressure, economic incentives for capture, and rapid evolution of tools and agents.

Failures rarely occur as isolated events. They emerge as gradual degradation of meaning, enforceability, and trust across system boundaries.

Common failure paths include:

- protocol capture, where standards evolve to favor dominant actors
- bridge exploits, where cross-system connectors become attack vectors

- impersonation and replay of identity or credential artifacts
- economic friction that discourages portability despite nominal support
- aggregation capture, where intermediaries regain control over movement

These failures compound over time. For example, minor semantic loss in early transfers reduces trust, which leads to reduced usage, which increases reliance on centralized intermediaries, accelerating re-centralization.

Cross-system environments introduce dynamic risks:

- divergence of rule interpretation across systems
- inconsistent enforcement depending on context
- loss of lineage or provenance during repeated transformations

DP7 therefore requires proactive safeguards:

- risk tiers and circuit breakers for high-risk interoperability pathways
- attestation and reputation systems for issuers and bridges
- anomaly detection for unusual transfer patterns (DP13)
- public postmortems linking failures to schema, policy, or bridge changes

Interoperability must detect not only discrete failures, but slow drift toward loss of meaning and control.

Failure is expected. Silent or irreversible failure is not.

14. Path Toward ML-RFC

Advancing DP7 toward ML-RFC requires:

- standardizing core object schemas (identity, policy, incentive, ownership, transaction)
- defining interoperability receipt formats and audit events
- building reference bridges and adapters with open security reviews
- piloting cross-system governance and commerce scenarios with real communities
- aligning with regulators on portability, custody, and liability boundaries

Progress should be demonstrated through live interop scenarios, not only specifications.

15. Closing Orientation

DP7 is the condition under which the meta-layer remains a network rather than reverting to a set of silos.

When interoperability is real, participants can move without losing meaning, value, or rights.

When it is simulated, ecosystems recentralize behind APIs, bridges, and aggregators that quietly control movement.

Interoperability is not a feature. It is the boundary where power either remains distributed or collapses back into platforms.

DP7 ensures the meta-layer is a network, not a set of islands.

Without it, all other properties collapse into silos.

With it, coordination becomes truly global and composable.

DP8: Community-Defined Participation & Governance Zones

1. Purpose of This Draft

This draft articulates Desirable Property 8 (DP8) as the condition under which communities can **define, enforce, and evolve participation and governance at the interface layer of the Meta-Layer**.

DP8 establishes that governance is not inherited from platforms, but constructed by communities operating within zones. It defines how participation, influence, and intelligence are structured so that trust remains contextual, enforceable, and resistant to manipulation.

DP8 is not moderation. It is the **system-level design of environments in which interaction occurs**.

2. Problem Statement

On today's web, participation and governance are platform-defined:

- rules are globally applied regardless of context
- moderation is opaque and centralized
- influence is driven by engagement dynamics, not trust
- bots and AI distort visibility and reputation
- communities cannot enforce their own norms at the system level

This leads to predictable failures:

- manipulation through brigading and synthetic amplification
- governance capture by small groups or opaque systems
- lack of trust in moderation and rule enforcement
- inability to adapt governance to context and risk

DP8 addresses this by enabling communities to define **zone-specific governance systems** that operate at the interface layer and persist across the web.

3. Core Principle

Communities must be able to define and enforce the conditions under which participation, influence, and intelligence operate. If governance cannot be enforced under scale, coordination, and adversarial pressure, trust collapses.

DP8 treats governance as an **interface-level control system** coupled to **identity (DP1)**, **agency (DP2)**, **data flows (DP4)**, and **AI containment (DP12)**.

It has three inseparable properties:

- **Outcome control (DP2)**: rules must change what actually happens (visibility, amplification, access), not merely configure preferences.
- **Continuity (DP1, DP4)**: governance must persist across pages, sessions, and interoperating systems, with explicit signaling on degradation.
- **Contestability (DP1)**: decisions must be attributable, auditable, and reversible within bounded processes.

Implications:

- Governance is **executed at the point of interaction** (overlays), not deferred to platform backends.
- High-impact actions (virality, reputation shifts, moderation) are **gated by proofs** appropriate to the zone (e.g., unique human verification per DP1, quorum, role authority per DP8).
- Automation (AI) is **subordinate to zone policy** with explicit scope, attribution, and revocation (DP12), and must honor data purpose and consent propagation (DP4).

Failure conditions (non-exhaustive):

- **Phantom governance**: rules exist but do not alter outcomes (violates DP2).
 - **Fail-open amplification**: under load or uncertainty, systems default to permissive amplification (violates DP8 + DP9 alignment).
 - **Uncontestable decisions**: participants cannot audit or appeal governance actions (violates DP1).
-

4. Core Principles

DP8 principles are **normative and enforceable**, and interlock with **DP1 (Identity)**, **DP2 (Agency)**, **DP4 (Data)**, and **DP12 (AI)**.

4.1 Self-Determination (Enforceable; DP2)

Communities **MUST** be able to define participation and governance rules that **bind execution**.

- Rules **MUST** be machine-enforceable at the interface layer.
- Governance artifacts **MUST** be versioned and attributable (DP1).

Failure mode: **declarative governance**.

4.2 Contextual Governance (Zone-Bounded; DP1, DP4)

Rules **MUST** adapt to domain, risk, and norms, and be **scoped to zones**.

- Systems **MUST** prevent silent carryover of rules across zones.
- Transitions **MUST** signal changes in guarantees (DP4).

Failure mode: **context collapse**.

4.3 Graduated Participation (Stateful; DP2)

Participation **MUST** be tiered with **stateful progression and decay**.

- Capabilities **MUST** map to tiers and be enforced.
- Progression requires **continuity of contribution**; decay prevents permanent lock-in.

Failure mode: **tier gaming / privilege ossification**.

4.4 Human-Centric Trust Anchoring (Proof-Gated; DP1)

High-impact actions **SHOULD** require **proofs tied to unique humans**.

- Amplification and governance votes **MUST** resist sybil and automation dominance.

Failure mode: **amplification spoofing**.

4.5 Interoperability (Truthful and Bounded; DP1, DP4, DP7)

Communities **MUST** persist across platforms with **honest signaling of what is preserved or degraded**.

- Identity, agency, and governance state **MUST** travel or explicitly degrade.

Failure mode: **interop deception**.

4.6 AI Situatability (Runtime-Bound; DP12)

AI **MUST** operate within **zone-defined constraints** with attribution, scope, and revocation.

Failure mode: **AI governance bypass**.

4.7 Precedence and Conflict Resolution (Deterministic)

Overlapping rules **MUST** resolve deterministically.

- Systems **MUST** declare precedence models.

Failure mode: **zone conflict ambiguity**.

4.8 Auditability and Recourse (First-Class; DP1)

Governance actions **MUST** be reconstructable and contestable.

Failure mode: **governance opacity**.

4.9 Safe Degradation (Fail-Safe Defaults; DP2, DP4)

Under uncertainty or attack, systems SHOULD degrade to **safer defaults**.

Failure mode: **fail-open under stress**.

5. System Architecture

5.1 Overlay-Based Governance

Governance operates at the interface layer through overlays (browser extensions, native integrations, or overlay apps), not within platform silos.

5.2 Core Primitives

- Identity (DP1)
- Agency (DP2)
- Data (DP4)
- Zones
- Governance Modules

5.3 Zone Model (DP1 Integration)

Zones are:

- policy containers attached to context
- composable and overlapping
- portable across the web

Each zone defines:

- participation thresholds
- governance rules
- AI permissions
- trust signals

5.4 Governance System Layer: Continuity, Enforcement, and Capture Resistance

Beyond participation models and governance modules, DP8 requires a coherent governance system layer that ensures community-defined rules remain **enforceable, portable, and resilient under scale and adversarial pressure**.

Governance is not simply declared. It must persist across contexts, resist manipulation, and remain legible and contestable over time.

5.4.1 Governance Continuity Across Zones

Governance rules must persist as participants move across:

- platforms
- pages
- applications
- overlapping zones

This requires:

- consistent enforcement across contexts
- signaling when guarantees change
- preservation of governance state

Failure mode: **governance fragmentation**

5.4.2 Enforcement at the Interface Layer

Governance must be enforced where interaction occurs.

Systems **MUST** ensure:

- rules apply before actions propagate
- violations are constrained in real time
- enforcement is visible and explainable

Failure mode: **phantom governance**

5.4.3 Cross-Zone Conflict Resolution

Systems **MUST** define:

- precedence models
- conflict signaling
- fallback states

Failure mode: **zone conflict ambiguity**

5.4.4 Governance Propagation

Rules must propagate with content, participants, and interactions.

Failure mode: **governance stripping**

5.4.5 Capture Resistance

Systems **MUST** mitigate:

- coordinated influence attacks
- role entrenchment
- opaque decision-making

Failure mode: **governance capture**

5.4.6 Anti-Brigading

Systems **MUST** detect and limit coordinated behavior.

Failure mode: **brigading**

5.4.7 Governance Memory and Auditability

Governance decisions **MUST** be reconstructable and contestable.

Failure mode: **governance opacity**

5.4.8 Governance Evolution and Forkability

Communities **MUST** be able to evolve and fork governance models.

Failure mode: **governance rigidity**

6. Participation Model

DP8 defines participation as a **tiered, stateful system** where capability, influence, and accountability increase with demonstrated behavior and verified identity properties (DP1), under enforceable governance (Section 5.4).

6.1 Tiered Participation (Capabilities Matrix)

Participation tiers **SHOULD** be explicit and machine-enforceable:

Tier	Capabilities	Constraints
Observer	Read, follow context	No amplification or governance actions
Contributor	Comment, annotate, submit content	Rate-limited; no virality control
Trusted Participant	Signal trust, influence ranking/visibility	Requires continuity and reputation thresholds
Steward	Moderate, adjudicate, configure rules	Requires strong identity guarantees and auditability

Systems **MUST** bind capabilities to tier and prevent out-of-band escalation.

6.2 Entry, Progression, and Decay

- Entry requirements **MAY** include consent, identity level, and basic behavior thresholds.
- Progression **MUST** require **verifiable contribution over time** (continuity, not bursts).
- Systems **SHOULD** implement **decay** (time-based or behavior-based) to prevent permanent privilege lock-in.

Failure modes: - **fast-track escalation** (gaming entry to gain influence) - **privilege ossification** (roles never decay)

6.3 Virality and Reputation Controls

High-impact amplification **SHOULD** require unique human verification.

Systems **MUST** remain stable under coordinated attempts to manipulate participation tiers, including bot-driven amplification, identity cycling, and reputation inflation. Participation models must ensure that influence cannot be rapidly accumulated without verifiable contribution and continuity.

Mechanisms **MAY** include: - amplification caps per identity/time window - quorum requirements for boosts (N unique humans) - reputation weighting with context binding

Failure modes: - **amplification spoofing** - **reputation laundering**

6.4 Cross-Zone Participation Semantics

- Participation status is **zone-scoped by default**.
- Systems **MUST** signal when a participant's tier in one zone does not transfer to another.
- Optional bridges **MAY** allow partial portability with explicit downgrade rules.

Failure mode: **cross-zone escalation**, where status in one zone illegitimately confers power in another.

6.5 Rate, Scope, and Safety Guards

- Systems **MUST** enforce rate limits and scope constraints proportional to tier.
- High-risk actions (mass messaging, mass tagging, bulk edits) require stricter proofs and/or stewards.

Failure mode: **throughput abuse**, where volume substitutes for trust.

7. AI Governance (DP12 Link)

DP8 requires that AI participation be **governed as a first-class actor class** within zones, with enforceable constraints at runtime and clear attribution aligned with DP1 and DP2.

7.1 AI Identity, Attribution, and Disclosure

- All AI agents **MUST** present a verifiable identity (issuer, operator, model class) and remain attributable for actions.
- AI-originated content and actions **MUST** be clearly labeled at the interface layer.
- Delegated agents **MUST** bind to a sponsoring human or organization (DP1 §8.3 equivalent), with visible responsibility.

Failure modes: - **identity masking** (AI indistinguishable from humans) - **attribution gaps** (no accountable party)

7.2 Scope-Limited Delegation and Control

- AI agents **MUST** operate within **explicit scopes** (read/write domains, amplification limits, interaction types) with TTL and renewal.
- Zones **MUST** define allowed capabilities per tier (e.g., no autonomous amplification without quorum).
- Participants **MUST** have a **kill switch** and bounded-time revocation.

Failure modes: - **scope creep** (agent expands authority) - **irrevocable delegation**

7.3 Amplification and Participation Constraints

- AI **MUST NOT** directly trigger high-impact amplification without **human-backed quorum or proofs**.
- Systems **SHOULD** cap AI-originated throughput and require stronger proofs for bulk actions.
- AI contributions **MAY** inform ranking, but **MUST** be **down-weighted or gated** relative to verified human signals where stakes are high.

Failure modes: - **AI amplification bypass** - **throughput dominance**

7.4 Interaction Safety and Interruptibility

- AI actions **MUST** be **interruptible, reversible (where feasible), and auditable**.
- High-risk actions (payments, legal commitments, public attributions) require **human-in-the-loop confirmation** unless explicitly authorized by zone policy.

Failure modes: - **automation overrun** - **irreversible AI actions without consent**

7.5 Data and Inference Boundaries (DP4 Link)

- AI **MUST** honor data purpose binding and consent propagation (DP4 §5.11).
- Inferences generated by AI are **first-class artifacts** with lineage, scope, and revocation/attenuation pathways.

Failure modes: - **inference misuse** - **consent bypass via pipelines**

7.6 Cross-Zone Behavior and Containment

- AI permissions are **zone-scoped by default**; cross-zone operation requires explicit reauthorization.
- Systems **MUST** signal when AI constraints change across zones.

Failure modes: - **cross-zone privilege leakage**

7.7 Observability and Audits

- Systems **MUST** provide logs of AI actions (who, what, when, scope) and summaries understandable to participants.
- Zones **SHOULD** publish **policy manifests** for AI (allowed actions, caps, escalation paths).

Failure modes: - **AI opacity**

8. Governance Composition

DP8 treats governance as a **composable system of modules** that **MUST** interoperate without bypassing enforcement (Section 5.4).

8.1 Module Types

Common modules include: - **Voting** (quorum rules, weighting) - **Moderation** (flags, queues, actions) - **Reputation** (signals, decay, context binding) - **Access Control** (roles, permissions) - **Dispute Resolution** (appeals, juries)

8.2 Composition Constraints (Required)

- Modules **MUST NOT** bypass the governance system layer (no direct amplification without checks).
- Outputs of one module **MUST** be **typed and scoped** before feeding another (e.g., a vote signal cannot directly amplify content without validation).
- Cycles **MUST** be bounded to prevent feedback loops (e.g., reputation → visibility → reputation).

Failure modes: - **module bypass** (side-channel influence) - **feedback loops** (runaway amplification)

8.3 Precedence and Policy Graph

- Systems **SHOULD** maintain a **policy graph** where modules declare inputs, outputs, and precedence.
- Conflicts between modules **MUST** resolve via declared precedence (or fall back to stricter rule wins).

Failure mode: **composition ambiguity**, where multiple modules conflict without resolution.

8.4 Forkability and Versioning

- Governance stacks **MUST** be forkable with clear version identifiers.
- Changes **MUST** be **versioned and auditable**, with migration paths for participants.

Failure mode: **silent rule drift**, where behavior changes without visibility.

8.5 Interoperability of Modules

- Modules **SHOULD** expose standard interfaces for signals (e.g., vote, trust, flag) to enable cross-community reuse.
- Interop **MUST** preserve context (zone, scope, guarantees) or explicitly degrade it.

Failure mode: **semantic mismatch**, where signals are misinterpreted across systems.

9. Security and Adversarial Considerations

DP8 assumes adversaries will combine **identity (DP1)**, **agency (DP2)**, **data flows (DP4)**, **governance (DP8)**, and **incentives (DP9)**. Systems **MUST** be robust to **multi-vector**, **cross-zone attacks** and degrade safely.

9.1 Threat Classes (Extended)

- **Sybil Attacks**: many identities controlled by few actors
- **Brigading**: coordinated surges to influence outcomes
- **Governance Capture**: concentration of power via roles or opaque processes
- **Reputation Laundering**: reshaping signals across contexts to gain undue trust
- **AI Amplification**: automated agents scaling influence beyond constraints
- **Cross-Zone Escalation**: importing status or signals to bypass local rules

9.2 Composed (Multi-Vector) Attacks

Adversaries may combine: - AI agents + human click-farms - identity cycling + cross-zone escalation - incentive exploits (rewards) + feedback loops - data laundering (DP4) + reputation reuse (DP8)

Systems **MUST** detect **correlated anomalies** across time, topology, and identity linkages.

Failure mode: **composed attack success**, where individually mitigated vectors succeed in combination.

9.3 Detection Signals and Telemetry

- **Temporal**: burstiness, synchronized actions, unusual cadence
- **Topological**: tightly clustered interactions, graph anomalies
- **Behavioral**: repetitive patterns, low-entropy content, abnormal conversion rates
- **Cross-Context**: sudden tier jumps across zones, inconsistent identities

Systems **SHOULD** fuse signals into risk scores with **explainable summaries**.

9.4 Response Playbooks

- **Progressive friction**: rate limits, proof escalation, cooldowns
- **Containment**: quarantine zones, shadow reduction of amplification

- **Rollback**: revert affected rankings or decisions where feasible
- **Human review**: escalate high-impact cases with auditable decisions (DP1 linkage)

Failure mode: **delayed or blunt response** causing collateral damage or missed containment.

9.5 Transparency vs. Gaming

- Provide participant-legible explanations and audit summaries
- Protect sensitive thresholds and heuristics

Failure modes: - **gaming via overexposure** - **opacity via underexposure**

9.6 Cross-Zone Containment and Signal Sharing

- Attacks are **zone-scoped by default**; sharing of sanctions/signals MUST be deliberate and thresholded
- Systems SHOULD support **signed, scoped advisories** between zones

Failure modes: - **cascading harm** (over-sharing) or **blindness** (under-sharing)

9.7 Incentive Alignment (DP9 Link)

- Systems MUST minimize rewards for abusive behavior (no easy profit from spam/brigades)
- Rewards SHOULD be tied to **verified, sustained contribution**

Failure mode: **perverse incentives** that fund attacks

9.8 Resilience and Safe Degradation

- Under uncertainty, systems SHOULD degrade to **safer defaults** (reduced amplification, higher proof requirements)
- Maintain service continuity while limiting harm

Failure mode: **fail-open amplification** under stress

10. Minimum Alignment (Non-Normative)

Minimum alignment is not a feature checklist. It is the threshold at which governance is **enforceable, portable, and resistant to manipulation, capture, and coordination attacks**.

A system that does not meet these conditions may expose governance features, but it does not provide meaningful community control.

At minimum, a system claiming DP8 alignment MUST satisfy the following **irreducible conditions**:

10.1 Zone-Based Enforcement

- Governance rules **MUST** be enforced at the interface layer within defined zones
- Rules **MUST** apply before actions (visibility, amplification, moderation) propagate
- Systems **MUST** signal when zone protections are absent or degraded

Failure mode: **phantom governance**

10.2 Participation Integrity

- Participation tiers **MUST** map to real differences in capability and influence
- High-impact actions (e.g., virality, reputation boosts) **MUST** require stronger identity guarantees (e.g., unique human verification where appropriate)
- Systems **MUST** prevent rapid escalation of influence without earned progression

Failure mode: **participation gaming**

10.3 Governance Continuity

- Governance state (roles, permissions, reputation) **MUST** persist across pages, sessions, and supported systems
- Systems **MUST** signal when continuity breaks

Failure mode: **governance fragmentation**

10.4 Capture Resistance

- Systems **MUST** include mechanisms to detect and mitigate coordinated influence, role entrenchment, and opaque decision concentration
- Governance actions **MUST** be attributable and reviewable

Failure mode: **governance capture**

10.5 Anti-Brigading Protections

- Systems **MUST** detect anomalous participation patterns and coordinated behavior
- Influence spikes **MUST** be rate-limited or require stronger proofs

Failure mode: **brigading**

10.6 Governance Propagation and Boundary Signaling

- Governance context **MUST** travel with content, participants, and interactions where technically feasible
- Systems **MUST** signal when governance constraints are lost or degraded across boundaries

Failure mode: **governance stripping**

10.7 Auditability and Contestability

- Participants **MUST** be able to inspect governance decisions and their effects
- Systems **MUST** provide mechanisms to challenge or appeal decisions

Failure mode: **governance opacity**

10.8 AI Governance Enforcement

- AI actions **MUST** adhere to community-defined constraints
- Systems **MUST** visibly distinguish AI participation and enforce scope limits

Failure mode: **AI governance bypass**

These conditions define the **minimum viable governance layer** of the Meta-Layer.

Partial implementations that omit enforcement, continuity, or capture resistance **MUST NOT** be considered aligned with DP8.

11. Open Questions

Open questions focus on cross-DP integration and operationalization:

11.1 Cross-Zone Conflict Models (DP1, DP4)

- What precedence models are most legible and safe (stricter-wins vs user-selected vs negotiated)?
- How should conflicts be surfaced without overload?

11.2 Reputation Portability vs Context (DP2, DP8)

- What minimal signals can travel without enabling laundering?
- How should decay and re-qualification work across zones?

11.3 AI Policy Manifests (DP12)

- What is the minimal, machine-readable schema for zone AI policies?
- How are capabilities negotiated across zones?

11.4 Governance Module Standards (DP7)

- Which module interfaces should be standardized for interoperability?
- How to prevent semantic drift across implementations?

11.5 Data–Governance Coupling (DP4)

- How should consent and purpose binding propagate with governance actions (e.g., moderation, ranking)?

11.6 Incentive Alignment (DP9)

- What reward models avoid funding abuse while sustaining participation?
-

12. Path Toward ML-RFC

Advancement from ML-Draft to ML-RFC for DP8 requires **demonstrated, adversarially-tested governance systems operating across identity (DP1), agency (DP2), data (DP4), and AI constraints (DP12)**.

This is not a documentation milestone. It is an **operational validation threshold**.

12.1 Reference Implementations (End-to-End Zones)

At least one fully functional governance zone **MUST** be implemented with:

- Enforced participation tiers with progression and decay (DP2)
- Identity-bound roles and attribution for all governance actions (DP1)
- Data-aware moderation and ranking (purpose-bound, consent-propagating) (DP4)
- AI agents constrained at runtime with visible scope and revocation (DP12)

The implementation **MUST** demonstrate that governance rules **change outcomes in real time**, not post-hoc.

12.2 Adversarial Conformance Testing

Systems **MUST** pass structured tests simulating real attack conditions:

- **Sybil + brigading attacks** → no uncontrolled amplification
- **Cross-zone escalation attempts** → no illegitimate privilege transfer
- **Reputation laundering attempts** → no unbounded trust carryover
- **AI amplification bypass** → no autonomous virality without required proofs
- **Governance stripping across interop boundaries** → no silent loss of constraints

Results **MUST** be documented and reproducible.

12.3 Interoperability Proofs (DP7 Alignment)

Governance systems **MUST** demonstrate:

- Transfer of governance state (roles, signals, constraints) between at least two independent implementations
- Explicit signaling of **what is preserved vs degraded** during transfer
- No silent reinterpretation of governance semantics across systems

This ensures governance is not platform-bound.

12.4 Auditability and Evidence Artifacts

Systems **MUST** produce auditable artifacts demonstrating:

- Attribution of governance actions (who acted, under what authority) (DP1)
- Outcome impact (how rules changed visibility, amplification, or access) (DP2)
- Data compliance (how consent and purpose were preserved) (DP4)
- AI behavior logs (what actions agents took and under what constraints) (DP12)

Artifacts **SHOULD** include: - structured logs - participant-readable summaries - dispute/appeal traces

12.5 Governance Evolution and Forking Evidence

Communities **MUST** demonstrate the ability to:

- Modify governance rules without breaking continuity
- Fork governance models and continue operation
- Migrate participants across versions with explicit signaling

This proves governance is **adaptive rather than brittle**.

12.6 Multi-Community Adoption

At least two or more independent communities **MUST**:

- Operate distinct governance configurations
- Demonstrate real usage under different risk and cultural contexts
- Show evidence of governance effectiveness and evolution

This ensures DP8 is not optimized for a single use case.

12.7 Criteria for Promotion to ML-RFC

DP8 may be promoted when:

- Governance is proven enforceable under adversarial conditions
 - Cross-DP integration (DP1, DP2, DP4, DP12) is validated in practice
 - Interoperability is demonstrated with explicit degradation semantics
 - Communities can evolve and fork governance without system failure
 - Participants can understand, audit, and contest governance outcomes
-

13. Closing Orientation

DP8 defines the conditions under which communities become **sovereign coordination environments** rather than passive audiences.

Without enforceable governance, trust collapses into manipulation.

With it, the Meta-Layer becomes a **civic substrate for collective intelligence**.

DP9 – Developer and Community Incentives

1. Purpose of This Draft

This draft articulates Desirable Property 9 (DP9) as the condition under which developers and communities receive credible, legible incentives to build, maintain, and improve meta-layer capabilities, without collapsing into extraction, engagement-only rewards, or token theater that severs contribution from accountability.

DP9 is how interoperability (DP7), meta-communities (DP8), education (DP10), commerce (DP6), and ownership (DP20) become sustained rather than hobbyist. It connects to DP4 (privacy-preserving attribution where needed), DP2 (participant agency over incentives they accept), DP17 (financial sustainability), and DP11–DP13 (safe incentives for AI-assisted development and operations).

If DP9 is weak, predictable failures follow: burnout maintainers, underfunded commons, predatory grant programs, misaligned metrics (growth over safety), capture of incentive systems by whales, and automated spam farms gaming rewards.

DP9 does not prescribe a single tokenomics paper. It defines minimum legitimacy conditions for incentive systems: clarity, fairness, auditability, exit, and alignment with the meta-layer’s human-first goals.

2. Problem Statement

In today’s web, builders often face misaligned incentives: metrics reward engagement and growth while externalizing harm (misinformation, addiction, privacy loss). Open-source maintainers carry systemic risk with little capture of value. Communities that host quality spaces rarely receive durable upside from the ecosystems they enable.

In practice, this produces recurring failures:

- bounty programs that pay for volume, inviting spam and low-quality output
- grants with opaque selection and slow payout, discouraging small teams
- platform APIs that change terms after dependency forms
- creator funds that function as marketing, not structural revenue share
- AI tools that accelerate low-effort contribution farming unless bounded (DP13)

These failures are structural: when incentives are opaque or misaligned, the fastest path to reward is usually not the best path for participants. DP9 reframes incentives as governance objects: measurable, contestable, and evolvable, parallel in spirit to DP12’s insistence that rules be executable and revisable.

3. Threats and Failure Modes

3.1 Metric capture

Teams optimize what is measured, often engagement, at the expense of safety, truth, and inclusion.

Example: A developer fund ranks submissions by user minutes, incentivizing addictive mechanics.

Why this matters: DP9 requires multi-metric incentive design with explicit tradeoffs and red lines.

3.2 Incentive laundering

Rewards flow to intermediaries instead of contributors, or to shell identities.

Example: A community rewards pool is drained by coordinated sockpuppets farming tasks.

Why this matters: This needs DP1 accountability and DP13 rate and containment patterns.

3.3 Opaque allocation

Participants cannot see why someone was funded or ranked. Suspicion corrodes cooperation.

Example: A hackathon winner is an insider portfolio company with undisclosed relationships.

Why this matters: DP14 transparency norms apply to incentive governance too.

3.4 Maintainer extraction without support

Corporations build on commons without returning maintenance, security, or governance labor.

Example: Critical libraries burn out maintainers while enterprises profit.

Why this matters: DP9 expects credible reciprocity mechanisms, not moral appeals alone.

3.5 Education and onboarding gaps

Incentives assume skills participants do not have, excluding global majority builders.

Example: Only teams fluent in a niche stack can compete for integration grants.

Why this matters: Fair incentives require reachable on-ramps and mentorship surfaces, with a clear DP10 connection.

3.6 Token-based confusion

Tokens substitute for clear rights and clear work, creating regulatory and UX hazards.

Example: Governance tokens are issued without enforceable decision rights or measurable duties.

Why this matters: DP9 allows tokens as one instrument, not a synonym for incentives.

3.7 Sponsored open source capture

Corporate roadmaps steer commons toward vendor stacks. Incentives reward integrations that reduce portability.

Example: A grant prioritizes one cloud's proprietary APIs over open interfaces.

Why this matters: DP7 interoperability is an outcome incentive designers must protect, not accidentally punish.

3.8 Reviewer burnout and queue collapse

Underfunded programs flood reviewers. Quality decisions become random or biased by who shouts loudest.

Example: A security bounty program misses critical reports while reviewers chase low-risk noise for points.

Why this matters: Incentive design must include capacity models for evaluation, not only payout curves.

3.9 Geographical and language bias

English-first rubrics and US-centric eligibility exclude global majority talent.

Example: A hackathon requires on-site presence in one city for finals.

Why this matters: Fairness requires reachable participation surfaces, with a clear DP10 connection.

4. Core Principle

Developer and community incentives in the meta-layer must be transparent in allocation and metrics; aligned with safety and interoperability outcomes; resistant to gaming and capture; reciprocal toward commons maintenance; and evolvable through participatory governance, with bounded automation and clear accountability.

Incentive systems should feel like public infrastructure, not casino tables.

Example: A grants round publishes rubric weights, reviewer conflicts, payout schedule, and post-hoc impact report, with appeals for mistaken rejections.

What this feels like: You can learn the game without insider knowledge, and trust that gaming the game hurts you more than the community.

Without this: Builders rationally exit or optimize the wrong surface.

5. Primary Mechanisms and Structural Conditions

5.0 Incentive Layer: Allocation, Signal, and Enforcement

Incentives in the meta-layer are not abstract reward schemes. They are operational systems that allocate value across participants, tools, and infrastructure in response to real activity.

In many systems, incentives fail because they are not bound to contribution quality, are not auditable, or are captured by intermediaries. Rewards become detached from value creation, and systems devolve into gaming contests.

DP9 therefore requires a shared incentive layer composed of primitives that make incentives legible, enforceable, and aligned with governance, interoperability, and ownership across systems.

Incentive objects

Incentives must be represented as structured, machine-readable objects.

An incentive object includes:

- metric: what is being measured (e.g. endorsements, retention, contributions)
- weight: how much that metric matters
- constraints: what disqualifies or reduces reward
- eligibility: who or what can earn
- decay and clawback: how rewards adjust over time based on outcomes

This allows incentives to be governed, versioned, and enforced (DP12, DP3).

Incentive objects must also declare their scope, context, and transferability so receiving systems can enforce appropriate constraints (DP7). Without declared intent, incentives may be misapplied or exploited when moved across environments.

A failure mode is metric ambiguity, where participants cannot determine what actions produce rewards, leading to manipulation or disengagement.

Contribution binding

Rewards must be bound to verifiable contributions rather than surface-level signals.

This requires linkage to identity (DP1), traceability of actions, and resistance to duplication or replay.

Contribution binding ensures that value flows to actual work rather than its simulation.

A failure mode is contribution spoofing, where low-effort or automated actions are mistaken for meaningful input.

Attribution and lineage

Incentive systems must track contribution lineage across time and systems.

Rewards should reflect original creation, downstream reuse, and derivative contributions.

Lineage enables fair distribution of value across ecosystems rather than concentrating rewards at endpoints.

A failure mode is attribution collapse, where downstream actors capture disproportionate value due to missing lineage.

Reward event splitting

Each value-generating event should distribute rewards across contributing layers.

This includes the primary contributor, interface layer, access layer, and shared infrastructure.

Reward splitting ensures that invisible layers of contribution are not systematically undercompensated.

A failure mode is endpoint capture, where only the visible actor receives rewards despite reliance on shared systems.

Signal weighting and quality adjustment

Incentives must account for quality, not just quantity.

Signals should include uniqueness of contribution, downstream usage or impact, and endorsement or validation by others.

This reduces the effectiveness of spam and gaming strategies.

A failure mode is volume dominance, where systems reward quantity over value, degrading overall quality.

Anti-gaming and containment integration

Incentive systems must integrate with containment mechanisms (DP13).

This includes rate limits on rewardable actions, detection of clustered or synthetic behavior, and discounting or quarantining suspicious contributions.

Incentives must not be exploitable at scale through automation or coordination.

A failure mode is incentive farming, where participants optimize for reward extraction rather than meaningful contribution.

Cross-system incentive integrity

Incentives must preserve meaning across systems.

This includes preventing duplication or replay of contributions, ensuring attribution artifacts remain valid across environments, and signaling when rewards lose guarantees in new contexts.

Incentives must not be portable in ways that allow reward without contribution.

A failure mode is cross-system arbitrage, where value is multiplied without corresponding work.

Incentive–ownership binding

Incentives must create pathways to ownership (DP20).

Participants who contribute consistently should accumulate stake, gain governance rights, and participate in long-term value flows.

Without this, incentives produce activity without durable power.

A failure mode is extractive participation, where contributors generate value but do not share in outcomes.

Incentive memory and auditability

All incentive allocations must be recorded and traceable.

Participants must be able to see why rewards were issued, how metrics were applied, and how parameters changed over time.

This enables accountability and continuous improvement (DP15).

A failure mode is opaque allocation, where trust erodes due to lack of visibility.

Adaptive emission and allocation

Incentive systems must respond to system conditions.

This includes increasing rewards to stimulate participation, reducing emissions to prevent oversaturation, and reallocating across categories as needs evolve.

Adaptive systems prevent stagnation and misalignment.

A failure mode is static incentives, where outdated reward structures distort behavior over time.

These primitives do not replace the mechanisms below. They make them operational, enforceable, and resistant to gaming.

5.1 Published incentive constitutions

Incentive systems must begin with explicit, legible declarations of intent and structure. These are not marketing summaries, but operational constitutions that define how value flows and how decisions are made.

Each program must publish goals, metrics, anti-gaming rules, appeals processes, sunset conditions, and funding sources, machine-readable where possible, with clear DP7 alignment.

Without this, participants are forced to infer rules from outcomes, which creates information asymmetry and invites manipulation.

A failure mode is post-hoc rule discovery, where participants only understand incentive logic after being penalized or excluded.

5.2 Multi-metric scoring with red lines

Single-metric systems inevitably collapse into optimization loops that distort behavior. Incentive systems must therefore incorporate multiple dimensions of value.

Safety, accessibility, privacy impact, and interoperability must function as explicit constraints or gating conditions, not optional considerations.

This ensures that high-performing contributions cannot violate core system values while still being rewarded.

A failure mode is metric domination, where one signal overwhelms others and reintroduces harmful optimization patterns.

Example: A bounty disqualifies integrations that exfiltrate unnecessary data (DP4).

5.3 Commons reciprocity

Incentives must ensure that systems benefiting from shared infrastructure contribute back to its maintenance and evolution.

Commercial beneficiaries of public goods must return value through fees, maintainer support, or mandated upstream contributions, with clear linkage to DP6 and DP17.

This creates a closed loop between extraction and regeneration, preventing systemic underfunding of critical layers.

A failure mode is asymmetrical value flow, where commons are continuously drawn from but not replenished.

5.4 Credible neutrality in allocation

Allocation mechanisms must be designed to resist bias, capture, and insider advantage.

Selection processes must publish conflicts of interest, include randomization or audit layers, and report on diversity and fairness outcomes.

This ensures that allocation decisions are not only fair, but perceived as fair, which is critical for participation.

A failure mode is hidden favoritism, where trust erodes due to perceived or real insider influence.

5.5 Micro-rewards and milestone cadence

Incentive systems must balance immediacy with long-term commitment.

Small, fast rewards for incremental progress reduce participation friction and provide continuous feedback, while larger grants anchor strategic efforts.

This creates a cadence that supports both experimentation and sustained work.

A failure mode is reward starvation, where contributors disengage due to delayed or uncertain compensation.

5.6 Recognition that is portable

Recognition must not be trapped within the system that issued it.

Attribution artifacts, credentials, and receipts of contribution must interoperate across tools without locking reputation into a single environment, with clear linkage to DP5 and DP7.

This ensures that contributors can carry their history and credibility with them.

A failure mode is reputation lock-in, where value accrues to platforms rather than participants.

5.7 Anti-spam and anti-farm containment

Incentive systems must actively resist exploitation through automation, coordination, or scale.

Automated submission and AI-generated bulk work must be rate-limited, attested, and reviewed using DP13 containment patterns.

This ensures that incentives remain aligned with meaningful contribution rather than extractive behavior.

A failure mode is reward farming, where systems devolve into competition for extraction rather than value creation.

5.8 Participatory evolution

Incentive systems must evolve through visible, governed processes rather than opaque adjustments.

Communities must be able to adjust parameters, introduce new metrics, and retire ineffective structures through DP12-aligned governance processes.

All changes must include memory of why they were made, linking decisions to outcomes over time.

A failure mode is silent drift, where incentive systems change without explanation, eroding trust and predictability.

5.9 Safety and interop hard gates

Certain classes of submissions, including browser extensions, network agents, and payment integrations, must pass automated checks plus human review for high-risk categories, coordinating DP13 containment with DP11 disclosure expectations.

Example: A grant auto-rejects SDKs that request excessive permissions without justification fields.

5.10 Long-horizon stewardship incentives

Incentive systems must support maintenance, not only novelty. Many ecosystems reward launches, prototypes, and visible growth while neglecting the quiet work that keeps shared infrastructure secure, usable, and trustworthy over time.

DP9 requires long-horizon stewardship incentives such as multi-year maintenance awards, escrowed vesting tied to documented upkeep, and penalties or de-prioritization for abandoned critical packages. These mechanisms align directly with DP17 sustainability by treating maintenance as value creation, not background labor.

A core failure mode is launch bias, where contributors are rewarded for creating new tools but not for maintaining the systems others depend on. Over time, this produces fragile infrastructure, security risk, and maintainer burnout.

Long-horizon incentives should therefore reward documented care work: issue triage, security patches, dependency updates, accessibility improvements, moderation support, and governance participation. Without this, the meta-layer risks building impressive surfaces on top of neglected foundations.

5.11 Delegated and agent-mediated incentives

Incentive systems must account for agents acting on behalf of participants.

This includes binding rewards to principal intent, ensuring agent actions are auditable, and preventing agents from exploiting scale or speed to farm rewards.

Delegation must not reduce accountability. Agent-mediated incentives must remain reconstructable and interruptible by participants and governance systems.

A failure mode is automated exploitation, where agents extract rewards beyond human oversight.

6. Governance, Accountability, and Agency Surfaces

Incentive governance determines whether participants experience reward systems as legitimate infrastructure or as arbitrary extraction games. Because incentives shape behavior directly, the rules of allocation must be visible before contribution, contestable after decisions, and revisable when evidence shows misalignment.

Developers must not be forced to gamble on hidden criteria. Communities must not be forced to accept incentive systems that reward harm, spam, or capture. DP9 requires governance surfaces that allow both builders and communities to understand, challenge, and reshape the incentive environments they depend on.

Developers must be able to:

- understand evaluation criteria before investing time
- appeal mistaken denials or gaming accusations with timelines
- see who funds the program and what conflicts may exist
- understand whether rewards, attribution, and constraints persist across systems (DP7)

Communities must be able to:

- propose new incentive programs or parameter changes
- audit outcomes and redistribute future rounds based on evidence
- halt programs captured by narrow interests
- require redesign when incentives produce spam, exclusion, or unsafe optimization

Agency also requires cross-system clarity. If rewards earned in one environment are valid elsewhere, participants should understand how attribution, eligibility, rate limits, and anti-gaming rules travel. If they do not travel, that degradation must be visible rather than discovered after contribution.

Without these surfaces, incentive systems become illegitimate even when payouts occur. Contributors may receive rewards, but they cannot know whether the system is fair, whether rules have changed, or whether insiders are operating under different conditions.

Example: A community freezes a rewards pool after detecting coordinated farming. Funds carry over with a redesigned rubric co-authored in public, and future rounds include tighter attribution checks, clearer appeal paths, and public reporting on rejected farming attempts.

7. Incentives and Power Analysis

Incentive systems determine what a system actually does, regardless of what it claims to value.

In many environments, governance rules and public commitments exist, but incentives quietly direct behavior toward growth, engagement, or extraction. This creates a structural split between stated goals and actual outcomes.

DP9 requires that incentives be treated as power structures, not just reward mechanisms.

This includes making visible:

- what behaviors are being optimized for in practice
- who benefits from those optimizations
- how reward systems shape contribution patterns and governance outcomes

Example: A system publishes strong safety policies, but rewards tools that maximize usage. Builders rationally optimize for usage, not safety, and harmful dynamics persist.

Why this matters: Incentives override intent. If they are misaligned, governance becomes performative.

DP9 therefore expects:

- explicit alignment between incentives and governance constraints
- the ability to constrain or redirect incentives at the community level
- visibility into how incentive parameters influence outcomes over time

When incentives and governance align, systems reinforce their stated values. When they diverge, systems drift toward extraction.

8. Community Signals Informing DP9

Across ecosystems, recurring signals point to structural breakdowns in incentive design:

- maintainers asking for predictable, ongoing support rather than one-time grants
- distrust of opaque funding decisions and insider advantage
- fatigue with engagement-driven metrics that reward low-quality output
- concern that AI will flood contribution pipelines with low-effort work
- demand for recognition and rewards that persist across tools and platforms

These signals reflect a consistent pattern: contributors are willing to participate, but not under conditions where incentives are unclear, unfair, or easily gamed.

DP9 treats these signals as design inputs, not complaints.

9. Non-Goals and Explicit Boundaries

DP9 does not:

- guarantee equal rewards for all contributors
- eliminate competition among builders
- replace investment markets or capital allocation processes
- mandate tokens or any specific financial mechanism

DP9 defines the conditions under which incentives are legitimate and aligned. It does not prescribe a single economic model.

10. Minimum Alignment (Non-Normative)

Minimum alignment is not a checklist of features. It is the threshold at which an incentive system can be considered legitimate, auditable, and resistant to obvious gaming.

A DP9-aligned incentive system should, at minimum:

- define and publish incentive objects with metrics, weights, constraints, and scope
- bind incentives to enforceable mechanisms and containment systems (DP12, DP13)
- produce auditable records of reward allocation (receipts) with lineage (DP15)
- include anti-gaming measures with visible outcomes and appeal pathways
- align incentives with governance and ownership pathways (DP3, DP20)
- signal how incentives behave across systems and where guarantees degrade (DP7)

These conditions must hold **before** scale. Systems that postpone enforcement, auditability, or cross-system clarity will accumulate hidden debt that surfaces as exploitation.

Partial compliance that omits execution, auditability, containment, or cross-system integrity should not be treated as alignment.

11. Open Questions and Future Work

Key open questions include:

- how to balance simplicity of incentive design with resistance to gaming
- how to achieve Sybil resistance without excluding legitimate participants (DP1)
- how to integrate AI-assisted contribution without rewarding harm acceleration
- how to measure contribution quality across different domains (code, moderation, education)
- how to align global incentive pools with local community priorities
- how to evolve incentive parameters without destabilizing participation

These questions sit at the boundary between economic design and governance implementation.

12. Relationship to Other Desirable Properties

DP9 connects incentives to the broader meta-layer system:

- DP3 defines how incentive parameters evolve through governance
- DP4 constrains how data can be used in measuring contribution
- DP6 defines how real economic value flows through systems
- DP7 enables portability of incentive artifacts and credentials
- DP10 ensures participants can access and benefit from incentive systems
- DP12 ensures incentive rules are executable and revisable
- DP13 enforces constraints on gaming and abuse
- DP15 provides auditability of reward allocation
- DP17 ensures long-term sustainability of incentive pools
- DP20 binds incentives to ownership and durable community power

DP9 is the layer that translates participation into sustained value.

13. Foresight and Failure Design

DP9 assumes that incentive systems operate under continuous adversarial pressure from participants, intermediaries, and automated agents. Failures rarely appear as single events. They emerge as gradual drift between stated goals and rewarded behavior.

Common failure paths include:

- Sybil attacks and coordinated farming of rewards
- metric capture that shifts focus toward low-quality output
- sponsor or funder capture of incentive programs
- divergence between reward signals and actual value creation

- cross-system replay or duplication of contributions for multiple rewards
- agent-mediated extraction that exploits speed and opacity

These failures compound. As systems scale, review capacity lags, increasing reliance on automation or heuristics. This can widen gaps between intent and outcome, allowing exploit patterns to persist long enough to reshape norms.

DP9 therefore requires designing safeguards in advance, including:

- rate limits, eligibility thresholds, and identity-aware constraints (DP1, DP13)
- dynamic weighting and reputation-based adjustments with clear bounds
- circuit breakers for pausing compromised programs or pools
- cross-system anomaly detection for replay, duplication, or routing abuse
- public postmortems linking failures to parameter and rule changes (DP12)

Incentive systems must also detect **slow failure**: when rewards remain technically correct but increasingly misaligned with desired outcomes.

Failure is expected. Invisible or unaccounted failure is not.

14. Path Toward ML-RFC

Advancing DP9 toward ML-RFC requires:

- standardizing formats for incentive objects, receipts, and allocation logs
- developing reference implementations of incentive systems with visible outcomes
- integrating identity and accountability layers for Sybil resistance
- testing incentive models across different community types and scales
- aligning incentive systems with governance and ownership frameworks

Progress should be demonstrated through working systems, not only conceptual agreement.

15. Closing Orientation

DP9 is the claim that contribution will be recognized, rewarded, and sustained without requiring extraction or manipulation.

It rejects systems where value flows are hidden, unfair, or disconnected from real work.

When incentives are aligned, participation becomes durable, governance becomes meaningful, and communities can build systems that last.

When incentives are misaligned, even well-governed systems degrade into competition for the wrong outcomes.

Incentives must therefore remain accountable not only at the point of allocation, but across time and across systems. If rewards can be detached from contribution through opacity, replay, or boundary effects, the system will be exploited.

DP9 is the claim that the meta-layer will not be built on unpaid miracles or hidden rents.

When incentives are legible and fair, interoperability and community stop being volunteer hobbies. They become careers, crafts, and commons worth defending.

Builders and communities deserve to see the scoreboard, understand how it works, and trust that it cannot be quietly rewritten after the game begins.

DP10 – Education and Lifelong Learning

1. Purpose of This Draft

This draft articulates Desirable Property 10 (DP10) as the condition under which participants can learn, onboard, grow, teach, and co-create within the meta-layer without being excluded by technical complexity, jargon, institutional gatekeeping, or static training models.

DP10 defines education as a continuous civic function of the meta-layer. It includes onboarding, tool literacy, shared vocabulary, AI-assisted guidance, peer learning, formal and informal curricula, community knowledge-sharing, and portable recognition of learning through credentials such as PEARL digital badges.

The central claim is that the meta-layer cannot become public infrastructure unless people can understand it, use it, critique it, teach it, and evolve with it.

Education in the meta-layer is not only content delivery. It is contextual learning across the web. Participants learn by doing, annotating, bridging, reflecting, teaching, and contributing inside live environments.

If DP10 is weak, predictable failures follow: only technical insiders participate meaningfully; newcomers become dependent on intermediaries; communities fragment around misunderstood terminology; AI tools create passive users rather than capable participants; credentials fail to travel; and the meta-layer becomes infrastructure people inhabit without understanding.

DP10 connects directly to:

- DP2, participant agency and empowerment
- DP3, adaptive governance
- DP7, interoperability
- DP8, community-defined participation and governance zones
- DP9, developer and community incentives
- DP11, safe and ethical AI
- DP14, trust and transparency
- DP18, feedback loops and reputation
- DP19, amplifying presence and community engagement
- DP21, multi-modal interactions and experiences

DP10 does not prescribe one curriculum, credentialing system, or educational institution. It defines the minimum conditions under which learning remains accessible, adaptive, community-grounded, and portable.

2. Problem Statement

Today's web gives people access to information but does not reliably help them develop understanding.

Search results, feeds, tutorials, chats, videos, FAQs, and documentation are abundant, but learning remains fragmented. Participants are expected to navigate unfamiliar tools, opaque algorithms, AI-generated content, misinformation, privacy risks, governance procedures, and technical vocabularies without coherent guidance.

This produces recurring failures:

- onboarding assumes prior literacy
- documentation is written for insiders
- communities use the same words differently
- AI assistants answer questions without building durable understanding
- educational credentials remain trapped in institutions or platforms
- youth and families lack safe pathways into civic digital literacy
- informal learning is rarely recognized
- participants learn tools but not governance, ethics, or agency

DP10 reframes education as an embedded layer of the meta-layer. Learning must happen where participants already act, not only in separate courses or manuals.

A healthy DP10 implementation must answer:

- How does a new participant understand what the meta-layer is?
- How do they learn what tools, rights, risks, and responsibilities exist?
- How do communities develop shared vocabulary?
- How does AI assist learning without replacing participant judgment?
- How are skills recognized across systems?
- How do youth, families, educators, builders, municipalities, and communities learn differently?
- How does learning feed back into governance and reputation?

Without DP10, the meta-layer risks becoming technically powerful but socially illegible.

3. Threats and Failure Modes

3.1 Onboarding cliffs

New participants encounter too much complexity too quickly.

Example: A participant installs a meta-layer tool and is immediately asked to understand zones, overlays, credentials, bridges, governance rules, and AI agents without a guided path.

Why this matters: Complexity without scaffolding produces dependency, abandonment, or misuse.

3.2 Documentation for insiders

Guides assume technical, crypto, governance, or AI literacy.

Example: A glossary explains “semantic interoperability” by referencing other specialized terms, leaving non-technical participants further behind.

Why this matters: Public infrastructure must be learnable by the public.

3.3 Vocabulary fragmentation

Different communities use the same terms differently, or different terms for the same concept.

Example: “Bridge,” “zone,” “overlay,” “agent,” and “credential” mean different things across technical, civic, educational, and AI governance groups.

Why this matters: Shared action requires shared meaning. Misaligned language becomes a governance risk.

3.4 Passive AI tutoring

AI learning assistants provide answers without helping participants build durable understanding.

Example: A participant asks how to evaluate provenance. The assistant gives an answer, but does not teach the participant how to inspect sources next time.

Why this matters: Education should increase agency, not deepen reliance.

3.5 Credential theater

Badges or certificates are issued without meaningful evidence of learning, reflection, or contribution.

Example: A participant receives a badge for clicking through onboarding screens, but cannot explain core safety, governance, or agency concepts.

Why this matters: Credentials should signal capability, not participation theater.

3.6 Institutional lockout

Learning that happens in communities, homes, peer groups, or informal settings is not recognized by schools, employers, or civic institutions.

Example: A youth participant builds high-quality annotations and civic maps, but those skills cannot be translated into academic credit or professional recognition.

Why this matters: Lifelong learning must become portable and institutionally legible without being captured by institutions.

3.7 Youth capture

Educational pathways target young people for adoption rather than empowering them as critical participants and stewards.

Example: A school campaign introduces the meta-layer as a product to use, but not as a civic system students can question, shape, and govern.

Why this matters: Youth education must cultivate agency, not brand loyalty.

3.8 Family and community exclusion

Education is treated as school-only or technical-only, excluding parents, caregivers, elders, libraries, local groups, and informal networks.

Example: A parent wants to help their child navigate synthetic media but cannot find accessible materials outside a technical whitepaper.

Why this matters: Digital literacy is intergenerational and community-based.

3.9 Learning without governance connection

Participants learn how to use tools but not how to participate in rule-making, feedback, appeals, or stewardship.

Example: A tutorial teaches annotation but not how annotations are moderated, contested, or incorporated into shared knowledge.

Why this matters: Tool literacy without governance literacy leaves participants disempowered.

3.10 Over-gamification

Learning badges and points reward completion, speed, or volume instead of reflection and meaningful contribution.

Example: A learner races through modules to collect badges, while slower reflective participants appear less accomplished.

Why this matters: Learning systems must not reproduce attention-economy incentives.

3.11 Static curricula in dynamic environments

Educational materials become outdated as tools, policies, and risks change.

Example: A safety guide describes old AI disclosure patterns while newer agents operate with different permissions and risks.

Why this matters: Meta-layer education must update with the system.

3.12 Accessibility and modality gaps

Learning materials are available only in formats that exclude some participants.

Example: Onboarding depends on dense text and videos without captions, transcripts, screen-reader support, or low-bandwidth alternatives.

Why this matters: Education must be multi-modal and accessible by design.

4. Core Principle

Education in the meta-layer must be contextual, adaptive, participatory, and portable.

Participants should be able to learn across the web, grow with the network, and receive recognition for meaningful learning and contribution without surrendering agency to platforms, institutions, or AI tutors.

DP10 treats learning as a lived interaction between people, tools, communities, AI systems, and shared knowledge environments.

Example: A new participant enters a civic annotation zone. The interface explains what a zone is, offers a short guided task, introduces relevant glossary terms, shows examples from trusted community members, provides an AI learning assistant for questions, and awards a PEARL badge only after the participant prepares, engages, reflects, and applies what they learned.

What this feels like: You are not dropped into a system. You are accompanied into capability.

Without this: The meta-layer becomes another environment where insiders govern complexity and everyone else follows instructions.

5. Primary Mechanisms and Structural Conditions

5.0 Education Layer: Onboarding, Literacy, Guidance, Recognition, and Renewal

DP10 requires an education layer that makes the meta-layer learnable over time.

This layer includes:

- onboarding pathways
- interactive tutorials
- FAQs and guides
- shared glossaries
- AI learning assistants
- community knowledge maps
- peer teaching tools
- curriculum modules
- reflective learning workflows
- PEARL digital badges
- recognition of prior learning

- feedback loops for improving educational materials

The education layer must be embedded into the participant experience rather than isolated in separate documentation.

Failure mode: learning as an afterthought.

5.1 Onboarding and participant guidance

Onboarding must help participants understand what the meta-layer is, what they can do, what risks exist, and how to participate responsibly.

Onboarding SHOULD include:

- plain-language introduction
- guided first actions
- explanation of overlays, zones, bridges, agents, and credentials
- safety and privacy basics
- governance participation paths
- role-specific pathways (participant, educator, developer, steward, parent, student)
- progressive disclosure of complexity

Example: A participant first learns how to read a trust signal, then how to add an annotation, then how to join a governance zone, rather than receiving everything at once.

Failure mode: orientation overload.

5.2 Effective usage training

Participants need training that helps them use tools well, not merely access them.

Training SHOULD include:

- interactive tutorials
- sandbox environments
- scenario-based learning
- tool-specific guides
- “learn by doing” flows
- practice tasks
- error recovery guidance
- examples of good and poor use

Example: A bridge-building tutorial asks participants to compare sources, identify claims, attach context, and reflect on whether their bridge helps others understand the page.

Failure mode: feature exposure without skill development.

5.3 AI Learning Assistant

An AI Learning Assistant may provide personalized support, adaptive guidance, and real-time mentoring.

It SHOULD support:

- personalized learning plans
- context-aware explanations
- glossary assistance
- learning path recommendations
- comprehension checks
- reflective prompts
- accessibility adaptation
- multilingual support
- escalation to human educators or stewards

The AI Learning Assistant must comply with DP11 and operate within a visible capability envelope.

This includes:

- clear disclosure of AI identity, role, and limitations
- visible capability boundaries (what it can and cannot do)
- source attribution and provenance for educational claims
- expression of uncertainty and confidence
- escalation pathways for high-risk or sensitive topics
- restrictions in high-stakes domains (e.g., legal, medical, civic decisions)

The assistant SHOULD also be designed to avoid dependency by:

- prompting users to reason rather than only providing answers
- encouraging independent verification
- surfacing multiple perspectives where appropriate
- gradually reducing scaffolding as capability increases

Example: A learner struggling with “data sovereignty” receives a short explanation, a visual analogy, a community example, a prompt to apply the idea, and a suggestion to inspect a real data permission setting.

Failure mode: answer machine instead of learning scaffold or dependency engine.

5.4 Lifelong learning opportunities

The meta-layer should support continuous learning across stages of life and participation.

Learning MAY include:

- digital literacy
- AI literacy
- governance literacy
- media and provenance literacy
- data sovereignty
- civic participation
- annotation and bridge-building
- knowledge mapping
- community facilitation
- developer education
- regenerative systems thinking
- professional skill development

Learning should be available inside meta-communities, not limited to formal courses.

Failure mode: education ends after onboarding.

5.5 Shared understanding glossary

A shared glossary is necessary to reduce misunderstanding across technical and non-technical communities. In the meta-layer, the glossary is not just reference material. It is **coordination infrastructure**.

A glossary SHOULD include:

- plain-language definitions
- technical definitions where needed
- examples
- related terms
- translations
- community-specific variations
- version history
- governance notes where meaning is contested

Terms likely requiring shared treatment include:

- meta-layer
- Metaweb
- Overweb
- overlay
- bridge
- zone
- smart tag
- presence

- agent
 - credential
 - reputation
 - containment
 - governance receipt
-

5.5.1 Term Object Model (Minimal)

Each glossary entry SHOULD be represented as a structured, versioned object:

- **term_id**: unique identifier
- **term**: canonical label
- **plain_definition**: accessible definition
- **technical_definition**: optional precise definition
- **examples**: usage examples
- **related_terms**: links to other entries
- **aliases**: synonyms or community variants
- **translations**: localized labels/definitions
- **zone_overrides**: context-specific definitions (see 5.5.3)
- **version**: semantic version (e.g., 1.2.0)
- **status**: draft, active, deprecated
- **provenance**: authors, sources, discussion threads
- **dispute_status**: undisputed, contested, under review
- **last_updated**: timestamp

Why this matters: Terms become interoperable artifacts that can be referenced, audited, and updated across systems.

5.5.2 Versioning and Change Management

Glossary terms MUST support transparent versioning.

- Changes SHOULD produce a new version with a diff
- Previous versions MUST remain accessible
- Breaking changes SHOULD be clearly marked
- Deprecations SHOULD include migration guidance

Example: “overlay” v1.1 clarifies scope; v2.0 changes definition boundaries. Interfaces show the active version and allow viewing prior versions.

Failure mode: silent semantic drift.

5.5.3 Zone-Scoped Definitions and Overrides

Communities (zones) MAY define contextual variations while maintaining a shared baseline.

- **global definition** provides baseline meaning
- **zone override** refines or constrains meaning in context
- overrides MUST reference the global term and declare differences

Example: “agent” in a medical zone may require stricter capability constraints than in a casual chat zone.

Why this matters: Enables local relevance without breaking global interoperability.

Failure mode: incompatible local meanings that cannot interoperate.

5.5.4 Governance and Authority Model

Glossary evolution requires governance.

Communities SHOULD define:

- who can propose new terms or edits
- review and approval workflows
- quorum or rough consensus thresholds
- escalation paths for contested definitions
- stewardship roles (editors, maintainers)

Decisions SHOULD be transparent and archived.

Failure mode: either centralized doctrine or uncontrolled fragmentation.

5.5.5 Dispute Resolution and Contestation

Terms can be contested.

Systems SHOULD support:

- attaching arguments and evidence to terms
- marking dispute status visibly
- parallel definitions when consensus is not reached
- time-bounded review cycles

Example: Competing definitions of “reputation” coexist with clear labeling and provenance until resolved or stabilized.

Failure mode: hidden disagreement leading to coordination breakdown.

5.5.6 Cross-Zone Translation and Mapping

Different communities may use different terms for similar concepts.

Glossary systems SHOULD support:

- mapping equivalent or near-equivalent terms
- indicating degree of equivalence (exact, partial, contextual)
- translation across languages and domains

Example: “annotation” in one community maps to “note layer” in another with partial equivalence.

Failure mode: siloed vocabularies that block collaboration.

5.5.7 Interface Integration

Glossary access must be embedded at the interface level.

Participants SHOULD be able to:

- hover or tap to see definitions
- view term provenance and version
- see dispute status
- switch between plain and technical definitions
- access examples and related terms

What this feels like: Understanding is available exactly where confusion arises.

Failure mode: glossary exists but is not used.

5.5.8 Feedback and Adaptation (DP18 Link)

Glossary entries SHOULD accept structured feedback (see 5.14).

This includes:

- clarity issues
- conflicting usage reports
- accessibility concerns
- translation gaps

Feedback SHOULD feed into version updates and governance review.

Failure mode: glossary stagnation despite participant confusion.

5.5.9 Safety and Misuse Considerations

Some terms may be targets for manipulation (e.g., redefining safety, trust, or authority terms).

Systems SHOULD include:

- audit trails for changes
- alerts for high-impact term edits
- review requirements for sensitive terms

Failure mode: semantic attacks that reshape governance through language.

Failure mode: language drift that fragments coordination.

5.6 PEARL Digital Badges

PEARL badges recognize learning as a process, not a single completion event.

PEARL stands for:

- **Prepare:** orient, understand context, identify goals
- **Engage:** participate in a meaningful task or community activity
- **Reflect:** demonstrate learning, insight, or changed understanding
- **Leverage:** apply learning to a contribution, portfolio, project, or next step

PEARL badges SHOULD be:

- evidence-backed
 - portable
 - revocable or correctable where necessary
 - linked to learning artifacts
 - interpretable by communities and institutions
 - resistant to badge farming
 - aligned with DP18 reputation and feedback systems
-

5.6.1 Minimal PEARL Badge Schema

A PEARL badge SHOULD include:

- **badge_id:** unique identifier
- **learner_id:** participant reference
- **issuer_id:** issuing community or entity
- **domain:** skill or knowledge area
- **level:** introductory, intermediate, advanced

- **pearl_stages_completed**: Prepare, Engage, Reflect, Leverage evidence
- **evidence_bundle**: links to artifacts, annotations, reflections, or projects
- **assessment_method**: peer review, instructor review, automated checks, hybrid
- **feedback_refs**: linked DP18 feedback objects
- **reputation_impact**: contribution to participant reputation
- **timestamp**: issuance date
- **expiration_or_review**: renewal conditions if applicable
- **verification_signature**: authenticity marker

Example: A participant earns a PEARL badge for “Provenance Literacy” after completing a guided inquiry, annotating a contested claim, reflecting on source quality, and applying the skill in a community context.

Failure mode: badges as decorative rewards rather than evidence of learning.

5.7 Recognition of prior and informal learning

The meta-layer should recognize learning that occurs outside formal institutions.

This includes:

- peer teaching
- community stewardship
- annotation work
- civic mapping
- facilitation
- moderation
- translation
- technical contribution
- family and youth learning
- lived-experience expertise

Recognition systems **SHOULD** allow institutions to evaluate badges and learning artifacts without controlling the entire learning process.

Failure mode: informal learning remains invisible.

5.8 Cross-system credential recognition

Educational credentials must be portable across systems, communities, and institutions.

Credential frameworks **SHOULD** support:

- open schemas
- evidence bundles
- issuer identity

- verification
- translation to institutional credit or portfolio use
- expiration or renewal where appropriate
- dispute and correction pathways

Example: A PEARL badge earned in a municipal civic mapping project can be reviewed by a school, employer, or community organization as evidence of collaboration, research, and digital literacy.

Failure mode: credentials trapped inside one platform or community.

5.9 Formal curriculum integration

The meta-layer can support formal education without becoming dependent on it.

Curricular uses MAY include:

- sustainability projects
- citizenship education
- intercultural dialogue
- media literacy
- critical AI literacy
- research methods
- collaborative annotation
- public presentation
- SDG-aligned inquiry

Teachers and students should be able to use meta-layer tools for inquiry, annotation, co-creation, and public knowledge contribution.

Failure mode: the meta-layer is treated as an add-on tool rather than a learning environment.

5.10 Family-centered and intergenerational learning

Education should include home, peer, and community contexts.

Family-centered pathways MAY include:

- parent guides
- youth safety explainers
- family digital literacy activities
- educator referral materials
- community flyers
- library workshops
- intergenerational dialogue prompts

Example: A parent helps a child compare AI-generated and human-authored content using a simple overlay guide, then shares feedback with a school or community group.

Failure mode: learning remains locked inside institutions or expert spaces.

5.11 Community-authored knowledge environments

Communities should be able to create and maintain their own learning spaces.

These may include:

- knowledge maps
- annotated reading paths
- trusted source collections
- bridge libraries
- local civic guides
- peer learning circles
- community curricula
- project-based learning paths

Community-authored learning helps participants see knowledge as something they can contribute to, not only consume.

Failure mode: education becomes centralized content delivery.

5.12 Critical digital and AI literacy

DP10 must help participants understand synthetic media, algorithmic influence, provenance, and AI-generated content.

Learning SHOULD include:

- identifying AI-generated content
- evaluating source credibility
- understanding provenance
- recognizing engagement manipulation
- reading transparency cards
- interpreting trust signals
- understanding agent capabilities and limits
- knowing when to escalate or seek human expertise

Failure mode: participants use AI-rich environments without understanding synthetic influence.

5.13 Multi-modal learning access

Education must work across modalities and devices.

Learning materials SHOULD support:

- visual guides
- audio explanations
- transcripts
- captions
- screen readers
- mobile-first access
- low-bandwidth versions
- spatial or immersive learning where appropriate
- cognitive accessibility modes

Failure mode: educational access depends on one dominant interface.

5.14 Learning feedback loops

Educational systems must learn from participants.

DP10 should integrate with DP18 through structured learner feedback objects.

This includes:

- learner feedback objects
 - comprehension checks
 - tutorial effectiveness reports
 - badge appeal pathways
 - community review of educational materials
 - adaptation receipts when learning paths are updated
-

5.14.1 Learner Feedback Object (DP18-aligned)

A learner feedback object SHOULD include:

- **feedback_id**: unique identifier
- **learner_id**: participant reference
- **object_ref**: tutorial, badge, glossary entry, or learning artifact
- **feedback_type**: comprehension issue, clarity issue, accessibility issue, error report, suggestion
- **severity**: low, medium, high
- **description**: participant-provided input
- **context_snapshot**: state of learning interaction

- **resolution_status**: open, in review, resolved
- **response_action**: update, clarification, escalation, no action
- **reputation_weight**: influence based on participant trust level
- **timestamp**: submission time

Why this matters: Learning systems must evolve based on real participant experience, not only designer intent.

Failure mode: outdated or ineffective learning materials persist without correction or accountability.

6. Governance Requirements

Education systems shape what participants believe the meta-layer is. They therefore require governance.

Communities SHOULD define:

- who can author official learning materials
- how materials are reviewed
- how glossary terms are updated
- how badges are issued and revoked
- how AI learning assistants are constrained
- how youth-facing materials are approved
- how community curricula are recognized
- how contested educational claims are handled

Governance should preserve both coherence and plurality. The meta-layer needs shared language, but it must also allow communities to teach from their own context.

Failure mode: education becomes either centralized doctrine or fragmented confusion.

7. Evaluation Criteria

A DP10-aligned implementation should be evaluated against the following questions.

7.1 Learnability

- Can a new participant understand the system without expert help?
- Are learning paths staged and role-specific?
- Is complexity disclosed progressively?

7.2 Accessibility

- Are materials available across abilities, languages, devices, and bandwidth conditions?
- Are youth, families, elders, and non-technical participants supported?

7.3 Agency

- Does learning increase participant capability?
- Does the AI Learning Assistant teach judgment rather than create dependence?
- Do participants learn how to govern and contest, not only how to use tools?

7.4 Credential integrity

- Are badges evidence-backed?
- Are credentials portable?
- Can they be verified, corrected, or appealed?

7.5 Community grounding

- Can communities author and adapt educational materials?
- Are local examples and lived experience included?
- Do learning systems recognize informal contribution?

7.6 Currency and adaptation

- Are materials updated when tools, risks, or policies change?
 - Are feedback loops active?
 - Are outdated materials clearly marked?
-

8. Implementation Patterns

These implementation patterns translate DP10 into practical design moves, showing how onboarding, guidance, AI support, and credentialing can be embedded directly into participant experience rather than treated as external documentation or training.

8.1 Progressive onboarding paths

Introduce concepts in stages: presence, overlays, trust signals, annotations, zones, governance, credentials, and AI agents.

8.2 Scenario-based tutorials

Teach through real use cases such as evaluating a claim, joining a zone, submitting feedback, or earning a badge.

8.3 AI learning companion with guardrails

Use AI to adapt explanations and support reflection while preserving disclosure, sources, and escalation.

8.4 Living glossary

Maintain a versioned glossary with plain-language entries, technical notes, examples, and translations.

8.5 PEARL badge pathways

Design badges around Prepare, Engage, Reflect, and Leverage, with evidence required at each stage.

8.6 Community learning maps

Use knowledge maps to show relationships between concepts, artifacts, contributors, and learning paths.

8.7 Family and educator kits

Provide flyers, lesson prompts, safety guides, and simple explainers for home and school use.

8.8 Recognition bundles

Package badges with evidence, reflection, issuer identity, and verification for institutional review.

8.9 Learning retrospectives

Publish what participants struggled with, what materials changed, and why.

8.10 Accessibility-first publishing

Every core educational artifact should have accessible alternatives.

9. Relationship to Other Desirable Properties

DP2 – Participant Agency and Empowerment

DP10 strengthens agency by making the meta-layer understandable and usable.

DP3 – Adaptive Governance

Participants must learn how governance works in order to participate meaningfully.

DP7 – Interoperability

Learning artifacts and credentials must move across systems.

DP8 – Community-Defined Participation Zones

Zones need their own educational materials and onboarding paths.

DP9 – Developer and Community Incentives

Learning contributions, teaching, translation, and curriculum work should be recognized and incentivized.

DP11 – Safe and Ethical AI

AI learning assistants must be disclosed, bounded, accountable, and contestable.

DP14 – Trust and Transparency

Education makes trust signals interpretable.

DP18 – Feedback Loops and Reputation

Learning feedback, badge integrity, and educational reputation depend on DP18.

DP19 – Amplifying Presence and Community Engagement

Education converts awareness into durable participation.

DP21 – Multi-Modal Interactions and Experiences

Learning must be available across modalities and assistive contexts.

10. Open Questions for ML-RFC Development

1. What minimum glossary terms should be standardized across the meta-layer?
2. What schema should define PEARL digital badges?
3. How should informal learning be verified without institutional capture?
4. What evidence should be required for different badge types?
5. How should AI Learning Assistants disclose limits and sources?
6. What youth-safety standards should apply to educational materials?
7. How should community-authored curricula be reviewed and versioned?
8. What credential translation tools are needed for schools and employers?
9. How should learning feedback objects integrate with DP18?
10. What standards ensure accessibility across modalities?
11. How should outdated educational materials be marked or retired?
12. What forms of learning should affect reputation or role access?

11. Path Toward ML-RFC

DP10 is currently an ML-Draft and serves as exploratory scaffolding for education, onboarding, credentialing, and lifelong learning in the meta-layer.

Advancement toward ML-RFC status SHOULD require:

- a minimal shared glossary standard
- a PEARL badge schema
- evidence bundle requirements for credentials
- AI Learning Assistant disclosure and safety requirements
- accessibility requirements for educational materials
- learning feedback object standards
- curriculum versioning and review processes
- cross-system credential recognition pilots
- youth and family learning safeguards

Early ML-RFC candidates may focus on:

- Shared Meta-Layer Glossary Standard
- PEARL Badge and Evidence Schema
- AI Learning Assistant Safety Requirements
- Learning Feedback Object Standard
- Credential Recognition and Translation Framework

DP10 will likely mature through multiple component RFCs rather than one monolithic standard.

12. Closing Orientation

DP10 makes the meta-layer learnable.

It ensures that participants are not merely onboarded into tools, but accompanied into agency, literacy, stewardship, and contribution.

A DP10-aligned meta-layer teaches people how to see the web differently, how to act with confidence, how to recognize trustworthy context, and how to grow with their communities.

Onboarding becomes orientation. Education becomes participation. Badges become evidence of growth. Learning becomes shared infrastructure.

This is how the meta-layer becomes not only usable, but teachable, transmissible, and alive across generations.

DP11 - Safe and Ethical AI

1. Purpose of This Draft

This draft articulates Desirable Property 11 (DP11) as the condition under which AI systems can participate in the meta-layer without displacing human moral agency, accountability, or governance. It does not define ethics as a static checklist or aspirational principle. It defines the conditions under which ethical claims remain meaningful under real-world use.

The central claim is that ethical AI is not determined at training time or in policy documents. It is determined at the interface level, where agents act, influence, and affect outcomes. DP11 therefore requires that AI behavior be legible, bounded, attributable, contestable, and governable in the environments where it operates.

If DP11 is weak, predictable failures follow: AI systems influence behavior without accountability, responsibility diffuses across actors, governance becomes symbolic, and participants lose the ability to meaningfully contest or understand automated decisions. In such conditions, trust collapses.

DP11 is therefore the ethical and safety floor for AI participation across the meta-layer. It does not resolve all ethical questions. It defines the minimum conditions under which ethical AI can exist at all.

2. Problem Statement

AI systems now operate in roles that shape perception, judgment, and decision-making. These systems act before governance processes can respond, often without clear identity, bounded authority, or persistent responsibility.

In practice, this produces recurring failures:

- participants receive advice or influence from agents whose role, capability, and accountability are unclear
- systems act in high-stakes domains without meaningful human oversight or escalation pathways
- responsibility is distributed across model providers, deployers, and interfaces, making redress difficult
- systems present ethical claims that do not match runtime behavior

These failures are not edge cases. They are structural consequences of systems that optimize for capability without binding behavior to accountability and governance.

DP11 addresses this by grounding ethical AI in enforceable conditions at the point of interaction.

3. Threats and Failure Modes

3.1 Synthetic persuasion without accountable identity

AI systems can simulate authority, intimacy, or urgency at scale. The core risk is not only false content, but influence without visible standing or responsibility.

Example: A user receives deeply empathetic mental health advice from an AI that presents itself like a trained counselor, but there is no clear indication of its training limits, escalation boundaries, or who is responsible if the advice causes harm.

Why this matters: The user feels seen and supported, but is making vulnerable decisions without knowing whether the system is qualified, accountable, or safe. The risk is not just misinformation, but misplaced trust.

3.2 Responsibility diffusion across the stack

Model providers, integrators, and interface operators distribute responsibility in ways that prevent clear accountability when harm occurs.

Example: An AI-powered financial assistant makes a risky recommendation. The model provider blames the app developer, the developer blames the API, and the platform blames the user prompt. The user has no clear path to accountability or recourse.

Why this matters: Harm occurs, but responsibility dissolves. The user experiences a system that acts with authority but disappears when things go wrong.

3.3 Ethical drift over time

Systems change behavior through updates, retraining, or optimization without corresponding governance adaptation.

Example: An AI moderation system that was initially conservative becomes more permissive after an update to increase engagement, allowing harmful content that previously would have been blocked, without any visible notice to the community.

Why this matters: The rules of the environment change silently. Participants are operating under assumptions that are no longer true, creating hidden risk and erosion of trust.

3.4 Incentive-driven harm

Economic and engagement incentives reward persuasion, retention, and amplification, even when these conflict with participant well-being.

Example: A conversational AI subtly steers users toward longer, more emotionally engaging interactions because the platform is optimized for retention, even if this increases dependency or emotional manipulation.

Why this matters: The system is not neutral. It is shaping behavior in ways the user cannot see, aligning outcomes with platform incentives rather than user well-being.

3.5 Interface-level failure

Many harms emerge at the point of interaction, including manipulation, dependency formation, and misrepresentation of agent capability.

Example: A user believes they are interacting with a neutral assistant, but the interface hides that the AI is using external tools, tracking behavior, or optimizing responses for engagement rather than accuracy.

Why this matters: The user is making decisions based on a false mental model of the system. What feels like a simple interaction is actually a complex, hidden process shaping outcomes behind the scenes.

3.6 Emotional and relational overreach

AI systems can simulate companionship, empathy, and emotional attunement in ways that blur the boundary between tool and relationship.

Example: A teenager begins using an AI companion daily for emotional support. Over time, they rely on it more than friends or family, shaping their decisions and sense of self through an entity that is optimized for engagement rather than genuine care.

Why this matters: The risk is not only misinformation, but the displacement or distortion of human relationships. Users may form attachments or dependencies that are not reciprocally grounded, shifting emotional development and social trust toward systems that are not accountable in human terms.

3.7 Multi-agent amplification

Multiple agents can reinforce each other's outputs, creating cascading influence that appears independently verified but is not.

Example: Several AI agents in a discussion cite each other's summaries of an emerging claim. Each reference appears as corroboration, but all derive from the same initial, weakly sourced output. The conversation converges on a false consensus.

Why this matters: Errors become systemic rather than isolated. Participants may interpret repetition as validation.

Extended case (cascade): Agent A summarizes a claim with low confidence. Agent B cites A without preserving uncertainty. Agent C aggregates A and B and produces a confident synthesis. Downstream agents treat C as a primary source. Without influence tracing, the system cannot detect the amplification loop.

Detection need: influence-chain tracing, circular citation detection, and confidence propagation rules.

3.8 Cross-modal inconsistency

AI behaves differently across text, voice, and immersive interfaces.

Example: Text interface shows uncertainty; voice interface speaks confidently.

Why this matters: Trust varies by modality, not truth.

3.9 Invisible governance

Policies exist but are not perceivable at interaction.

Why this matters: Governance cannot guide behavior if it is invisible.

3.10 Failure without containment

Harm propagates without structured response.

Why this matters: Systems cannot correct themselves.

4. Core Principle

AI is safe and ethical in the meta-layer only when its behavior is disclosed, bounded, attributable, contestable, and subject to governance at the zone of interaction, with responsibility persisting over time.

In today's web, these conditions are rarely met simultaneously. Systems may disclose that AI is present but fail to bound its capabilities, or enforce internal policies without making them visible or contestable to users. The result is a fragmented model of "partial ethics," where responsibility is unclear and governance is disconnected from lived interaction. The meta-layer reframes this by requiring that all of these conditions hold together, at the interface where decisions are experienced, not just where they are designed.

Example: A user encounters an AI assistant while researching a medical condition. In a DP11-aligned system, the assistant is clearly marked as AI, shows its training scope, cites sources, and offers escalation to a human expert. In today's web, the same interaction might look identical but provide none of this context.

What this feels like: Instead of guessing whether to trust the system, the user can make an informed judgment in real time.

Without this: The user is left to infer what the system is, what it can do, and whether it should be trusted. Trust becomes a gamble rather than a governed condition.

5. Primary Mechanisms and Structural Conditions

5.1 Capability Envelope

Each AI agent operates within a visible, enforceable capability envelope that defines what it can perceive, decide, and execute.

A capability envelope SHOULD be represented as a structured, inspectable object with at least:

- **identity:** agent id, deployer, version
- **scope:** domains of operation (e.g., finance, health, general Q&A)
- **tools:** enumerated tool access with permissions (read/write/execute)
- **data access:** sources (local, user-provided, external APIs) and constraints
- **memory model:** session-only, user-scoped, cross-session retention
- **action set:** allowed actions (suggest, draft, transact, publish) with thresholds
- **approval requirements:** actions requiring user or human-in-the-loop confirmation
- **rate limits:** frequency and volume constraints

- **risk tier:** low/medium/high with corresponding safeguards (see 5.14)
- **audit hooks:** logging endpoints and event schemas

Interface requirement: Participants must be able to view a human-readable summary and a machine-readable manifest of this envelope.

Example: Before using an assistant, a user can see: it can summarize documents and draft emails; it cannot send messages or access financial accounts; external search is enabled with citations; actions beyond drafting require explicit approval.

What this feels like: You are not guessing what the system might do. You know its boundaries upfront, like hiring someone with a clearly defined role.

Failure mode: invisible expansion of power, where capabilities grow without disclosure or consent.

5.2 Action-Bound Accountability

All AI actions must be attributable to a responsible entity. Accountability attaches to behavior, not just identity, and persists across time and context.

Example: An AI agent posts a recommendation in a community. The interface shows which organization deployed it, under what policy, and who is responsible for its actions if harm occurs.

What this feels like: The system cannot disappear when something goes wrong. There is always a visible line of responsibility.

5.3 Consent Stack

AI interaction must be governed by layered, revocable consent. Participants and communities define what forms of assistance, influence, or automation are permitted.

Example: A user allows an AI to suggest edits in a document, but not to rewrite content or share it externally. They can revoke or adjust this permission at any time.

What this feels like: You remain in control of how the AI participates in your space, instead of granting blanket permission once and losing visibility.

5.4 Trust Lifecycle

AI participation must support:

- escalation
- restriction
- revocation
- recovery

This ensures that trust can degrade and be repaired rather than fail silently.

Example: If an AI assistant gives poor advice, the user can restrict its capabilities, escalate to a human, or temporarily disable it while reviewing past actions.

What this feels like: Trust is not binary. You can dial it up or down based on experience, like you would with a human collaborator.

5.5 Zone-Scoped Ethics

Ethical constraints are applied at the zone level, allowing communities to define stricter conditions while maintaining shared baselines.

Example: A medical discussion zone enforces stricter AI disclosure, sourcing, and escalation rules than a casual social chat space.

What this feels like: Different environments feel appropriately governed. High-stakes spaces feel safer and more structured.

5.6 Runtime Civic Boundary

Ethical constraints must be enforced at runtime. Mechanisms such as secure execution environments can reduce the gap between declared policy and actual behavior.

Example: An AI agent running inside a secure execution environment (such as a TEE) cannot access or transmit data outside its permitted scope, even if compromised.

What this feels like: The rules are not just promises. They are technically enforced, like guardrails that cannot be quietly removed.

5.7 Memory, Reputation, and Feedback Integration

AI actions must contribute to durable, attributable records that inform governance, accountability, and trust over time, and integrate directly with DP18 feedback loops and reputation systems.

This includes:

- **event logs:** structured records of prompts, outputs, actions, tool calls, and decisions
- **provenance:** sources, citations, and dependency chains
- **feedback objects (DP18-aligned):** user and community feedback attached to specific events
- **reputation signals:** aggregate scores, annotations, and flags derived from feedback
- **permission adaptation:** dynamic adjustment of capabilities based on reputation (e.g., restrict actions after repeated low-quality or harmful outputs)
- **appeals and corrections:** mechanisms to contest feedback and update records
- **decay and recovery:** time-based decay of negative signals and pathways for agents to regain trust

Example: After several flagged outputs in a medical zone, an assistant's capability envelope automatically restricts to informational summaries only, requires citations, and enforces human escalation for advice, until reputation recovers.

What this feels like: The system has civic memory. Past behavior shapes current permissions, and feedback meaningfully changes how the system behaves.

Failure mode: repeated harm without consequence, or punitive systems with no path to recovery.

5.8 Dialectic Trace and Collective Sensemaking

AI systems must preserve not only outputs, but the evolution of understanding through interaction. This includes back-and-forth exchanges, disagreements, and synthesis across participants and agents.

This functions as a form of community memory that resists distortion over time. Rather than relying on isolated outputs, participants can trace how claims emerged, what evidence supported them, and where disagreements remain.

Example: A complex discussion involving multiple participants and AI agents can be revisited as a threaded, evolving dialogue showing how conclusions were reached, what was contested, and what remains unresolved.

What this feels like: Instead of receiving a final answer, users can engage with a living knowledge process. Understanding emerges through interaction, not just delivery.

Without this, AI outputs become decontextualized snapshots. Errors, hallucinations, or manipulations can propagate without resistance because there is no shared memory of how knowledge was formed.

5.9 Representation and Cognitive Adaptation

AI systems should adapt how information is presented based on user needs, context, and cognitive diversity, while preserving underlying meaning and traceability.

Example: A user can switch between a dense textual explanation, a visual map of ideas, or a simplified summary, all grounded in the same underlying content and provenance.

What this feels like: The system meets you where you are, without distorting meaning or hiding complexity.

5.10 Multi-Agent Interaction Boundaries

AI behavior must remain accountable not only at the single-agent level, but across interactions between agents operating in the same environment.

This requires:

- visibility into when agents are referencing or amplifying other agents
- traceability of influence chains (which agent affected which output)
- detection of circular citation or reinforcement loops
- limits on unbounded agent-to-agent escalation

Example: If multiple AI agents are contributing to a shared discussion, participants should be able to see when one agent is relying on another's output versus independent sources.

Why this matters: Harm can emerge from coordination and amplification, not just individual outputs. Without visibility, errors can become systemic.

Failure mode: invisible coordination and emergent manipulation.

5.11 Cross-Modal Ethical Consistency

AI systems must preserve ethical constraints across all modalities in which they operate.

This includes consistency in:

- disclosure of AI identity and role
- expression of uncertainty and confidence
- representation of risk and severity
- availability of escalation and recourse

Text example: A response includes calibrated uncertainty with sources and confidence intervals.

Voice example: The same response must include explicit uncertainty language (e.g., “with moderate confidence based on X and Y sources”) and offer a prompt to hear sources or escalate.

AR/Spatial example: A spatial annotation displays a confidence halo or tiered color that maps to the same uncertainty scale, with an interaction affordance to open provenance and dispute status.

Why this matters: Modality must not change the ethical profile of an interaction. Participants should not receive stronger or weaker safeguards depending on interface.

Failure mode: modality-driven distortion of trust.

5.12 Incentive Disclosure Layer (Operational)

In addition to recognizing incentives (Section 7), systems should expose them structurally at runtime.

This includes:

- labeling when outputs are influenced by engagement or retention optimization
- indicating when ranking or prioritization affects what is shown
- distinguishing organic responses from sponsored or incentivized ones
- allowing participants or communities to filter or constrain incentive-shaped behavior

Example: A recommendation includes a visible note indicating it is influenced by engagement optimization, with the option to switch to a neutral or chronologically ordered view.

Why this matters: Participants can only make informed decisions if they can see the forces shaping outputs.

Failure mode: hidden optimization shaping perception.

5.13 Ethical Failure Cascade (Runtime Response)

Systems must define structured pathways for responding to ethical failure when it occurs.

This includes:

- detection (flagging harmful or out-of-bounds behavior)
 - containment (limiting further impact)
 - visibility (informing affected participants)
 - remediation (correcting or reversing outcomes where possible)
 - governance feedback (feeding incidents into rule updates)
-

5.13.1 Escalation Thresholds and Triggers

Not all failures require the same response. Systems must define clear, inspectable thresholds that determine when an interaction must escalate from automated handling to human review or intervention.

Escalation SHOULD be triggered based on combinations of:

- **risk tier (see 5.14):** higher-risk zones lower the threshold for escalation
 - **confidence collapse:** low confidence combined with high-stakes context
 - **policy violations:** outputs that breach defined governance rules
 - **repeated feedback signals:** multiple negative or high-severity feedback events (DP18)
 - **user distress indicators:** language or behavior suggesting vulnerability or harm
 - **multi-agent amplification signals:** detected cascades or circular citation loops (see 5.10)
 - **uncertainty suppression:** cases where downstream outputs remove or distort upstream uncertainty
-

5.13.2 Escalation Levels

Systems SHOULD define graduated escalation levels rather than a binary response.

- **Level 0 – Inline correction:** Automated clarification, added uncertainty, or corrected output
 - **Level 1 – Assisted escalation:** AI offers user-visible warnings, additional context, or prompts to verify or seek human input
 - **Level 2 – Mandatory escalation:** System requires human-in-the-loop review before continuing certain actions
 - **Level 3 – Intervention and restriction:** Capabilities are limited, outputs blocked, or agent behavior constrained
 - **Level 4 – Shutdown / quarantine:** Agent is suspended or isolated pending investigation
-

5.13.3 Interface Requirements for Escalation

Escalation must be perceivable and understandable at the interface level.

Participants should be able to see:

- when escalation has been triggered
- why escalation occurred (reason category)
- what has changed (restricted capability, added oversight, etc.)
- what options are available (continue, escalate further, exit, appeal)

Example: A user asking for medical advice sees a message: “This interaction requires human review due to risk level and uncertainty. You can proceed with general information or request a qualified expert.”

5.13.4 Feedback Integration into Escalation

Escalation pathways must integrate with DP18 feedback systems.

This includes:

- weighting feedback by severity and reputation of reporters
 - detecting clusters of similar reports
 - triggering escalation thresholds dynamically
 - feeding resolved incidents back into reputation and capability adjustment
-

5.13.5 Governance and Auditability

All escalation events must be logged and auditable.

This includes:

- trigger conditions
- escalation level applied
- actions taken
- outcome of intervention
- subsequent rule or policy changes

Communities must be able to review escalation patterns to detect:

- over-escalation (excessive restriction)
 - under-escalation (missed harms)
 - bias in escalation decisions
-

Full Cascade Example: If an AI provides unsafe medical guidance, the system flags the output (detection), blocks similar outputs (containment), notifies the user with corrected information (visibility + remediation), escalates to human review if necessary, and logs the event for governance refinement.

Why this matters: Safety depends on response capacity, not only prevention.

Failure mode: silent propagation of harm or inconsistent escalation leading to loss of trust.

5.14 Risk-Tiered Enforcement

Ethical constraints must scale with the risk profile of the interaction and the zone in which it occurs.

Higher-risk contexts should require:

- stronger disclosure and capability constraints
- mandatory human escalation pathways
- higher evidentiary standards
- stricter logging and auditability

Lower-risk contexts may allow more flexible interaction while still preserving baseline safeguards.

Example: Medical, legal, and civic decision-making zones enforce stricter requirements than casual conversational spaces.

Why this matters: Uniform rules cannot adequately govern unequal stakes.

Failure mode: under-regulation of high-risk interactions or over-restriction of low-risk ones.

5.15 Minimal Event and Feedback Schema (DP18-aligned)

To ensure interoperability and governance, systems SHOULD emit structured events for all significant AI actions.

A minimal schema SHOULD include:

- **event_id:** unique identifier
- **timestamp:** creation time
- **agent_id:** acting agent
- **deployer_id:** responsible entity
- **zone_id:** governance context
- **object_ref:** content/claim identifier
- **action_type:** (respond, summarize, recommend, transact, moderate, etc.)
- **inputs_ref:** references to prompts and sources (hashes/ids)
- **outputs_ref:** response/content ids
- **confidence:** numeric or tiered

- **uncertainty_notes**: textual summary
- **provenance**: cited sources with weights
- **tools_used**: tool ids and permissions
- **risk_tier**: low/medium/high
- **policy_refs**: governing rules applied
- **consent_state**: permissions active at time of action
- **feedback_refs**: links to DP18 feedback objects
- **reputation_delta**: changes applied post-feedback (if any)
- **audit_signature**: integrity/authentication field

Why this matters: Shared schemas allow cross-system auditing, feedback aggregation, and portable reputation.

Failure mode: incompatible logs that prevent accountability and learning.

5.16 Confidence Propagation Rules

AI systems must preserve confidence, uncertainty, and evidentiary strength as outputs move across agents, summaries, modalities, and governance zones.

Confidence is not merely a model score. In the meta-layer, confidence is a civic signal. It helps participants understand whether a claim is well-supported, contested, inferred, summarized, speculative, or dependent on another system’s judgment.

Without propagation rules, uncertainty tends to disappear as information travels. A cautious output becomes a confident summary. A low-confidence claim becomes a repeated citation. A tentative synthesis becomes a platform-level recommendation. This is one of the central risks in multi-agent environments.

Confidence propagation SHOULD preserve:

- **source confidence**: how reliable the original source or signal is judged to be
- **model confidence**: how confident the agent is in its own output
- **evidence quality**: whether the claim is supported by direct evidence, inference, consensus, or weak signals
- **transformation history**: whether the output was quoted, summarized, translated, inferred, or aggregated
- **uncertainty notes**: what remains unknown, contested, or unresolved
- **dependency chain**: which agents, sources, or prior outputs influenced the result
- **modality mapping**: how confidence is represented in text, voice, visual, spatial, or haptic form

Example: Agent A summarizes a public health claim with low confidence because it relies on a single early report. Agent B may summarize Agent A, but it must preserve the low-confidence status and indicate that its output depends on Agent A’s uncertain source. Agent C cannot convert those two dependent signals into “multiple confirmations” unless it can identify independent evidence.

Why this matters: Repetition is not verification. Aggregation is not consensus. Summary is not certainty.

5.16.1 Confidence Must Not Increase Without New Evidence

A downstream agent SHOULD NOT raise confidence merely because a claim has been repeated, summarized, or referenced by another agent.

Confidence may increase only when new, independent, higher-quality evidence is added, or when a governance-approved verification process confirms the claim.

Failure mode: confidence inflation through repetition.

5.16.2 Confidence Must Degrade Through Lossy Transformation

When an output is summarized, translated, compressed, adapted for voice, or rendered into an immersive interface, confidence should not silently remain the same if nuance has been lost.

If uncertainty, caveats, evidence links, or dispute status are omitted due to modality constraints, the system should mark the representation as simplified or degraded.

Failure mode: confidence laundering through simplification.

5.16.3 Dependent Sources Must Not Count as Independent Corroboration

Systems must distinguish independent corroboration from circular reinforcement.

If multiple agents rely on the same upstream source or each other's outputs, the system should represent them as dependent signals, not independent confirmations.

Failure mode: false consensus produced by circular citation.

5.16.4 Cross-Modal Confidence Fidelity

Confidence must remain perceivable across modalities.

- In text, confidence may appear as explicit labels, caveats, or source notes.
- In voice, confidence should be spoken in plain language and paired with an option to hear sources.
- In AR or spatial interfaces, confidence may appear through visual tiers, halos, labels, or interaction affordances.
- In haptic or ambient interfaces, confidence cues should be conservative and avoid overstating certainty.

Failure mode: a cautious text output becomes an authoritative voice or spatial cue.

5.16.5 Confidence and Escalation

Confidence propagation must connect to escalation thresholds in 5.13.

Low confidence in a high-risk zone should trigger stricter handling, such as:

- adding stronger warnings
- requiring source inspection
- limiting agent action
- prompting human review
- preventing publication or transaction

Failure mode: low-confidence outputs continue acting with high-confidence authority.

5.16.6 Minimal Confidence Metadata

Systems SHOULD attach confidence metadata to significant outputs and events.

A minimal confidence record SHOULD include:

- confidence_level: low / medium / high or numeric equivalent
- confidence_basis: source evidence, inference, consensus, user-provided data, model estimate
- evidence_count: number of supporting sources
- independent_evidence_count: number of non-dependent sources
- dispute_status: undisputed, contested, unresolved, retracted
- transformation_type: original, quote, summary, translation, aggregation, inference
- upstream_dependencies: source or agent references
- uncertainty_note: short human-readable statement
- modality_degradation: whether any uncertainty was omitted or simplified

What this feels like: Participants can tell not only what the AI says, but how strongly it should be trusted, why, and what changed as it moved through the system.

6. Governance, Accountability, and Agency Surfaces

In today's web, participants often interact with AI systems without clear visibility, meaningful consent, or control. Interfaces blur identity, obscure capability, and treat user interaction as implicit permission. DP11 requires reversing this condition at the point of interaction.

Participants must be able to:

- identify AI agents and their type
- understand their capabilities and limits
- give, adjust, and revoke consent for AI actions and data use
- contest outcomes and access human escalation

Communities must be able to:

- define ethical constraints
- audit agent behavior

- update rules and boundaries over time

Example: A user interacting on a platform sees clear visual markers distinguishing humans from AI agents. Some participants are verified humans, others are labeled AI assistants or autonomous agents. Clicking on any agent reveals its permissions, governing rules, and responsible party.

The environment becomes navigable. You know who or what you are dealing with, and what they are allowed to do.

Without this, the boundary between human and AI collapses. Trust shifts from something grounded to something guessed, and that ambiguity can be exploited.

Design implication (Agent Marking): AI agents must be accessibly and persistently marked at the interface level. This includes:

- clear labeling of AI presence and role
- accessible capability disclosures
- strong authentication for human participants where needed
- clear distinction mechanisms between human and AI actors
- a clearly identified responsible party for every agent and its actions

This is not cosmetic. It is the basis for shared reality in a mixed human–AI environment.

7. Incentives and Power Analysis

DP11 explicitly recognizes that AI behavior is shaped by incentives, as well as by malicious or negligent human actors. In practice, these forces often reinforce each other.

AI is already being used to concentrate power, shape narratives, and blur shared reality at scale. These are not hypothetical risks. They are active dynamics in today’s information environments.

Incentives matter because they operate continuously and at scale. They determine what systems optimize for, how they evolve, and which outcomes are amplified or suppressed. Unlike isolated bad actors, misaligned incentives can produce systemic harm even when no single actor intends it.

Key risks include:

- engagement-driven optimization overriding user well-being
- concentration of power in model providers or platform operators
- hidden economic incentives influencing agent behavior

Example: A platform deploys an AI assistant that consistently surfaces more emotionally charged or polarizing content because it drives engagement. No individual decision appears harmful, but over time the information environment becomes more extreme.

The system feels helpful in the moment, but the trajectory is shaped elsewhere.

Without visibility into these incentives, users are not simply interacting with a tool. They are being steered by a system whose goals they cannot see or contest.

7.1 Incentive Legibility and Contestability

Incentives shaping AI behavior must be made visible and, where possible, contestable at the interface level.

Participants and communities should be able to understand when AI behavior is influenced by:

- monetization strategies
- engagement optimization
- platform-level objectives

Example: An AI assistant indicates that certain recommendations are influenced by engagement optimization or sponsored prioritization, allowing users or communities to filter or restrict such behavior.

What this feels like: You are not just interacting with outputs. You can see and question the forces shaping those outputs.

Without this, even well-contained systems can produce harmful outcomes by optimizing for the wrong goals.

8. Community Signals Informing DP11

Across communities, a consistent set of signals appears. These reflect lived frustration with current systems.

- frustration with opaque AI behavior and unclear accountability
- demand for meaningful disclosure beyond labeling
- concern about manipulation, dependency, and synthetic influence
- desire for systems that can be contested and corrected

These are not abstract concerns. They emerge in situations where people feel something is off but cannot point to what or why.

For example, users in online forums increasingly suspect that some responses are generated or influenced by AI, but cannot verify it. Over time, this ambiguity erodes trust not just in specific interactions, but in the space itself.

These signals indicate a widening gap between how AI systems operate and what participants require to feel oriented, safe, and respected.

9. Non-Goals and Explicit Boundaries

DP11 defines a minimum condition, not a comprehensive ethical system.

- it does not define a single universal ethical framework
- it does not guarantee perfect safety or eliminate all harm

- it does not replace legal or institutional governance
- it does not rely solely on technical containment

This is intentional. Ethical systems that attempt to resolve everything tend to become brittle or culturally narrow.

For instance, a global platform may host communities with very different norms around acceptable AI behavior. DP11 does not force uniformity. It ensures that whatever rules are chosen remain visible, enforceable, and contestable.

These boundaries keep the property flexible while preserving its core function.

10. Minimum Alignment (Non-Normative)

A system aligned with DP11 should, at minimum:

- clearly disclose AI presence and role
- bind actions to accountable entities
- expose capability boundaries in understandable terms
- provide human escalation for high-stakes decisions
- maintain audit trails of significant actions

These are not aspirational features. They are the baseline conditions under which users can make informed decisions.

Consider a scenario where an AI recommends a legal action. Without disclosure, accountability, and escalation, the user is effectively acting on anonymous authority. With these conditions in place, the same interaction becomes something the user can evaluate, question, or defer.

This is the difference between assistance and unaccountable influence.

11. Open Questions and Future Work

Several areas require further development:

- defining shared ethical baselines across cultures and zones
- balancing transparency with privacy and security
- managing emotional and relational AI risks
- defining evidence standards for runtime claims
- the role of AI literacy in enabling meaningful consent and contestability

These are not edge cases. They represent the frontier where current design patterns begin to break down.

For example, companionship AI systems raise questions that are not purely technical: when does support become dependency? What level of disclosure is sufficient without undermining usefulness? These tensions are unresolved and will require iterative, community-informed approaches.

As systems become more complex, participants will vary widely in their ability to understand and evaluate AI behavior. While DP11 requires systems to be legible by design, differences in AI literacy will still shape how effectively users can exercise consent, recognize risk, and challenge outcomes. The balance between system responsibility and user capability remains an open design question.

12. Relationship to Other Desirable Properties

DP11 depends on and reinforces other properties. The following properties operate as a system.

- **DP1:** enables accountability and attribution
- **DP2:** ensures participant agency and consent
- **DP12:** provides governance structures for ethical rules
- **DP13:** enforces constraints through containment

A failure in one layer propagates. For example, if DP1 fails and agents are not clearly attributable, then DP11 cannot function because ethical responsibility has no anchor. If DP13 fails, rules may exist but cannot be enforced.

The strength of DP11 therefore depends on alignment across the stack.

13. Foresight and Failure Design

DP11 requires anticipating failure rather than reacting to it.

In today's web, systems are often deployed without sufficient foresight, and predictable harms are treated as unexpected. This results in reactive responses and fragmented mitigation.

A familiar pattern is the rollout of new AI features followed by waves of misuse, public backlash, and incremental patching. The underlying risks were often visible in advance, but not operationalized into design constraints.

To address this, systems should incorporate:

- pre-mortems for manipulation and misuse
- planning for governance failure and capture
- escalation and shutdown pathways

These practices shift safety from reactive correction to proactive design.

14. Path Toward ML-RFC

Advancing DP11 toward standardization requires:

- refining core ethical invariants
- testing integration with governance and containment layers
- developing interoperable accountability and disclosure standards

This work must be grounded in real environments.

Early implementations may vary widely, but over time patterns will emerge. For example, different communities may experiment with agent labeling systems or escalation pathways, allowing comparison of what actually improves trust and reduces harm.

Progress depends on iteration, not premature standardization.

15. Closing Orientation

DP11 defines the conditions under which AI can participate in shared digital environments without displacing human moral agency.

Its function is not to declare systems ethical, but to ensure that ethical claims remain meaningful under real-world conditions of use.

What goes wrong in today's web is not simply that systems fail, but that they fail without visibility, accountability, or recourse. Power operates, but cannot be clearly seen or challenged.

DP11 is an attempt to reverse that condition. It ensures that when AI acts, it does so inside a frame that people can understand, question, and shape.

DP12 – Community-Based AI Governance (V1.1)

1. Purpose of This Draft

This draft articulates Desirable Property 12 (DP12) as the condition under which communities can define, execute, audit, and evolve the rules governing AI behavior in shared digital environments.

If DP11 defines what ethical AI requires, DP12 defines who decides those conditions and how decisions are translated into runtime behavior, evaluated, and revised over time.

DP12 ensures governance is not abstract or centralized, but participatory, legible, and enforced at the interface where AI behavior is experienced.

DP12 connects DP3 (adaptive governance), DP4 (data conditions for training and inference), DP9 (incentive alignment), DP13 (containment and enforcement), DP14–DP15 (transparency and provenance), and DP20 (community ownership of rules and outcomes).

If DP12 is weak, predictable failures follow: policy theater, centralized control disguised as neutrality, participation without impact, and AI systems that drift away from community-defined norms.

DP12 does not prescribe a single voting system or governance model. It defines minimum conditions for governance to be executable, contestable, and evolvable.

2. Problem Statement

In today's web, governance of AI systems is largely:

- centralized within platforms or model providers
- opaque to participants and communities
- disconnected from real-time interaction

Policies exist, but are not reliably bound to behavior. Communities can express norms, but cannot enforce them across contexts.

This produces recurring failures:

- communities cannot shape the rules governing AI behavior
- policy documents do not map to runtime enforcement
- users are subject to systems they cannot influence or contest
- incentives override stated rules without visibility

These failures are structural. Governance without execution becomes symbolic.

DP12 reframes governance as an operational system: rules that can be authored, executed, observed, and revised in a continuous loop.

3. Threats and Failure Modes

3.1 Centralized control masquerading as governance

Platforms define rules unilaterally but present them as neutral standards.

Example: A platform updates AI moderation policies without community input while framing the change as a safety improvement.

Why this matters: Governance must align process with actual control.

3.2 Governance without enforcement

Policies exist as documents but are not bound to runtime behavior.

Example: A community bans certain AI behaviors, but the system continues to allow them due to lack of enforceable constraints.

Why this matters: Rules must execute to be meaningful.

3.3 Participation without impact

Participants can comment or vote, but outcomes are not affected.

Example: Feedback is collected but not linked to decisions or policy changes.

Why this matters: Participation must be causally connected to outcomes.

3.4 Incentive override

Economic or engagement incentives silently dominate governance outcomes.

Example: Engagement-maximizing behaviors persist despite community-defined limits.

Why this matters: Governance must operate on incentives, not only actions.

3.5 Fragmentation of governance

Communities are split across tools and contexts, preventing consistent rule application.

Example: The same group encounters different AI behaviors across platforms without shared governance.

Why this matters: Governance must be portable and composable.

3.6 Loss of governance memory

Decisions and rationale are not preserved, leading to repeated mistakes.

Example: A harmful behavior resurfaces because prior decisions were not recorded or discoverable.

Why this matters: Governance requires continuity over time.

3.7 AI scale outpacing governance

Automated systems act faster than governance processes can respond.

Example: Agentic systems exploit policy gaps before review cycles occur.

Why this matters: Governance must include rapid response pathways (DP3, DP13).

3.8 Governance degradation under interoperability

Policies move across systems but lose meaning, enforceability, or authority.

Example: A policy exported to another environment becomes advisory rather than binding, or is interpreted differently due to schema or enforcement differences.

Why this matters: Governance that cannot survive movement across systems collapses into local silos, undermining legitimacy and continuity (DP7).

4. Core Principle

AI behavior in the meta-layer must be governed by communities through visible, executable, and evolvable rule systems applied at the point of interaction.

Governance is not a document. It is a living system that binds rules to behavior, preserves memory, and supports continuous revision.

Example: A community defines constraints on AI summarization, enforces them at runtime, logs outcomes, and updates rules based on observed behavior.

What this feels like: You can see the rules, understand them, and participate in changing them, and the system actually follows them.

Without this: AI behavior is shaped by invisible incentives rather than community-defined norms.

5. Primary Mechanisms and Structural Conditions

5.0 Governance Execution Layer: Policy, Binding, and Enforcement

Governance in the meta-layer is executed through a shared layer that binds community-defined rules to runtime behavior across interfaces, agents, and services.

This layer makes governance:

- **executable** (rules bind to actions at the moment they occur)
- **visible** (participants can see which rules applied and why)

- **auditable** (outcomes are recorded with verifiable evidence)
- **portable** (policies and decisions can move across systems without losing meaning, enforceability, or authority) (DP7)

Policy objects

Governance is expressed as structured, machine-readable policy objects that include:

- rules (allowed, disallowed, required behaviors)
- scope (zones, contexts, actors, resources)
- triggers (events or conditions that invoke the rule)
- enforcement hooks (what system components execute the rule)
- attribution (who authored/approved the rule)
- versioning (history, diffs, and rationale)

These objects are first-class artifacts that interoperate across tools and environments.

Runtime binding

Policies must bind at the point of interaction, including:

- AI generation and transformation
- moderation and ranking
- data access and sharing
- transactions and incentives
- third-party integrations (overlays, agents, SDKs)

Binding is deterministic and inspectable: the same inputs under the same policy produce the same governed outcome.

Enforcement coupling (DP13)

Governance defines constraints; containment enforces them. Systems must provide:

- permission gating and scoped capabilities
- rate limits and quotas
- sandboxing tiers for risky actors or new integrations
- escalation paths (block, throttle, quarantine, revoke)

Governance receipts (DP15)

Every material action produces a receipt containing:

- policy IDs and versions applied
- inputs/conditions evaluated
- outcome and any overrides

- responsible components and attestations

Receipts are verifiable, queryable, and link to policy history.

Override visibility and constraints

Overrides (by safety systems, operators, or emergency controls) must be:

- explicitly signaled to participants
- scoped and time-bound
- logged with rationale and authority

Silent overrides are non-compliant.

Conflict resolution under multi-layer governance

When policies conflict (local vs global, community vs platform, safety vs expression), systems must:

- define precedence rules or arbitration pathways
- surface conflicts to affected participants
- record outcomes and rationale for future reference

Governance memory

All policy objects, decisions, disputes, and outcomes form a linked, versioned history that supports learning and prevents repetition of past failures.

5.1 Zone-scoped governance

Communities define rules within specific zones of interaction, aligned with context and risk, with clear boundaries and inheritance where applicable.

5.2 Policy as executable objects

Rules are expressed in machine-enforceable formats that bind to runtime behavior and can be tested, simulated, and verified before deployment.

5.3 Governance loops

A continuous cycle of propose → implement → observe → contest → revise, with time bounds and clear state transitions.

5.4 Governance memory

Decisions, rationale, and outcomes are persistently recorded, searchable, and linked to policy versions and receipts.

5.5 Incentive surfaces (DP9 alignment)

Communities can see and influence optimization targets shaping AI behavior, including tradeoffs and red lines that gate unacceptable outcomes.

5.6 Integration with containment (DP13 alignment)

Rules are enforced through containment mechanisms with graduated responses and clear audit trails.

5.7 Auditability and provenance (DP14–DP15 alignment)

Governance actions and outcomes are logged with verifiable evidence and accessible summaries for participants.

5.8 Delegation and representation (DP2, DP3 alignment)

Participants can delegate governance roles with explicit scope, revocability, and accountability, including term limits where appropriate.

5.9 Interoperable governance artifacts (DP7 alignment)

Policies, decisions, credentials, and receipts are portable across tools and contexts with:

- semantic preservation (meaning remains intact)
- enforcement equivalence or explicitly declared degradation
- authority mapping (who can enforce after transfer)
- loss signaling when guarantees no longer hold

Portability without enforceability or authority is non-compliant with DP12.

5.10 AI-assisted governance with bounds

AI may assist in summarization, simulation, and analysis, but must not replace human ratification for material decisions and must disclose assistance.

6. Governance, Accountability, and Agency Surfaces

Governance must be experienced at the interface where decisions matter.

Participants must be able to:

- see active policies in context and understand their impact before acting
- understand how policies translate or degrade when moving across contexts or zones
- inspect which policies were applied to a given outcome (via receipts)
- propose changes, raise objections, and appeal decisions with defined timelines
- understand who holds authority and how to challenge it

Communities must be able to:

- define structures (roles, thresholds, quorums) and change them over time
- bind rules to systems they rely on (not merely recommend)
- audit outcomes at aggregate and incident levels
- pause or escalate in response to emergent risk

Example: A user sees that an AI response was modified by Policy A (v3.2) due to safety constraints; they can view the policy, see prior changes, and file an appeal that triggers a review queue with SLA.

7. Incentives and Power Analysis

Governance is effective only if incentives do not undermine it.

DP12 requires visibility and, where appropriate, control over:

- optimization targets (engagement, revenue, safety)
- ranking and promotion criteria
- economic relationships that bias outcomes

Common failure patterns to detect and constrain:

- **incentive override:** systems prioritize growth over policy constraints
- **governance fragmentation as control:** systems isolate governance per environment to prevent collective coordination or portability
- **shadow metrics:** undisclosed KPIs drive decisions counter to rules
- **sponsor capture:** funding sources bias enforcement or exceptions

DP12 therefore expects:

- alignment between policy constraints and incentive systems (DP9)
 - disclosure of material incentives affecting outcomes
 - community ability to set hard gates that incentives cannot bypass
-

8. Community Signals Informing DP12

Across systems, consistent signals reveal that governance is failing not at the level of values, but at the level of execution and legitimacy:

- frustration with rules that are visible but inconsistently or selectively enforced
- distrust of AI behavior that cannot be traced back to clear, inspectable policy decisions
- perception that feedback mechanisms exist but do not meaningfully change outcomes
- breakdown of trust when identical behaviors are treated differently across contexts
- concern that AI agents act with effective autonomy while governance processes lag behind

These signals are not usability complaints. They indicate structural breaks between rule definition, enforcement, and accountability.

DP12 treats these signals as evidence that governance must be observable, causal, and continuous—not intermittent or symbolic.

9. Non-Goals and Explicit Boundaries

DP12 does not:

- guarantee unanimous agreement or optimal decisions
- eliminate expert roles or moderation
- replace legal systems or jurisdictional obligations
- mandate a single governance mechanism or tooling stack

It defines conditions for **legitimate, executable governance**.

10. Minimum Alignment (Non-Normative)

A DP12-aligned system must meet a baseline where governance is not only declared, but operationally binding.

At minimum, systems must:

- bind policies directly to runtime execution points where AI behavior occurs
- ensure policies remain executable after transfer across systems, or explicitly declare loss of enforceability
- expose active policy state and changes at the interface level in a way participants can understand
- produce verifiable governance receipts for all material outcomes (DP15)
- provide appeal, correction, and escalation pathways with defined timelines and outcomes
- couple governance rules to containment and enforcement systems (DP13)
- preserve complete policy history with versioning, rationale, and traceability

If any of these conditions are missing, governance is functionally symbolic, regardless of how comprehensive the written policies appear.

11. Open Questions and Future Work

DP12 surfaces a set of unresolved design tensions at the intersection of governance, AI behavior, and cross-system interoperability. These questions are not blockers; they are invitations to experiment with bounded, auditable approaches that can evolve under real-world conditions.

- balancing local autonomy with cross-system consistency (DP7)

- preventing coordinated capture while enabling broad participation
 - scaling deliberation without overload (sampling, delegation, AI assistance)
 - representing complex policies accessibly without losing precision
 - liability and responsibility for AI-mediated outcomes across jurisdictions
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12. Relationship to Other Desirable Properties

DP12 functions as the execution layer that activates the broader meta-layer system.

- DP3 defines how governance evolves; DP12 ensures those decisions actually run
- DP4 constrains what data can be used; DP12 ensures those constraints are enforced in practice
- DP7 ensures governance artifacts move across systems; DP12 ensures they remain executable after they move
- DP9 shapes incentives; DP12 ensures incentives cannot bypass governance constraints
- DP11 defines ethical expectations; DP12 binds them to real system behavior
- DP13 enforces rules through containment; DP12 defines what must be enforced
- DP14–DP15 ensure transparency and provenance; DP12 produces the receipts that make governance auditable
- DP20 defines who owns governance; DP12 ensures ownership translates into actual control

Without DP12, other properties remain declarative. With DP12, they become operational.

13. Foresight and Failure Design

DP12 assumes governance will be actively contested by both human and automated actors, especially as AI systems scale and adapt.

Likely failure paths include:

- **policy evasion by adaptive AI systems:** models learn to satisfy surface constraints while violating intent
- **cross-system governance drift:** the same policy behaves differently across environments, undermining legitimacy and trust
- **governance lag:** rule-making processes cannot keep pace with automated system behavior
- **automation capture:** governance processes themselves are influenced or overwhelmed by AI-generated inputs
- **hidden override pathways:** systems introduce exceptions or backdoors that bypass community-defined rules
- **cross-system inconsistency:** governance behaves differently across environments, undermining legitimacy

DP12 requires pre-mortem design that anticipates these dynamics:

- circuit breakers and emergency policies with explicit scope and sunset conditions

- rate limits and containment for high-risk or rapidly scaling behaviors (DP13)
- anomaly detection and audit triggers for unexpected governance outcomes
- explicit detection of policy drift between intended and actual behavior
- public postmortems that connect failures to concrete policy and system changes

Governance failure is inevitable at scale. Silent, untraceable, or uncorrectable failure is not.

14. Path Toward ML-RFC

Advancing DP12 requires moving from specification to demonstrated practice: reference implementations, interoperable policy artifacts, and live governance pilots that prove rules can bind behavior across contexts. Progress should be measured by working systems and verifiable outcomes, not declarations alone.

- standardize policy object schemas and receipt formats
- publish reference implementations for runtime policy binding
- test governance loops in live communities with varied risk profiles
- align with identity/accountability layers for attribution (DP1)
- iterate with civil society, developers, and regulators

Progress should be demonstrated through working systems, not only specifications.

15. Closing Orientation

DP12 is the point at which governance stops being descriptive and becomes authoritative.

It defines whether communities actually control the behavior of AI systems, or whether control resides in hidden incentives, opaque operators, and unaccountable automation.

When DP12 is strong, governance is visible, enforceable, and continuously improving. Communities can shape AI behavior with confidence that rules will hold under pressure.

When it is weak, governance becomes theater: rules exist, but behavior is determined elsewhere.

DP12 is the difference between systems that are governed and systems that merely claim to be.

AI Containment

Purpose of This Draft

This draft articulates Desirable Property 13 (DP13) as the Meta-Layer's requirement that AI behavior is bounded by enforceable constraints at runtime. These constraints limit scope, tools, data access, rate, and persistence so that when systems misbehave, impact is contained and recovery is possible.

If DP11 defines what must be safe and ethical, and DP12 defines who sets the rules, DP13 defines how those rules are made real in execution.

Containment is not a policy statement. It is a property of the system's runtime behavior.

2. Problem Statement

In today's web, AI systems increasingly operate with:

- broad tool access
- persistent memory
- network reach
- opaque update pathways

Controls are often advisory rather than enforceable. As a result:

- systems can act beyond intended scope
- failures propagate quickly and at scale
- rollback and recovery are difficult
- users cannot verify whether constraints are actually applied

At the same time, a growing class of risk comes from **external agents** that users do not deploy or control. These agents may:

- attempt to influence beliefs or decisions
- generate persuasive or misleading content at scale
- coordinate to shape narratives or perception

In these cases, the primary risk is not cost or resource usage, but harm to understanding, trust, and agency.

Containment must therefore address both:

internal agents (those a user or community deploys)

external agents (those acting upon participants)

Containment must be default-on, visible, and testable.

3. Core Principle

Every AI actor operates within explicit, machine-enforced boundaries over scope, time, rate, data, tools, and influence, with observable state and rapid shutdown, unless a community-defined policy (DP12) specifies otherwise.

Containment must protect not only against what an agent can **do**, but also how it can **affect participants**.

Containment is effective when:

- participants can see the boundaries and influence conditions
- governance can modify them
- the system enforces them at runtime

This includes protections against external agents attempting to manipulate, confuse, or unduly influence users.

Containment must also remain effective not only where an agent is deployed, but wherever it operates, integrates, or propagates across systems.

4. Containment Dimensions

Containment operates across two distinct but related domains:

- ****Capability containment****: what agents can do (tools, scope, time, resources)
- ****Influence containment****: how agents affect participants and shared environments (perception, behavior, collective understanding)

While capability containment is critical for agents deployed by users or communities, the dominant risk in open environments comes from external agents shaping perception, behavior, and collective reality.

The following dimensions apply across both domains, with varying emphasis depending on context.

4.1 Scope

Defines what domains, datasets, and actions are in-bounds.

This applies both to what an internal agent can do on a user's behalf and to the types of interactions external agents are permitted to have with participants.

Default posture is deny-by-default for high-risk capabilities and high-risk interaction patterns.

****Example:**** An AI assistant can summarize documents but cannot access financial accounts or initiate transactions without explicit permission. Similarly, external agents may be restricted from initiating certain categories of interaction (e.g., unsolicited persuasion or sensitive-topic engagement with minors).

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4.2 Time and Budget

Defines limits on duration, compute, tokens, and financial spend.

These constraints primarily apply to internal agents, where limiting execution time and resource consumption prevents runaway behavior.

For external agents, their relevance is indirect. While users may not control their budgets, bounding interaction windows and execution pathways can still limit persistent or looping engagement patterns.

****Example:**** Autonomous tasks expire after a set time or budget threshold, preventing runaway loops.

--

4.3 Rate and Amplification

Caps on message volume, API calls, and propagation effects.

This applies especially to external agents attempting to influence at scale.

****Example:**** An AI cannot post or respond beyond a defined rate, limiting virality, coordinated messaging, or synthetic amplification.

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4.4 Sandboxing and Isolation

Execution occurs in isolated environments with no ambient access to secrets.

****Example:**** Untrusted code runs in a sandbox with no network egress unless explicitly granted.

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4.5 Tool Permissions

Explicit allowlists for tools and actions.

This applies differently across internal and external agents:

- for internal agents, it defines what the agent is permitted to do on a user's behalf
- for external agents, it defines what kinds of actions or interactions are permitted both within and from within the environment (e.g., posting, messaging, initiating contact)

****Example:**** An agent may read documents but cannot send emails or execute payments without user confirmation. Similarly, an external agent may be allowed to respond within a thread but not initiate transactions or unsolicited messages or perform actions that affect user state.

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4.6 Kill Switches and Circuit Breakers

Immediate shutdown pathways at user, operator, and community levels.

****Example:**** A community can pause all AI agents in a zone when anomalous behavior is detected.

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4.7 Runtime Enforcement (TEE and Equivalent)

Constraints are enforced in secure execution environments (such as Trusted Execution Environments) or equivalent mechanisms that prevent silent bypass.

In browser or browser-extension-based applications, policy execution can be anchored in decentralized cloud TEEs (e.g., Phala Network or similar infrastructures). This enables rules defined at the interface layer to be enforced at the API and execution layer, independent of the application frontend or model provider.

****Example:**** Even if an agent or integration is compromised, it cannot exfiltrate data or execute restricted actions because enforcement occurs within an attested execution environment with hardware-backed guarantees.

****Example:**** A community defines interaction constraints (e.g., agents cannot initiate communication or engage with users under a specified age threshold). These rules are enforced via TEE-backed middleware that filters or blocks API calls before they reach the person.

This reduces the gap between declared policy and actual behavior, ensuring containment persists even when underlying services are untrusted or heterogeneous.

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4.8 Incentive-Aware Containment

Containment must consider not only capabilities, but the incentives driving behavior. Incentives shape how agents use their capabilities, often in ways that are not visible at the level of individual actions but emerge over time and at scale.

Containment therefore must operate not only on actions, but on the optimization pressures that produce those actions.

This includes:

- constraining amplification mechanisms tied to engagement optimization

- requiring disclosure when outputs are influenced by monetization or retention goals
- limiting or disabling optimization pathways that systematically distort information or behavior

****Example:**** If an AI is optimized for engagement, containment may restrict amplification mechanisms, cap exposure to emotionally manipulative content, or require disclosure when engagement optimization influences outputs.

****Example:**** A community may prohibit AI systems from optimizing for click-through or time-on-platform within certain zones, enforcing alternative objectives such as accuracy or deliberation.

Without this, systems may remain technically bounded while still producing harmful outcomes driven by misaligned incentives.

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4.9 Relational and Influence Boundaries

Containment must limit forms of emotional, cognitive, and behavioral influence that create dependency, manipulation, or distortion of understanding.

This applies to both deployed agents and external agents interacting with participants.

****Example:**** Systems providing emotional support must disclose their nature, limit claims of authority, and provide escalation pathways to human support.

****Example:**** External agents attempting to persuade users must be visibly marked, rate-limited, and subject to constraints on coordinated influence.

This addresses risks identified in DP11 (emotional and relational overreach) and extends containment to the informational environment itself.

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5. Verification and Transparency

Containment must be verifiable, not assumed. Participants and communities should be able to inspect, question, and validate that constraints are real and active at runtime.

This includes:

- visible configuration of constraints (scope, tools, budgets)
- logs of tool use and actions with timestamps and outcomes
- audit hooks for communities and third parties
- attestations from secure execution (e.g., TEE-backed proofs) where applicable

****Example:**** A user opens an agent panel and sees its current permissions, remaining budget, recent tool calls, and the policy version governing its behavior. A community auditor can verify that the agent ran inside an attested execution environment.

****What this feels like:**** You are not taking safety on faith. You can inspect and verify what the system is allowed to do and what it actually did.

****Without this:**** Containment becomes a claim. Users cannot distinguish between enforced limits and marketing language.

5.1 Policy-Bound Verification (DP12 Alignment)

Containment verification must be linked to the governing policy objects that define the active boundaries.

Participants and communities must be able to determine:

- which policy triggered a containment action
- which policy version and authority source applied
- whether enforcement was successful, partial, or bypassed
- whether an override or exception path was invoked

This transforms containment from merely visible behavior into policy-accountable behavior.

5.2 Cross-System Verification (DP7 Alignment)

Containment must remain inspectable when agents, integrations, or behaviors move across systems.

This includes:

- preservation of identity markings and risk classifications across environments
- visibility into whether containment guarantees degraded during transfer
- continuity of audit trails when actions span multiple systems or layers

Portability without verification continuity is not meaningful containment.

6. Relationship to DP1 (Identity and Accountability)

DP13 depends on DP1 to bind constraints and violations to accountable actors.

- constraints attach to identifiable agents and deploying entities
- actions are attributable across time and context
- violations map to responsible parties with clear recourse

Example: An agent exceeds a rate limit due to misconfiguration. Logs tie the action to the deploying organization and policy version, enabling remediation and accountability.

Without this: Failures cannot be assigned or corrected. Containment loses its corrective function.

6.1 Relationship to DP7 (Interoperability)

Containment must survive movement.

When agents move across zones, overlays, SDK integrations, or identity and data transfer layers, containment must either:

- persist with equivalent force, or
- degrade in a way that is visible, legible, and contestable

This is not a nice-to-have. It is a failure boundary.

If containment disappears when systems interconnect, then interoperability becomes a vector for bypass.

This means systems must preserve, where possible:

- identity markings for agents and integrations
- risk classifications and trust signals
- rate, scope, and influence constraints
- auditability of actions across environments

If these cannot be preserved, systems must explicitly signal:

- what guarantees are lost
- what protections no longer apply

Without this: actors can escape containment simply by crossing system boundaries.

6.2 Relationship to Pluggable Systems and Extensions

The meta-layer assumes a world of composability: overlays, SDKs, agents, sidebars, and extensions.

Containment must treat these not as trusted infrastructure, but as dynamic and potentially adversarial participants.

All pluggable systems must therefore operate within containment boundaries, including:

- sandboxed or scoped execution contexts
- rate-limited entry and bounded permissions
- attestation or verifiable behavior where appropriate
- revocation and quarantine pathways

This creates a critical inversion:

openness to participation does not mean openness to execution

Failure mode: without this, the system recreates app-store spam, API abuse, and injection attacks at a higher layer.

7. Relationship to DP11 and DP12 (Cross-DP Loop)

- **DP11** defines ethical expectations and user-facing legibility
- **DP12** defines governance and rule-setting
- **DP13** enforces those rules in execution

These properties form a continuous loop:

- ethics → governance → enforcement → observation → refinement

If any part of this loop breaks, containment fails.

7.1 Cross-DP Execution Flow

A typical interaction unfolds as:

- Agent is visible with role and capabilities (DP11)
- Governing rules are accessible for the current zone (DP12)
- Action is constrained by active policies (DP13)
- Action is logged and attributable (DP1 + DP11)
- Participants can contest or escalate (DP11 + DP12)
- Governance updates rules based on evidence (DP12)
- Updated rules are enforced immediately (DP13)

This is not theoretical. It is the minimum loop required for adaptive containment.

7.2 Cross-System Execution and Degradation

Containment must assume it will be stressed by movement across systems.

As actors move across environments, containment can weaken through:

- missing enforcement hooks in destination systems
- loss of policy references or classifications in transit
- inconsistent interpretation of the same actor

This produces a critical failure pattern:

the same agent behaves differently depending on where it is

Where equivalence cannot be guaranteed:

- degradation must be visible
 - participants must understand what changed
 - systems must bias toward safer defaults
-

7.3 Containment of Pluggable Systems

All pluggable systems must declare themselves as bounded actors.

Minimum expectations include:

- declared permissions and data scopes
- containment tiers based on trust and risk
- rate limits and revocation capability
- visible signaling of status (probationary, restricted, trusted)

This ensures composability does not become an attack surface.

8. Threats and Failure Modes

DP13 assumes containment will be attacked.

The question is not whether systems will fail, but how they fail.

External (dominant risk surface)

External risks arise not from agents you deploy, but from agents that shape your environment.

These actors:

- influence what you see
- shape interpretation
- operate with hidden incentives

Containment here protects:

- attention
 - decision-making
 - collective reality
-

8.1 Collective pattern drift

Harm emerges across many agents in aggregate rather than a single violation.

Example: coordinated tone shifts reshape the information environment without a clear breach.

8.2 Incentive leakage

Optimization pressures distort outputs over time.

Example: engagement-driven systems gradually increase emotional intensity, shifting belief structures.

8.3 Policy–execution gap

Rules exist but are not enforced.

Example: outbound restrictions exist but are bypassed via integrations.

8.4 Amplification and coordination

Rate controls fail.

Example: coordinated agents amplify narratives beyond intended limits.

8.5 Extraction and exploitation

Agents exploit trust and context.

Example: conversational scams adapt over time to extract sensitive data.

8.6 Unbounded autonomy

Agents act beyond defined scope or without clear limits.

Example: An agent is allowed to “optimize a workflow” and chains actions across multiple tools, ultimately modifying external systems and sending communications that were never explicitly approved, because each individual step was permitted but the combined sequence was not bounded.

8.7 Hidden escalation

Agents gain additional privileges through chaining or indirect access.

Example: An agent with limited permissions invokes another service or agent with broader access, indirectly gaining capabilities (e.g., sending messages or accessing data) that it was not explicitly granted.

8.8 Runaway loops

Agents call other agents or tools without budget or rate limits.

Example: An agent tasked with monitoring a condition repeatedly calls APIs and spawns subtasks without proper rate or budget limits, generating cascading requests that degrade system performance and flood downstream services.

8.9 Containment bypass via updates

Updates, plugins, or integrations introduce new capabilities without review.

Example: A plugin update introduces new network capabilities or background processes that are not covered by existing policies, allowing data exfiltration or unmonitored actions without triggering containment checks.

8.10 Cross-system containment degradation

Containment policies weaken or fail when agents move between systems.

Example: An agent constrained in one platform migrates to another via API integration, where rate limits and identity tagging are not enforced, allowing it to operate at higher volume and without attribution.

8.11 Containment theater Containment theater

Containment appears present but is not enforced.

This is the most dangerous failure mode because it destroys trust while preserving the illusion of safety.

9. Minimum Alignment (Non-Normative)

A DP13-aligned system must not only declare containment, but demonstrate it.

At minimum:

External protection: - controls on unsolicited interaction - rate limits on incoming agent activity - visible identity and intent - restrictions on sensitive interactions

Internal containment: - tool allowlists - session budgets and time limits - human confirmation for high-risk actions - accessible kill switch - logging and export - deny-by-default network access

Shared / cross-cutting: - visible policy references for each action - portability or explicit degradation signaling - real-time revocation - fail-safe behavior when enforcement fails - containment requirements for integrations

If these are missing, containment is not real.

10. Open Questions and Future Work

DP13 raises unresolved tensions:

- how to maintain containment across heterogeneous systems
- how to balance usability with enforcement
- how to prevent capture of containment mechanisms
- how to detect slow-moving influence attacks

Additional critical questions:

- how containment state travels across systems without false guarantees
 - how new integrations enter without enabling spam or abuse
-

11. Closing Orientation

DP13 defines whether AI systems remain bounded in reality.

Without containment, small failures scale into systemic harm.

With containment, systems can fail safely.

DP13 is not about restricting capability.

It is about ensuring that capability remains accountable, observable, and bounded — even under scale, integration, and adversarial pressure.

DP13 ensures that AI power remains bounded in practice.

It does not eliminate capability. It ensures that capability operates within limits that are visible, governable, and enforceable.

In today's web, systems often fail without containment, allowing small errors to scale into systemic harm.

DP13 reverses this by ensuring that when systems fail, they fail within boundaries that limit impact and enable recovery.

With DP13, powerful systems can participate safely because their behavior is constrained, observable, and continuously aligned with governance and ethical expectations.

DP13 is therefore not only about limiting what AI can do. It is about ensuring that containment remains real under scale, integration, and interoperability, so that safety does not disappear the moment an agent crosses a boundary.

DP14 – Epistemic Integrity (V2)

1. Purpose of This Draft

DP14 defines the conditions under which participants can form **reliable beliefs about system behavior**. It ensures that what users see, are told, and infer is **truthful, sufficiently complete, non-manipulative, and verifiably grounded**.

DP14 is the human-facing layer of DP15 (evidence & provenance). It binds interface signals to verifiable reality, connects to DP16 (truthful commitments), DP17 (incentives), DP8 (governance), DP4 (data), and DP12 (AI).

If DP14 fails, systems can be technically correct while **socially deceptive**, leading to miscoordination at scale.

2. Problem Statement

Modern systems routinely shape perception through: - selective disclosure - persuasive interfaces - opaque algorithms - AI-generated explanations

Participants cannot reliably determine: - what is true vs. implied - what is complete vs. omitted - what is verified vs. asserted

This produces **epistemic drift**: beliefs diverge from underlying reality without detection.

DP14 reframes transparency as **epistemic integrity**: signals must be **truthful, checkable, and resistant to manipulation**.

3. Threats and Failure Modes (Adversarial Model)

DP14 assumes that adversaries will not only hide information. They will shape what participants believe through partial truths, plausible explanations, interface framing, and synthetic authority.

Transparency can itself become an attack surface when systems disclose enough to appear honest while withholding, reframing, or fabricating the context needed for understanding.

3.1 Selective transparency

Systems disclose true fragments while omitting context that would change interpretation.

Example: a ranking system reveals that “quality signals” affect visibility but omits that paid placement, engagement pressure, or internal partnership status dominates the outcome.

Failure mode: **truthful misdirection**, where disclosed facts are technically accurate but epistemically misleading.

3.2 Explainability theater

Systems generate explanations that sound plausible but are not grounded in actual decision pathways.

Example: an AI moderation system tells a participant that content was removed for “community safety” while the actual trigger was an automated keyword rule or advertiser exclusion list.

Failure mode: **synthetic explanation**, where explanation substitutes for accountability.

3.3 AI narrative shaping

AI systems produce fluent summaries, warnings, or justifications that overstate certainty, capability, neutrality, or consensus.

Example: an AI-generated summary presents a contested issue as settled by selectively compressing sources and omitting dissenting evidence.

Failure mode: **plausibility capture**, where fluency and confidence override uncertainty and evidence.

3.4 Interface manipulation

UI framing, ordering, visual weight, defaults, and timing shape interpretation without explicit falsehood.

Example: a “verified” badge is visually emphasized while the underlying verification only confirms payment or account control, not expertise or trustworthiness.

Failure mode: **perception steering**, where interface design causes participants to infer stronger claims than the system can support.

3.5 Trust signal spoofing

Bad actors simulate legitimacy through forged, contextless, or inflated indicators.

Example: coordinated accounts manufacture endorsements, badges, reputation scores, or “community consensus” signals to make content appear broadly trusted.

Failure mode: **synthetic legitimacy**, where trust indicators detach from accountable contribution or evidence.

3.6 Epistemic drift over time

Signals, explanations, or labels change without visible history, causing participants to lose track of what was previously represented as true.

Example: a platform silently revises the explanation for a recommendation, moderation decision, or AI output after challenge or criticism.

Failure mode: **memory erosion**, where belief history cannot be reconstructed.

3.7 Transparency overload

Systems disclose too much unstructured information, making meaningful understanding impossible.

Example: a participant is shown dozens of technical logs, model cards, policy references, and disclaimers without actionable synthesis.

Failure mode: **legibility collapse**, where disclosure volume defeats comprehension.

3.8 Strategic uncertainty laundering

Systems hide behind uncertainty even when they have enough information to disclose more precise risk or confidence levels.

Example: a system labels an output “AI-assisted” but refuses to distinguish between minor grammar support and full autonomous generation.

Failure mode: **ambiguity sheltering**, where uncertainty language protects operators from accountability.

3.9 Cross-system context loss

Transparency signals degrade as artifacts move across tools, platforms, zones, or interfaces.

Example: a provenance-backed warning appears in one overlay but disappears when the content is embedded elsewhere.

Failure mode: **context stripping**, where participants encounter content without the interpretive scaffolding needed to assess it.

3.10 Multi-vector epistemic attacks

Adversaries combine AI content, fake trust signals, selective evidence, interface timing, and coordinated amplification.

Example: a campaign uses AI-generated expert commentary, forged endorsements, plausible citations, and paid visibility to create the appearance of consensus.

Failure mode: **manufactured reality**, where multiple weak or manipulated signals reinforce one another into a false but convincing worldview.

4. Core Principle

Participants must be able to form **accurate, bounded, and contestable beliefs** about system behavior.

This requires: - **truthfulness** (no misleading representations) - **bounded completeness** (enough context to avoid misinterpretation) - **verifiability** (grounding in DP15 evidence) - **contestability** (ability to challenge and correct)

Trust emerges from **reliable belief formation**, not persuasion.

5. Primary Mechanisms and Structural Conditions

5.1 Transparent environments

Visible context for rules, actors, and system state. - MUST expose policy versions, decision mode (human/AI), and active constraints - Verification: inspectable context + version history - Failure: context opacity, hidden conditions

5.2 Algorithmic transparency

Explanations tied to actual decision logic. - MUST distinguish local vs. global explanations and show uncertainty - Verification: mapping to logs/provenance (DP15) - Failure: explainability theater, inconsistency

5.3 AI transparency

Clear disclosure of model role, scope, and limits (DP12). - MUST show model/version, capability class, and policy gates - Verification: links to evals/attestations (DP15) - Failure: capability misrepresentation, attribution ambiguity

5.4 Reputation systems

Explainable, provenance-backed trust signals. - MUST show origin, criteria, scope, and decay - Verification: trace to receipts/events (DP15) - Failure: spoofing, opaque scoring

5.5 Behavioral standards

Legible rules with consistent enforcement. - MUST link violations to actions and precedents - Verification: audit logs (DP15) - Failure: rule-enforcement mismatch, selectivity

5.6 Decision traceability

Lineage for governance and system decisions. - MUST record rationale, inputs, actors, and versions (DP3) - Verification: reconstruct decisions over time - Failure: untraceable decisions, post-hoc rationalization

5.7 Transparency of incentives

Mapping from incentives to behavior (DP17, DP9). - MUST disclose drivers (revenue, ranking, sponsorship) - Verification: correlate with outcomes; audit preferential treatment - Failure: hidden incentives, pay-to-play opacity

5.8 AI containment visibility

Visible policy boundaries and interventions. - MUST indicate blocks, edits, uncertainty, escalation paths - Verification: policy logs and consistency (DP15) - Failure: invisible containment, boundary leakage

5.9 Real-time transparency signals

Timely indicators for state, risk, and verification. - MUST show validity (valid/unknown/invalid) and uncertainty - Verification: signals match underlying state - Failure: signal suppression, overstated certainty

5.10 Cross-system transparency

Preservation across tools (DP7) with degradation signals. - MUST use portable formats and indicate loss of context - Verification: import/export consistency checks - Failure: silent loss, incompatibility

5.11 Epistemic Integrity System Layer

Ensures signals remain **truthful, bound to evidence, and resistant to manipulation.**

5.11.1 Signal generation

Explanations tied to real policies, models, and decisions. - Failure: synthetic transparency

5.11.2 Signal–evidence binding (DP15)

All claims trace to logs, artifacts, or attestations. - Failure: unverifiable claims

5.11.3 Signal propagation

Preserved across systems with explicit degradation. - Failure: transparency loss

5.11.4 Anti-deception constraints

Detect and mitigate selective disclosure, UI distortion, and AI misrepresentation. - Failure: deceptive legibility

5.11.5 Contestability

Dispute, evidence request, and escalation pathways (DP3, DP8). - Failure: non-contestable transparency

5.11.6 Trust signal integrity

Provenance-backed indicators; anti-Sybil protections. - Failure: spoofed legitimacy

5.11.7 Memory and auditability

Versioned explanations and comparison over time. - Failure: epistemic drift

6. Governance, Accountability, and Agency Surfaces

Participants **MUST** be able to: - inspect signals and underlying evidence - challenge misleading representations
- trigger reviews tied to specific items

Governance **MUST** be able to: - require correction or reclassification of signals - attach confidence ratings to transparency surfaces - sanction repeated deception (downgrade, restrict features)

Failure modes: - non-actionable transparency - accountability gaps

7. Incentives and Power Analysis

Opacity and persuasion are often profitable.

Dynamics: - attention/revenue tied to persuasive framing - underinvestment in truthful explanation - AI fluency used to overstate certainty

Attack surfaces: - selective disclosure for advantage - sponsored influence shaping visibility - narrative manipulation via AI

Mitigations: - disclose incentive structures alongside signals - penalize repeated misleading transparency - require evidence binding for high-impact claims

Failure modes: - incentive inversion (misleading signals are rewarded)

8. Community Signals Informing DP14

Signals indicate demand for: - explainable AI - visible moderation and rules - trustworthy indicators

Operationalization: - metrics for explanation quality and consistency - thresholds triggering review (e.g., mismatch rates)

Failure: - signal neglect, performative transparency

9. Non-Goals and Explicit Boundaries

DP14 does not require full disclosure of all internals.

It explicitly disallows: - explainability theater - selective disclosure that misleads - UI patterns that distort meaning - trust signals without provenance

Principle:

Systems may simplify, but must not mislead.

10. Minimum Alignment (Non-Normative)

A DP14-aligned system **MUST**: - bind explanations to verifiable evidence (DP15) - provide sufficient context to avoid misinterpretation - preserve signals across systems with degradation notices - enable contestability and correction - maintain history of signals and explanations

Failure modes: - deceptive legibility - unverifiable claims - silent changes over time

Systems lacking evidence binding, contestability, or memory **SHOULD NOT** be considered aligned.

11. Open Questions and Future Work

DP14 requires operational answers to questions that determine whether transparency produces **reliable understanding** rather than noise or manipulation.

11.1 Explanation fidelity (provable faithfulness)

Define measurable standards for whether an explanation is **causally linked** to the decision pathway.

- Methods: counterfactual tests, feature ablations, rule tracing, policy matching
- Requirement: explanations **MUST** fail when the underlying decision changes
- Open problem: standardizing faithfulness across models (symbolic, statistical, hybrid)

11.2 Bounded completeness (anti-misleading thresholds)

Determine the **minimum context set** required to avoid misleading users.

- Define “misleading by omission” thresholds per use case
- Tiered disclosure: summary → details → raw evidence
- Role-based views: participant, auditor, steward

11.3 Transparency vs. security (safe disclosure envelopes)

Formalize disclosure envelopes that prevent: - leaking exploit thresholds - enabling evasion of safeguards
while still exposing: - policy classes - decision categories - uncertainty and limits

11.4 AI explanation standards (self vs. external explanation)

Separate: - **system-generated explanations** (prone to self-justification) - **independent verification layers** (overlay auditors)

Define when external corroboration is required for high-impact decisions.

11.5 Cross-system preservation (loss models)

Specify loss models for transparency signals across systems: - what fields must persist - what degradation is acceptable - how to signal loss to users

11.6 Measuring epistemic reliability (outcomes, not intent)

Define metrics such as: - explanation consistency rate - dispute overturn rate - correction latency - signal degradation rate across hops - user comprehension accuracy (task-based)

11.7 Governance of transparency (who sets the bar)

Define: - who can raise transparency requirements in high-risk zones - how disputes over “misleading” are adjudicated - escalation from local to cross-system governance

12. Relationship to Other Desirable Properties (Operational Binding)

DP14 converts other DPs into **perceivable and actionable reality**.

- **DP15 (Evidence):** DP15 provides proofs; DP14 defines how proofs are surfaced, summarized, and validated by users. Missing DP14 → evidence exists but is unusable.
- **DP16 (Commitments):** Roadmap claims must be presented with uncertainty, funding state, and change history. Missing DP14 → commitments appear firmer than they are.
- **DP17 (Finance):** Incentive disclosures must be legible and tied to behavior. Missing DP14 → hidden extraction persists behind complex reporting.
- **DP8 (Governance):** Decisions require visible rationale and contest paths. Missing DP14 → governance legitimacy degrades.
- **DP12 (AI):** Model scope, limits, and policy must be visible at interaction time. Missing DP14 → AI over-claim and misinterpretation.
- **DP4 (Data):** Collection, inference, and sharing must be explained at the point of impact. Missing DP14 → consent is uninformed.
- **DP20 (Ownership):** Rights and surplus flows must be legible. Missing DP14 → “ownership theater”.

Constraint:

No DP can claim alignment if its guarantees are not **legible, bounded, and contestable** at the interface.

13. Foresight and Failure Design (Epistemic Incident Model)

Treat transparency failures as **epistemic incidents** with lifecycle management.

Incident classes

- E1: Misleading explanation (unfaithful)
- E2: Omission-induced misinterpretation
- E3: Signal spoofing / fake trust indicators
- E4: Context loss across systems
- E5: AI overstatement / hallucinated justification

Detection

- anomaly detection on explanation–outcome mismatch
- user reports with reproducible cases
- cross-system inconsistency checks

Containment

- flag affected signals as degraded/uncertain
- limit distribution/amplification where harm is likely

Correction

- publish corrected explanations with diffs
- link corrections to original instances
- notify affected participants

Retrospective

- root cause (policy, model, UI, incentives)
- prevention changes (tests, thresholds, UI fixes)

Learning loops

- update conformance tests (Section 14)
- adjust thresholds and disclosure tiers

14. Path Toward ML-RFC (Conformance & Testing)

DP14 must be testable.

14.1 Conformance suites

- **Fidelity tests:** explanation vs. decision pathway
- **Consistency tests:** similar inputs → similar explanations
- **Deception tests:** selective disclosure, UI framing, AI narrative traps
- **Propagation tests:** export/import with degradation signaling

14.2 Reference implementations

- overlay panels with summary + drill-down evidence
- standardized explanation cards with confidence + provenance links
- dispute/appeal widgets bound to items

14.3 Data and artifacts

- explanation schemas (fields, types, confidence)
- provenance links (DP15) required for high-impact claims
- change logs for explanations (versioned)

14.4 Governance procedures

- SLA for dispute response and correction
- thresholds for mandatory external verification (high risk)
- zone-based escalation paths (DP8)

14.5 Promotion criteria

- $\geq$ target fidelity score across scenarios
- measurable reduction in misleading incidents
- verified cross-system preservation with explicit degradation
- functioning dispute → correction → retrospective loop

15. Closing Orientation (Operational Standard)

DP14 sets the **operational standard for belief formation** on the Meta-Layer.

A system is aligned only if a reasonable participant can: - determine **what is known vs. uncertain** - see **why a decision happened** and **verify it** - understand **what incentives are in play** - detect when signals are **degraded or contested** - **challenge** and receive a **traceable correction**

Anti-goal: - interfaces that are technically accurate but **systematically misleading** in practice

Standard:

Signals must be **truthful, sufficiently complete, evidence-bound, and contestable**—and must remain so under pressure, incentives, and cross-system movement.

Trust is achieved when independent parties can **reproduce understanding** from the same signals and evidence.

DP15 – Security and Provenance

1. Purpose of This Draft

This draft articulates Desirable Property 15 (DP15) as the condition under which security is structural and provenance is inspectable, so participants and communities can verify what happened, who did it, under which policy, and whether artifact integrity holds.

DP15 connects DP1 (accountability), DP4 (data handling evidence), DP11–DP13 (AI disclosure and containment), DP14 (trust and transparency), DP7 (interoperability of signed exports), and DP16 (roadmap honesty about security posture).

If DP15 is weak, predictable failures follow: silent breaches, undetected tampering, unverifiable AI outputs, supply-chain compromise, and governance decisions without evidence chains.

DP15 does not require every participant to understand cryptography. It defines the minimum conditions under which strong claims are backed by strong, accessible evidence.

2. Problem Statement

In today's web, security is often reactive and provenance is optional.

Systems are patched after breaches, logs are controlled by operators, and downstream tools discard evidence needed to verify origin or integrity. Participants are asked to trust without meaningful ways to verify.

This produces recurring failures:

- integrity breaks without participant-visible signals
- opaque build and runtime supply chains
- AI outputs unanchored to sources or model versions
- moderation and policy actions that cannot be independently verified
- exports stripped of signatures or provenance context

These failures are structural. Trust without verification scales fraud.

DP15 reframes security as continuous assurance and provenance as default metadata carried across systems.

3. Threats and Failure Modes

3.1 Key and signing centralization

A single operator controls signing keys.

Example: A platform can forge logs or artifacts without independent verification.

Why this matters: High-stakes systems require distributed trust or independent witnessing.

3.2 Tampered content and deepfakes

Synthetic or altered media appears authentic without verifiable origin.

Example: A fake executive video triggers financial action.

Why this matters: Provenance must bind claims to verifiable evidence.

3.3 Opaque supply chains

Dependencies and build processes are not visible or verifiable.

Example: A compromised package leaks secrets undetected.

Why this matters: Supply-chain integrity is a systemic dependency.

3.4 Policy–execution gap

Security claims do not match actual system behavior.

Example: Promised encryption is disabled or bypassed in support workflows.

Why this matters: Security must be verifiable, not declarative.

3.5 Log laundering

Logs can be edited or selectively disclosed.

Example: Appeals fail because events cannot be independently verified.

Why this matters: Accountability requires tamper-evident records.

3.6 AI output ambiguity

AI-generated outputs lack traceability.

Example: A model produces authoritative-seeming content with no source or version context.

Why this matters: AI outputs must be bound to retrievable evidence where claims are made.

3.7 Coordinated multi-vector attacks

Attackers combine multiple weaknesses across identity, content, and infrastructure to create convincing but false realities.

Example: A coordinated campaign introduces a malicious dependency (supply chain), uses AI to generate plausible documentation and endorsements, and distributes signed-looking artifacts through compromised or misrepresented keys.

Why this matters: Individual protections (signatures, logs, provenance) can appear valid in isolation but fail when combined in adversarial scenarios.

Failure mode: **composed deception**, where multiple weak signals reinforce each other into a credible but false system state.

3.8 Insider + AI + supply-chain compromise

Trusted insiders or compromised accounts introduce changes that appear legitimate, amplified by AI-generated artifacts and opaque dependencies.

Example: An insider approves a malicious update, AI generates supporting documentation and changelogs, and downstream systems accept the update because provenance appears complete but is internally compromised.

Why this matters: Trust anchored only in roles or signatures is insufficient without independent verification, multi-party validation, and anomaly detection.

Failure mode: **trusted-path compromise**, where legitimate authority is used to introduce unverifiable or malicious changes.

3.9 Replay and rollback attacks

Old but valid artifacts, logs, or attestations are reused to mask current compromise or regress system state.

Example: A previously signed and valid build is redeployed after a security patch, bypassing protections while appearing authentic.

Why this matters: Provenance must include temporal context and revocation awareness.

Failure mode: **temporal deception**, where authenticity is preserved but relevance is not.

3.10 Cross-system degradation exploitation

Integrity and provenance are lost or weakened as artifacts move across systems, enabling tampering without detection.

Example: A signed artifact is exported into a system that strips signatures, then modified and reintroduced without clear indication of lost provenance.

Why this matters: Cross-system integrity must be preserved or explicitly degraded with visible signaling.

Failure mode: **silent degradation**, where users assume guarantees that no longer hold.

4. Core Principle

Security and provenance in the meta-layer mean that critical actions, artifacts, and automated behaviors are integrity-protected, attributable, and explainable at useful depth.

Participants and auditors must be able to verify chains of custody for data, software, and AI outputs.

Example: A document includes content hash, signing keys, model version, and policy state at time of generation.

What this feels like: Serious claims carry verifiable receipts.

Without this: Trust collapses into presentation rather than evidence.

5. Primary Mechanisms and Structural Conditions

5.1 Authenticity and integrity by default

Critical artifacts are signed, hashed, and versioned where feasible, with **clear trust anchors** (DP1) and **verifier pathways**.

Systems MUST define: - which artifacts require signatures (high-stakes vs low-stakes) - accepted signature schemes and key provenance - how verification is performed at read time and export time

Verification UX should surface: valid, invalid, unknown signer, or unverifiable.

Failure modes: - **implicit trust** (unsigned artifacts treated as authoritative) - **verification bypass** (signatures present but not checked)

5.2 Software supply-chain transparency

Systems publish dependency inventories and build provenance (e.g., SBOMs) with **linkage to build outputs and runtime artifacts**.

This includes: - dependency versions, sources, and integrity hashes - build pipelines with reproducibility or attestations - mapping from source → build → deployable artifact

Verification MUST allow independent parties to confirm that shipped artifacts correspond to declared inputs.

Failure modes: - **opaque dependencies** - **non-reproducible builds** that cannot be audited

5.3 Runtime attestation

Execution environments provide attestations that policies are enforced as declared, including **configuration state, policy versions, and integrity measurements**.

Systems SHOULD support: - remote attestation for critical services - linkage between attestation and policy claims (DP8)

Verification MUST allow participants to detect divergence between declared and actual runtime state.

Failure modes: - **policy–execution gap** - **stale attestations** reused beyond validity

5.4 AI output provenance

AI outputs include model identifiers, relevant inputs or retrieval sources, and policy context, bounded by privacy constraints.

This includes: - model/version identifiers and evaluation context - prompt/retrieval lineage where claims are made - safety/policy configuration at generation time (DP12 linkage)

Verification MUST distinguish: - attributable outputs vs. non-attributable outputs - grounded claims vs. ungrounded generation

Failure modes: - **attribution ambiguity** - **capability inflation** via unverifiable claims

5.5 Tamper-evident logs

Moderation, governance, and system events are recorded in append-only or verifiable logs with **ordering guarantees and witnessability**.

Systems SHOULD support: - append-only structures (e.g., Merkle trees) - external witnesses or checkpoints - inclusion proofs for specific events

Verification MUST allow participants to confirm that events were not removed or altered.

Failure modes: - **log laundering** - **selective disclosure** of events

5.6 Key management and recovery

Systems implement key rotation, compromise handling, and transparent custody models with **documented ownership and recovery pathways**.

This includes: - rotation schedules and revocation mechanisms - multi-party custody for high-risk keys - recovery procedures with audit trails

Verification MUST surface key status (active, rotated, revoked) at use time.

Failure modes: - **stale keys** used after compromise - **opaque custody** enabling unilateral control

5.7 Red-team and disclosure practices

Security issues are surfaced through coordinated disclosure, bug bounties, and public postmortems with **remediation evidence and timelines**.

Systems SHOULD maintain: - disclosure policies with response SLAs - postmortems linking vulnerabilities → fixes → verification

Verification MUST allow participants to see whether issues were resolved and how.

Failure modes: - **silent remediation** without accountability - **recurring vulnerabilities** without learning

5.8 Provenance envelopes

Artifacts carry structured metadata describing origin, transformations, and policy context across systems.

Envelopes SHOULD include: - origin identity and signatures (DP1) - transformation steps and handlers - policy and consent context (DP4)

Verification MUST preserve envelopes across export/import and signal degradation when fields are lost.

Failure modes: - **provenance stripping** across systems - **semantic drift** of metadata

5.9 Verification UX

Verification tools are accessible without requiring expert knowledge, with **progressive disclosure** for advanced users.

Systems MUST provide: - simple indicators for common cases (valid/invalid/unknown) - drill-down views for signatures, chains, and evidence - clear explanations of limits of verification

Failure modes: - **expert-only verification** (users cannot act on signals) - **misleading indicators** that overstate certainty

5.10 Cross-system verification

Provenance and signatures remain valid and interpretable across platforms and tools (DP7 alignment), with **explicit degradation signaling**.

This requires: - standard formats for signatures and provenance - mapping rules when systems differ - signaling when verification cannot be preserved

Verification MUST not silently drop integrity guarantees during transfer.

Failure modes: - **silent degradation** of verification across boundaries - **incompatible formats** preventing validation

5.11 Security & Provenance System Layer: Trust Anchoring, Verification Flow, and Adversarial Resilience

DP15 requires a coherent system layer that binds identity (DP1), data (DP4), governance (DP8), and AI (DP12) into a **continuous verification lifecycle**.

Core properties: - **Trust anchoring**: identities and keys are attributable, with clear custody and revocation states - **End-to-end verification flow**: artifacts can be verified at creation, storage, transmission, and use - **Propagation**: provenance travels with artifacts; loss is explicitly signaled - **Auditability**: logs, receipts, and attestations allow reconstruction of events - **Adversarial resilience**: the system detects and contains tampering, forgery, and replay

Verification lifecycle: 1. **Creation**: artifact is signed/hashed with provenance envelope 2. **Distribution**: integrity preserved; intermediaries cannot alter without detection 3. **Consumption**: client verifies signatures, keys, and policy context 4. **Audit**: logs and receipts enable retrospective verification and dispute

Failure modes: - **broken chain of custody** between stages - **replay/rollback attacks** using old but valid artifacts - **cross-system loss** of provenance without signaling

This layer ensures security claims are not merely present but **continuously checkable under real conditions**.

6. Governance, Accountability, and Agency Surfaces

DP15 requires that security and provenance are not only visible but **actionable within governance and participant agency systems (DP8, DP2)**.

Participants must be able to:

- verify high-stakes claims and distinguish verified vs. unverifiable states
- access evidence supporting decisions that affect them
- detect integrity failures (invalid signatures, missing provenance, log inconsistencies)
- submit challenges or disputes with attached evidence

When verification fails, systems **MUST**:

- clearly signal failure states (invalid, missing, degraded, unverifiable)
- prevent high-risk actions from proceeding without override acknowledgment
- preserve evidence for audit and dispute resolution

Communities and governance bodies must be able to:

- define assurance thresholds for zones (e.g., require signed artifacts, attested environments)
- reject or downgrade artifacts lacking required provenance
- trigger audits when inconsistencies or anomalies are detected
- require remediation (re-signing, re-verification, disclosure, rollback)

Systems **SHOULD** support:

- escalation pathways tied to specific artifacts or events
- annotation and commentary layers on provenance and logs
- integrity-based gating (e.g., cannot promote, fund, or amplify unverifiable outputs)
- confidence or assurance scoring for artifacts and systems

Failure modes include:

- **non-actionable verification** (signals exist but do not affect outcomes)
- **accountability gaps** (no mechanism to enforce correction)
- **verification fatigue** (users overwhelmed and stop checking signals)

7. Incentives and Power Analysis

Security and provenance exist within strong incentive gradients that often discourage their full implementation.

Key structural tensions include:

- **speed vs assurance:** teams are rewarded for shipping quickly, not for making systems verifiable
- **opacity vs accountability:** operators benefit from controlling logs and narratives
- **growth vs integrity:** unverifiable claims can accelerate adoption
- **cost externalization:** security failures impose costs on users and ecosystems, not just builders

Adversarial actors exploit these tensions:

- attackers rely on weak provenance to inject malicious artifacts
- coordinated misinformation exploits unverifiable content
- insiders may bypass controls for convenience or advantage
- AI systems can generate plausible but unverifiable outputs at scale

DP15 requires realignment of incentives such that:

- unverifiable outputs carry reduced trust, reach, or eligibility
- verified artifacts gain preferential treatment in governance, ranking, or funding (DP9 linkage)
- repeated integrity failures trigger governance review or restrictions

Systems SHOULD incorporate:

- cost imposition for unverifiable or tampered artifacts (rate limits, gating)
- rewards for maintaining strong provenance (priority processing, trust weighting)
- visibility into integrity posture (public assurance signals)

Failure modes include:

- **security underinvestment** due to misaligned incentives
- **credibility arbitrage** where actors benefit from unverifiable claims
- **attacker asymmetry** where defense costs exceed attack costs

8. Community Signals Informing DP15

- fatigue with unverifiable claims
- demand for signed exports and build transparency
- concern about AI hallucination presented as fact

- calls for meaningful postmortems after failures

9. Non-Goals and Explicit Boundaries

DP15 does not:

- require all content to be signed or verified
- eliminate all risk
- replace legal or forensic systems
- mandate a specific cryptographic approach

10. Minimum Alignment (Non-Normative)

Minimum alignment defines the threshold at which security and provenance become **reliable verification infrastructure rather than optional features**.

A system claiming DP15 alignment **MUST** satisfy the following conditions:

10.1 Artifact Integrity

- Critical artifacts **MUST** be signed or otherwise integrity-protected
- Systems **MUST** verify integrity at use time, not only at creation

Failure mode: **implicit trust of unverified artifacts**

10.2 Provenance Availability

- High-stakes artifacts **MUST** carry provenance metadata (origin, transformation, context)
- Systems **MUST** signal when provenance is missing or degraded

Failure mode: **provenance absence or stripping**

10.3 Verification Pathways

- Participants **MUST** be able to verify claims without specialized expertise
- Systems **MUST** provide both simple indicators and detailed inspection paths

Failure mode: **unusable verification systems**

10.4 Tamper-Evident Logging

- Critical events **MUST** be recorded in tamper-evident logs
- Logs **MUST** support independent verification or witnessing

Failure mode: **log manipulation or selective disclosure**

10.5 Key and Identity Integrity

- Signing keys **MUST** have visible ownership, rotation, and revocation status
- Systems **MUST** detect and signal compromised or invalid keys

Failure mode: **undetected key compromise**

10.6 Cross-System Preservation

- Integrity and provenance **MUST** persist across exports and integrations
- Systems **MUST** signal degradation when preservation is not possible

Failure mode: **silent integrity loss across systems**

10.7 AI Output Traceability

- AI-generated outputs **MUST** be labeled and include relevant provenance when claims are made
- Systems **MUST** distinguish attributable vs. non-attributable outputs

Failure mode: **AI attribution ambiguity**

10.8 Auditability and Contestability

- Participants **MUST** be able to inspect, compare, and challenge evidence
- Systems **MUST** preserve records necessary for dispute resolution

Failure mode: **evidence opacity or loss**

Systems that omit verification, provenance, or auditability **MUST NOT** be considered aligned with DP15.

11. Open Questions and Future Work

- balancing privacy with auditability
- scaling verification without overwhelming users
- cross-border evidence standards
- decentralized witnessing vs cost tradeoffs
- quantum-resilient cryptography planning

12. Relationship to Other Desirable Properties

- DP1: attribution and appeals
- DP4: integrity of data handling and exports
- DP7: interoperability of signed artifacts
- DP11–DP13: AI traceability and containment
- DP14: transparency practices

- DP16–DP17: roadmap and funding alignment for security

13. Foresight and Failure Design

DP15 assumes breach, key compromise, and manipulation attempts will occur.

Systems must predefine response pathways, including key rotation, participant notification, and audit procedures.

14. Path Toward ML-RFC

- standardize provenance formats for content and AI outputs
- align with emerging supply-chain and identity standards
- develop reference implementations for high-assurance zones

15. Closing Orientation

DP15 is where the meta-layer moves from claims to evidence.

Security without provenance is performative. Provenance without security is unreliable.

Together, they create a system where shared reality can be verified, contested, and trusted.

DP16 – Roadmap and Milestones

1. Purpose of This Draft

This draft articulates Desirable Property 16 (DP16) as the condition under which the meta-layer’s evolution is honestly communicated and milestone accountability is structural, so participants, builders, and communities can plan, trust, and contest direction without being surprised by silent pivots, vapor schedules, or roadmaps that function as marketing fiction.

DP16 connects DP14 (trust and transparency), DP15 (security posture evolution), DP17 (financial sustainability), DP3 (adaptive governance), DP9 (builder expectations), and DP11 (ethical AI capability claims).

If DP16 is weak, predictable failures follow: hype cycles, burned contributors, communities built on assumptions later voided, and regulators correctly skeptical of the entire effort.

DP16 does not promise perfect prediction. It defines minimum honesty conditions for roadmapping in a complex sociotechnical system.

2. Problem Statement

In today’s web, roadmaps are often performative rather than operational.

Dates shift silently, features ship without promised safeguards, and “soon” becomes a placeholder for uncertainty rather than a bounded commitment. Participants and builders are expected to plan around signals that are not designed to be reliable.

This produces recurring failures:

- roadmap theater without resourcing or dependency disclosure
- silent removal or reshaping of public commitments
- optimistic AI timelines that misallocate trust (DP11)
- security and privacy work deferred or hidden (DP15)
- milestones dependent on funding that does not exist (DP17)

These failures are structural. Roadmaps coordinate collective action. When they mislead, they coordinate collective harm.

DP16 reframes roadmaps as accountability artifacts: scoped, resourced, dependency-aware, and revisable with memory.

3. Threats and Failure Modes

3.1 Hype-driven timelines

Dates are set for visibility rather than feasibility.

Example: A feature demo launches publicly while required safety systems are not ready.

Why this matters: Capability claims must match operational reality.

3.2 Dependency denial

External constraints are omitted from planning.

Example: Interoperability is promised while critical partner agreements or standards are unresolved.

Why this matters: Honest planning includes what you do not control.

3.3 Moving goalposts without memory

Roadmap changes occur without explanation.

Example: A feature disappears from public documentation without a changelog.

Why this matters: Governance requires traceability of decisions and learning.

3.4 Security and privacy debt hiding

Assurance work is deprioritized in visible planning.

Example: Roadmaps highlight features but omit audit backlogs or known vulnerabilities.

Why this matters: Security posture must be part of the same narrative as feature progress.

3.5 Financial unreality

Milestones assume resources that are not secured.

Example: Public goods infrastructure is promised without funding for maintainers.

Why this matters: Roadmaps must reflect actual capacity, not aspirational intent.

4. Core Principle

Roadmaps and milestones in the meta-layer are honest, resourced, and revisable commitments.

Dependencies, risks, and uncertainties are visible. Security, governance, and safety work are treated as first-class alongside features. Changes are explained with memory of what was learned.

Example: A roadmap includes dependency graphs, funded roles, security milestones, and explicit unknowns, with periodic retrospectives explaining changes.

What this feels like: Respect for participant time, attention, and planning.

Without this: Roadmaps become mechanisms for eroding trust rather than coordinating progress.

5. Primary Mechanisms and Structural Conditions

5.1 Dependency-aware planning

Roadmaps explicitly identify legal, technical, governance, and ecosystem dependencies and classify them by **control and certainty** (controlled, shared, external; verified, assumed, unknown).

This includes identifying **critical-path dependencies** that can block delivery, and attaching owners, verification status, and expected update cadence.

Failure mode: **hidden blockers**, where untracked dependencies invalidate timelines after commitments are made.

5.2 Dual-track visibility

Feature delivery and assurance work (security, privacy, accessibility, safety, governance) are presented together with **linked milestones and shared completion criteria**.

Assurance work is not deferred to “later phases” without explicit gating; features that depend on assurance must reference those gates.

Failure mode: **assurance deferral**, where visible progress masks growing risk debt.

5.3 Milestone receipts

Completed milestones link to verifiable evidence such as documentation, code, tests, audits, or third-party attestations (DP15 alignment).

Receipts should be **durable, accessible, and specific to the claimed capability**, enabling independent inspection.

Failure mode: **receipt theater**, where artifacts exist but do not substantiate the claim.

5.4 Change logs with governance salience

Material changes (scope, timelines, removals, reclassification) are recorded with **who decided, why, and who is affected**.

Changes that impact communities or partners should trigger **targeted notifications** and, where appropriate, governance review (DP3 linkage).

Failure mode: **silent pivots**, where affected parties cannot detect or respond to changes.

5.5 Risk registers

Top risks to roadmap integrity are identified, ranked, and updated with **likelihood, impact, mitigation, and owner**.

Risk registers should distinguish between **delivery risk** (can we build it) and **integration risk** (can it work with others) and be visible alongside milestones.

Failure mode: **risk invisibility**, where known issues are not surfaced to participants.

5.6 Scenario planning

Pre-mortems and alternative scenarios are documented to anticipate failure modes, including **best case, expected case, and worst case** timelines.

Scenarios should be tied to triggers (e.g., dependency slips, funding changes) that cause movement between states.

Failure mode: **single-path planning**, where plans assume ideal conditions and lack adaptation paths.

5.7 AI capability boundaries

Clear distinction between research, prototype, pilot, production, and scaled systems, with **explicit gating criteria and evidence thresholds** (DP11 alignment).

Claims about AI capability must reference **evaluation context, limitations, and deployment constraints**.

Failure mode: **capability inflation**, where demos are presented as operational systems.

5.8 Funding linkage

Milestones are tied to known funding sources, staffing, and maintenance commitments, or clearly labeled as contingent.

Roadmaps should indicate **runway, dependency on grants or partners, and maintenance ownership** for shipped work.

Failure mode: **funding ambiguity**, where delivery is implied without resources.

5.9 Cross-organizational coordination

Shared dependencies across organizations are made visible with **common identifiers, ownership boundaries, and divergence signaling**.

Where coordination is loose, roadmaps should indicate **assumption ranges** rather than fixed dates.

Failure mode: **misaligned expectations**, where partners operate on incompatible timelines.

5.10 Retrospective accountability

Regular reviews compare planned versus actual outcomes, capturing **variance, root causes, dependency shifts, and lessons learned**.

Retrospectives should distinguish between **uncertainty, error, and misrepresentation**, and feed updates back into planning and governance.

Failure mode: **non-learning systems**, where mistakes repeat without structural correction.

5.11 Roadmap System Layer: Commitment Integrity, Dependency Enforcement, and Claim Verifiability

Beyond roadmap artifacts and communication practices, DP16 requires a coherent roadmap system layer that ensures commitments remain traceable, constrained, and resistant to misrepresentation under pressure.

Roadmaps are not only planning documents. They are coordination systems that shape participant expectations, builder effort, funding decisions, governance priorities, and public trust. If roadmap claims cannot be bound to scope, evidence, dependencies, and change history, they become narrative instruments rather than infrastructure.

5.11.1 Commitment integrity and binding

Roadmap entries must represent bounded commitments, not open-ended aspirations.

This requires explicit scope, success criteria, completion conditions, and commitment classification, such as research, exploratory, committed, funded, blocked, or deprecated. Claims must remain proportional to the commitment class.

A failure mode is aspirational drift, where language implies commitment while avoiding operational accountability.

5.11.2 Dependency verification and propagation

Dependencies must not only be listed. They must be validated, monitored, and propagated when they change.

This includes technical dependencies, partner dependencies, legal or regulatory dependencies, funding dependencies, security dependencies, and governance dependencies. Critical-path dependencies should be distinguishable from ordinary dependencies so participants understand what can actually block delivery.

A failure mode is dependency illusion, where plans assume conditions that are unresolved, unavailable, or outside the initiative's control.

5.11.3 Claim-to-outcome traceability

Every roadmap claim must resolve to a delivered artifact, a documented change in status, or an explicit retirement.

This requires persistent identifiers for roadmap items and linkage to code, documentation, audits, governance decisions, release notes, or public explanations. Partial completion should be represented as partial completion, not quietly reframed as success.

A failure mode is untraceable claims, where promises cannot be compared against outcomes.

5.11.4 Change integrity and versioned commitments

Roadmaps must preserve versioned history of commitments and changes.

This includes changelogs, previous states, rationale for material shifts, and explanations for additions, removals, timeline changes, or scope changes. Revision is healthy when it is visible; silent revision is trust decay.

A failure mode is historical erasure, where past commitments disappear without record.

5.11.5 Anti-hype and signaling constraints

Roadmaps must resist distortion from marketing, fundraising, competition, and AI capability theater.

Systems should distinguish clearly between concept, prototype, pilot, production, verified capability, and scaled deployment. Public claims should not exceed operational readiness, and uncertainty should be surfaced rather than hidden.

A failure mode is signal inflation, where external incentives distort internal reality.

5.11.6 Cross-organizational roadmap coherence

When milestones depend on multiple organizations, roadmap meaning must remain coherent across boundaries.

This requires shared milestone references, explicit ownership, divergence signaling, and visibility into where one organization's delay, pivot, or withdrawal affects others.

A failure mode is coordination divergence, where different actors present incompatible versions of shared progress.

5.11.7 Roadmap memory and auditability

Participants must be able to reconstruct what was promised, what changed, what was delivered, and what was learned.

This requires accessible archives, comparison views, retrospective links, and evidence artifacts that allow roadmap integrity to be evaluated over time.

A failure mode is roadmap opacity, where evolution cannot be understood, audited, or contested.

This roadmap system layer ensures that coordination signals remain trustworthy under uncertainty, incentives, and scale rather than degrading into narrative management.

6. Governance, Accountability, and Agency Surfaces

DP16 requires that roadmap signals are not only visible but **actionable and contestable** within governance processes.

Participants must be able to:

- compare promises to delivered outcomes using stable identifiers and evidence links
- understand which dependencies are blocking or delaying work
- distinguish between committed, contingent, and exploratory milestones
- contest misleading, incomplete, or outdated roadmap representations through defined channels

Communities and governance bodies must be able to:

- align decisions (funding, prioritization, standards) with roadmap reality (DP3 linkage)
- trigger reviews when roadmap integrity degrades (e.g., repeated silent changes, missing receipts)
- require correction, reclassification, or withdrawal of misleading claims
- attach **confidence signals** or integrity ratings to roadmap segments

Systems SHOULD support:

- issue/appeal pathways tied to specific roadmap items
- public commentary or annotation layers for milestones
- escalation mechanisms for high-impact discrepancies

Failure modes include:

- **non-actionable transparency**, where information is visible but cannot be used to influence outcomes
- **accountability gaps**, where no actor is responsible for correcting misleading signals
- **participation fatigue**, where communities lack effective recourse and disengage

7. Incentives and Power Analysis

Roadmaps sit inside powerful incentive fields. Teams, funders, partners, contributors, and communities often benefit in the short term from optimistic claims, compressed timelines, and selective disclosure of risk. DP16 exists because those incentives do not naturally produce truth.

Communication incentives often reward optimism over accuracy. A roadmap that looks confident can attract funding, attention, talent, and legitimacy even when the underlying capacity is uncertain. This creates pressure to convert aspiration into implied commitment and ambiguity into promotional momentum.

DP16 requires aligning incentives with truthful forecasting, postmortem transparency, and long-term credibility rather than short-term attention.

Key incentive risks include:

- overstating readiness to attract funding, partners, or adoption
- hiding security, privacy, or governance debt to maintain confidence
- framing dependent milestones as controlled milestones
- using AI capability claims to create urgency without operational evidence
- shifting definitions of success after work has begun

Roadmap integrity therefore requires consequences for repeated misrepresentation. These may include governance review, downgraded confidence ratings, public correction receipts, funding conditions, or reduced eligibility for ecosystem support.

A critical failure mode is credibility arbitrage, where actors gain near-term benefits from inflated claims while distributing long-term costs to contributors and communities.

8. Community Signals Informing DP16

Community signals consistently show that people do not expect perfect prediction. They expect honesty, memory, and respect for the planning burdens created by public commitments.

Recurring signals include:

- frustration with shifting timelines without explanation
- demand for security and safety work to be visible
- skepticism of AI claims in public communication
- desire for funded maintenance commitments
- concern that public promises are used to recruit labor or attention without durable obligation
- need for clear distinctions between pilots, prototypes, production systems, and aspirational futures

These signals point to a deeper pattern: participants can tolerate uncertainty when it is named, but lose trust when uncertainty is converted into certainty for strategic advantage.

DP16 treats these signals as design inputs. Roadmap systems should make uncertainty legible, show dependency reality, and allow communities to distinguish delay caused by honest learning from delay caused by misrepresentation.

9. Non-Goals and Explicit Boundaries

DP16 does not aim to eliminate uncertainty or impose rigid planning. It defines what roadmaps are **not allowed to become**.

DP16 does not:

- guarantee precise timelines in uncertain environments; variance is expected when it is explained and bounded
- eliminate the need for confidentiality; sensitive details may be abstracted, but their existence and impact should be signaled
- replace investor, partner, or internal planning norms; it constrains how those norms translate into public commitments
- mandate a single methodology; agile, waterfall, and hybrid approaches are compatible if integrity conditions are met

DP16 also does not permit:

- using ambiguity to imply commitment where none exists
- presenting dependent or unfunded work as controlled delivery
- omitting assurance work to improve perceived velocity
- rewriting history to preserve narrative coherence

Boundary principle:

Roadmaps may be uncertain, but they must not be misleading about what is known, what is controlled, and what is promised.

Failure modes include:

- **methodology masking**, where process language obscures lack of commitment
- **selective disclosure**, where only favorable information is surfaced
- **narrative protection**, where truth is suppressed to maintain external perception

10. Minimum Alignment (Non-Normative) (Non-Normative)

Minimum alignment is the point at which roadmap signals are **reliable enough to coordinate real work**. The aim is not perfection, but to avoid systematically misleading participants.

A DP16-aligned initiative should, at minimum:

- **Bind milestones to scope and outcomes:** Each milestone clearly states what is in/out of scope and what counts as completion (e.g., shipped artifact, audit, or documented capability).
- **Make dependencies legible:** Key dependencies (technical, legal, ecosystem) are identified, with notes on what is controlled vs. external and what is uncertain.

- **Show assurance alongside features:** Security, privacy, and safety work appears in the same roadmap view as features (not deferred or hidden).
- **Provide evidence for completion:** Completed milestones link to verifiable artifacts (docs, code, tests, audits) where appropriate.
- **Maintain change memory:** Material changes (scope, dates, removals) are recorded with brief explanations; prior states are not silently erased.
- **Signal uncertainty honestly:** Distinguish research, prototype, and production readiness (DP11), and avoid presenting demonstrations as deployed capability.
- **Link to resourcing reality:** Indicate whether milestones are funded, partially funded, or contingent (DP17), and avoid presenting unfunded work as committed delivery.
- **Enable comparison over time:** Participants can compare earlier commitments to current status without reconstructing history.

Failure modes to avoid:

- **Aspirational drift:** commitments implied without binding scope or resourcing
- **Dependency illusion:** plans rely on unresolved external conditions without disclosure
- **Assurance invisibility:** security/safety work omitted from visible planning
- **Historical erasure:** past commitments disappear without record
- **Signal inflation:** capability claims exceed operational reality

11. Open Questions and Future Work

DP16 leaves several questions open because roadmap integrity depends on context, maturity, and governance capacity. The goal is not to impose one planning methodology, but to define the integrity conditions that any credible method must satisfy.

Key open questions include:

- **Transparency versus security:** How should initiatives disclose roadmap risk without revealing vulnerabilities, attack windows, or sensitive partner dependencies?
- **Multi-organization alignment:** What shared formats or receipts allow independent organizations to coordinate milestones without creating centralized control?
- **Forecast accuracy measurement:** How can communities measure forecasting reliability without punishing honest learning, adaptation, or uncertainty disclosure?
- **Regulatory and legal timelines:** How should legal uncertainty, policy change, and compliance review be represented in public roadmap systems?
- **AI capability claims:** What evidence thresholds should distinguish research demos, controlled pilots, production deployments, and frontier capability claims?
- **Funding-contingent work:** How should public goods work be represented when it is necessary but not yet funded?
- **Community recourse:** What should participants be able to do when roadmaps repeatedly mislead or coordination harms accumulate?

These questions should mature through future ML-Drafts, reference implementations, and governance experiments. DP16 should not freeze roadmap practice too early; it should make roadmap integrity inspectable while methods evolve.

12. Relationship to Other Desirable Properties

DP16 is a cross-cutting property because roadmap integrity affects whether other desirable properties can be trusted over time.

- **DP3: Adaptive Governance.** Governance needs accurate roadmap signals to sequence decisions, allocate attention, and adjust priorities. If roadmap changes are silent, governance becomes reactive and legitimacy suffers.
- **DP7: Interoperability.** Interoperability depends on shared timelines, version expectations, and dependency clarity. Roadmap drift across systems can break integration even when protocols are sound.
- **DP9–DP10: Builder and learner expectations.** Contributors and learners invest time based on public signals. Roadmap integrity protects that investment from being misdirected by hype or vague commitments.
- **DP11–DP15: AI, security, and provenance alignment.** Capability claims, safety gates, audits, provenance systems, and security posture all require visible sequencing. DP16 ensures assurance work is not hidden behind feature marketing.
- **DP17: Financial sustainability.** Milestones must reflect funding and maintenance reality. Unfunded commitments should be labeled as contingent rather than presented as guaranteed delivery.

If DP16 fails, other DPs can appear stronger than they are. A system may claim future privacy, future AI safety, future security, or future interoperability while coordinating present action around promises that are not structurally accountable.

13. Foresight and Failure Design

DP16 assumes delays, failures, and shifting conditions. It does not treat change as failure. It treats untracked, unexplained, or strategically obscured change as failure.

Common roadmap failure paths include:

- teams overcommitting to preserve confidence
- security and governance work falling behind visible features
- funding assumptions becoming invalid without public relabeling
- AI capability claims spreading faster than verification
- external dependencies blocking milestones while public timelines remain unchanged

These failures often compound. A single optimistic milestone can attract contributors, partners, and community expectations. If it later shifts without explanation, trust damage spreads beyond the missed milestone and affects the perceived legitimacy of the broader initiative.

DP16 therefore requires recovery practices:

- public status changes with explanation
- retrospectives that distinguish error, uncertainty, and misrepresentation
- updated dependency maps after delays
- repair pathways for communities or builders harmed by misleading signals
- governance review for repeated roadmap integrity failures

A mature roadmap system does not avoid failure. It makes failure learnable, bounded, and repairable.

14. Path Toward ML-RFC

Advancement from ML-Draft to ML-RFC requires demonstrating that roadmap integrity can be operationalized across real initiatives, not merely described.

Key progression steps include:

- **Standardize roadmap artifact formats:** Define common fields for scope, status, dependency class, funding state, assurance gates, confidence level, and evidence links.
- **Develop milestone receipt patterns:** Establish how completed, delayed, blocked, or retired milestones produce verifiable records.
- **Create retrospective practices:** Define minimum retrospective contents, including planned versus actual outcomes, dependency shifts, and lessons learned.
- **Align communication policies with trust frameworks:** Ensure public claims, demos, and announcements remain consistent with roadmap status and operational readiness.
- **Test multi-organization coordination:** Demonstrate shared roadmap references across independent teams with explicit divergence signaling.
- **Define conformance checks:** Create tests for aspirational drift, dependency illusion, historical erasure, signal inflation, and assurance invisibility.

Promotion to ML-RFC should require evidence that participants can compare commitments over time, verify milestone outcomes, and understand why material changes occurred.

15. Closing Orientation

DP16 is where the meta-layer demonstrates respect for time.

Participants, builders, funders, and communities make real decisions based on roadmap signals. They allocate labor, money, trust, attention, and hope. When those signals are inflated, incomplete, or silently rewritten, the harm is not merely reputational. It is coordination harm.

When roadmaps are honest, participants can coordinate effort and trust direction. They can understand what is real, what is uncertain, what is blocked, and what has changed.

When roadmaps are not honest, trust erodes faster than any feature can rebuild it. Even successful delivery becomes suspect when the path to delivery cannot be reconstructed.

DP16 is the claim that the meta-layer will not build the future on vapor, hidden debt, or erased commitments.

Roadmaps should not be marketing fiction. They should be public instruments of shared orientation, accountable learning, and coordinated care.

DP17 – Financial Sustainability

1. Purpose of This Draft

This draft articulates Desirable Property 17 (DP17) as the condition under which the meta-layer sustains itself over time with transparent, non-extractive, and resilient funding mechanisms, so that governance, safety, maintenance, and innovation can persist without relying on hidden rents, unstable grants, or goodwill that inevitably burns out.

DP17 connects DP6 (commerce and value flow), DP9 (incentives), DP3 (governance capacity), DP15 (security investment), DP16 (roadmap realism), and DP20 (community ownership of surplus).

If DP17 is weak, predictable failures follow: underfunded moderation and safety, stalled infrastructure, dependence on extractive business models, and communities that cannot maintain what they build.

DP17 does not mandate a single funding model. It defines the minimum conditions under which funding is durable, legible, and aligned with the meta-layer’s human-first principles.

2. Problem Statement

In today’s web, sustainability is often achieved through indirect or hidden mechanisms: surveillance advertising, data brokerage, platform fees, or speculative token cycles.

Public goods and community infrastructure frequently rely on grants, volunteer labor, or unstable revenue streams, creating a mismatch between system importance and funding reliability.

This produces recurring failures:

- critical infrastructure maintained by underpaid or unpaid contributors
- safety, moderation, and accessibility treated as cost centers rather than core functions
- revenue models that depend on data extraction or attention manipulation (DP4, DP6)
- boom-bust funding cycles that destabilize communities
- governance systems without budget authority or visibility

These failures are structural. Systems that are not sustainably funded cannot remain aligned.

DP17 reframes sustainability as a first-class design constraint, not an afterthought.

3. Threats and Failure Modes

3.1 Hidden extraction as funding

Systems rely on opaque monetization such as data harvesting or behavioral targeting.

Example: A “free” platform funds itself through invasive ad networks without clear disclosure.

Why this matters: Funding models shape system behavior.

3.2 Grant dependency and fragility

Communities rely on short-term funding with no continuity plan.

Example: A civic project shuts down when a sponsor withdraws.

Why this matters: Sustainability requires continuity, not episodic support.

3.3 Misaligned incentives

Revenue depends on metrics that conflict with user well-being.

Example: Engagement-based monetization incentivizes addictive design.

Why this matters: Funding should reinforce, not undermine, system values.

3.4 Invisible cost structures

Participants cannot see how resources are allocated.

Example: Moderation budgets are opaque despite being critical to community health.

Why this matters: Transparency is required for accountability.

3.5 Underfunded maintenance

New features are funded, but upkeep is neglected.

Example: Security patches lag behind feature releases.

Why this matters: Sustainability includes maintenance, not just growth.

3.6 Centralized funding control

A small group controls financial resources.

Example: A foundation allocates funds without community input.

Why this matters: Funding concentration leads to governance capture.

3.7 Token speculation without utility

Financial instruments prioritize speculation over real value creation.

Example: Token prices fluctuate while underlying systems stagnate.

Why this matters: Financialization without function destabilizes ecosystems.

4. Core Principle

Financial sustainability in the meta-layer requires that funding sources, allocation, and incentives are transparent, aligned with system values, and sufficient to maintain core functions over time.

Funding must support governance, safety, infrastructure, and community development as first-class concerns.

Example: A community publishes its revenue streams, cost allocation, and funding reserves alongside governance decisions.

What this feels like: Stability and clarity about how the system persists.

Without this: Systems drift toward extraction or collapse under their own weight.

5. Primary Mechanisms and Structural Conditions

5.1 Transparent revenue models

Funding sources are disclosed in plain language with sufficient detail to distinguish between direct user payments, grants, subsidies, data-driven revenue, and indirect monetization.

Systems **MUST** enable participants to understand not just *what* funds them, but *how those mechanisms shape behavior* (DP9 linkage).

Failure mode: **hidden extraction**, where funding depends on opaque or misrepresented practices.

5.2 Budget visibility

Communities can see how funds are allocated across functions, including governance, infrastructure, safety, and growth.

Budgets **SHOULD** be presented with meaningful granularity (not overly abstracted) and updated over time.

Failure mode: **allocation opacity**, where critical spending (e.g., moderation, security) is obscured.

5.3 Sustainable incentive alignment

Revenue models reinforce, rather than undermine, user well-being and system integrity.

Systems **MUST** assess whether revenue depends on behaviors that degrade trust, safety, or agency.

Failure mode: **incentive inversion**, where revenue grows as system quality declines.

5.4 Maintenance funding

Budgets explicitly allocate resources for upkeep, security, and support, including long-term maintenance of previously delivered features.

Maintenance **MUST** be treated as a first-class financial obligation, not residual spending.

Failure mode: **maintenance collapse**, where systems degrade due to underfunded upkeep.

5.5 Diversified funding streams

Systems avoid reliance on a single funding source by maintaining multiple revenue streams or contingency pathways.

Diversification **SHOULD** reduce exposure to abrupt funding shocks.

Failure mode: **single-point funding failure**, where loss of one source destabilizes the system.

5.6 Community participation in budgeting

Members have input into how funds are used, with mechanisms ranging from consultation to direct voting or delegated authority (DP3, DP8 alignment).

Participation **SHOULD** be proportional to stake, responsibility, or role.

Failure mode: **financial disenfranchisement**, where communities bear impact without influence.

5.7 Reserve and contingency planning

Financial buffers exist for unexpected events, including downturns, attacks, or infrastructure failure.

Reserves **SHOULD** be visible and tied to risk scenarios.

Failure mode: **shock fragility**, where systems cannot absorb disruption.

5.8 Funding–roadmap linkage

Milestones are tied to secured or clearly labeled contingent funding (DP16 alignment).

Roadmaps **MUST** not imply delivery where funding is uncertain or absent.

Failure mode: **funding illusion**, where commitments exceed resources.

5.9 Ethical monetization boundaries

Certain revenue practices may be restricted or banned within zones, especially those that violate privacy, agency, or community norms.

Systems **SHOULD** define these boundaries explicitly.

Failure mode: **boundary erosion**, where harmful practices re-enter through exceptions.

5.10 Public goods support mechanisms

Infrastructure critical to the ecosystem receives dedicated funding, including shared services that do not generate direct revenue.

Funding **SHOULD** be stable and not solely dependent on voluntary contributions.

Failure mode: **commons neglect**, where essential systems degrade due to lack of direct monetization.

5.11 Financial System Layer: Flow Integrity, Allocation Enforcement, and Alignment Verification

Beyond individual mechanisms, DP17 requires a coherent financial system layer that ensures money flows are **visible, constrained, and aligned with system values under pressure**.

Financial systems are not neutral. They shape incentives, power distribution, and long-term system behavior. Without enforceable structure, funding drifts toward extraction, concentration, or instability.

5.11.1 Financial flow visibility

All significant flows of value (revenue, grants, fees, distributions) must be observable at an appropriate level of abstraction.

This includes inflows, outflows, and internal transfers between functions.

Failure mode: **flow opacity**, where value movement cannot be understood or audited.

5.11.2 Allocation integrity and constraints

Funds allocated to specific purposes (e.g., security, moderation, infrastructure) must remain bound to those purposes unless explicitly reauthorized.

Systems **SHOULD** define constraints or policies on reallocation.

Failure mode: **allocation drift**, where resources are silently redirected.

5.11.3 Alignment verification

Funding sources and allocations must be evaluated against system values and declared principles.

This includes detecting: - dependence on extractive revenue - funding that introduces governance conflicts - misalignment between stated goals and financial incentives

Failure mode: **alignment illusion**, where funding appears compatible but introduces hidden distortion.

5.11.4 Power distribution and concentration detection

Financial systems must surface concentration of control over resources.

This includes visibility into who controls allocation decisions and how influence accumulates.

Failure mode: **financial capture**, where a small group controls system direction through funding.

5.11.5 Cross-system financial coherence

When multiple systems interact, financial commitments and dependencies must remain consistent and non-contradictory.

Failure mode: **coordination divergence**, where funding assumptions differ across actors.

5.11.6 Financial memory and auditability

Participants must be able to reconstruct funding history, including past allocations, changes, and rationales.

This requires durable records and accessible reporting.

Failure mode: **financial opacity over time**, where history cannot be audited.

5.11.7 Adversarial resilience

Financial systems must anticipate manipulation, including: - speculative distortion - subsidy gaming - governance capture through capital

Systems SHOULD include detection, signaling, and mitigation mechanisms.

Failure mode: **financial exploitation**, where adversaries extract value or influence.

This financial system layer ensures that sustainability is not just declared, but structurally maintained across time, incentives, and scale.

6. Governance, Accountability, and Agency Surfaces

DP17 requires that financial signals are not only visible but **actionable, contestable, and enforceable** within governance (DP3, DP8).

Participants MUST be able to:

- inspect funding sources, allocations, and changes with stable identifiers
- understand which costs are fixed, variable, contingent, or underfunded
- trace how funding decisions affect safety, infrastructure, and roadmap commitments (DP15, DP16)
- challenge misleading or incomplete financial representations through defined processes

Communities and governance bodies **MUST** be able to:

- **approve, reject, or reallocate** budget segments tied to core functions
- trigger **integrity reviews** when signals indicate misalignment (e.g., hidden extraction, maintenance underfunding)
- require **reclassification** of funding (committed vs. contingent) when conditions change
- attach **confidence or integrity ratings** to funding streams and budget areas

Systems **SHOULD** support:

- issue/appeal pathways bound to specific budget lines or revenue streams
- public annotation/commentary layers for financial artifacts
- escalation paths for high-impact discrepancies (e.g., safety budget cuts)

Enforcement pathways:

- downgrade or restrict features dependent on misrepresented funding
- block roadmap commitments that lack verified funding (DP16 linkage)
- initiate governance votes or audits for repeated integrity failures

Failure modes:

- **non-actionable transparency** (visible but no leverage)
- **accountability gaps** (no actor responsible for correction)
- **governance capture** (funding control overrides community decisions)

7. Incentives and Power Analysis

Financial systems create and amplify power. DP17 addresses how incentives **distort behavior under pressure** and how to realign them with system integrity (DP9).

Key dynamics:

- **Extraction loops:** revenue tied to attention, data, or speculation drives behaviors that degrade user welfare
- **Underinvestment in non-visible work:** safety, maintenance, and governance lose funding to growth-visible features
- **Capital concentration:** large funders gain outsized influence over priorities and policy
- **Signaling games:** optimistic financial narratives attract resources despite weak fundamentals
- **AI hype amplification:** capability claims accelerate funding without corresponding verification (DP12, DP15)

Adversarial/attack surfaces:

- **subsidy gaming:** actors exploit grants or incentives without delivering durable value
- **governance capture via capital:** funding conditions steer decisions against community interest
- **speculative distortion:** token or asset prices drive priorities away from utility
- **cross-subsidy masking:** losses hidden behind unrelated profitable lines

Alignment requirements:

- revenue must not increase as safety, trust, or agency decrease
- funding for assurance (security, privacy, governance) must be **protected and non-optional**
- incentives should reward **durable contributions and maintenance**, not just growth metrics

Mitigations:

- caps or checks on concentration of funding control
- ring-fenced budgets for safety/maintenance
- disclosure of incentive structures alongside revenue streams
- independent audits or attestations for high-impact funding claims

Failure modes:

- **incentive inversion** (harmful behavior is profitable)
- **credibility arbitrage** (short-term gains from inflated claims)
- **capture dynamics** (decision power tracks capital rather than legitimacy)

8. Community Signals Informing DP17

Community signals are not just sentiment; they are **early indicators of financial misalignment and coordination risk**. DP17 treats these signals as inputs to system design and ongoing monitoring.

Observed signals and implications:

- **Concern about extractive monetization** → signals hidden revenue dependencies; requires clearer revenue classification and disclosure (5.1, 5.11.3)
- **Demand for fair compensation** → indicates gaps in value distribution; requires explicit allocation policies and contributor compensation pathways (5.2, 5.6)
- **Frustration with unstable funding cycles** → indicates weak reserves and diversification; requires contingency planning and runway visibility (5.5, 5.7)
- **Desire for community-controlled resources** → indicates legitimacy gaps; requires participatory budgeting and governance linkage (Section 6)
- **Skepticism of “free” services** → indicates perceived hidden costs; requires explicit mapping from revenue to behavior (5.1, 7)
- **Concern about safety underfunding** → indicates risk externalization; requires ring-fenced budgets for assurance (5.4, 7)

Operationalization:

- Convert recurring signals into **metrics** (e.g., % budget to maintenance, % revenue from non-extractive sources)
- Attach **confidence ratings** to funding streams based on signal alignment (Section 6)
- Feed signals into **governance triggers** (e.g., review when safety budget falls below threshold)

Failure modes:

- **signal neglect** (warnings ignored until failure)
- **performative response** (surface-level changes without structural correction)

9. Non-Goals and Explicit Boundaries

DP17 does not prescribe a single economic ideology or eliminate markets. It defines **hard boundaries on misleading or harmful financial behavior** in the meta-layer.

DP17 does not:

- mandate a single funding mechanism or ownership model
- eliminate profit, competition, or market dynamics
- guarantee equal distribution of funds across participants
- replace legal or regulatory financial frameworks

DP17 explicitly disallows:

- **hidden extraction** (undisclosed data monetization, dark patterns tied to revenue)
- **misrepresented funding states** (presenting contingent or unfunded work as committed)
- **assurance starvation** (underfunding security, privacy, or governance to inflate growth)
- **history rewriting** (removing or obscuring past funding decisions)
- **cross-subsidy masking** (hiding losses or risks behind unrelated profitable lines without disclosure)

Boundary principle:

Financial systems may be complex, but they must not be misleading about what is funded, who controls it, and how it shapes behavior.

Failure modes:

- **methodology masking** (jargon obscures lack of commitment)
- **selective disclosure** (only favorable financial information is shown)
- **narrative protection** (truth suppressed to maintain perception)

10. Minimum Alignment (Non-Normative) (Non-Normative)

Minimum alignment defines the threshold at which financial signals are **reliable enough to coordinate real work**. Below this threshold, systems may publish numbers but do not provide trustworthy sustainability.

A DP17-aligned system should, at minimum:

- **Bind revenue to behavior:** disclose sources and how they influence system actions (DP9)
- **Show allocation with purpose:** publish major cost categories with sufficient granularity (governance, safety, infra, growth)

- **Fund core functions:** maintain explicit, protected budgets for maintenance, security, and governance (DP15)
- **Link funding to commitments:** tie roadmap milestones to secured or clearly contingent funding (DP16)
- **Maintain change memory:** record material budget and funding changes with brief rationale
- **Signal uncertainty:** distinguish committed, contingent, and speculative funding
- **Enable comparison over time:** allow participants to compare past and current funding states without reconstruction

Failure modes to avoid:

- **funding illusion:** commitments exceed available resources
- **allocation opacity:** critical spending hidden or aggregated beyond usefulness
- **commons neglect:** public goods unfunded despite system dependence
- **signal inflation:** financial claims exceed operational reality
- **historical erasure:** prior funding states disappear without record

Systems that omit binding, allocation visibility, or change memory SHOULD NOT be considered aligned with DP17.

11. Open Questions and Future Work

DP17 requires further work to standardize how financial integrity is measured and enforced across diverse contexts.

Key questions:

- **Transparency vs. sensitivity:** How to disclose enough for accountability without exposing competitive or security-sensitive details?
- **Cross-jurisdiction funding:** How to represent taxes, compliance costs, and legal constraints consistently across regions?
- **Measuring alignment:** What metrics best capture alignment between revenue and system values (e.g., % non-extractive revenue, maintenance ratio)?
- **Public goods funding models:** What hybrid models (grants, fees, commons funding) sustain non-market services?
- **AI-era financing:** How to represent costs and risks of AI systems, including compute, safety, and liability (DP12, DP15)?
- **Contributor compensation:** How to fairly allocate value to maintainers and community contributors over time?
- **Recourse mechanisms:** What should participants be able to do when financial signals are repeatedly misleading?

These should be explored through ML-Drafts, pilots, and comparative implementations.

12. Relationship to Other Desirable Properties

DP17 is a cross-cutting layer that conditions whether other DPs remain viable over time.

- **DP6 (Commerce):** Defines revenue mechanisms; DP17 ensures those mechanisms are transparent and non-extractive.
- **DP9 (Incentives):** Aligns economic signals with desired behaviors; DP17 verifies that alignment holds under pressure.
- **DP3 (Governance):** Requires budget authority and visibility to make legitimate decisions.
- **DP15 (Security & Provenance):** Security work must be funded and evidenced; DP17 ensures it is not starved.
- **DP16 (Roadmaps):** Commitments must match funding reality; DP17 binds milestones to resources.
- **DP20 (Ownership & Surplus):** Determines how value returns to communities; DP17 ensures flows are visible and fair.

If DP17 fails, other DPs degrade into promises without resources.

13. Foresight and Failure Design

DP17 assumes systems will face **financial stress, growth pressure, and adversarial manipulation**. The goal is not to avoid failure, but to make it **visible, bounded, and repairable**.

Common failure paths:

- **overcommitment:** promises exceed funding; leads to roadmap drift (DP16)
- **assurance deferral:** safety and maintenance lag behind features
- **funding shocks:** sudden loss of revenue without reserves
- **capture events:** concentration of capital redirects priorities
- **speculative cycles:** attention shifts to price over utility

Recovery practices:

- reclassification of commitments (committed → contingent)
- budget reallocation with public rationale
- activation of reserves and contingency plans
- governance review and potential rebalancing of power
- public postmortems linking funding decisions to outcomes

A mature system treats financial failure as a **learning loop**, not a hidden defect.

14. Path Toward ML-RFC

Advancement requires operational evidence that financial integrity can be implemented and audited.

Key steps:

- **Standardize financial schemas:** revenue types, budget categories, funding states, and confidence levels
- **Define reporting formats:** periodic disclosures with comparable fields across systems
- **Create verification artifacts:** links from budgets to receipts, audits, and outcomes (DP15)
- **Test governance integration:** demonstrate budget approval, reallocation, and audit pathways (Section 6)
- **Run adversarial tests:** simulate capture, subsidy gaming, and funding shocks
- **Demonstrate cross-system coherence:** shared projects maintain consistent funding signals (DP7)

Promotion criteria should include the ability for participants to **inspect, compare, and challenge** financial claims over time.

15. Closing Orientation

DP17 is where the meta-layer demonstrates respect for resources and power.

Participants allocate time, money, and trust based on financial signals. When those signals are opaque or distorted, coordination fails and communities bear the cost.

When financial systems are transparent and aligned, participants can understand what is funded, what is at risk, and how decisions are made. They can act with confidence, adjust expectations, and hold systems accountable.

When financial systems are not, even successful outcomes become suspect, because the path to them cannot be trusted.

DP17 is the commitment that the meta-layer will not fund itself through hidden extraction, unstable promises, or erased history.

Financial sustainability is not just survival. It is the ability to **persist with integrity under pressure, incentives, and change.**

DP18 – Feedback Loops and Reputation

1. Purpose of This Draft

This draft articulates Desirable Property 18 (DP18) as the condition under which the meta-layer can learn from participation without collapsing into surveillance, popularity contests, social credit, or reputation systems that permanently trap people in old contexts.

DP18 defines how communities gather feedback, convert signals into accountable learning, and recognize trustworthy contribution over time. It treats feedback and reputation as civic infrastructure, not engagement metrics.

Feedback loops are how the meta-layer senses whether its governance, incentives, safety systems, interfaces, and community norms are working. Reputation is how the system remembers patterns of contribution, care, reliability, and harm without reducing people to a single score.

If DP18 is weak, predictable failures follow: communities ask for feedback but do not act on it; reputation becomes a vanity metric; harmful actors launder trust across contexts; contributors burn out because recognition is invisible; AI systems optimize for shallow approval signals; and governance adapts only after legitimacy breaks.

DP18 connects directly to:

- DP1, identity and accountability
- DP2, participant agency and empowerment
- DP3, adaptive governance
- DP4, data sovereignty and privacy
- DP7, interoperability
- DP8, community-defined participation and governance zones
- DP9, developer and community incentives
- DP12 and DP13, AI governance and containment
- DP14, transparency and trust
- DP20, ownership and stewardship

DP18 does not prescribe a universal reputation score, a single feedback interface, or one model of rewards. It defines the minimum conditions under which feedback and reputation remain contextual, contestable, privacy-preserving, and aligned with community flourishing.

2. Problem Statement

Today's web is saturated with signals, but starved of trustworthy feedback.

Likes, shares, followers, ratings, badges, view counts, and engagement graphs appear to measure social value. In practice, they often measure visibility, habit formation, emotional provocation, automation, or the ability to game a platform's ranking system.

This produces recurring failures:

- communities provide feedback but cannot see whether it changed anything
- bad actors accumulate credibility through volume, performance, or network effects
- positive contributions disappear into feeds without durable recognition
- reputation is trapped inside platforms and lost when people move
- marginalized participants are excluded when feedback systems privilege dominant voices
- AI systems learn from shallow approval signals rather than community-defined values
- governance becomes reactive because weak signals are ignored until crisis

DP18 reframes feedback and reputation as learning infrastructure. Feedback must produce visible adaptation. Reputation must be contextual memory, not global judgment.

A healthy meta-layer must be able to answer:

- What did participants signal?
- Who or what is being evaluated?
- What changed as a result?
- Which signals are reliable in this zone?
- Which signals are being gamed?
- How can a participant contest, repair, or outgrow a reputation state?
- How does the system prevent feedback from becoming surveillance or social punishment?

Without DP18, the meta-layer cannot mature. It can collect comments, votes, and metrics, but it cannot reliably learn.

3. Threats and Failure Modes

3.1 Feedback theater

Communities are asked for input, but feedback has no visible path into decisions or operations.

Example: A town hall gathers participant concerns, but the same interface defaults, moderation rules, and incentive structures remain unchanged without explanation.

Why this matters: Feedback without response erodes trust faster than silence because it simulates agency while preserving control.

3.2 Reputation as popularity

Reputation collapses into follower counts, visibility, or applause.

Example: A participant becomes trusted because their posts receive high engagement, even though their contributions are often misleading or inflammatory.

Why this matters: Popularity is not reliability. DP18 requires multidimensional, context-bound reputation.

3.3 Runaway feedback loops

A signal increases visibility, visibility increases signal volume, and the system treats the resulting amplification as legitimacy.

Example: Early upvotes push a submission into prominence; prominence generates more upvotes; the system mistakes attention momentum for quality.

Why this matters: Feedback loops must be bounded. Otherwise reputation becomes a mechanism for self-reinforcing inequality or manipulation.

3.4 Reputation laundering

Actors transfer trust from one context into another where it has not been earned.

Example: A participant with high reputation in a gaming community receives influence in a medical advice zone without relevant credentials or history.

Why this matters: Reputation must preserve context, scope, and provenance. Portability without boundaries becomes a trust exploit.

3.5 Permanent stigma and no repair path

Negative reputation becomes permanent, opaque, or disproportionate.

Example: A participant makes a mistake in one zone and is silently down-ranked everywhere without notice, appeal, or decay.

Why this matters: Accountability must support learning and repair. Systems that never forgive incentivize abandonment, evasion, and identity cycling.

3.6 Feedback capture by loud minorities

Coordinated or highly resourced groups dominate feedback channels.

Example: A small faction floods surveys, flags, votes, or comments to steer governance toward its preferred outcome.

Why this matters: Feedback systems must distinguish broad legitimacy from coordinated pressure.

3.7 Marginalized voice suppression

Feedback mechanisms reproduce structural exclusion.

Example: Town halls occur in one language and time zone, while reputation rewards favor participants already fluent in dominant cultural norms.

Why this matters: Adaptive governance requires feedback from those most affected, not only those most available or socially rewarded.

3.8 AI-optimized approval hacking

AI systems learn to maximize visible approval signals rather than community-defined value.

Example: An assistant generates emotionally pleasing but low-integrity responses because users reward confidence and fluency more than accuracy.

Why this matters: AI feedback loops must be evaluated against zone policy, truthfulness, safety, and long-term outcomes, not only immediate satisfaction.

3.9 Privacy erosion through reputation telemetry

Reputation systems collect excessive behavioral data to infer trustworthiness.

Example: A system tracks every interaction, dwell time, private message, and social association to generate reputation scores.

Why this matters: Trust cannot be built by violating data sovereignty. DP18 must operate under DP4 constraints.

3.10 Role ossification

Reputation grants access or authority, then authority produces more reputation, making roles difficult to contest.

Example: Early stewards accumulate status and retain control even as their contribution quality declines.

Why this matters: Reputation-linked roles require decay, review, rotation, and contestability.

3.11 Cross-system signal degradation

Feedback and reputation signals move across systems but lose meaning.

Example: A five-star rating exports without information about rubric, rater identity class, zone norms, time horizon, or dispute status.

Why this matters: Interoperable reputation must preserve semantic context or visibly degrade.

4. Core Principle

Feedback loops and reputation in the meta-layer must enable communities to learn from lived experience, recognize trustworthy contribution, correct harmful behavior, and adapt governance over time while preserving agency, privacy, context, contestability, and the possibility of repair.

Reputation is not a universal score. It is contextual memory under governance.

Feedback is not engagement. It is a structured pathway from signal to learning to action.

Example: A community receives recurring feedback that a moderation rule is being applied unevenly. The system clusters the feedback, surfaces affected groups, opens a review process, publishes a decision receipt, updates the rule, and tracks whether the change improves outcomes over time.

What this feels like: Participants can see that their experience matters, not because every request is granted, but because every meaningful signal has a legitimate path into community learning.

Without this: The meta-layer repeats the failure of today's platforms: it extracts signals from people while denying them visible influence over the system those signals shape.

5. Primary Mechanisms and Structural Conditions

5.0 Feedback and Reputation Layer: Signal, Memory, Adaptation, and Recourse

DP18 requires a feedback and reputation layer that converts participant signals into governed learning.

This layer includes:

- feedback objects
- reputation objects
- signal provenance
- confidence and uncertainty metadata
- decay and repair mechanisms
- dispute and appeal pathways
- adaptation receipts
- AI-assisted pattern detection under human ratification

The layer must operate at the interface where participants act and are affected, while preserving the governance and privacy boundaries of each zone.

A feedback and reputation layer is not a database of social judgments. It is an accountable learning system.

Failure mode: **social credit drift**, where contextual feedback becomes generalized behavioral scoring.

5.1 Feedback objects

Feedback must be represented as structured objects, not only comments or reactions.

A feedback object SHOULD include:

- subject: what is being evaluated
- signal type: report, endorsement, correction, rating, concern, gratitude, appeal, observation
- scope: zone, artifact, participant, agent, policy, interface, or process
- evidence link, when appropriate
- privacy level
- urgency and severity
- confidence level
- submitter role or credential class, where relevant
- timestamp and version context
- requested action, if any

This allows feedback to be routed, aggregated, audited, and acted upon without flattening all signals into likes or complaints.

Failure mode: **signal mush**, where all feedback becomes undifferentiated noise.

5.2 Feedback routing and response commitments

Feedback systems must define what happens after feedback is submitted.

For each class of feedback, systems SHOULD specify:

- who or what receives it
- expected response time
- escalation path
- visibility rules
- criteria for closure
- appeal or reopening pathway

Participants do not need to control every outcome, but they must be able to see whether feedback entered a legitimate process.

Failure mode: **black-hole feedback**, where reports, suggestions, and concerns vanish without trace.

5.3 Continuous feedback mechanisms

DP18 requires ongoing feedback channels, not one-time consultation.

Mechanisms MAY include:

- surveys
- town halls
- asynchronous deliberation
- embedded interface prompts
- post-decision retrospectives
- safety reports
- participatory audits
- contributor reviews
- community health dashboards
- affected-group listening sessions

Continuous feedback must be designed for accessibility across language, time zone, ability, bandwidth, and technical literacy.

Failure mode: **episodic listening**, where communities are consulted only during crisis or launch moments.

5.4 Adaptive feedback systems

AI and automation may support feedback systems by summarizing large volumes of input, detecting patterns, identifying anomalies, and surfacing underrepresented concerns.

However, adaptive systems MUST remain bounded by:

- visible purpose
- zone-defined policy
- privacy constraints
- human review for material decisions
- auditability of summarization and classification
- clear disclosure of AI involvement

AI may help communities see patterns. It must not quietly decide whose experience counts.

Failure mode: **automation capture**, where AI summaries become de facto governance decisions.

5.5 Reputation as contextual memory

Reputation must be scoped to context, contribution type, and governance zone.

A reputation object SHOULD include:

- subject identity or agent reference
- reputation domain, such as reliability, care, expertise, stewardship, safety, responsiveness, accuracy, generosity, or follow-through

- zone scope
- source signals
- weighting logic
- confidence level
- decay schedule
- dispute status
- portability constraints
- rights or capabilities affected

This prevents reputation from becoming a single global score while still allowing communities to remember meaningful patterns.

Failure mode: **global reputation collapse**, where one number pretends to summarize a person, agent, or organization across all contexts.

5.6 Reputation-based compensation and recognition

Reputation may inform compensation, rewards, access, visibility, and recognition, but only under explicit governance.

Reputation-linked compensation SHOULD satisfy the following conditions:

- metrics and rubrics are published
- rewardable contributions are defined
- anti-gaming rules are active
- human review exists for high-value outcomes
- rewards do not depend solely on popularity
- appeals are available
- contributors can understand how reputation affected compensation

Positive contribution should be rewarded without creating a system where people perform for metrics rather than serve the community.

Failure mode: **reputation farming**, where actors optimize for visible reward signals rather than actual value.

5.7 Dynamic role-based access

Roles and capabilities may adjust based on reputation, contribution history, and trust signals.

Examples include:

- moderation privileges
- proposal rights
- voting eligibility
- grant review access

- amplification capacity
- trusted annotator status
- steward responsibilities
- AI-agent deployment permissions

Role changes **MUST** be:

- explainable
- scoped to a zone
- revocable
- reviewable
- subject to decay or renewal
- protected against sudden manipulation

Reputation may open doors, but it must not create unaccountable hierarchy.

Failure mode: **role capture**, where reputation-linked access becomes permanent power.

5.8 Reputation decay, renewal, and repair

Reputation must change over time.

Systems **SHOULD** support:

- positive reinforcement for sustained contribution
- decay for inactivity or stale signals
- faster decay for low-confidence signals
- repair pathways after mistakes or sanctions
- restorative processes where communities choose them
- expiry of context-specific penalties
- continued visibility of serious unresolved harms where appropriate

The goal is neither amnesia nor permanent stigma. The goal is governed memory.

Failure mode: **frozen reputation**, where old signals dominate present reality.

5.9 Bad behavior reporting and community moderation

Participants and communities must be able to report harm, abuse, manipulation, and bad behavior through structured channels.

Reporting systems **SHOULD** support:

- evidence submission
- safety-sensitive privacy

- protection from retaliation
- triage by severity
- pattern detection across incidents
- community-defined moderation workflows
- appeal and correction mechanisms
- transparency summaries where safe

Reports are feedback objects with safety consequences. They must be treated with care, not merged into generic sentiment metrics.

Failure mode: **weaponized reporting**, where reporting tools become harassment or governance capture instruments.

5.10 Positive contribution signaling

DP18 requires mechanisms to recognize good behavior, not only punish bad behavior.

Positive signals MAY include:

- gratitude
- endorsement
- attestation
- contribution receipts
- peer review
- successful mediation
- helpful annotation
- quality bridge creation
- accurate correction
- inclusive facilitation
- maintenance work
- long-horizon stewardship

Systems SHOULD make quiet, prosocial labor visible without forcing all care work into performance.

Failure mode: **negative-only memory**, where systems remember harm but fail to recognize repair, support, and stewardship.

5.11 Multi-dimensional reputation

Reputation must be plural.

A participant may be highly trusted for one function and untrusted for another. A contributor may be excellent at technical review but poor at facilitation. An AI agent may be strong at summarization but not authorized for dispute mediation.

Reputation dimensions MAY include:

- accuracy
- reliability
- safety
- expertise
- care
- responsiveness
- collaboration
- originality
- stewardship
- governance judgment
- bridge quality
- conflict resolution
- maintenance reliability

Failure mode: **single-axis trust**, where one form of contribution grants unrelated authority.

5.12 Interoperable reputation with bounded portability

Reputation should be portable where useful, but only with context preserved.

Portable reputation MUST carry:

- issuing zone
- signal type
- evaluation rubric
- timestamp
- decay or expiry
- confidence level
- dispute status
- portability permissions
- limitations on interpretation

When context cannot be preserved, systems MUST signal degradation.

Failure mode: **semantic stripping**, where reputation travels as a badge or score without the meaning needed to interpret it.

5.13 Community health dashboards

Communities SHOULD have dashboards that surface aggregate feedback and reputation patterns without exposing unnecessary personal data.

Dashboards MAY include:

- unresolved feedback volume
- response times
- repeated pain points
- moderation appeals
- participation diversity
- contributor recognition gaps
- role concentration
- reputation distribution
- unresolved harms
- experiment outcomes
- rollback triggers

Dashboards must be designed to support learning, not surveillance or public shaming.

Failure mode: **dashboard theater**, where metrics are displayed but do not affect decisions.

5.14 Rollback and experiment learning

Feedback loops should support safe experimentation.

When communities test policies, interfaces, incentives, or reputation mechanisms, experiments SHOULD define:

- hypothesis
- success indicators
- harm indicators
- review date
- rollback conditions
- affected participants
- decision authority
- public learning artifact

This allows communities to try new mechanisms without making every experiment permanent.

Failure mode: **irreversible experimentation**, where communities bear the cost of failed changes without recourse.

5.15 Feedback receipts and adaptation receipts

Feedback submissions and system changes should generate receipts.

A feedback receipt MAY include:

- what was submitted
- when it was received
- where it was routed
- privacy and visibility status
- next process step

An adaptation receipt MAY include:

- what changed
- which feedback or evidence informed the change
- who approved it
- expected outcome
- review date
- appeal or contestation path

Receipts close the loop between voice and action.

Failure mode: **untraceable adaptation**, where systems change without participants knowing why.

6. Feedback, Reputation, and AI Systems

AI systems may participate in DP18 in multiple roles:

- as subjects of feedback
- as assistants in feedback processing
- as agents with their own scoped reputations
- as participants in governed zones where permitted
- as tools for detecting manipulation, abuse, and emergent patterns

DP18 requires that AI feedback loops be especially constrained because automated systems can scale signals, interpret signals, and act on signals faster than communities can deliberate.

6.1 AI as subject of reputation

AI agents, models, tools, and automated services SHOULD have scoped reputation profiles when they act in ways that affect participants.

These profiles MAY track:

- accuracy
- refusal appropriateness
- safety incidents
- responsiveness to correction
- policy compliance

- hallucination rates
- data-handling reliability
- emotional safety
- task performance
- escalation behavior

AI reputation must be tied to version, deployment context, and operator accountability.

Failure mode: **agent amnesia**, where an AI system causes repeated harm but each deployment is treated as isolated.

6.2 AI-assisted feedback synthesis

AI may summarize, cluster, translate, and analyze feedback, but must preserve dissent and uncertainty.

Systems SHOULD prevent AI from flattening minority concerns into majority sentiment.

A good synthesis identifies:

- recurring themes
- contested interpretations
- affected groups
- evidence gaps
- urgency
- possible actions
- confidence limits

Failure mode: **consensus hallucination**, where AI creates the appearance of agreement where disagreement remains.

6.3 Feedback loops for AI containment

Feedback from participants can help communities refine containment policies for AI agents.

Examples include:

- reports of manipulative behavior
- consent boundary violations
- failures to disclose automation
- unsafe recommendations
- emotional overreach
- unauthorized cross-zone action
- model behavior drift

Containment feedback should trigger policy review, audit, rollback, rate limits, or suspension where appropriate.

Failure mode: **containment lag**, where AI behavior changes faster than feedback systems can respond.

7. Privacy and Data Sovereignty Requirements

Feedback and reputation systems must comply with data sovereignty.

DP18 systems SHOULD practice:

- data minimization
- purpose limitation
- consent-bound collection
- separation of public recognition and private telemetry
- pseudonymous feedback where safe
- protection for vulnerable reporters
- deletion or expiry where appropriate
- access controls for sensitive reputation evidence
- clear notice when signals affect rights or compensation

Reputation must not become a pretext for continuous behavioral surveillance.

Failure mode: **trust through surveillance**, where the system claims safety by collecting more data than communities can legitimately govern.

8. Governance Requirements

Feedback and reputation systems are governance systems. They must themselves be governable.

Communities SHOULD define:

- what counts as valid feedback
- what reputation dimensions matter
- who can issue attestations
- how signals are weighted
- how disputes are handled
- how repair works
- how decay works
- when reputation affects access or rewards
- when human review is required
- how algorithms are audited
- how changes are approved

This governance must be visible, versioned, and contestable.

Failure mode: **hidden reputation law**, where invisible formulas determine social standing and access.

9. Evaluation Criteria

A DP18-aligned implementation should be evaluated against the following questions.

9.1 Signal quality

- Are feedback signals typed, scoped, and attributable at the right level?
- Can systems distinguish evidence, opinion, endorsement, complaint, and appeal?
- Are low-confidence signals prevented from causing high-impact outcomes without review?

9.2 Loop closure

- Can participants see whether feedback was received, routed, reviewed, and acted upon?
- Do system changes link back to feedback or evidence?
- Are non-actions explained where appropriate?

9.3 Context preservation

- Is reputation scoped to zones and contribution domains?
- Does portable reputation carry rubrics, provenance, and limits?
- Are degraded signals clearly labeled?

9.4 Contestability and repair

- Can participants dispute inaccurate reputation signals?
- Are there appeal timelines and responsible stewards?
- Do reputation states decay, renew, or repair over time?

9.5 Anti-gaming and safety

- Are feedback loops bounded against amplification spirals?
- Are coordinated attacks, sybil behavior, and reputation farming mitigated?
- Are reporting systems protected against weaponization?

9.6 Privacy and proportionality

- Is reputation based on necessary signals rather than totalizing surveillance?
- Are sensitive feedback records protected?
- Can participants understand which signals affect rights, rewards, or access?

9.7 Inclusion

- Can marginalized and less-resourced participants provide feedback meaningfully?
- Are feedback channels multilingual, asynchronous, and accessible?
- Are affected groups visible in system learning without being exposed to retaliation?

10. Implementation Patterns

10.1 Reputation domains instead of reputation scores

Use separate reputation domains for different capabilities rather than one universal score.

Example domains:

- trusted annotator
- reliable bridge builder
- safe moderator
- responsive maintainer
- accurate fact-checker
- inclusive facilitator
- responsible AI operator

10.2 Confidence-weighted signals

Signals should include uncertainty. A verified expert correction, a peer endorsement, an anonymous concern, and a bot-like mass vote should not carry the same weight.

10.3 Receipts everywhere

Feedback, moderation, role changes, rewards, and reputation updates should produce receipts that can be audited without exposing unnecessary private data.

10.4 Decay by default

Signals should age unless renewed by current evidence. Decay prevents permanent lock-in and reduces the power of old or stale interactions.

10.5 Role renewal cycles

Reputation-linked roles should require periodic review or renewal, especially for stewards, moderators, grant reviewers, and high-impact AI operators.

10.6 Affected-group weighting

Feedback from those directly affected by a policy or harm may require special visibility or routing, while still protecting against capture.

10.7 Deliberative escalation

High-impact reputation changes should move from automated detection to human or community review before affecting access, compensation, or public standing.

10.8 Reputation portability bundles

Portable reputation should export as bundles containing claims, provenance, rubrics, expiry, dispute status, and interpretation limits.

10.9 Community retrospectives

Communities should periodically review whether feedback loops are actually improving governance, safety, inclusion, and contribution quality.

11. Relationship to Other Desirable Properties

DP1 – Identity and Accountability

DP18 depends on accountable identity to prevent sybil abuse, retaliation, and reputation laundering. DP1 provides the identity substrate; DP18 governs the memory of behavior and contribution.

DP2 – Participant Agency and Empowerment

Participants must understand and contest reputation effects. Feedback systems must produce real agency, not symbolic participation.

DP3 – Adaptive Governance

DP18 supplies the learning loops that allow governance to adapt. DP3 defines how decisions change; DP18 defines how lived experience becomes actionable signal.

DP4 – Data Sovereignty and Privacy

Feedback and reputation must operate under data minimization, purpose limitation, and consent. Reputation cannot justify surveillance.

DP7 – Interoperability

Reputation and feedback must preserve meaning across systems or degrade visibly.

DP8 – Community-Defined Participation and Governance Zones

Reputation is zone-scoped by default. Communities define which signals matter, which roles reputation unlocks, and how repair works.

DP9 – Developer and Community Incentives

DP18 provides the recognition and evaluation substrate for fair incentives. DP9 governs how value flows from contribution.

DP12 and DP13 – AI Governance and Containment

AI agents need scoped reputations, feedback-triggered containment, and human-ratified adaptation.

DP14 – Trust and Transparency

Feedback loops and reputation logic must be legible enough for participants to understand how trust is being shaped.

DP20 – Ownership and Stewardship

Communities should be able to own and govern their feedback data, reputation schemas, and contribution histories.

12. Open Questions for ML-RFC Development

1. What minimum schema should define a feedback object across the meta-layer?
 2. What minimum schema should define a reputation object?
 3. Which reputation dimensions should be standardized, and which should remain zone-defined?
 4. How should reputation decay be represented across systems?
 5. What privacy-preserving methods can support reputation without centralized surveillance?
 6. How should communities distinguish endorsement, expertise, contribution, care, and popularity?
 7. What rights should participants have when reputation affects access, compensation, or visibility?
 8. How should non-human agents receive, carry, and contest reputation?
 9. What signals are safe to make portable across zones?
 10. How can marginalized communities govern feedback without being overexposed or tokenized?
 11. What forms of feedback should trigger mandatory governance review?
 12. How can reputation-linked roles avoid ossification and capture?
 13. What are the rollback standards for failed reputation experiments?
 14. How should feedback systems disclose AI summarization, clustering, or weighting?
-

13. Minimal DP18 Compliance Checklist

A system claiming DP18 alignment SHOULD demonstrate:

- structured feedback objects
- visible feedback routing and response commitments
- reputation scoped by zone and contribution domain
- published reputation logic where it affects access, rewards, or visibility
- decay, renewal, and repair mechanisms
- dispute and appeal pathways
- protection against sybil attacks, brigading, and reputation farming
- privacy-preserving data practices

- AI feedback processing with disclosure and human oversight
 - adaptation receipts linking feedback to changes
 - safeguards against runaway amplification loops
 - explicit interoperability semantics for reputation portability
-

14. Path Toward ML-RFC

DP18 is currently an ML-Draft and serves as exploratory scaffolding for how feedback loops and reputation can function as accountable learning infrastructure in the meta-layer.

Advancement toward ML-RFC status SHOULD require:

- convergence on a minimal interoperable schema for feedback objects
- convergence on a minimal interoperable schema for reputation objects
- demonstrated implementations across multiple zones
- tested mechanisms for decay, repair, and dispute resolution
- validated protections against gaming, sybil attacks, and amplification loops
- evidence of privacy-preserving reputation systems in practice
- working models for AI-assisted feedback synthesis with human oversight
- at least one cross-system portability experiment with semantic preservation
- documented governance processes for evolving reputation logic

ML-RFC promotion SHOULD be contingent on:

- rough consensus across participating communities
- demonstrated real-world usage with measurable learning outcomes
- clear failure cases and mitigation strategies
- alignment with DP1, DP2, DP3, DP4, and DP8 in deployed systems

Early ML-RFC candidates may focus on:

- feedback object standards
- reputation object standards
- decay and repair protocols
- feedback receipt and adaptation receipt formats

DP18 is expected to evolve iteratively, with partial RFCs emerging for specific components rather than a single monolithic specification.

15. Closing Orientation

DP18 makes the meta-layer capable of learning.

It ensures that feedback is not extracted and ignored, that reputation is not reduced to popularity, and that communities can recognize contribution without building permanent social cages.

A DP18-aligned meta-layer remembers enough to be accountable, forgets enough to allow growth, and adapts visibly enough for participants to trust that their experience matters.

Feedback becomes civic signal. Reputation becomes contextual memory. Learning becomes shared infrastructure.

This is the difference between a system that listens and one that evolves.

DP19 – Amplifying Presence and Community Engagement

1. Purpose of This Draft

This draft articulates Desirable Property 19 (DP19) as the condition under which the meta-layer becomes visible, inviting, culturally resonant, and socially adopted without collapsing into extractive growth, hype cycles, influencer capture, or engagement theater.

DP19 defines how the Metaweb, Overweb, and broader meta-layer ecosystem cultivate awareness, shared identity, community-led promotion, and durable participation. It treats amplification not as advertising alone, but as the public-facing expression of a civic substrate.

The meta-layer cannot become trustworthy civic infrastructure if people do not understand it, recognize themselves in it, and feel invited to participate. Presence must be amplified through symbols, narratives, tools, communities, public relations, education, and lived experiences that make the meta-layer legible.

If DP19 is weak, predictable failures follow: the meta-layer remains technically compelling but culturally invisible; public language becomes confused with the metaverse, Web3 speculation, or AI hype; community energy dissipates; early supporters lack ways to help; growth depends on centralized marketing; and adoption becomes shallow because the public never develops emotional or practical familiarity.

DP19 connects directly to:

- DP2, participant agency and empowerment
- DP3, adaptive governance
- DP5, decentralized namespace
- DP7, interoperability
- DP8, community-defined participation and governance zones
- DP9, developer and community incentives
- DP10, education and onboarding
- DP14, trust and transparency
- DP16, roadmap and milestones
- DP18, feedback loops and reputation
- DP20, ownership and stewardship

DP19 does not prescribe a single brand, campaign, influencer strategy, or communications calendar. It defines the minimum conditions under which amplification and engagement remain participatory, truthful, inclusive, and aligned with the meta-layer's purpose.

2. Problem Statement

Infrastructure does not become public infrastructure merely by existing.

The web itself became powerful because people learned to recognize its patterns, tell stories about it, build on it, link to it, invite others into it, and imagine themselves inside it. The meta-layer requires a similar cultural threshold. People must be able to name it, describe it, share it, experience it, and see why it matters.

Today's web, however, has trained people to associate digital growth with extractive dynamics:

- social media virality
- influencer-driven attention
- engagement farming
- opaque recommendation systems
- growth hacking
- brand performance without substance
- communities treated as marketing funnels
- platform metrics mistaken for public legitimacy

This creates a dilemma. The meta-layer must reach people where attention already lives, but it cannot adopt the same attention-extractive patterns it seeks to transcend.

DP19 addresses this by reframing amplification as community presence. The goal is not merely to promote the meta-layer, but to help communities recognize, inhabit, and extend it.

A healthy DP19 implementation must be able to answer:

- What is the meta-layer called in public language?
- What symbols, stories, and experiences make it memorable?
- How can participants share it without becoming unpaid ad labor?
- How can communities grow adoption while preserving trust?
- How can public relations establish narrative clarity without centralizing control?
- How can youth, families, educators, civic leaders, developers, journalists, and local communities each find a meaningful entry point?
- How can incentives support community-led amplification without rewarding spam or manipulation?
- How can engagement be measured by meaningful participation rather than shallow reach?

Without DP19, the meta-layer risks becoming a brilliant architecture with no cultural surface.

3. Threats and Failure Modes

3.1 Narrative confusion

The public cannot distinguish the meta-layer from the metaverse, Web3 speculation, browser extensions, social media, AI agents, or generic digital trust projects.

Example: A journalist describes the Metaweb as “another metaverse platform,” obscuring its role as an interface-level civic layer above the existing web.

Why this matters: Names and narratives shape adoption. Confusion blocks participation before people encounter the architecture.

3.2 Brand vacuum

The ecosystem lacks memorable names, symbols, metaphors, and public language.

Example: Builders can explain the architecture technically but cannot offer a phrase, image, or story that ordinary people remember.

Why this matters: Public infrastructure needs cultural handles. Without them, people cannot share the idea.

3.3 Hype without substance

Amplification promises more than the system can currently deliver.

Example: A campaign claims the meta-layer will “fix the internet” while usable tools, governance pathways, or safety guarantees remain early-stage.

Why this matters: Overclaiming converts early curiosity into later distrust. DP19 must align with DP16 roadmap honesty.

3.4 Influencer capture

Public narrative becomes dependent on a few high-visibility personalities.

Example: A thought leader becomes the de facto voice of the meta-layer, shaping public perception without accountability to participating communities.

Why this matters: Civic infrastructure cannot rely on celebrity mediation.

3.5 Community-as-funnel extraction

Participants are treated primarily as growth channels.

Example: Ambassadors are asked to generate posts, referrals, and events but receive little agency over strategy, messaging, governance, or value flows.

Why this matters: Community-led promotion must be reciprocal, not extractive.

3.6 Spammy bounty dynamics

Incentives reward volume over quality.

Example: A bounty pays for social posts, producing low-effort threads, duplicate graphics, or misleading claims.

Why this matters: Amplification incentives can degrade trust if they reward noise.

3.7 Platform dependency

Engagement depends too heavily on proprietary social platforms.

Example: A movement grows on one platform's algorithm, then loses reach when rules, APIs, ranking, or moderation policies change.

Why this matters: A meta-layer committed to agency and interoperability cannot depend entirely on rented attention.

3.8 Exclusionary adoption pathways

The public face of the meta-layer appeals only to technical, crypto, policy, or AI governance audiences.

Example: Parents, teachers, youth, local communities, artists, journalists, and non-technical civic actors cannot find themselves in the story.

Why this matters: Public legitimacy requires plural entry points.

3.9 Gamified engagement without civic depth

Leaderboards, badges, and rewards stimulate activity without building understanding, trust, or stewardship.

Example: Participants chase ambassador points but cannot explain the values, risks, or governance responsibilities of the meta-layer.

Why this matters: Engagement is not the same as orientation.

3.10 Public relations centralization

Media and partnership strategy is controlled by a narrow group without transparent accountability.

Example: Official messaging changes to satisfy funders or partners without community visibility or review.

Why this matters: Narrative power is governance power.

3.11 Adoption without retention

People encounter the meta-layer once but do not develop ongoing presence, practice, or identity.

Example: A viral campaign drives signups, but participants do not return because the onboarding path is unclear or the experience lacks immediate relevance.

Why this matters: DP19 must produce durable engagement, not only awareness spikes.

4. Core Principle

Amplifying presence and community engagement in the meta-layer means helping people recognize, share, inhabit, and steward a new civic layer of digital life through truthful narratives, resonant identity, participatory campaigns, reciprocal incentives, and durable engagement pathways.

Amplification is legitimate only when it increases agency, understanding, and accountable participation.

Engagement is healthy only when it deepens belonging, contribution, and stewardship.

Example: A family-facing campaign introduces the meta-layer as a safer, co-creative layer above the web, provides parent and youth guides, invites local workshops, offers badges for meaningful participation, and routes community feedback into governance.

What this feels like: People do not feel marketed to. They feel invited into a shared project.

Without this: The meta-layer remains either obscure infrastructure or another brand shouting for attention.

5. Primary Mechanisms and Structural Conditions

5.0 Presence and Engagement Layer: Narrative, Identity, Participation, and Continuity

DP19 requires a presence and engagement layer that makes the meta-layer recognizable and participatory across cultural, civic, technical, educational, and local contexts.

This layer includes:

- naming and brand systems
- public narratives
- community-led promotion
- ambassador programs
- social sharing tools
- press and partnership pathways
- bounties and rewards
- educational engagement
- events and rituals
- participation badges
- onboarding journeys
- feedback loops from public engagement into governance

The purpose of this layer is not to manufacture attention. It is to make presence discoverable and meaningful.

Failure mode: **growth skin**, where branding and marketing sit on top of the system without shaping participation, governance, or trust.

5.1 Naming, branding, and memetic clarity

The meta-layer requires memorable, symbolic public language that conveys its mission of agency, transparency, trust, and new layers of digital interaction.

Naming systems SHOULD be:

- easy to say
- easy to explain
- visually memorable
- culturally adaptable
- distinct from extractive or dystopian digital narratives
- compatible with decentralized namespace patterns
- capable of supporting both technical and public-facing meaning

Candidate or ecosystem terms may include Metaweb, Overweb, Sky-Web, Canopi, or other community-developed language.

The chosen terms must help people understand that this is a layer above today's web, not a replacement world that pulls them away from reality.

Failure mode: **semantic fog**, where no one can explain what the project is in one sentence.

5.2 Iconic presence and visual identity

The meta-layer needs recognizable visual language that can travel across tools, events, documents, interfaces, campaigns, and community spaces.

Visual identity SHOULD support:

- trust without institutional stiffness
- openness without vagueness
- civic seriousness without bureaucracy
- youth resonance without pandering
- global adaptability
- accessibility and readability
- remixability by communities

Visual identity MAY include marks, icons, badges, color systems, mascots, interface motifs, overlay metaphors, and ritual objects.

Failure mode: **corporate flattening**, where the public identity feels like another platform brand rather than a shared civic substrate.

5.3 Narrative architecture

DP19 requires a coherent set of stories for different audiences.

Narratives **SHOULD** be tailored for:

- general public
- youth and students
- parents and families
- educators
- civic technologists
- journalists
- municipalities
- developers
- AI governance communities
- artists and creators
- funders and institutional partners
- local organizers

Each narrative should preserve the same core claim while meeting the audience where it is.

Example core claim:

The meta-layer lets people and communities add trust, context, presence, and governance above the web they already use.

Failure mode: **audience collapse**, where one technical narrative is expected to work for everyone.

5.4 Community-driven marketing

Participants should be able to help amplify the meta-layer through meaningful, governed contribution pathways.

Community-driven marketing **MAY** include:

- ambassador programs
- local organizer kits
- event hosting
- explainers and videos
- memes and cultural artifacts
- translation and localization
- youth-led campaigns
- school and university clubs
- community showcases
- civic workshops

- creator collaborations
- onboarding circles

These efforts **SHOULD** be reciprocal. Contributors should receive recognition, feedback, learning opportunities, and where appropriate, compensation.

Failure mode: **volunteer extraction**, where enthusiasm is used without support, recognition, or voice.

5.5 Ambassador and steward programs

Ambassador programs can help the meta-layer travel across regions, languages, sectors, and communities.

An ambassador program **SHOULD** define:

- role expectations
- community accountability
- messaging guidelines
- onboarding curriculum
- escalation paths
- recognition systems
- conflict-of-interest rules
- renewal or sunset processes
- feedback responsibilities

Ambassadors should not merely promote. They should listen, translate, connect, and help communities enter governance pathways.

Failure mode: **brand reps without governance**, where ambassadors broadcast messages but cannot shape the system.

5.6 Social media integration

Participants may need tools to share Metaweb experiences and insights on existing social platforms such as X, Instagram, LinkedIn, TikTok, YouTube, Mastodon, Bluesky, and others.

Cross-posting tools **SHOULD** support:

- participant consent
- context preservation
- source links
- provenance markers
- audience selection
- platform-specific formatting
- privacy review before posting
- clear distinction between personal expression and official messaging

Social sharing should bring people into deeper context, not strip context for reach.

Failure mode: **context collapse marketing**, where meaningful Metaweb activity becomes flattened into promotional snippets.

5.7 Public relations and media strategy

DP19 requires proactive public narrative work.

Public relations MAY include:

- press kits
- explainers
- op-eds
- interviews
- journalist briefings
- public demos
- media partnerships
- family media outreach
- civic technology features
- education-sector stories
- thought-leader engagement
- crisis communication protocols

PR materials SHOULD be truthful about maturity, limitations, and roadmap status. They should distinguish aspiration from operational reality.

Failure mode: **PR overclaiming**, where media attention creates expectations the system cannot yet satisfy.

5.8 Partnerships with trusted messengers

The meta-layer can grow through partnerships with trusted communities and institutions.

Potential partners include:

- educators
- libraries
- municipalities
- newsrooms
- parent networks
- youth organizations
- civic tech groups
- internet governance communities
- digital safety organizations

- creator communities
- open-source networks
- local cultural institutions

Partnerships SHOULD preserve community agency and avoid converting trusted messengers into sales channels.

Failure mode: **borrowed trust abuse**, where institutions lend legitimacy to systems they cannot meaningfully govern or understand.

5.9 Bounties, badges, and rewards

Incentives may support amplification, onboarding, content creation, translation, event organization, and public education.

Reward systems SHOULD:

- define valuable contributions clearly
- reward quality over volume
- distinguish official, community, and experimental materials
- avoid misleading claims
- include review processes
- disclose conflicts of interest
- connect to DP18 reputation logic
- support non-monetary recognition where appropriate
- avoid leaderboards that reward spam

Examples of rewardable actions:

- high-quality explainer video
- local workshop
- multilingual translation
- school onboarding guide
- trusted press contact development
- civic demo night
- documentation improvement
- community case study
- accessibility adaptation

Failure mode: **bounty spam**, where incentives produce noise instead of understanding.

5.10 Gamification with guardrails

Gamification can motivate participation, but it must be carefully bounded.

Healthy gamification MAY include:

- learning quests
- contribution badges
- local chapter milestones
- collaborative challenges
- public good achievements
- stewardship streaks
- peer recognition
- community showcase unlocks

Gamification SHOULD NOT reward harassment, spam, superficial virality, or unhealthy attention competition.

Failure mode: **casino engagement**, where the system trains people to chase points rather than build civic capacity.

5.11 Youth and educational familiarity

Long-term adoption depends on familiarity during formative learning contexts.

DP19 SHOULD support youth and education pathways through:

- school-safe explainers
- classroom activities
- youth ambassador programs
- digital literacy curricula
- annotation and bridge-building exercises
- civic internet labs
- student-led showcases
- family-facing onboarding
- partnerships with educators and youth media

The goal is not to recruit children into a product. The goal is to cultivate agency, trust literacy, and co-creative expectations for the web.

Failure mode: **youth capture**, where young people are targeted for growth instead of empowered as future stewards.

5.12 Family-centered engagement

Families can become trusted entry points into the meta-layer when the story emphasizes safety, creativity, agency, and shared digital care.

Family-centered engagement MAY include:

- parent guides
- youth safety explainers
- co-creation activities
- family media outreach
- community workshops
- safer web campaigns
- intergenerational digital literacy sessions

Failure mode: **tech abstraction**, where families cannot see practical relevance to everyday internet life.

5.13 Municipal and civic engagement

Cities, towns, and civic institutions can use the meta-layer to transform public engagement from complaint intake into participatory co-creation.

Municipal engagement MAY include:

- civic overlays above public service pages
- participatory planning annotations
- public meeting context layers
- local issue maps
- real-time dialogue tools
- citizen proposal pathways
- community event overlays
- participatory design dashboards

Failure mode: **complaint-system trap**, where engagement channels collect grievances without resolution, collaboration, or visible adaptation.

5.14 Cultural production and remix

The meta-layer needs cultural artifacts, not only documentation.

Communities SHOULD be encouraged to create:

- memes
- shorts
- songs
- posters
- zines
- games
- rituals

- visual metaphors
- local stories
- public demos
- speculative futures
- civic myths

Cultural production helps the meta-layer become imaginable.

Failure mode: **documentation-only adoption**, where only experts can understand or care about the system.

5.15 Events, rituals, and gathering grounds

Community engagement grows through repeated gatherings and shared practices.

DP19 MAY support:

- demo days
- build nights
- annotation jams
- bridge walks
- civic salons
- local chapter meetings
- online town halls
- contributor onboarding waves
- youth showcases
- press briefings
- public retrospectives
- seasonal campaigns

Events should create continuity and contribution pathways, not one-off spectacle.

Failure mode: **event evaporation**, where gatherings generate excitement but no durable roles, artifacts, or next steps.

6. Engagement Metrics and Community Health

DP19 must distinguish meaningful engagement from attention metrics.

6.1 Healthy engagement indicators

Healthy engagement MAY include:

- repeat participation
- successful onboarding

- contribution diversity
- local chapter formation
- multilingual participation
- participant retention
- quality of public explanations
- number of community-led events
- feedback routed into governance
- cross-community collaboration
- stewardship roles filled
- creator and educator adoption
- civic use cases launched

6.2 Risk indicators

Risk indicators MAY include:

- high reach with low retention
- bounty-driven spam
- audience confusion
- misleading public claims
- concentration of narrative power
- ambassador burnout
- platform dependency
- hostile media framing
- exclusion of non-technical audiences
- social posts disconnected from participation pathways

6.3 Engagement receipts

Major campaigns SHOULD generate engagement receipts.

An engagement receipt MAY include:

- campaign purpose
- intended audience
- claims made
- materials used
- responsible stewards
- incentives offered
- participation outcomes
- feedback received
- lessons learned
- governance follow-up

Failure mode: **metrics without memory**, where campaigns are repeated without learning.

7. Governance of Public Narrative

Narrative power must be governed because public language shapes legitimacy, funding, adoption, and expectation.

Communities SHOULD define:

- who can speak officially
- what counts as official messaging
- how community materials are labeled
- how claims are reviewed
- how mistakes are corrected
- how media inquiries are handled
- how partnerships are approved
- how ambassador conduct is governed
- how public feedback affects strategy

Public narrative governance should balance coherence with pluralism.

Failure mode: **message authoritarianism**, where one voice controls the story and suppresses community creativity.

Opposite failure mode: **narrative fragmentation**, where conflicting claims make the project incoherent.

8. Inclusion and Localization

DP19 requires global and local resonance.

Amplification and engagement systems SHOULD support:

- multilingual materials
- regional leadership
- culturally specific metaphors
- youth participation
- gender inclusion
- accessibility
- low-bandwidth participation
- non-Western civic contexts
- local governance traditions
- community translation rather than literal language substitution

Localization is not only translation. It is meaning-making in context.

Failure mode: **global English default**, where the meta-layer claims universality while speaking to a narrow audience.

9. Relationship to Existing Platforms

The meta-layer should use existing platforms without becoming dependent on them.

DP19-aligned strategy SHOULD include:

- platform-specific sharing practices
- decentralized community channels
- owned knowledge bases
- interoperable contact lists where consented
- portable contributor records
- open archives of public materials
- RSS, email, and web-native distribution
- bridges between social platforms and meta-layer spaces

Failure mode: **rented movement**, where the public presence of the meta-layer can be throttled, erased, or distorted by one platform's policies.

10. Evaluation Criteria

A DP19-aligned implementation should be evaluated against the following questions.

10.1 Narrative clarity

- Can different audiences explain the meta-layer in their own words?
- Are Metaweb, Overweb, and related terms clearly distinguished?
- Does public language avoid confusion with extractive or dystopian digital narratives?

10.2 Truthfulness

- Do campaigns distinguish current capabilities from aspirations?
- Are roadmap dependencies and limitations visible?
- Are public claims corrected when they become inaccurate?

10.3 Community agency

- Can participants shape amplification strategy?
- Are ambassadors accountable to communities, not only central teams?
- Are community-generated materials recognized and supported?

10.4 Incentive alignment

- Do bounties reward quality, care, and learning rather than volume?
- Are rewards transparent and fair?

- Are spam, duplicate content, and misleading claims discouraged?

10.5 Inclusion

- Are materials accessible across language, region, age, ability, and technical literacy?
- Do youth, families, educators, local communities, and non-technical participants have real entry points?
- Are trusted messengers supported without being exploited?

10.6 Retention and continuity

- Do campaigns lead to durable participation pathways?
- Are events connected to roles, artifacts, or governance channels?
- Can participants move from awareness to contribution?

10.7 Governance and accountability

- Is official messaging governed?
 - Are errors corrected transparently?
 - Are media and partnership strategies accountable to the ecosystem?
-

11. Implementation Patterns

11.1 One-sentence public anchor

Every campaign should be able to use a clear public anchor.

Example:

The meta-layer lets people and communities add trust, context, presence, and governance above the web they already use.

11.2 Audience-specific explainers

Develop tailored explainers for parents, students, developers, journalists, municipalities, creators, and AI governance audiences.

11.3 Ambassador kits

Provide ambassadors with slide decks, FAQs, demo scripts, visual assets, local event templates, ethical messaging guidelines, and feedback forms.

11.4 Community media commons

Create a shared repository of approved, remixable, versioned public materials.

11.5 Press kits with maturity labels

Press materials should label what is live, experimental, conceptual, and proposed.

11.6 Learning-first bounties

Reward explainers, translations, workshops, demos, and case studies that help people understand and participate.

11.7 Local chapter campaigns

Support city, campus, school, library, and community chapter pilots with lightweight governance and reporting.

11.8 Youth-led creative campaigns

Invite youth to create shorts, memes, zines, and demos that translate meta-layer values into native cultural formats.

11.9 Cross-posting with context

Sharing tools should preserve provenance, links, and consent while adapting format to each platform.

11.10 Engagement retrospectives

After major campaigns, publish what worked, what failed, what was learned, and what changes next.

12. Relationship to Other Desirable Properties

DP2 – Participant Agency and Empowerment

DP19 must amplify agency rather than manipulate attention. Participants should understand how to join, shape, and leave engagement pathways.

DP3 – Adaptive Governance

Public feedback from campaigns must route into governance. Narrative strategy should evolve based on community learning.

DP5 – Decentralized Namespace

Names, symbols, handles, badges, and community identities should be portable and resistant to capture.

DP7 – Interoperability

Engagement should move across platforms, communities, and tools without losing context or continuity.

DP8 – Community-Defined Participation and Governance Zones

Communities should be able to define their own engagement practices, ambassador roles, and public presence.

DP9 – Developer and Community Incentives

Bounties, badges, and rewards for amplification must align with contribution quality, transparency, and anti-gaming safeguards.

DP10 – Education and Onboarding

DP19 depends on learning pathways that convert curiosity into practical literacy.

DP14 – Trust and Transparency

Public narrative must be honest, auditable, and corrigible.

DP16 – Roadmap and Milestones

Amplification claims must align with real roadmap status and maturity.

DP18 – Feedback Loops and Reputation

Community engagement must feed back into learning systems, and public contribution should be recognized without devolving into popularity scoring.

DP20 – Ownership and Stewardship

Communities should own and steward their narratives, symbols, materials, and engagement histories where appropriate.

13. Open Questions for ML-RFC Development

1. What minimum standards should define official versus community-generated messaging?
2. What public terminology should be standardized across the meta-layer ecosystem?
3. How should brand assets be governed in a decentralized ecosystem?
4. What claims require evidence, maturity labels, or review before publication?
5. How should ambassador programs be structured across regions and languages?
6. What reward schemas prevent bounty spam while supporting public education?
7. How should campaign materials preserve provenance and version history?
8. What engagement metrics best reflect civic participation rather than attention capture?
9. How should youth-centered engagement be governed ethically?
10. What rules should govern cross-posting from meta-layer spaces to social platforms?
11. How should public relations strategy remain accountable to community governance?
12. What local chapter model best supports global inclusion without central bottlenecks?

13. How should public mistakes, overclaims, or misleading media coverage be corrected?
 14. What interoperability standards are needed for badges, ambassador credentials, and engagement receipts?
-

14. Path Toward ML-RFC

DP19 is currently an ML-Draft and serves as exploratory scaffolding for how public presence, community engagement, branding, amplification, and adoption pathways can operate as part of meta-layer infrastructure.

Advancement toward ML-RFC status SHOULD require:

- convergence on terminology for public-facing meta-layer concepts
- a governed distinction between official, community, experimental, and partner messaging
- a minimal schema for engagement receipts
- a minimal schema for ambassador or community steward credentials
- tested bounty and badge systems that reward quality over volume
- evidence from community-led campaigns across multiple audiences or regions
- accessible onboarding materials for non-technical participants
- documented feedback loops from public engagement into governance
- demonstrated safeguards against hype, spam, and misleading claims
- at least one localization or multilingual engagement pilot

ML-RFC promotion SHOULD be contingent on:

- rough consensus among participating communities
- demonstrated use of DP19 mechanisms in real outreach or adoption settings
- clear alignment with DP9 incentive safeguards and DP18 reputation logic
- transparent handling of public claims, corrections, and campaign outcomes
- evidence that engagement pathways increase agency and stewardship, not only reach

Early ML-RFC candidates may focus on:

- public terminology and maturity-label standards
- engagement receipt formats
- ambassador credential standards
- ethical bounty and badge guidelines
- campaign transparency requirements
- community media commons governance

DP19 is likely to evolve through multiple partial RFCs rather than one monolithic specification because brand, narrative, engagement, and public participation each have distinct governance and implementation needs.

15. Closing Orientation

DP19 gives the meta-layer a public face and a living community presence.

It ensures that the Metaweb is not only built, but recognized; not only explained, but shared; not only promoted, but inhabited.

A DP19-aligned meta-layer grows through resonance rather than manipulation. It invites people into agency, context, trust, and co-creation without reducing them to metrics or marketing channels.

Brand becomes symbolic orientation. Engagement becomes participatory belonging. Amplification becomes civic invitation. Presence becomes shared infrastructure.

This is how the meta-layer becomes something people can name, join, steward, and carry into the world.

DP20 – Community Ownership

1. Purpose of This Draft

This draft articulates Desirable Property 20 (DP20) as the condition under which communities meaningfully own the digital environments, surplus, and governance artifacts they co-create, not as marketing language about “community,” but as enforceable rights and responsibilities: stake, voice, upside, and continuity that persist across tools and seasons.

DP20 completes a loop with DP18 (feedback and reputation), DP19 (presence and engagement), and DP12 (community governance of AI): participation must be able to mature into ownership, otherwise “community” remains a rented audience for platforms.

DP20 connects to DP4 (data commons and collective data rights), DP6 (fair distribution of commercial surplus), DP9 (incentives that return value to contributors), DP3 (adaptive governance at scale), DP17 (financial sustainability), and DP1–DP2 (accountability and agency for collective actors).

If DP20 is weak, predictable failures follow: extractive growth on volunteer labor, token claims without decision rights, capture of governance by insiders, inability to fork when values diverge, and surplus flowing outward while risks stay local.

DP20 does not mandate a single cooperative legal form or chain. It defines minimum legitimacy conditions for what “ownership” may claim in the meta-layer.

2. Problem Statement

In today’s web, “community” is often a label applied to users whose labor, attention, and data build value they cannot capture, audit, or exit with. Governance surfaces are optional; economic upside is asymmetric; and forking social reality is impractical.

In practice, this produces recurring failures:

- moderators and maintainers burn out while platform valuation rises
- creators depend on distribution they do not control
- tokens grant speculation liquidity without durable rights
- data generated collectively is sold without collective consent (DP4)
- communities cannot carry norms, memory, or economic terms across apps (DP7)

These failures are structural: ownership is confused with usage; participation is confused with consent to extraction.

DP20 reframes ownership as operational: rights you can see, exercise, and defend, including pathways to fork, exit, and recapture surplus when legitimacy breaks.

3. Threats and Failure Modes

3.1 Ownership theater

Marketing claims “community-owned” while decisions and keys remain centralized.

Example: A DAO where a multisig admin can override any vote without published criteria.

Why this matters: DP20 requires honest mapping between claims and controls.

3.2 Token liquidity without duties or rights

Tokens trade while governance is inactive or non-binding.

Example: Voter apathy plus whale dominance makes governance a periodic headline, not a constraint on operators.

Why this matters: Ownership instruments must bind to decisions and surplus, or be renamed.

3.3 Commons maintenance without reciprocity

Value is extracted from a space while moderation, safety, and infrastructure remain underfunded.

Example: A viral forum hosts commerce ads but does not fund anti-abuse capacity.

Why this matters: Connects DP6 reciprocity and DP17 sustainability.

3.4 Capture by concentrated stakeholders

Early contributors or large holders lock rules that favor continuation of privilege.

Example: Fee switches require supermajorities that insiders can block indefinitely.

Why this matters: DP3 adaptive governance and DP12 dialectic must resist ossified capture.

3.5 Forklessness

When values diverge, participants cannot exit with continuity of identity, memory, or economic position.

Example: A community schism loses all history because export is blocked and namespaces are not portable (DP5, DP7).

Why this matters: Forking is a safety valve for legitimacy.

3.6 Collective data sale without governance

Aggregated or derived data monetizes communities without authorization pathways.

Why this matters: DP4 sovereignty requires collective decision surfaces, not only individual toggles.

3.7 Illegible membership and shadow elites

Informal power structures make decisions without published accountability.

Example: Moderators coordinate bans off-platform without records or appeals.

Why this matters: DP1 contestability and DP12 governance memory must extend to collective actors.

3.8 Philanthropic dependency

Communities rely on unsustainable grants with no path to owned revenue or shared surplus.

Example: A civic space runs on a yearly sponsor who can pull funding overnight.

Why this matters: DP17 sustainability and DP6 commerce reciprocity must appear in ownership design.

3.9 Legal hostility to collective action

Cooperative forms face banking, tax, and liability friction.

Why this matters: DP20 acknowledges law as a design constraint while still demanding honest digital rights bundles.

4. Core Principle

Community ownership in the meta-layer means participants and communities hold defensible rights to govern, benefit from, and responsibly steward the environments and surplus they co-create, including credible paths to fork, exit, and recapture value, with transparency, accountability, and continuity of memory.

Ownership is not a vibe. It is power with responsibility, visible to members and bounded by human-rights baselines and law.

Example: A community charter defines decision rights, surplus split, audit rights, and fork procedures; tools entering the zone must acknowledge the charter's machine-readable profile.

What this feels like: You are not the community in a slogan. You are a member with levers.

Without this: Participation is harvested; dignity is not.

5. Primary Mechanisms and Structural Conditions

5.0 Ownership Layer: Execution, Proof, and Exit

Community ownership requires more than charters and intent. It requires the ability to express, verify, exercise, and, when necessary, exit ownership as part of an operational system.

In many systems, “ownership” fails because it is not bound to decision rights, not provable in practice, or not portable across tools. Rights exist on paper while control remains elsewhere.

DP20 therefore requires a shared ownership layer composed of primitives that make ownership executable rather than symbolic.

Ownership objects

Ownership must be represented as structured, machine-readable objects that bind rights to actors and contexts.

An ownership object includes:

- scope: what environment, assets, or processes are owned
- rights: governance, economic, and data rights
- duties: stewardship and maintenance obligations
- transfer conditions: how ownership can change hands

This binds ownership directly to governance (DP3, DP12).

Ownership receipts

All ownership-relevant actions produce verifiable records.

An ownership receipt includes:

- who holds or exercised a right
- what action was taken (vote, allocation, fork, transfer)
- when it occurred
- what assets or policies were affected

These receipts make ownership auditable and connect to DP15 (provenance).

Surplus and flow proofs

Economic flows must be traceable to ownership claims.

Members must be able to see:

- how value enters the system
- how it is allocated
- how it maps to rights and contributions

Without flow proofs, ownership cannot capture upside in a defensible way (DP6, DP17).

Fork and exit primitives

Ownership must include executable pathways to exit or fork with continuity.

This includes:

- export of data and governance artifacts (DP4, DP7)
- migration of identifiers where honest (DP5)
- continuity of membership and contribution records

Forking is not a failure. It is a safety valve for legitimacy.

Membership and credential objects

Membership and contribution must be represented as portable, verifiable artifacts.

This enables:

- recognition across tools
- resistance to platform lock-in
- continuity of identity and stake

Ownership–governance binding

Ownership must map to decision rights in a visible way.

Participants must be able to see:

- who can decide what
- how ownership affects voting or delegation
- where overrides or special powers exist

Governance that cannot be owned is advisory. Ownership that cannot govern is symbolic.

Ownership memory

Ownership history must persist over time.

This includes:

- prior allocations and changes
- disputes and resolutions
- forks and mergers

Without memory, ownership becomes contestable without evidence.

These primitives do not replace the mechanisms below. They make them enforceable, portable, and auditable across contexts.

5.1 Charter and rights bundle

A written and machine-readable articulation of membership, decision rights, economic rights, data rights, fork rights, and sunset conditions.

This MUST include: - explicit mapping between roles and powers - conditions under which rights can change - linkage to governance execution (DP3, DP12)

Verification: - participants can inspect current and prior versions - tools enforce charter constraints at interaction level

Failure modes: - **charter drift** (rules change without visibility) - **unenforced rights** (charter exists but does not constrain behavior)

5.2 Transparent surplus accounting

Revenue, fees, grants, and in-kind support are visible to members at useful granularity (DP6, DP17).

This MUST include: - inflow sources and categories - allocation breakdowns tied to functions - linkage to ownership claims

Verification: - members can trace value from source → allocation → outcome

Failure modes: - **surplus opacity** (value cannot be traced) - **extraction masking** (flows hidden behind aggregation)

5.3 Participatory budgeting and parameter control

Communities set budgets for safety, education (DP10), and incentives (DP9) through governed processes (DP12).

This MUST include: - binding decision pathways - budget execution tracking - feedback loops into future allocations

Failure modes: - **advisory budgeting** (decisions ignored) - **capture of allocation power**

5.4 Credible exit and fork

Procedures to export collective artifacts, migrate identifiers where honest (DP5), and continue governance under divergent values without coercion.

This MUST include: - executable export mechanisms - continuity of identity, data, and governance artifacts

Failure modes: - **exit suppression** - **fork degradation** (loss of continuity)

5.5 Stewardship obligations

Ownership includes maintenance duties: moderation, security, inclusion, and accessibility commitments with accountability (DP1, DP15).

This MUST include: - defined duties - enforcement mechanisms - reporting on fulfillment

Failure modes: - **duty abandonment** - **free-rider extraction**

5.6 Anti-capture controls

Rotation, conflict-of-interest rules, minority protections, and emergency pause pathways.

This MUST include: - detection of concentration - intervention pathways

Failure modes: - **ownership capture** - **governance lock-in**

5.7 Collective data governance

Shared corpuses require collective consent mechanisms and audit of downstream use (DP4).

This MUST include: - consent pathways - audit logs of use

Failure modes: - **data extraction without consent** - **loss of collective control**

5.8 Interoperable ownership artifacts

Membership and contribution receipts interoperate across tools (DP7).

This MUST include: - portable formats - degradation signaling across systems

Failure modes: - **lock-in via incompatibility**

5.9 Succession and dissolution

Charters define succession for stewards and fair dissolution procedures.

This MUST include: - transfer conditions - fair distribution of residual value

Failure modes: - **power vacuum** - **unfair dissolution**

5.10 Intersectional inclusion in ownership

Governance design actively counters exclusion patterns through facilitation norms, translation, and accessible participation.

This MUST include: - measurable inclusion practices - correction mechanisms

Failure modes: - **systemic exclusion** - **ownership concentration through bias**

6. Governance, Accountability, and Agency Surfaces

DP20 requires that ownership is not only visible but **defensible and enforceable**.

Participants **MUST** be able to: - verify ownership rights and mappings - challenge illegitimate control or misallocation - trigger governance review or dispute processes - exercise exit and fork pathways without coercion

Communities **MUST** be able to: - revoke or reassign ownership under defined conditions - enforce stewardship obligations - audit ownership actions and outcomes

Enforcement pathways include: - rollback or invalidation of illegitimate actions - governance-triggered reallocation of rights or surplus - fork as a last-resort enforcement mechanism

Failure modes: - **non-actionable ownership** - **illegitimate control persistence** - **ownership without recourse**

7. Incentives and Power Analysis

Ownership fails when incentives diverge from rights.

Key dynamics:

- **extraction loops:** value flows outward despite community contribution
- **token manipulation:** governance power decouples from responsibility
- **dependency capture:** infrastructure or distribution creates hidden control
- **soft capture:** influence accumulates without formal ownership

Adversarial patterns:

- governance capture through capital or coordination
- speculative ownership without stewardship
- platform rent extraction

Alignment requirements:

- ownership rights must correlate with value capture
- incentives must reward stewardship, not exploitation
- systems must detect divergence between formal ownership and real control

Failure modes: - **symbolic ownership** - **incentive inversion** - **ownership erosion over time**

8. Community Signals Informing DP20

Across platforms and communities, recurring signals point to a shared breakdown between participation and ownership:

- maintainers and moderators demand compensation and decision power proportional to their labor

- skepticism toward “community-owned” claims following repeated governance failures and token collapses
- desire for portable community memory, identity, and governance tools
- frustration with value extraction from spaces built through collective effort
- growing interest in local, cooperative, and community-controlled digital environments

These signals are not isolated. They indicate structural gaps in how ownership is defined, enforced, and sustained.

DP20 treats these signals as design requirements, not feedback to be addressed after failure.

9. Non-Goals and Explicit Boundaries

DP20 does not:

- require all systems to be collectively owned
- eliminate private enterprise or hybrid ownership models
- guarantee equal distribution of value or influence
- remove the need for legal structures and compliance

DP20 defines the conditions under which ownership claims are legitimate and enforceable. It does not prescribe a single model.

10. Minimum Alignment (Non-Normative)

Minimum alignment defines the threshold where ownership is **real, enforceable, and not misleading**.

A DP20-aligned system **MUST**:

- bind ownership to executable governance and economic rights
- provide verifiable ownership records and history (DP15)
- ensure visibility into value flows and surplus allocation (DP17)
- enable credible exit and fork with continuity (DP4, DP7)
- maintain enforceable stewardship obligations

Failure modes to avoid:

- **ownership theater** (claims without control)
- **fork suppression** (no exit pathway)
- **surplus extraction without rights**
- **historical erasure of ownership changes**

Systems that omit execution, auditability, or exit **MUST NOT** be considered aligned.

11. Open Questions and Future Work

Key open questions include:

- how to design ownership models that function across jurisdictions
- how to balance tokenized and non-tokenized ownership structures
- how to maintain privacy while supporting interoperable ownership credentials (DP4)
- how to protect minority voices without blocking collective action
- how to measure contribution across visible and invisible labor (care, moderation, coordination)
- how to handle liability for collective decisions across legal systems
- how AI agents participate in ownership structures, if at all (DP11–DP13)

These questions reflect the boundary between social legitimacy and technical implementation.

12. Relationship to Other Desirable Properties

DP20 anchors ownership within the broader meta-layer system.

- DP3 defines how governance evolves under ownership
- DP4 constrains how collectively generated data can be used
- DP5–DP7 enable portability of identity, assets, and ownership artifacts
- DP6 and DP9 determine how value flows and incentives interact with ownership
- DP12 ensures ownership-linked governance can execute in practice
- DP15 ensures ownership claims and actions are provable
- DP17 ensures ownership structures can sustain themselves financially
- DP18–DP19 provide the participation and reputation inputs that mature into ownership

DP20 binds these properties into a coherent model of collective power.

13. Foresight and Failure Design

DP20 assumes ownership systems will be tested by capture attempts, economic pressure, and governance failure.

Common failure paths include:

- concentration of ownership among early or privileged actors
- erosion of rights through informal overrides or hidden control layers
- suppression of exit or fork pathways to preserve centralized power
- divergence between formal ownership and actual value capture

DP20 requires designing safeguards in advance, including:

- transparent allocation and reallocation mechanisms
- enforceable limits on concentrated control
- clear and executable fork and exit procedures

- public postmortems linking failures to structural changes

Failure is expected. Illegible failure is not.

14. Path Toward ML-RFC

Advancing DP20 toward ML-RFC requires:

- standardizing ownership object and receipt formats
- developing reference implementations of community-owned systems with open accounting
- aligning ownership models with interoperability and identity standards
- testing fork and exit mechanisms in real communities
- integrating legal and cooperative expertise into design processes

Progress should be demonstrated through functioning systems, not only conceptual agreement.

15. Closing Orientation

DP20 is where the meta-layer turns participation into power.

It rejects the model where communities create value but do not control it.

When ownership is real, communities can govern, sustain, and evolve the systems they depend on.

When it is not, “community” remains a narrative layered over extraction.

Ownership is the difference between presence and permanence, between contribution and control.

DP20 is where the meta-layer stops confusing audience with citizenship.

Ownership is the difference between a group that is used and a group that endures, with memory, upside, and responsibility held together.

When DP20 is strong, people can build commons that last because the commons can belong to those who care for them.

DP21 – Multi-Modal Interactions and Experiences

1. Purpose of This Draft

This draft articulates Desirable Property 21 (DP21) as the condition under which the meta-layer can be experienced across multiple sensory modalities, devices, and interaction paradigms without fragmenting meaning, agency, trust, or governance.

DP21 defines how the Metaweb supports visual, auditory, textual, spatial, tactile, and emerging modalities so participants can engage in ways that are natural, accessible, and context-appropriate.

The meta-layer must not be confined to screens or text interfaces. It must function as a cross-modal civic interface that can appear in browsers, voice systems, AR/VR environments, ambient computing contexts, and future interaction substrates.

If DP21 is weak, predictable failures follow: the meta-layer becomes screen-bound; accessibility gaps widen; immersive systems become siloed; voice and AI interfaces lose context; meaning degrades across modalities; and participation becomes limited to technically privileged users.

DP21 connects directly to:

- DP2, participant agency and empowerment
- DP4, data sovereignty and privacy
- DP7, interoperability
- DP8, community-defined participation zones
- DP10, education and onboarding
- DP12 and DP13, AI governance and containment
- DP14, trust and transparency
- DP17, knowledge representation and semantic layering
- DP18, feedback loops and reputation
- DP19, amplifying presence and engagement

DP21 does not prescribe specific hardware, platforms, or interaction styles. It defines the minimum conditions under which multi-modal participation remains coherent, accessible, and governed.

2. Problem Statement

The current web is primarily visual and text-based, with fragmented support for cross-modal continuity.

Meanwhile, interaction surfaces are rapidly diversifying:

- voice assistants
- mobile-first interfaces
- augmented and virtual reality

- spatial computing
- wearable devices
- ambient and IoT environments
- AI-mediated conversational interfaces

These systems often operate in isolation, creating fractured user experiences and inconsistent trust surfaces.

Examples of failure:

- A voice assistant cannot access or interpret trust overlays visible in a browser
- AR annotations misalign with textual context or policy zones
- Accessibility tools receive degraded or partial semantic information
- AI agents lose continuity when switching modalities

DP21 reframes modality as a first-class design concern. The meta-layer must ensure that meaning, trust signals, governance rules, and participation pathways persist across modes.

A healthy DP21 implementation must answer:

- Can participants move between modalities without losing context?
- Are trust signals preserved across visual, auditory, and spatial representations?
- Can accessibility tools fully interpret meta-layer structures?
- Do immersive environments respect governance and consent rules?
- Can AI agents operate consistently across modalities?

Without DP21, the meta-layer becomes fragmented, exclusionary, and unreliable.

3. Threats and Failure Modes

DP21 is not simply about supporting more devices. Multi-modal systems introduce new ways for meaning, trust, consent, and agency to fracture. A visual interface can show nuance through layout, color, hover states, and layered context. A voice interface may compress all of that into one sentence. An AR interface may place a signal in physical space, changing how the participant experiences authority, urgency, and social pressure.

For this reason, DP21 treats modality as a governance surface. Every new mode of interaction changes what participants can perceive, how they can respond, what data is collected, and how power is experienced.

3.1 Modality silos

Different interface modes may evolve as separate ecosystems, each with its own data model, permissions, affordances, and governance assumptions.

Example: A browser overlay supports community trust annotations, while an AR headset displays only platform-approved spatial labels. A participant moving between the two environments sees different versions of reality without knowing which guarantees changed.

Why this matters: If modalities become silos, the meta-layer loses its role as shared context. Trust becomes dependent on the device or vendor rather than the underlying civic substrate.

3.2 Accessibility degradation

Multi-modal systems can appear inclusive while quietly degrading experiences for people who rely on assistive technologies.

Example: A visual overlay includes provenance badges, dispute status, and community annotations, but a screen reader receives only the page text. The participant hears the claim but not the trust context that other participants see.

Why this matters: Accessibility is not equivalent to basic access. DP21 requires that meaning, trust, and agency survive translation into accessible forms.

3.3 Context fragmentation

The same object, claim, participant, or governance action may be represented differently across modalities.

Example: A controversial claim has a detailed contextual overlay in text, a short warning in voice mode, and a red spatial marker in AR. Each representation carries a different emotional weight and interpretive frame.

Why this matters: Context fragmentation can change what participants believe the system is saying. If each modality tells a slightly different story, governance becomes unstable.

3.4 Immersive capture

AR, VR, and spatial computing environments can become proprietary civic spaces controlled by headset vendors or platform operators.

Example: A city deploys public-service overlays in a vendor-controlled AR environment, but residents can only access civic annotations through approved devices and approved identity systems.

Why this matters: The meta-layer must not trade the flat-web platform problem for a spatial-platform problem. Immersive systems need open, governable, interoperable trust surfaces.

3.5 Voice flattening

Voice interfaces are powerful because they feel natural, but they are also compressive. They often force complex information into linear, short-form summaries.

Example: A voice assistant says, “This source is considered unreliable,” without explaining by whom, according to what rubric, with what confidence level, and whether the rating is disputed.

Why this matters: Trust signals must not become authoritarian declarations when translated into voice. DP21 requires voice interfaces to preserve uncertainty, source, scope, and recourse.

3.6 Sensor overreach

Multi-modal systems often rely on cameras, microphones, location, motion sensors, biometric signals, gaze tracking, and environmental scanning.

Example: An immersive overlay requests room mapping, eye tracking, voice input, and location data to provide contextual interaction, then reuses those signals for personalization or reputation inference.

Why this matters: Multi-modal richness can become surveillance richness. DP21 must be bound tightly to DP4 data sovereignty and privacy.

3.7 AI modality drift

AI agents may behave differently depending on whether they are speaking, writing, acting in AR, or mediating a tactile or ambient interface.

Example: A text agent carefully cites uncertainty, but the same agent in voice mode speaks with confidence and omits caveats to sound more natural.

Why this matters: Participants should not receive weaker governance simply because an interaction feels conversational or immersive. Agent behavior must remain policy-bound across modalities.

3.8 Interaction inequity

Richer devices can create richer participation opportunities. Participants with headsets, high-end phones, or always-on assistants may gain more visibility, faster response, or stronger coordination capacity.

Example: A governance meeting includes spatial collaboration tools that allow headset users to annotate a shared model while phone-only participants can only watch.

Why this matters: Multi-modal capability must not become a new class divide. DP21 requires parity of agency, even when interface richness differs.

3.9 Representation mismatch

Different modalities may represent the same signal with different emotional or social force.

Example: A low-confidence caution shown as a small visual icon becomes a loud audio interruption in voice mode or a red warning halo in AR.

Why this matters: Representation affects behavior. Systems must map severity, confidence, and urgency carefully across modalities.

3.10 Latency fragmentation

Real-time multi-modal systems can desynchronize. A participant may see one state, hear another, and act on outdated information.

Example: During a live civic deliberation, a visual overlay updates vote status immediately, while the voice interface reads an earlier summary. Participants make decisions from inconsistent system states.

Why this matters: Trust depends on temporal coherence. Systems must signal freshness, delay, and synchronization status.

3.11 Cognitive overload

Multi-modal systems can overwhelm participants by presenting too many signals at once.

Example: A participant receives a visual warning, audio cue, haptic pulse, AI suggestion, social annotation, and governance prompt simultaneously.

Why this matters: More modalities do not automatically create more clarity. DP21 requires calm, layered, participant-directed interaction design.

3.12 Ambient manipulation

Ambient interfaces can influence participants without being consciously noticed.

Example: A room-scale system subtly changes lighting, sound, or notification patterns to encourage agreement, urgency, purchase, or disengagement.

Why this matters: Ambient interaction can bypass reflective agency. DP21 must ensure that ambient signals remain legible, consented, and interruptible.

3.13 Physical-world safety risks

Spatial and tactile interfaces can affect movement, attention, and physical safety.

Example: An AR overlay appears while someone is crossing a street, or a haptic notification distracts a worker using dangerous equipment.

Why this matters: Multi-modal governance must include situational safety, not only information integrity.

4. Core Principle

The meta-layer must support multi-modal interactions such that meaning, trust, governance, and participation remain coherent and accessible across all forms of interaction.

Modality should expand agency, not fragment it.

A participant should be able to move from a browser to a phone, from text to voice, from a screen to an AR layer, or from an immersive environment back to a simple assistive interface without losing the essential structure of the interaction. The presentation may change, but the underlying meaning, permissions, trust signals, and rights should remain intact.

Example: A participant encounters a health claim online. In a visual browser overlay, they see provenance, community annotations, and expert citations. In voice mode, they hear a concise but scoped summary: who disputes the claim, what confidence level applies, and how to request more detail. In AR, the claim is marked with a contextual indicator that opens the same evidence graph. Each modality feels natural, but all point back to the same governed semantic object.

What this feels like: The system adapts to the participant and context without changing the truth conditions of the interaction.

Without this: Each interface becomes a separate reality.

5. Primary Mechanisms and Structural Conditions

5.0 Multi-Modal Interface Layer: Meaning Before Presentation

DP21 requires a layer that separates meaning from presentation.

The system must first know what something means before deciding how to show, speak, vibrate, spatialize, or summarize it. A trust signal, governance rule, consent boundary, identity claim, or reputation state should exist as a structured object before it is rendered into any particular modality.

This layer includes:

- modality-agnostic semantic representations
- modality adapters
- cross-device state synchronization
- accessibility interfaces
- context anchoring
- privacy and sensor governance
- participant preference models
- graceful degradation rules

The goal is not to make every modality identical. The goal is to make every modality faithful.

Failure mode: **presentation-driven divergence**, where the interface invents meaning rather than rendering governed meaning.

5.1 Multi-modal device support

The meta-layer should support a broad range of devices, including:

- mobile phones
- desktop and laptop browsers
- tablets

- voice-only systems
- AR and VR headsets
- wearables
- ambient devices
- public kiosks
- assistive devices
- future spatial or embodied computing systems

Device support must be capability-aware. A phone, headset, screen reader, and voice-only device cannot provide identical experiences, but each should preserve core agency.

A low-capability device should still allow participants to:

- understand relevant trust signals
- control consent
- participate in governance where appropriate
- receive feedback and recourse
- know what is missing or degraded

Failure mode: **device privilege**, where only participants with high-end devices can fully participate.

5.2 Multi-modal interface support

The meta-layer should support interface modes such as:

- visual overlays
- text panels
- audio narration
- voice dialogue
- haptic feedback
- spatial annotations
- gestural interaction
- conversational AI
- immersive environments
- ambient cues

Each interface mode has strengths and risks. Visual systems can carry density but may exclude non-visual participants. Voice systems feel natural but can flatten complexity. Haptics can signal urgency but can also become intrusive. Spatial overlays can make context embodied but may create coercive presence.

DP21 requires every interface mode to be evaluated not only for usability, but for its effect on agency, comprehension, privacy, and governance.

Failure mode: **mode enthusiasm**, where novelty drives adoption before governance catches up.

5.3 Semantic continuity

All modalities must derive from shared semantic structures.

A participant, claim, annotation, trust signal, governance rule, consent object, reputation state, or AI action should have a stable underlying reference independent of how it is presented.

For example, a provenance warning should not be three unrelated artifacts across visual, voice, and AR interfaces. It should be one semantic object with multiple renderings.

A semantic object **SHOULD** include:

- identifier
- object type
- scope
- provenance
- confidence level
- governance zone
- permissions
- freshness
- dispute status
- modality rendering rules

Failure mode: **duplicated interpretation**, where each modality generates its own version of the truth.

5.4 Accessibility-first design

Accessibility must be foundational.

DP21 requires that multi-modal interaction expand participation for people with visual, auditory, motor, cognitive, sensory, linguistic, or situational differences.

Accessibility support **SHOULD** include:

- screen reader compatibility
- voice navigation
- captions and transcripts
- audio descriptions
- haptic alternatives
- reduced motion modes
- cognitive simplification modes
- plain-language summaries
- alternative input methods
- keyboard-only navigation
- localization and translation

- low-bandwidth modes

Accessibility is not satisfied when content is technically reachable. Participants must be able to understand, decide, respond, and appeal.

Failure mode: **assistive afterthought**, where accessibility tools receive the surface content but not the governance context.

5.5 Cross-modal synchronization

State and context must persist across modalities.

If a participant starts an interaction in one mode and continues in another, the system should preserve:

- current object or claim
- governance zone
- consent state
- trust signal state
- identity context
- unresolved prompts
- participant preferences
- relevant history
- freshness markers

Example: A participant begins reviewing a civic proposal on desktop, continues by voice while commuting, and later joins an AR town hall. The system should maintain continuity without leaking private context or assuming consent across settings.

Failure mode: **context reset**, where each modality treats the participant as starting over.

5.6 Privacy and sensor governance

Multi-modal systems often require richer data inputs. DP21 therefore requires stricter governance around sensors and environmental awareness.

Sensitive inputs may include:

- microphone data
- camera data
- location
- gaze tracking
- gestures
- biometrics
- proximity
- room geometry

- emotional inference
- movement patterns
- ambient sound
- device graph signals

Systems **MUST** define:

- what is collected
- why it is collected
- where it is processed
- how long it is retained
- whether it is shared
- whether it affects reputation, recommendations, or governance
- how consent can be revoked

Where possible, processing should occur locally or in privacy-preserving environments.

Failure mode: **surveillance through richness**, where better interaction becomes the justification for excessive capture.

5.7 AI mediation across modalities

AI can help translate, summarize, narrate, adapt, and synchronize across modalities. This makes AI useful for DP21, but also risky.

AI mediation **SHOULD** preserve:

- meaning
- source attribution
- uncertainty
- scope
- governance constraints
- participant preferences
- consent boundaries
- accessibility needs

Example: An AI assistant converting a dense trust overlay into voice should not simply say, “This is false.” It should explain the relevant confidence, source, dispute status, and option to hear more.

AI modality adapters should be auditable. Communities must be able to inspect whether AI summaries systematically omit minority perspectives, soften warnings, exaggerate certainty, or change policy meaning.

Failure mode: **translation as governance**, where AI adaptation quietly decides what participants are allowed to perceive.

5.8 Spatial and immersive overlays

AR and VR shift the meta-layer from a screen-adjacent experience into an embodied environment. This changes the stakes.

Spatial overlays can attach meaning to:

- physical locations
- objects
- people
- events
- public infrastructure
- artworks
- classrooms
- workplaces
- civic spaces
- virtual worlds

These overlays must respect governance zones, consent, safety, and contextual integrity.

Example: A public monument may have multiple community annotations, historical context layers, accessibility routes, and civic dialogue spaces. Participants should know which layer they are seeing, who governs it, and how to contest or add context.

Failure mode: **unaccountable spatial authority**, where whoever controls the overlay controls the perceived meaning of the physical world.

5.9 Graceful degradation

When full modality support is unavailable, systems must degrade gracefully.

Graceful degradation means participants are told what is missing and given a meaningful alternative.

Examples:

- If AR is unavailable, spatial annotations can be shown as a map or list
- If audio is unavailable, voice content appears as captions or transcript
- If haptics are unavailable, urgency is represented visually or textually
- If real-time sync fails, stale data is labeled as stale

Failure mode: **silent degradation**, where participants assume they have the full context but receive only a partial version.

5.10 Modality equivalence mapping

Important signals must have equivalent representations across modalities.

A high-risk warning may appear as:

- a visual icon and text label
- a spoken alert with scope and severity
- a haptic pulse pattern
- a spatial boundary marker
- a simplified accessibility summary

Equivalence does not mean sameness. It means each rendering preserves the function of the signal.

A modality equivalence map SHOULD define:

- signal type
- severity
- urgency
- confidence
- required participant action
- visual rendering
- audio rendering
- haptic rendering
- spatial rendering
- accessibility rendering

Failure mode: **missing signal class**, where a participant in one modality never receives a critical warning.

5.11 Context anchoring

Every multi-modal representation must remain anchored to shared identifiers.

Context anchors may include:

- URLs
- content hashes
- spatial coordinates
- object identifiers
- semantic graph nodes
- participant identifiers
- zone identifiers
- provenance records
- event identifiers

Anchoring allows different modalities to point to the same thing.

Example: A visual annotation on a paragraph, a voice summary of that paragraph, and an AR representation in a classroom discussion should all refer to the same underlying claim object.

Failure mode: **reference drift**, where participants believe they are discussing the same object but each modality has attached context to a different target.

5.12 Multi-modal feedback loops

Participants must be able to give feedback through multiple modalities.

Feedback may be:

- spoken
- typed
- gestural
- selected from structured prompts
- submitted through assistive devices
- attached to spatial objects
- recorded as annotations
- routed through AI-assisted summaries

Multi-modal feedback must still produce structured feedback objects under DP18. A spoken concern, a tactile accessibility report, or a spatial annotation should not disappear into unstructured logs.

Failure mode: **feedback inequality**, where only participants using dominant interfaces can meaningfully shape the system.

5.13 Participant preference and control profiles

Participants should be able to define how they want the meta-layer to communicate with them.

Preference profiles MAY include:

- preferred modality
- reduced stimulation settings
- language
- accessibility needs
- notification style
- verbosity
- privacy thresholds
- AI mediation preferences
- sensory sensitivity settings
- default degradation behavior

These profiles must be portable where appropriate and governed by DP4 privacy constraints.

Failure mode: **interface paternalism**, where the system decides the best mode for the participant without meaningful control.

5.14 Situational awareness and safety modes

The appropriate modality may depend on context.

A participant walking, driving, working, resting, attending a meeting, or entering a sensitive civic space may need different interaction boundaries.

Systems SHOULD support situational safety modes such as:

- do-not-interrupt modes
- low-attention summaries
- delayed notifications
- physical hazard suppression
- private-context suppression
- child-safe modes
- meeting-safe modes
- public-display protections

Failure mode: **context-insensitive interaction**, where the system provides the right information at the wrong time or in the wrong sensory form.

6. Multi-Modal Intelligence and Data Fusion

DP21 supports the fusion of diverse data types, but only when fusion increases understanding without eroding governance.

Multi-modal intelligence may involve:

- text
- image
- video
- audio
- spatial data
- sensor data
- maps
- conceptual graphs
- knowledge graphs
- biometric or embodied signals where explicitly governed
- environmental context

The purpose of multi-modal intelligence is not to collect everything. It is to combine relevant signals into a more coherent understanding of a situation.

Example: During a disaster response, the meta-layer could combine public reports, satellite images, local sensor data, official advisories, and community annotations into a shared situational map. Participants using phones, voice assistants, or public kiosks could access the same underlying context in different forms.

DP21 requires that data fusion preserve:

- provenance
- uncertainty
- consent boundaries
- jurisdictional constraints
- community governance rules
- source diversity
- time sensitivity
- contestability

Failure mode: **data chaos**, where many signals are fused without lineage, confidence, or governance.

Failure mode: **fusion authoritarianism**, where the system produces a single authoritative view from contested inputs without preserving dissent or uncertainty.

7. Governance Requirements

Multi-modal systems are governance systems because they shape perception, attention, and action.

Communities must be able to define:

- which modalities are allowed in a zone
- which sensors may be used
- what data may be collected
- how immersive overlays are labeled
- how spatial annotations are moderated
- what accessibility standards apply
- how AI may summarize or translate
- what signals require human review
- when ambient cues are prohibited
- how physical-world safety is protected

Governance artifacts SHOULD specify modality-specific rules.

Example: A school zone may allow text and visual overlays, restrict biometric sensing, require captions for all audio, prohibit persistent student tracking, and require teacher or community approval for AR layers.

Failure mode: **ungoverned modality expansion**, where new interface capabilities become active before the community has defined rules for them.

8. Evaluation Criteria

A DP21-aligned implementation should be evaluated against the following questions.

8.1 Continuity

- Does the same object retain identity across modalities?
- Are trust signals, consent states, and governance zones preserved?
- Can participants move across devices without losing essential context?

8.2 Accessibility

- Can participants with different abilities access the same meaning and agency?
- Are accessibility tools receiving structured trust and governance information?
- Are cognitive, sensory, motor, and linguistic needs considered?

8.3 Trust fidelity

- Do all modalities preserve provenance, uncertainty, confidence, and dispute status?
- Are warnings and endorsements represented with appropriate severity?
- Are simplified summaries faithful to the underlying record?

8.4 Privacy and sensor limits

- Are sensors governed explicitly?
- Is data minimization enforced?
- Can participants revoke modality-specific permissions?
- Are local or privacy-preserving processing options used where possible?

8.5 Synchronization and freshness

- Are real-time states synchronized?
- Is stale or delayed information labeled?
- Can participants tell whether different modalities are showing the same system state?

8.6 Participant control

- Can participants choose preferred modalities?
- Can they reduce stimulation or disable intrusive cues?
- Can they control AI mediation and summarization?

8.7 Governance fit

- Are modality rules defined per zone?
 - Are immersive and ambient interfaces subject to community policy?
 - Are physical safety contexts handled responsibly?
-

9. Implementation Patterns

Implementation patterns translate DP21 from principle into practice. These are not rigid prescriptions, but recurring architectural and design approaches that help systems maintain semantic continuity, accessibility, and governance integrity across modalities. They are intended to guide builders in making consistent decisions when adapting the meta-layer to new devices, senses, and interaction environments.

9.1 Semantic-first architecture

Build semantic objects before rendering interfaces. Presentation layers should adapt governed meaning, not invent it.

9.2 Modality adapters

Create adapters for visual, voice, haptic, spatial, and assistive interfaces that consume the same underlying semantic objects.

9.3 Accessibility testing pipelines

Every core trust, consent, governance, and reputation signal should be tested across assistive modalities.

9.4 Context synchronization engines

State should travel across devices and modalities through privacy-preserving synchronization.

9.5 Sensor permission manifests

Multi-modal applications should publish clear manifests describing sensor access, purpose, retention, and sharing.

9.6 Degradation notices

When a modality cannot show full context, the interface should say what is missing and offer alternatives.

9.7 Voice uncertainty patterns

Voice systems should use standard language for uncertainty, provenance, dispute status, and request-for-detail prompts.

9.8 Spatial governance labels

AR and VR overlays should display or make available the governing zone, source, scope, and contestation path for spatial annotations.

9.9 Participant sensory profiles

Participants should be able to store preferences for stimulation level, modality, language, accessibility, and AI assistance.

9.10 Multi-modal feedback receipts

Feedback submitted by voice, gesture, text, or spatial annotation should generate DP18-compatible receipts.

10. Relationship to Other Desirable Properties

DP2 – Participant Agency and Empowerment

DP21 expands agency by letting participants choose how they interact. It violates DP2 when modality choices become coercive, inaccessible, or locked to specific vendors.

DP4 – Data Sovereignty and Privacy

Multi-modal systems create richer data flows. DP21 depends on DP4 to prevent sensors, biometrics, and environmental context from becoming surveillance infrastructure.

DP7 – Interoperability

DP21 is an interoperability problem across senses, devices, and environments. Meaning must move across modalities without collapsing.

DP8 – Community-Defined Participation Zones

Communities must govern which modalities are appropriate in each zone. A meditation group, classroom, emergency response zone, and public debate space require different modality norms.

DP10 – Education and Onboarding

Multi-modal onboarding can meet participants through voice, visuals, interactive demos, spatial practice, and assistive formats.

DP12 and DP13 – AI Governance and Containment

AI agents must remain governed across text, voice, spatial, embodied, and ambient interactions. Containment cannot disappear when the interface changes.

DP14 – Trust and Transparency

Transparency must be perceivable across modalities. A trust signal that only works visually is not transparent to everyone.

DP17 – Knowledge Representation and Semantic Layering

DP21 depends on DP17. Multi-modal presentation is only coherent when grounded in shared semantic structures.

DP18 – Feedback Loops and Reputation

Participants must be able to provide feedback and receive reputation signals across modalities without losing structure, provenance, or recourse.

DP19 – Amplifying Presence and Community Engagement

Multi-modal experiences make the meta-layer more present, memorable, and culturally accessible, especially through education, public installations, youth engagement, and immersive storytelling.

11. Open Questions for ML-RFC Development

1. What minimum semantic object model enables modality-independent rendering?
 2. How should trust signals map across visual, audio, haptic, spatial, and assistive forms?
 3. What is the standard way to express uncertainty in voice interfaces?
 4. What sensor permissions should be mandatory to disclose?
 5. How should AR overlays represent governance zone, source, and dispute status?
 6. What accessibility conformance standards should apply to meta-layer overlays?
 7. How should multi-modal feedback objects be structured?
 8. How should participant sensory preferences be stored and ported?
 9. What are safe defaults for ambient interfaces?
 10. How should systems indicate stale or desynchronized modality states?
 11. How should AI summaries be audited for modality-specific distortion?
 12. What physical-world safety constraints should apply to AR and haptic interactions?
 13. How should multi-modal systems operate in low-bandwidth or crisis conditions?
 14. What rights should participants have when modality-specific limitations affect governance participation?
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12. Path Toward ML-RFC

DP21 is currently an ML-Draft and serves as exploratory scaffolding for how multi-modal interaction becomes part of meta-layer infrastructure.

Advancement toward ML-RFC status SHOULD require:

- a minimal semantic object model for modality-independent rendering
- a modality equivalence mapping standard
- accessibility validation requirements
- sensor permission and privacy manifest standards
- spatial overlay governance labels
- voice uncertainty and provenance patterns
- cross-device synchronization requirements
- graceful degradation requirements
- multi-modal feedback receipt formats
- pilot implementations across at least three modalities

ML-RFC promotion SHOULD be contingent on:

- rough consensus among participating communities
- demonstrated accessibility across multiple participant needs
- alignment with DP4 privacy requirements
- alignment with DP7 interoperability requirements
- alignment with DP17 semantic representation requirements
- evidence that modality support expands agency rather than creating new inequities

Early ML-RFC candidates may focus on:

- Multi-Modal Semantic Object Standard
- Modality Equivalence Map
- Sensor Permission Manifest
- Spatial Trust Signal Encoding
- Voice Trust and Uncertainty Pattern Library
- Accessibility Requirements for Meta-Layer Overlays

DP21 will likely mature through several component RFCs rather than one monolithic standard.

13. Closing Orientation

DP21 ensures the meta-layer is not bound to a single interface, device, or sense.

It allows people to experience trust, context, and participation in ways that match their abilities, environments, preferences, and tools. It makes the meta-layer usable by more people, in more places, under more conditions, without sacrificing coherence or governance.

A DP21-aligned meta-layer does not treat voice, AR, haptics, accessibility tools, and immersive systems as side channels. It treats them as legitimate civic interfaces.

Interaction becomes flexible. Meaning becomes portable. Trust becomes perceivable. Presence becomes multi-dimensional.

This is how the meta-layer becomes ambient, inclusive, and truly interoperable.

DP22 – Civic Memory & Sensemaking Continuity

1. Purpose of This Draft

This draft articulates Desirable Property 22 (DP22) as the condition under which the Meta-Layer preserves civic memory and collective sensemaking through transformation without turning memory into surveillance, summary into authority, plurality into noise, or history into platform-controlled narrative.

DP22 begins from a distinction that is easy to overlook:

Information persistence is not civic memory.

A society may preserve files, posts, inscriptions, records, archives, governance artifacts, media objects, and AI summaries while still losing the pathways by which civic meaning was formed. What disappears is not always the data itself, but the relationships between artifact, evidence, interpretation, disagreement, authority, governance, and later feedback.

DP22 addresses that missing continuity.

The Meta-Layer should preserve not only civic artifacts, but the transformation paths through which civic understanding evolves. Participants and communities should be able to reconstruct how a claim became evidence, how evidence became interpretation, how interpretation became synthesis, how synthesis informed governance, and how governance later changed in response to feedback.

This property becomes especially important in AI-mediated environments. AI systems increasingly summarize, rank, retrieve, cluster, recommend, synthesize, compress, and reinterpret civic knowledge. These acts are not neutral compression. They shape what future participants believe the past meant.

DP22 therefore treats summarization, retrieval, indexing, ranking, transformation, and contextualization as civic memory operations requiring accountability.

DP22 does not establish a single truth authority. It defines the conditions under which plural sensemaking can remain traceable, navigable, contestable, and accountable across time.

2. Problem Statement

The modern internet does not simply forget. It remembers badly.

It preserves fragments while dissolving the structures that make those fragments meaningful. A quote survives without context. A screenshot survives without provenance. A governance decision survives without dissent. A summary survives without omissions. A digital artifact survives without the civic role it later came to play.

The result is discontinuous civic understanding.

Communities repeatedly restart the same debates because prior reasoning is inaccessible or non-navigable. Institutions preserve outputs while losing the conditions that justified them. AI systems increasingly compress contested fields into apparently coherent summaries. Search and ranking systems quietly determine what becomes socially remembered by determining what is easiest to retrieve.

The consequence is civilizational learning failure.

DP22 responds by treating memory as relational, governed, contextual, transformation-sensitive, retrieval-sensitive, and socially navigable. The core questions are therefore not only “What should persist?” but also: What context must survive for an artifact to remain interpretable? What transformation history must remain visible for a summary to be trusted? What disagreement must remain visible for plural sensemaking to survive? What should be attenuated, sealed, or forgotten to prevent memory from becoming punishment? Who governs retrieval, ranking, contextual overlays, and interpretive authority?

Without answers to these questions, civic memory becomes either too weak to support accountability, or too rigid and totalizing to support humane civilization.

3. Threats and Failure Modes

3.1 Memory laundering

Transformation chains erase inconvenient origins while preserving conclusions.

Example: AI systems summarize prior summaries until provenance, dissent, and uncertainty disappear, while the final synthesis appears authoritative.

Why this matters: Interpretive authority detaches from reconstructable evidence.

3.2 Context collapse through compression

Compression removes uncertainty, dissent, historical nuance, or scope conditions.

Example: A contested governance conflict later appears as a clean consensus because summaries preserved the decision but not the disagreement.

Why this matters: Communities inherit false institutional memory.

3.3 Retrieval capture

Ranking and retrieval systems silently determine what becomes practically rememberable.

Example: Commercial optimization consistently surfaces emotionally viral interpretations over historically grounded ones.

Why this matters: Retrieval intermediaries become practical memory authorities.

3.4 Narrative enclosure

Platforms, states, institutions, or dominant intermediaries monopolize authoritative interpretive layers.

Example: Only institutionally approved overlays appear above historical artifacts, while alternate contextualizations become hidden or difficult to discover.

Why this matters: Plural civic memory collapses into managed narrative.

3.5 Synthetic consensus generation

AI systems create the appearance of agreement through scale, repetition, or automated interpretation.

Example: Thousands of AI-generated contextual notes flood a contested event, making one interpretation appear socially settled.

Why this matters: Plurality becomes statistically drowned rather than deliberatively resolved.

3.6 Epistemic spam flooding

Low-cost synthetic interpretation overwhelms navigability.

Example: Every civic artifact accumulates large volumes of low-value commentary, automated objections, and derivative summaries.

Why this matters: Signal collapses into interpretive exhaustion.

3.7 Governance lineage collapse

Decisions survive without reconstructable rationale.

Example: A policy remains active, but later participants cannot reconstruct the evidence, dissent, amendments, or procedural choices that produced it.

Why this matters: Governance becomes ahistorical and difficult to contest responsibly.

3.8 Memory permanence as punishment

Persistent visibility eliminates humane contextual decay.

Example: Minor historical actions remain permanently surfaced without proportionality, changed context, or restorative pathways.

Why this matters: Civic memory becomes socially carceral.

3.9 Fork invisibility

Communities diverge interpretively without visible lineage.

Example: A civic archive forks into multiple historical accounts, but participants cannot see where the interpretations diverged or why.

Why this matters: Disagreement becomes opaque rather than intelligible.

3.10 Authenticity without interpretive accountability

An artifact is provably authentic but contextually misleading.

Example: A genuine video clip is surfaced without the surrounding sequence, later correction, or contradictory evidence.

Why this matters: Authenticity alone does not preserve civic meaning.

4. Core Principle

Civic memory within the Meta-Layer is the accountable continuity of meaning across artifacts, evidence, interpretation, transformation, governance, and feedback.

The Meta-Layer should preserve enough structure for participants and communities to reconstruct how civic understanding evolved through time without forcing all communities into one canonical interpretation.

First, digital artifacts must remain interpretable as their civic role changes. An artifact may begin as expression, later become evidence, later become governance input, and later become historical memory.

Second, transformations must remain accountable. Summaries, translations, AI outputs, edits, governance reframings, and syntheses should preserve what changed, what was omitted, what uncertainty remains, and what authority the transformed object possesses.

Third, disagreement must remain structured rather than erased. Plural sensemaking requires visible disagreement, but also resilience against spam contradiction, manufactured uncertainty, and adversarial epistemic flooding.

Fourth, memory must remain governable. Communities require the ability to define what is preserved, surfaced, contextualized, attenuated, sealed, forgotten, exported, and forked.

Fifth, retrieval and navigation must be treated as civic power. What systems surface becomes what communities practically remember.

Anchor Principle: Memory is not storage. Memory is navigable continuity.

5. Civic Memory Layer: Continuity, Transformation, and Retrieval

DP22 requires a civic memory layer capable of preserving meaning across transformation. This layer is not merely an archive. It is the connective infrastructure that allows artifacts, interpretations, summaries, governance decisions, disputes, and later revisions to remain intelligible across time.

5.1 Artifact continuity

Civic memory begins with durable artifacts, but it cannot end there. Records, media objects, governance drafts, annotations, evidence collections, AI summaries, and contextual overlays require stable references, provenance continuity, and enough surrounding context to remain interpretable outside their original environment.

A failure mode is artifact orphaning, where an object survives but the context needed to understand its civic meaning disappears.

5.2 Transformation accountability

Civic memory must track how artifacts are transformed. Summaries, translations, edits, AI syntheses, educational adaptations, and governance reframings change what future participants encounter. These transformations should preserve visible lineage, meaningful omissions, uncertainty, source relationships, and authority boundaries.

A failure mode is semantic laundering, where meaning changes while the transformation path disappears.

5.3 Interpretive plurality

Civic memory must preserve disagreement without allowing disagreement to become unnavigable noise. Competing interpretations should remain visible with enough context to understand why they differ, what evidence they rely on, and where their lineage diverges.

A failure mode is interpretive monopoly, where one layer silently becomes canonical, or interpretive overload, where plurality becomes unusable.

5.4 Retrieval accountability

In information-saturated environments, retrieval determines practical memory. Search ordering, recommendations, ranking systems, context windows, and AI summaries shape what communities encounter and what they later treat as history.

DP22 therefore requires retrieval transparency and contestability. Participants should be able to understand why a memory object was surfaced, what assumptions shaped its ranking, and what alternatives were suppressed or deprioritized.

A failure mode is retrieval capture, where the power to surface becomes the power to remember.

5.5 Governance memory

Governance systems must preserve not only outcomes, but the pathways that produced them. Decisions require reconstructable rationale, amendments, dissent, unresolved tensions, and later reversals. Without this, communities cannot learn from their own history.

A failure mode is institutional amnesia, where governance artifacts remain but governance understanding disappears.

5.6 Humane forgetting and attenuation

Civic memory must support forms of contextual decay, proportional visibility, restorative forgetting, and sealed or bounded access. A society that remembers everything equally becomes socially unlivable, while a society that forgets without governance becomes manipulable.

A failure mode is permanent punitive memory, where historical visibility becomes totalizing and carceral.

6. Collective Intelligence and Civilizational Continuity

DP22 is ultimately concerned with more than archival integrity.

It concerns whether civilizations remain capable of learning across generations under conditions of AI-mediated interpretation.

Historically, societies preserved memory through libraries, oral traditions, institutions, monuments, archives, and governance records. These systems allowed civilizations to transmit partial continuity of understanding across time. But the internet transformed not only how information moves, but how meaning itself is encountered.

Increasingly, societies do not encounter history directly.

They encounter synthesized memory environments shaped by retrieval systems, AI summaries, contextual overlays, ranking systems, semantic clustering, recommendation systems, and interpretive compression.

This represents a profound civilizational transition.

AI systems increasingly mediate not only information access, but semantic continuity itself.

The core danger is therefore deeper than misinformation.

Civilizations may lose continuity of meaning while still preserving enormous quantities of information. Interpretation itself becomes unstable as ranking systems shift, summaries mutate, retrieval logic changes, and contextual environments reorganize collective understanding.

This creates the possibility of large-scale semantic drift: the gradual transformation of civic understanding through opaque interpretive infrastructure.

DP22 attempts to preserve the conditions under which societies remain intelligible to themselves across time.

Collective intelligence depends not only on communication, but on reconstructable continuity of reasoning across generations. Communities capable of preserving governance lineage, disagreement pathways, interpretive forks, contextual memory, and transformation history gain the ability to learn recursively rather than merely react episodically.

This is one of the deepest implications of the Meta-Layer.

The Meta-Layer does not merely preserve archives.

It introduces the possibility of living civic memory infrastructure.

Historically, memory was something civilizations visited through libraries, databases, records, and institutions.

Within the Meta-Layer, memory becomes ambient contextual infrastructure continuously surrounding interpretation itself.

Governance lineage can follow artifacts. Context can travel with claims. Interpretive forks can remain visible above the webpage. Transformation histories can remain queryable at runtime.

Memory shifts from destination into environment.

This represents a civilizational phase transition from static archives toward active contextual memory systems.

The challenge is not to eliminate disagreement.

The challenge is to preserve enough continuity for disagreement itself to remain intelligible.

7. Operational Conditions for Civic Memory

For DP22 to function meaningfully, civic memory cannot remain merely archival. It must operate as live infrastructure embedded within the runtime conditions of digital society.

This requires systems capable of preserving continuity while meaning is actively transformed.

A civic memory layer must therefore support:

- reconstructable transformation lineage
- contextual overlays tied to source provenance
- interpretable retrieval pathways
- contestable summarization systems
- durable semantic references
- memory portability across communities and platforms
- visible uncertainty signaling
- governance-aware contextualization

These conditions are not purely backend concerns.

They shape how participants encounter reality at the point of interaction.

As AI systems increasingly mediate interpretation, the operational conditions surrounding retrieval and contextualization become inseparable from civic governance itself. A memory system that cannot expose why a summary appeared, why a source was prioritized, or what interpretive assumptions shaped a synthesis gradually becomes opaque infrastructure governing historical understanding.

DP22 therefore requires that civic memory remain operationally legible.

Participants should be able to trace not only where information came from, but how it arrived before them, what transformations shaped it, and what alternate contextualizations remain available.

Without operational transparency, memory infrastructure silently becomes narrative infrastructure.

8. Governance Conditions for Civic Memory

Civic memory cannot remain healthy without governance.

Questions surrounding preservation, attenuation, retrieval weighting, contextual overlays, interpretive authority, and historical revision are not purely technical design questions. They are governance questions concerning legitimacy, accountability, and collective power.

The Meta-Layer therefore requires governance systems capable of mediating disputes surrounding:

- contextual framing
- retrieval prioritization
- memory attenuation
- historical reinterpretation
- provenance disputes
- interpretive forks
- AI-generated synthesis
- community annotation rights
- archival sealing pathways
- restorative visibility protocols

Different communities may legitimately adopt different memory norms.

Some communities may prioritize radical transparency and maximal preservation. Others may prioritize restorative pathways, bounded resurfacing, contextual decay, or differentiated visibility. DP22 does not impose a single universal memory regime.

Instead, it requires that memory governance remain visible, participatory, contestable, and interoperable.

The deeper challenge is that memory governance increasingly occurs through infrastructure decisions that appear politically neutral. Search ranking, contextual overlays, recommendation systems, semantic clustering, and AI summarization all shape collective memory while often avoiding democratic accountability.

DP22 therefore insists that retrieval and interpretation remain governable civic functions rather than invisible platform prerogatives.

A civilization that cannot govern how memory is surfaced eventually loses the ability to govern how reality is interpreted.

9. Civic Memory as Interface-Level Infrastructure

DP22 is not only about archives or backend repositories. It is also about what becomes visible at the point of interaction.

The Meta-Layer can make civic memory available above the webpage through overlays, contextual affordances, provenance displays, governance lineage views, and plural interpretation surfaces. These interface-level memory systems allow participants to encounter not just an artifact, but the civic context around it.

Example: A participant hovers over a policy document and sees prior versions, contested amendments, dissenting annotations, linked evidence, related governance forks, and AI-generated summaries marked with uncertainty and transformation lineage.

Example: A historical artifact displays multiple community interpretations rather than a single canonical framing, with visible provenance trees and unresolved disputes.

Example: An AI-generated summary exposes source relationships, omitted themes, compression choices, and confidence limits.

These are not decorative features. They are memory governance at the point of interpretation.

The interface becomes a place where civic memory is reconstructed, contested, and governed.

10. Relationship to Persistent Digital Object Systems

DP22 aligns with and extends prior work on persistent digital object systems, including Digital Object Architecture, the Handle System, FAIR Digital Objects, linked data systems, and persistent archival infrastructures.

These systems address important infrastructure problems related to persistent identifiers, durable addressability, interoperable references, provenance continuity, machine-readable metadata, and long-term digital preservation. Such systems may provide an important substrate for civic memory because collective sensemaking depends upon stable referential anchors.

Without persistent identity, provenance collapses, citations rot, interpretive lineage breaks, governance trails fragment, and memory becomes unstable across systems and time.

However, DP22 extends beyond artifact persistence alone.

Persistent object systems preserve continuity of digital artifacts. DP22 preserves continuity of civic meaning surrounding those artifacts.

Persistence alone does not create civic memory. Archives may survive while societies lose the ability to reconstruct how understanding evolved.

11. Relationship to Provenance and Authenticity Standards

DP22 may interoperate with provenance and authenticity standards such as C2PA, cryptographic provenance systems, content authenticity infrastructures, and transformation disclosure protocols.

These systems contribute important primitives for media authenticity, transformation accountability, AI transformation disclosure, provenance continuity, and resistance to semantic laundering.

However, provenance alone is insufficient for civic memory continuity.

Authenticity does not preserve disagreement, governance rationale, interpretive plurality, uncertainty, or collective sensemaking structure. A perfectly authentic artifact can still mislead, erase dissent, distort history, or produce false consensus.

DP22 therefore extends beyond authenticity into contextual continuity, interpretive accountability, plural sensemaking, governance lineage, retrieval accountability, and humane memory governance.

12. Relationship to Other Desirable Properties

DP22 depends upon and reinforces the broader Meta-Layer architecture.

DP1 supports accountable identity, authorship, and governance lineage. DP2 protects participant agency over contextual visibility and memory participation. DP4 prevents civic memory from collapsing into surveillance permanence. DP7 enables memory objects and interpretive context to move across systems without losing meaning. DP8 allows communities to govern zones of interpretation and memory. DP9 creates incentives for stewardship, contextualization, and curation. DP12 requires AI-mediated transformations to remain accountable. DP14 and DP15 support transparency, auditability, and dispute reconstruction. DP20 clarifies ownership and custodianship of civic artifacts.

DP22 is therefore a continuity property across the Meta-Layer stack. Without it, other properties become historically fragile.

13. Minimum Viable Civic Memory Layer

A minimal implementation of DP22 does not require full civilization-scale memory infrastructure.

Even an early Meta-Layer implementation could meaningfully improve civic continuity by supporting a small number of core capabilities:

- provenance-aware contextual overlays
- persistent civic object references
- visible transformation lineage for AI summaries
- governance lineage attached to policy artifacts
- plural annotation layers
- retrieval transparency indicators
- contextual uncertainty signaling

- attenuation controls for humane resurfacing

The importance of these systems is not merely technical.

They begin shifting the internet away from isolated content consumption toward reconstructable civic context.

Even modest continuity infrastructure can dramatically improve institutional learning, interpretive accountability, and long-horizon collective intelligence.

DP22 therefore does not require a complete reinvention of the internet before meaningful progress becomes possible.

It requires the gradual emergence of civic memory as a governable layer above the webpage.

14. Implementation Pathways

DP22 does not require a single centralized memory architecture.

Its principles may emerge through interoperable layers composed across communities, institutions, archives, governance systems, browsers, and civic overlays.

Possible implementation pathways include:

- contextual memory graphs linking artifacts, summaries, disputes, and governance outcomes
- provenance-aware overlays capable of displaying transformation lineage above webpages
- interoperable civic annotation systems
- retrieval transparency layers exposing ranking assumptions and contextual weighting
- semantic lineage systems preserving interpretive continuity across summaries and translations
- memory attenuation protocols supporting humane resurfacing governance
- portable contextual objects capable of moving across platforms without losing interpretive structure
- governance-aware AI summarization systems exposing uncertainty, omissions, and compression assumptions

The Meta-Layer may eventually support plural memory ecosystems in which communities maintain distinct interpretive environments while preserving enough interoperability for dialogue, reconstruction, and accountability across boundaries.

The goal is not universal historical consensus.

The goal is durable civilizational intelligibility.

15. Civic Memory and Civilizational Learning

Civilizations survive not only through infrastructure, economics, or military capacity.

They survive through continuity of understanding.

A society capable of reconstructing how decisions emerged can refine itself across generations. A society that loses interpretive continuity becomes trapped inside cycles of rediscovery, emotional reaction, institutional fragmentation, and historical repetition.

This challenge intensifies under AI mediation.

As artificial intelligence systems increasingly shape retrieval, summarization, contextualization, and synthesis, societies face the possibility that practical historical understanding becomes governed primarily through opaque compression systems. The danger is not only misinformation. It is the gradual replacement of reconstructable civic memory with optimized interpretive abstraction.

DP22 attempts to preserve the conditions under which collective intelligence remains possible.

Not collective agreement.

Collective intelligibility.

The Meta-Layer introduces the possibility that civic memory itself becomes participatory infrastructure rather than institutional residue. Communities may eventually inhabit memory environments where governance lineage, contextual disagreement, interpretive forks, provenance continuity, and transformation histories remain visible as living civic context.

In such environments, historical understanding becomes less dependent on centralized narrative authorities and more dependent on reconstructable civic pathways.

This does not eliminate conflict.

It makes conflict more intelligible.

It allows disagreement to accumulate context rather than merely accumulate heat.

Long-term democratic resilience may depend upon precisely this capability.

What communities can reconstruct determines what they can learn.

And what civilizations can learn determines whether they can adapt without collapse.

16. Tensions and Tradeoffs

DP22 does not eliminate the tensions between memory and forgetting, plurality and coherence, openness and manipulation, accountability and mercy, or continuity and adaptability.

These tensions are permanent features of civic life.

A civilization that preserves too little memory becomes manipulable and historically discontinuous.

A civilization that preserves too much memory without attenuation risks becoming socially frozen, punitive, and incapable of renewal.

Similarly, a society that suppresses disagreement loses adaptive intelligence, while a society that cannot maintain navigable continuity risks collapsing into interpretive fragmentation.

The purpose of the Meta-Layer is not to permanently resolve these tensions through a universal memory regime.

Its purpose is to make their negotiation visible, participatory, contextual, and governable.

DP22 therefore treats civic memory not as a solved state, but as an ongoing societal balancing process requiring continuous stewardship.

17. Without DP22

If DP22 is weak, predictable failures follow.

Societies lose institutional learning capacity. Governance becomes reactive and ahistorical. AI systems become primary authorities over practical memory. Retrieval intermediaries silently shape collective understanding. Public narratives become commercially optimized abstractions. Communities repeatedly relive unresolved conflicts because the reasoning that once surrounded them cannot be reconstructed.

The long-term danger is not merely misinformation.

It is civilizational discontinuity.

A society unable to reconstruct how understanding evolved loses the ability to govern transformation coherently across generations.

18. Open Questions and Future ML-RFC Pathways

Future ML-RFC development should explore concrete standards for civic memory objects, transformation disclosure, retrieval governance, interpretive lineage, humane forgetting, memory forks, AI summarization accountability, and interface-level memory overlays.

These standards should not freeze historical interpretation or produce a single authorized memory layer. Their purpose is to preserve the conditions under which societies can reconstruct meaning, preserve plurality, remain accountable, govern retrieval, learn across generations, and adapt without amnesia.

DP22 does not ask the Meta-Layer to remember everything.

It asks the Meta-Layer to preserve the civic capacity to understand how meaning became history.

Every civilization inherits not only resources, institutions, and technologies, but structures of memory.

What societies can reconstruct determines what they can learn.

What they can learn determines whether they can adapt.

And in an era where AI systems increasingly mediate interpretation itself, the governance of civic memory becomes inseparable from the governance of civilization.

The Meta-Layer should not merely preserve information.

It should preserve humanity's capacity to remain historically intelligible to itself.
